

Sectors

Technology

1. Software Development & Engineering

This is the core of building applications and systems.

- **Frontend Development:** Focuses on the user interface (UI) and user experience (UX) of a website or app (what the user sees and interacts with).
- **Backend Development:** Focuses on the server, database, and application logic that power the frontend (what the user *doesn't* see).
- **Full-Stack Development:** A hybrid role that handles both frontend and backend tasks.
- **Mobile Development:** Specializes in building applications for mobile devices (e.g., iOS Developer, Android Developer).
- **Game Development:** Focuses on building video games, including graphics, physics, and game logic.
- **Software Engineering in Test (SET) / QA Automation:** Writes code specifically to test other software and build automated testing systems.

2. Data & Artificial Intelligence

This category is focused on managing, interpreting, and leveraging data.

- **Data Analytics:** Focuses on interpreting historical data to find trends and answer business questions (What happened? Why?).
- **Data Science:** Uses statistics, machine learning, and programming to make predictions and build models (What *will* happen?).
- **Data Engineering:** Builds and maintains the "pipelines" and infrastructure that collect, store, and prepare large volumes of data for analysis.
- **AI / Machine Learning Engineering:** Specializes in designing and building complex artificial intelligence and predictive models.
- **Database Administration (DBA):** Manages, secures, and maintains the health and performance of a company's databases.

3. Cybersecurity

This category is dedicated to protecting data, networks, and systems from threats.

- **Cybersecurity Analyst (or InfoSec Analyst):** The front line of defense; monitors networks, investigates alerts, and responds to security incidents.
- **Penetration Tester (or "Ethical Hacker"):** Paid to legally hack into systems to find vulnerabilities before malicious actors do.
- **Security Engineering:** Designs and builds secure network and system architectures.
- **Governance, Risk & Compliance (GRC):** Focuses on policy, law, and standards (like GDPR, HIPAA, ISO 27001) to ensure the company is following security rules.

4. IT Infrastructure & Cloud

This is the backbone that supports all other technology functions.

- **Cloud Engineering:** Manages and builds infrastructure on cloud platforms like Amazon Web Services (AWS), Microsoft Azure, or Google Cloud (GCP).
- **DevOps Engineering:** Bridges the gap between Software Development (Dev) and IT Operations (Ops), focusing on automation, deployment, and efficiency (CI/CD).
- **Network Engineering / Administration:** Manages the internal and external networks (Wi-Fi, routers, switches, LAN/WAN) that allow computers to communicate.
- **Systems Administration:** Manages the company's servers, operating systems (like Linux or Windows Server), and core IT functions.
- **IT Support / Help Desk:** Provides direct technical support to employees, troubleshooting hardware, software, and access issues.

5. Product & Project Management

These roles organize the *work* and define the *why* and *what* to build.

- **Technical Project Manager:** Organizes the timeline, resources, and budget to ensure a technology project is completed on time.
- **Product Manager:** Decides *what* gets built. They represent the customer, define product features, and align the technology with business goals.
- **Scrum Master / Agile Coach:** A specific role that facilitates the Agile development process, removes obstacles, and coaches the team.
- **Business Analyst (Technical):** Works as a liaison between business stakeholders and the tech team to translate business needs into technical requirements.

6. Design & User Experience

This category focuses on the human side of technology.

- **UX Designer (User Experience):** Researches user behavior to design a product that is logical, intuitive, and easy to use.
- **UI Designer (User Interface):** Focuses on the visual elements of a product (buttons, colors, typography, layout) to make it beautiful and functional.
- **UX Researcher:** Conducts interviews, surveys, and usability tests to understand user needs and pain points.

Front End

1. What does HTML stand for?

- (A) HyperText Markup Language
 - (B) HighText Machine Language
 - (C) HyperText and Links Markup
 - (D) HyperTool Multi-Language
 - (E) Home Tool Markup Language
 - *Correct Answer: (A)*
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2. What is the primary purpose of CSS?

- (A) To add interactivity and behavior to a website.
 - (B) To define the structure and content of a web page.
 - (C) To style the visual presentation of a web page.
 - (D) To manage server-side operations and databases.
 - (E) To optimize the website for search engines.
 - *Correct Answer: (C)*
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3. Which programming language is primarily used to add behavior and interactivity to websites?

- (A) HTML
 - (B) CSS
 - (C) Python
 - (D) JavaScript
 - (E) SQL
 - *Correct Answer: (D)*
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4. Which HTML tag is used to create a hyperlink (a clickable link)?

- (A) `<link>`
 - (B) `<a>`
 - (C) `<href>`
 - (D) `<p>`
 - (E) `<ul>`
 - *Correct Answer: (B)*
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5. In CSS, how would you select all elements with a class name of "highlight"?

- (A) `#highlight`
 - (B) `highlight`
 - (C) `.highlight`
 - (D) `*highlight`
 - (E) `<highlight>`
 - *Correct Answer: (C)*
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6. Which HTML tag is used to define the main content of an HTML document (the part visible to the user)?

- (A) `<head>`
 - (B) `<body>`
 - (C) `<html>`
 - (D) `<title>`
 - (E) `<content>`
 - *Correct Answer: (B)*
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7. What CSS property is used to change the text color of an element?

- (A) `text-color`
 - (B) `font-color`
 - (C) `background-color`
 - (D) `color`
 - (E) `text-style`
 - *Correct Answer: (D)*
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8. What is the purpose of the `alt` attribute on an `<img>` tag?

- (A) To set the alignment of the image.
 - (B) To provide alternative text if the image cannot be displayed.
 - (C) To link the image to another URL.
 - (D) To set the title of the image, which appears on hover.
 - (E) To define the source URL of the image.
 - *Correct Answer: (B)*
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9. In JavaScript, what keyword is used to declare a variable that cannot be reassigned?

- (A) `var`
 - (B) `let`
 - (C) `const`
 - (D) `static`
 - (E) `final`
 - *Correct Answer: (C)*
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10. Which of these is NOT a core component of the CSS Box Model?

- (A) Margin
 - (B) Border
 - (C) Padding
 - (D) Content
 - (E) Display
 - *Correct Answer: (E)*
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11. What does a `<!DOCTYPE html>` declaration do?

- (A) It comments out the entire HTML document.
 - (B) It tells the browser that the document is an HTML5 page.
 - (C) It links a JavaScript file to the document.
 - (D) It creates the main heading for the page.
 - (E) It is an old, optional tag that is no longer used.
 - *Correct Answer: (B)*
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12. Which HTML tag is considered "semantic," meaning it describes its content's meaning?

- (A) `<div>`
 - (B) `<span>`
 - (C) `<article>`
 - (D) `<b>`
 - (E) `<i>`
 - *Correct Answer: (C)*
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13. How do you add an inline comment in JavaScript?

- (A) ``
 - (B) `/* This is a comment */`
 - (C) `// This is a comment`
 - (D) `# This is a comment`
 - (E) `' This is a comment`
 - *Correct Answer: (C)*
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14. What is the term for designing a website to look good on all devices, from desktops to mobile phones?

- (A) Responsive Design
 - (B) Fixed Design
 - (C) Adaptive Design
 - (D) Fluid Design
 - (E) Mobile-First Design
 - *Correct Answer: (A)*
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15. Which CSS property controls the spacing *inside* an element's border?

- (A) `margin`
 - (B) `border-spacing`
 - (C) `spacing`
 - (D) `padding`
 - (E) `outline`
 - *Correct Answer: (D)*
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16. In CSS, which selector has the *highest* specificity (priority)?

- (A) An ID selector (e.g., `#my-id`)
 - (B) A class selector (e.g., `.my-class`)
 - (C) An element selector (e.g., `p`)
 - (D) A universal selector (e.g., `*`)
 - (E) An attribute selector (e.g., `[type='text']`)
 - *Correct Answer: (A)*
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17. What is the correct HTML tag for the largest heading?

- (A) `<h6>`
 - (B) `<heading>`
 - (C) `<head>`
 - (D) `<h1>`
 - (E) `<main>`
 - *Correct Answer: (D)*
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18. What is the 'DOM' in the context of frontend development?

- (A) Digital Object Model
 - (B) Data Object Model
 - (C) Document Object Model
 - (D) Dynamic Object Manipulation
 - (E) Domain Object Mainframe
 - *Correct Answer: (C)*
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19. Which HTML tag is used to create an *unordered* list (i.e., a bulleted list)?

- (A) `<ol>`
 - (B) `<ul>`
 - (C) `<li>`
 - (D) `<list>`
 - (E) `<dl>`
 - *Correct Answer: (B)*
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20. What is the correct way to link an external CSS stylesheet in an HTML file?

- (A) Inside the `<body>` with `<style src="style.css">`
- (B) Inside the `<head>` with `<link rel="stylesheet" href="style.css">`
- (C) Inside the `<head>` with `<script src="style.css">`
- (D) Inside the `<body>` with `<link rel="stylesheet" href="style.css">`
- (E) Inside the `<head>` with `<style link="style.css">`
- *Correct Answer: (B)*

Level 2

1. What is the practical difference between `==` and `===` in JavaScript?

- (A) `==` compares for value only, while `===` compares for both value and type.
 - (B) `==` is for numbers and `===` is for strings.
 - (C) `==` assigns a value, while `===` compares a value.
 - (D) `==` is the old way and `===` is the new, faster way.
 - (E) There is no practical difference; they do the same thing.
 - *Correct Answer: (A)*
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2. How would you select an element with the ID `main-title` using JavaScript?

- (A) `document.querySelector('.main-title')`
 - (B) `document.getElementsByClassName('main-title')`
 - (C) `document.getElementById('main-title')`
 - (D) `document.querySelector('main-title')`
 - (E) `document.getElement('main-title')`
 - *Correct Answer: (C)*
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3. What does the CSS property `box-sizing: border-box;` do?

- (A) It makes the `border` and `padding` part of the element's total `width` and `height`.
 - (B) It adds a border around the element's margin.
 - (C) It automatically calculates the `width` and `height` based on the content.
 - (D) It centers the element vertically and horizontally.
 - (E) It removes all `padding` and `margin` from the element.
 - *Correct Answer: (A)*
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4. Which CSS selector would target only the `<p>` elements that are *direct children* of a `<div>`?

- (A) `div p`
- (B) `div > p`
- (C) `div + p`
- (D) `div.p`
- (E) `div ~ p`
- *Correct Answer: (B)*

5. How do you add a 'click' event listener to a button with the ID **btn** in JavaScript?

- (A) `btn.onClick = function() { ... }`
- (B) `const btn = document.getElementById('btn');`
`btn.addEventListener('click', functionName);`
- (C) `const btn = document.getElementById('btn');`
`btn.listen('click', functionName);`
- (D) `<button id="btn" onclick="functionName()">Click</button>`
- (E) `const btn = document.querySelector('btn');`
`btn.addEvent('click', functionName);`
- *Correct Answer: (B)*

6. What is the difference between **display: block;** and **display: inline;** in CSS?

- (A) **block** elements are for text, and **inline** elements are for images.
- (B) **block** elements start on a new line and take up the full available width, while **inline** elements do not.
- (C) **block** elements can have **margin**, but **inline** elements cannot.
- (D) **block** elements cannot be nested, while **inline** elements can.
- (E) **block** elements are static, while **inline** elements can be moved with **position**.
- *Correct Answer: (B)*

7. Given the JavaScript **const fruit = ['Apple', 'Banana', 'Cherry'];**, how would you access 'Banana'?

- (A) `fruit(1)`
- (B) `fruit.1`
- (C) `fruit[1]`
- (D) `fruit[2]`
- (E) `fruit.get(1)`
- *Correct Answer: (C)*

8. What is the purpose of the **method** attribute on an HTML **<form>** tag?

- (A) It specifies the URL where the form data will be sent.

- (B) It defines the JavaScript function to run on submission.
 - (C) It specifies the HTTP method (e.g., `GET` or `POST`) to use when submitting the form.
 - (D) It defines which input fields are required.
 - (E) It sets the visual styling and layout of the form.
 - *Correct Answer: (C)*
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9. Which CSS property would you use to create space *outside* an element's border?

- (A) `padding`
 - (B) `margin`
 - (C) `border-spacing`
 - (D) `outline`
 - (E) `box-shadow`
 - *Correct Answer: (B)*
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10. What will this JavaScript code do?

`document.getElementById('header').style.color = 'blue';`

- (A) It will change the background color of the 'header' element to blue.
 - (B) It will select the 'header' element and change its text color to blue.
 - (C) It will add a blue border to the 'header' element.
 - (D) It will check if the 'header' element's color is blue and return true or false.
 - (E) The code will fail because `.style.color` is not valid syntax.
 - *Correct Answer: (B)*
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11. Which of these is the correct syntax for a 'template literal' in JavaScript, used for embedding variables in a string?

- (A) `"Hello, " + name + "!"`
 - (B) `'Hello, {name}!'`
 - (C) ``Hello, ${name}!``
 - (D) `('Hello, ' + name + '!')`
 - (E) ``Hello, [name]!``
 - *Correct Answer: (C)*
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12. In CSS, what does `position: absolute;` do to an element?

- (A) It centers the element on the page.
 - (B) It removes the element from the normal document flow and positions it relative to its nearest *positioned* ancestor.
 - (C) It fixes the element to the viewport, so it doesn't move when scrolling.
 - (D) It makes the element's position relative to its own default position in the flow.
 - (E) It makes the element a block-level element.
 - *Correct Answer: (B)*
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13. Given `const user = { name: 'Alex', age: 30 };`, how would you access Alex's age?

- (A) `user['age']`
 - (B) `user(age)`
 - (C) `user.get('age')`
 - (D) `user[age]`
 - (E) `user[1]`
 - *Correct Answer: (A)*
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14. Which CSS selector targets an `<input>` element with a `type` attribute set to `text`?

- (A) `input.text`
 - (B) `input[type='text']`
 - (C) `input:text`
 - (D) `input#text`
 - (E) `text(input)`
 - *Correct Answer: (B)*
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15. What is the purpose of the `alt` attribute in an `<img>` tag?

- (A) It provides a title for the image that appears on hover.
 - (B) It provides alternative text for screen readers and when the image fails to load.
 - (C) It defines the URL or path to the image file.
 - (D) It sets the alignment (left, right, center) of the image.
 - (E) It specifies a lower-resolution version of the image to load first.
 - *Correct Answer: (B)*
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16. How do you write a 'fat arrow' function in JavaScript?

- (A) `function myFunction() { ... }`
 - (B) `const myFunction => () { ... }`
 - (C) `const myFunction = () => { ... }`
 - (D) `const myFunction = function() => { ... }`
 - (E) `myFunction: () => { ... }`
 - *Correct Answer: (C)*
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17. In CSS, what is the difference between `visibility: hidden;` and `display: none;`?

- (A) There is no difference; both hide the element.
 - (B) `display: none;` hides the element, while `visibility: hidden;` deletes it from the DOM.
 - (C) `display: none;` removes the element from the layout, while `visibility: hidden;` hides it but leaves its space.
 - (D) `visibility: hidden;` is for text, and `display: none;` is for images.
 - (E) `display: none;` is permanent, but `visibility: hidden;` can be toggled with JavaScript.
 - *Correct Answer: (C)*
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18. Which CSS pseudo-class is used to style an element when the user's mouse is over it?

- (A) `:active`
 - (B) `:focus`
 - (C) `:hover`
 - (D) `:on`
 - (E) `:mouse`
 - *Correct Answer: (C)*
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19. What is the correct HTML tag to use for a self-contained piece of content, like a blog post or a news article, that could be distributed independently?

- (A) `<div>`
- (B) `<section>`

- (C) `<article>`
 - (D) `<aside>`
 - (E) `<main>`
 - *Correct Answer: (C)*
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20. How would you change the text inside an element with the ID `message` to "Hello World"?

- (A) `document.getElementById('message').value = "Hello World";`
- (B) `document.getElementById('message').innerHTML = "Hello World";`
- (C) `document.getElementById('message').text = "Hello World";`
- (D) `document.querySelector('#message').setAttribute('text', 'Hello World');`
- (E) `document.getElementById('message').write("Hello World");`
- *Correct Answer: (B)*

Level 3.

1. How do you center a child element both vertically and horizontally inside a parent container using CSS Flexbox?

- (A) `parent { display: flex; justify-content: center; align-items: center; }`
 - (B) `parent { display: flex; flex-direction: column; }`
 - (C) `parent { display: grid; } child { margin: auto; }`
 - (D) `parent { display: flex; vertical-align: middle; text-align: center; }`
 - (E) `parent { display: flex; align-content: center; flex-wrap: wrap; }`
 - *Correct Answer: (A)*
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2. You have an array `const numbers = [1, 2, 3, 4, 5]`. How do you create a *new* array containing only the even numbers?

- (A) `const evens = numbers.map(num => num % 2 === 0);`
- (B) `const evens = numbers.forEach(num => num % 2 === 0);`
- (C) `const evens = numbers.filter(num => num % 2 === 0);`
- (D) `const evens = numbers.find(num => num % 2 === 0);`

- (E) `let evens = []; for(num in numbers) { if(num % 2 === 0) { evens.push(num); } }`
 - *Correct Answer: (C)*
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3. What is the primary purpose of the `.then()` block when using the JavaScript `fetch` API?

- (A) To catch and handle any errors that occur during the request.
 - (B) To define the URL of the API endpoint.
 - (C) To handle the successful response *after* the asynchronous request completes.
 - (D) To send the data (payload) to the API.
 - (E) To run code immediately, before the `fetch` call is made.
 - *Correct Answer: (C)*
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4. Why is it generally better to use a `<button>` tag for a clickable action instead of a `<div>` with a JavaScript click event?

- (A) Because `<div>` elements cannot have click events.
 - (B) Because `<button>` is automatically focusable, can be triggered by 'Enter'/'Space', and is recognized by screen readers.
 - (C) Because `<button>` elements can be styled more easily with CSS.
 - (D) Because `<button>` is a non-semantic tag, and `<div>` is semantic.
 - (E) Because `<div>` click events are slower than `<button>` click events.
 - *Correct Answer: (B)*
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5. Which CSS Grid syntax creates a 3-column layout where the middle column is twice as wide as the two outer columns?

- (A) `grid-template-columns: 25% 50% 25%;`
 - (B) `grid-template-columns: 1fr 2fr 1fr;`
 - (C) `grid-columns: 1 2 1;`
 - (D) `grid-template-rows: 1fr 2fr 1fr;`
 - (E) `grid-template-columns: repeat(3, 1fr 2fr);`
 - *Correct Answer: (B)*
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6. You have an array `const names = ['alice', 'bob', 'charlie'];`. How do you create a new array `['ALICE', 'BOB', 'CHARLIE']`?

- (A) `const upper = names.map(name => name.toUpperCase());`
 - (B) `const upper = names.filter(name => name.toUpperCase());`
 - (C) `const upper = names.forEach(name => name.toUpperCase());`
 - (D) `const upper = names.toUpperCase();`
 - (E) `let upper = []; for(let i=0; i<names.length; i++) { names[i].toUpperCase(); }`
 - *Correct Answer: (A)*
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7. In CSS, what is the key difference between `position: absolute;` and `position: relative;`?

- (A) `relative` elements are removed from the document flow, while `absolute` elements are not.
 - (B) `absolute` elements are positioned relative to the viewport, while `relative` elements are positioned relative to the `<body>`.
 - (C) `relative` elements remain in the normal flow, while `absolute` elements are removed from the flow and positioned relative to their *nearest positioned ancestor*.
 - (D) `absolute` elements cannot be moved with `top/left`, while `relative` elements can.
 - (E) `relative` is for layout and `absolute` is for text.
 - *Correct Answer: (C)*
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8. In a `fetch` request, why does the `response.json()` method also require a `.then()` or `await`?

- (A) Because it needs to convert a JavaScript object into a JSON string.
 - (B) Because the `response` body arrives as a data stream and must be fully read and parsed asynchronously.
 - (C) Because it is an old method that is no longer used.
 - (D) Because it needs to add the data to `localStorage`.
 - (E) It does not require a `.then()`; only the initial `fetch` does.
 - *Correct Answer: (B)*
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9. Which CSS syntax correctly applies a red background *only* when the screen width is 800px or *less*?

- (A) `if (screen.width <= 800px) { body { background: red; } }`
 - (B) `@media query (width <= 800px) { body { background: red; } }`
 - (C) `@media screen (min-width: 800px) { body { background: red; } }`
 - (D) `@media screen (max-width: 800px) { body { background: red; } }`
 - (E) `body { @media (max-width: 800px) { background: red; } }`
 - *Correct Answer: (D)*
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10. What is the purpose of an **aria-label** attribute in HTML?

- (A) To provide an accessible name for an element when a visible text label is not present (e.g., an icon-only button).
 - (B) To provide a visible label for a form `<input>`.
 - (C) To define a new JavaScript variable.
 - (D) To hide an element from screen readers completely.
 - (E) To link to an external ARIA stylesheet.
 - *Correct Answer: (A)*
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11. Which of these CSS selectors has the *highest* specificity?

- (A) `.container p.text`
 - (B) `#main-content p`
 - (C) `body div p.text`
 - (D) `p.text.highlight`
 - (E) `div p`
 - *Correct Answer: (B)*
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12. You have 3 `<div>` elements inside a flex container. By default, they are in a row. How do you make them stack vertically?

- (A) `display: grid;`
- (B) `flex-wrap: wrap;`
- (C) `align-items: center;`
- (D) `flex-flow: vertical;`
- (E) `flex-direction: column;`

- *Correct Answer: (E)*
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13. What is the "spread operator" (...) used for in this code: `const arr2 = [...arr1];`?

- (A) It creates a reference, where `arr2` points to the *same* array as `arr1`.
 - (B) It creates a *new* array (`arr2`) containing a shallow copy of all items from `arr1`.
 - (C) It gets the last item from `arr1`.
 - (D) It converts `arr1` into an object.
 - (E) It deletes all items from `arr1`.
 - *Correct Answer: (B)*
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14. Which of the following values is "falsy" in JavaScript?

- (A) `"0"` (the string '0')
 - (B) `[]` (an empty array)
 - (C) `{}` (an empty object)
 - (D) `NaN` (Not a Number)
 - (E) `"false"` (the string 'false')
 - *Correct Answer: (D)*
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15. What is "object destructuring" in JavaScript?

- (A) The process of converting an object into a JSON string.
 - (B) A syntax for unpacking properties from an object into distinct variables (e.g., `const { name } = user;`).
 - (C) A way to delete properties from an object.
 - (D) A way to combine two objects into one.
 - (E) A method for finding errors in an object.
 - *Correct Answer: (B)*
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16. What is the security risk of using `.innerHTML` instead of `.textContent` to insert user-provided data?

- (A) It can lead to a Cross-Site Scripting (XSS) attack if the data contains malicious `<script>` tags.

- (B) It is much slower and can crash the browser.
 - (C) It can corrupt the browser's `localStorage`.
 - (D) It does not work in all browsers.
 - (E) There is no risk; `.innerHTML` automatically sanitizes all input.
 - *Correct Answer: (A)*
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17. How would you store a user's name, "Alice", in the browser's `localStorage`?

- (A) `localStorage.add('username', 'Alice');`
 - (B) `document.cookie = 'username=Alice';`
 - (C) `localStorage('username') = 'Alice';`
 - (D) `sessionStorage.setItem('username', 'Alice');`
 - (E) `localStorage.setItem('username', 'Alice');`
 - *Correct Answer: (E)*
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18. You have a JavaScript object: `const user = { name: 'Alice' };`. What must you do before storing it in `localStorage`?

- (A) Nothing, `localStorage.setItem('user', user);` will store the object correctly.
 - (B) Convert it to a JSON string using `JSON.stringify(user)`.
 - (C) Encrypt it using a password.
 - (D) Convert it to an array, as objects cannot be stored.
 - (E) Use `localStorage.setObject('user', user);` instead.
 - *Correct Answer: (B)*
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19. In CSS Flexbox, what does the property `flex: 1;` on a child element primarily do?

- (A) It sets the element's `max-width` to 100px.
 - (B) It allows the element to grow and take up any available free space in the flex container.
 - (C) It makes the element 1px wide.
 - (D) It sets the element's `opacity` to 1.
 - (E) It forces the element to be the first item in the container.
 - *Correct Answer: (B)*
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20. What is the purpose of the `await` keyword in an `async` function?

- (A) It is an old keyword that has been replaced by `.then()`.
- (B) It pauses the execution of the `async` function *until* a `Promise` is settled (resolved or rejected).
- (C) It immediately rejects a `Promise` and throws an error.
- (D) It converts a regular function into an `async` function.
- (E) It makes the JavaScript code run faster by using multiple threads.
- *Correct Answer: (B)*

Level 4

Expert & Frameworks.

1. What is JSX?

- (A) A database query language used by React.
 - (B) A CSS pre-processor for styling React components.
 - (C) A syntax extension for JavaScript that looks like HTML, used to describe UI structure.
 - (D) A JavaScript method for fetching API data.
 - (E) A built-in React component for handling user state.
 - *Correct Answer: (C)*
-

2. In React, what is the primary difference between "props" and "state"?

- (A) Props are used for functions; state is used for class components.
 - (B) Props are passed *into* a component from its parent, while state is managed *within* the component.
 - (C) Props can be changed by the child component, while state cannot.
 - (D) Props are for styling, while state is for logic.
 - (E) There is no difference; they are interchangeable.
 - *Correct Answer: (B)*
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3. What does the `useState` hook return?

- (A) A single value representing the current state.
- (B) An object with `value` and `setValue` properties.
- (C) An array containing the current state value and a function to update it.
- (D) A boolean indicating if the state has changed.
- (E) Just a function to update the state.
- *Correct Answer: (C)*

4. Why are "keys" important when rendering a list of elements using `.map()` in React?

- (A) To provide a unique `id` for CSS styling.
- (B) To help React identify which items have changed, been added, or been removed, optimizing rendering.
- (C) To set the order of the items in the list.
- (D) To make the list items clickable.
- (E) They are not important and can be omitted.
- *Correct Answer: (B)*

5. How do you conditionally render a component? For example, show a `<UserProfile>` component *only if* `isLoggedIn` is true?

- (A) `<UserProfile if={isLoggedIn} />`
- (B) `if (isLoggedIn) { <UserProfile /> }`
- (C) `{isLoggedIn && <UserProfile />}`
- (D) `isLoggedIn ? <UserProfile /> : null`
- (E) Both C and D are common, valid ways to do this.
- *Correct Answer: (E)*

6. What is the purpose of the `useEffect` hook?

- (A) To handle user click events.
- (B) To perform side effects in functional components, such as data fetching or DOM manipulation.
- (C) To define the component's initial state.
- (D) To create reusable functions within a component.
- (E) To conditionally style a component.
- *Correct Answer: (B)*

7. You have a `useEffect` hook: `useEffect(() => { ... }, [])`; What does the empty dependency array `[]` signify?

- (A) The effect will run on *every* re-render of the component.
- (B) The effect will run only *once*, when the component first mounts (like `componentDidMount`).

- (C) The effect will not run at all.
 - (D) The effect will run when the component unmounts.
 - (E) This syntax is invalid and will cause an error.
 - *Correct Answer: (B)*
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8. How do you pass a function called `handleClick` from a parent component `App` to a child component `Button`?

- (A) `<Button event={handleClick} />`
 - (B) `<Button onClick={handleClick} />`
 - (C) `<Button function={handleClick} />`
 - (D) `<Button>{handleClick}</Button>`
 - (E) You cannot pass functions as props.
 - *Correct Answer: (B)*
-

9. What is the "Virtual DOM" in React?

- (A) A browser feature that React hooks into.
 - (B) A in-memory representation of the real browser DOM, used by React to calculate the most efficient changes.
 - (C) Another name for the `useState` hook.
 - (D) A tool for debugging React applications.
 - (E) A secure, sandboxed DOM for running tests.
 - *Correct Answer: (B)*
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10. What is "lifting state up" in React?

- (A) Moving state from a child component to its parent component to share it between siblings.
 - (B) Storing your state in `localStorage` instead of in a component.
 - (C) Using the `useState` hook at the top of your function.
 - (D) Upgrading a class component's state to a functional component's hook.
 - (E) Moving state into a separate file.
 - *Correct Answer: (A)*
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11. What is the purpose of React Router?

- (A) It optimizes the build process of a React app.
 - (B) It is a state management library like Redux.
 - (C) It handles client-side routing, allowing navigation between different "pages" in a single-page application.
 - (D) It is a tool for writing unit tests for React components.
 - (E) It connects React to a backend server.
 - *Correct Answer: (C)*
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12. How would you update a `user` object in state without mutating the original state?

- (A) `const [user, setUser] = useState({ name: 'Alex' });`
`setUser(user.name = 'Bob');`
 - (B) `const [user, setUser] = useState({ name: 'Alex' });` `setUser({ ...user, name: 'Bob' });`
 - (C) `const [user, setUser] = useState({ name: 'Alex' });` `user.name = 'Bob';` `setUser(user);`
 - (D) `const [user, setUser] = useState({ name: 'Alex' });` `setUser({ name: 'Bob' });`
 - (E) `const [user, setUser] = useState({ name: 'Alex' });`
`setUser.name = 'Bob';`
 - *Correct Answer: (B)*
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13. What is a "controlled component" in React?

- (A) A component that is only visible to logged-in users.
 - (B) A component that has its own internal state.
 - (C) A form element (like `<input>`) whose value is controlled by React state.
 - (D) A component that has been unit tested.
 - (E) A component that is imported from an external library.
 - *Correct Answer: (C)*
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14. What is the primary use of the React Context API?

- (A) To fetch data from an external API.
- (B) To provide a way to pass data through the component tree without having to pass props down manually at every level.
- (C) To handle errors and crashes within the application.
- (D) To define the component's visual style.

- (E) To create a new React component.
 - *Correct Answer: (B)*
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15. How do you pass data from a child component *back up* to its parent component?

- (A) You can't; data only flows from parent to child.
 - (B) By modifying the `props` object directly in the child.
 - (C) By using the `useContext` hook.
 - (D) The parent passes a callback function (as a prop) to the child, and the child calls that function with the data as an argument.
 - (E) By setting the data in `localStorage`.
 - *Correct Answer: (D)*
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16. What is the purpose of `React.Fragment`, often written as `<> ... </>`?

- (A) It is a way to write comments in JSX.
 - (B) It is used to create a "portal" to another DOM node.
 - (C) It lets you group a list of children without adding an extra `<div>` to the DOM.
 - (D) It is a component that marks its children for lazy loading.
 - (E) It is a way to define state that is shared between components.
 - *Correct Answer: (C)*
-

17. What is a "prop drilling"?

- (A) The process of passing data from a parent component to a deeply nested child through many intermediate components.
 - (B) A debugging technique to "drill" into a component's props.
 - (C) A performance optimization for React.
 - (D) The act of using `useState` to manage props.
 - (E) An error that occurs when props are not defined.
 - *Correct Answer: (A)*
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18. What is the difference between `useEffect` and `useLayoutEffect`?

- (A) `useEffect` runs asynchronously *after* the browser has painted, while `useLayoutEffect` runs synchronously *before* the browser paints.
- (B) `useEffect` is for data fetching; `useLayoutEffect` is for styling.

- (C) `useEffect` can have a dependency array; `useLayoutEffect` cannot.
 - (D) `useLayoutEffect` is the old version of `useEffect`.
 - (E) There is no difference; they are aliases.
 - *Correct Answer: (A)*
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19. Given `<ProfileCard user={data} />`, how would the `ProfileCard` component access the `user` object?

- (A) `function ProfileCard(props) { return <h1>{props.user.name}</h1>; }`
 - (B) `function ProfileCard(user) { return <h1>{user.name}</h1>; }`
 - (C) `function ProfileCard() { return <h1>{this.props.user.name}</h1>; }`
 - (D) `function ProfileCard() { return <h1>{props.data.name}</h1>; }`
 - (E) `function ProfileCard({ data }) { return <h1>{data.name}</h1>; }`
 - *Correct Answer: (A)*
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20. What is the problem with this code? `function Counter() { const [count, setCount] = useState(0); function handleIncrement() { setCount(count + 1); setCount(count + 1); } ... }`

- (A) You cannot call `setCount` more than once in a function.
- (B) React "batches" state updates, so `count` will only increment by 1, not 2, because both calls use the same `count` value from the initial render.
- (C) This will cause an infinite loop.
- (D) `setCount` must be awaited (e.g., `await setCount(...)`).
- (E) This will increment the count by 2, but it is not good for performance.
- *Correct Answer: (B)*

Level 5: Strategic & Architectural

1. What is the primary business-level advantage of using TypeScript in a large-scale project?

- (A) It makes the website run faster in the browser.
- (B) It automatically writes unit tests for your code.
- (C) It reduces bugs and improves maintainability by enforcing type safety, making code easier to refactor.

- (D) It replaces the need for a module bundler like Webpack.
 - (E) It is the only way to use modern frameworks like React and Angular.
 - *Correct Answer: (C)*
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2. A user's Lighthouse report shows a poor "Largest Contentful Paint" (LCP) score. Which of the following is the *most likely* cause and solution?

- (A) The JavaScript bundle is too large, so it blocks the main thread.
 - (B) The main "hero" image is a very large, unoptimized file that is slow to load.
 - (C) There are too many `console.log()` statements in the code.
 - (D) The website is not using a Service Worker.
 - (E) The CSS files are not minified.
 - *Correct Answer: (B)*
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3. What is "tree shaking" in the context of a modern build process (e.g., Webpack, Vite)?

- (A) A debugging process for finding errors in the component tree.
 - (B) A process that automatically removes unused code (dead code) from the final JavaScript bundle.
 - (C) A method for re-organizing the file structure of a project.
 - (D) A security scan that looks for vulnerable dependencies.
 - (E) A way to "shake" the DOM to find performance bottlenecks.
 - *Correct Answer: (B)*
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4. When building a large application, when would you choose the Context API over a dedicated state library like Redux?

- (A) When you need to manage complex, high-frequency updates, like in a stock trading app.
 - (B) When you need to pass "global" data (like theme, user authentication) deeply down the component tree without prop drilling.
 - (C) When you need to store your state in `localStorage`.
 - (D) When you are working on a class-based component, as Redux is for functional components only.
 - (E) Context API is always better than Redux because it is built into React.
 - *Correct Answer: (B)*
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5. What is the primary philosophy of the "React Testing Library" (RTL)?

- (A) To test the internal implementation details of a component (e.g., "does this component have the correct state?").
 - (B) To test how the component behaves from a *user's perspective* (e.g., "can the user see this text?", "can the user click this button?").
 - (C) To test the component's performance and speed.
 - (D) To test the component's styles and visual appearance (snapshot testing).
 - (E) To test the component's JavaScript code logic, but not its JSX.
 - *Correct Answer: (B)*
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6. What is the primary security risk of storing a JWT (JSON Web Token) in `localStorage`?

- (A) It can be easily accessed by a Cross-Site Scripting (XSS) attack, allowing an attacker to steal the token.
 - (B) It takes up too much space and will slow the browser down.
 - (C) It will be deleted automatically when the user closes the browser tab.
 - (D) It cannot be shared between different browser tabs.
 - (E) It is not encrypted and can be read by anyone.
 - *Correct Answer: (A)*
-

7. In the context of Server-Side Rendering (SSR) in a React application, what is "hydration"?

- (A) The process of the server fetching all the necessary data *before* sending the HTML.
 - (B) The process of the browser "watering down" the JavaScript to make it run faster.
 - (C) The client-side JavaScript process of attaching event listeners to the static HTML sent by the server, making the page interactive.
 - (D) The process of storing assets in a browser cache.
 - (E) The process of compressing the HTML file on the server.
 - *Correct Answer: (C)*
-

8. What is the "micro-frontend" architectural pattern?

- (A) A pattern where you only use very small, lightweight frameworks.
- (B) A method of building the *entire* frontend application inside a single Web Component.
- (C) An architectural style where a large frontend application is split into smaller, independently deployable "micro-apps."
- (D) A CSS methodology for creating small, reusable component styles.
- (E) A pattern where all state is stored in a single, global "micro" object.
- *Correct Answer: (C)*

9. Why is it generally bad practice to use an array's **index** as a **key** when rendering a list in React, especially if the list is dynamic (can be re-ordered or filtered)?

- (A) Because it is not a valid data type for a key; it must be a string.
- (B) Because it can lead to incorrect state and rendering bugs when the list items change order.
- (C) Because it is much slower than using a unique ID.
- (D) Because it is not compliant with accessibility (WCAG) standards.
- (E) Because it will throw a fatal error in production builds.
- *Correct Answer: (B)*

10. What is a "Progressive Web App" (PWA)?

- (A) A web application that is built using a special, "progressive" version of JavaScript.
- (B) A website that is still in development (a "work in progress").
- (C) A web app that uses a Service Worker, a manifest file, and HTTPS to provide an "app-like" experience (e.g., offline capability, push notifications).
- (D) An application that progressively "gets harder" as the user interacts with it.
- (E) A marketing term for any modern, single-page application.
- *Correct Answer: (C)*

11. What problem does the **React.useMemo** hook solve?

- (A) It memoizes the *result* of an expensive calculation, preventing it from being re-run on every render unless its dependencies change.
- (B) It fetches data from an API and "memoizes" (caches) the response.
- (C) It memoizes an entire component, preventing it from re-rendering (that is **React.memo**).
- (D) It remembers the user's state even after they close the browser tab.
- (E) It allows you to write memos (comments) in your code that are visible in the React DevTools.
- *Correct Answer: (A)*

12. What is the main difference between a unit test and an integration test in a frontend application?

- (A) A unit test checks a single, isolated function or component; an integration test checks how multiple components work *together*.

- (B) Unit tests are written in JavaScript; integration tests are written in TypeScript.
 - (C) Unit tests run in the browser; integration tests run on the server.
 - (D) Unit tests are fast and automated; integration tests are slow and manual.
 - (E) A unit test checks for bugs; an integration test checks for performance.
 - *Correct Answer: (A)*
-

13. What is the BEM (Block, Element, Modifier) methodology in CSS?

- (A) A JavaScript framework for building user interfaces.
 - (B) A naming convention for CSS classes (e.g., `.card__title--primary`) designed to create scoped, maintainable styles.
 - (C) A build tool that bundles, extends, and minifies CSS.
 - (D) A set of pre-built CSS components, similar to Bootstrap.
 - (E) A testing framework for visual regression.
 - *Correct Answer: (B)*
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14. What is a "Cross-Site Request Forgery" (CSRF) attack, and how is it commonly mitigated?

- (A) An attack where malicious scripts are injected into your site; mitigated by escaping data.
 - (B) An attack where an attacker tricks a logged-in user into sending an unwanted request; mitigated by using anti-CSRF tokens.
 - (C) An attack where a user is redirected to a fake version of your website; mitigated by using HTTPS.
 - (D) An attack where the server is flooded with requests; mitigated by rate-limiting.
 - (E) An attack where `localStorage` is stolen; mitigated by using HTTP-only cookies.
 - *Correct Answer: (B)*
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15. In an `async` function, what is the *functional difference* between using `await` and using `.then()`?

- (A) `await` pauses the function execution, while `.then()` schedules a callback to run *after* the function has finished.
- (B) `await` is linear, "synchronous-looking" code, while `.then()` is a chain of callback functions. Both handle the promise.
- (C) `await` can be used on any function, while `.then()` only works on `fetch` requests.
- (D) `await` can handle errors, while `.then()` requires a separate `.catch()` block.
- (E) B and D are both correct.

- *Correct Answer: (E)*
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16. What is the "bundle size" of a frontend application, and why is it a critical performance metric?

- (A) The total number of files in the project's source code; it is not a performance metric.
 - (B) The physical size (in MB or KB) of the final JavaScript/CSS files sent to the user; larger bundles take longer to download, parse, and execute.
 - (C) The amount of memory the application uses in the browser; it is not related to download time.
 - (D) The number of components in the React component tree.
 - (E) The size of the `node_modules` folder.
 - *Correct Answer: (B)*
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17. What is an "End-to-End" (E2E) test, and how does it differ from a unit or integration test?

- (A) It is another name for a unit test.
 - (B) It tests a complete user workflow (e.g., "login, add item to cart, checkout") by simulating a *real user* in a *real browser*.
 - (C) It only tests the API "endpoints" and nothing on the frontend.
 - (D) It is a manual test done by a QA team.
 - (E) It tests the "end" of the code (the last function) but not the beginning.
 - *Correct Answer: (B)*
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18. Why would you use a `debounce` function on an event listener (e.g., for a search bar's 'keyup' event)?

- (A) To force the event to fire immediately, improving responsiveness.
 - (B) To limit how often the function is executed (e.g., "wait until the user has stopped typing for 300ms") to prevent excessive API calls.
 - (C) To cancel the event completely and prevent it from firing.
 - (D) To add a "bouncing" animation to the user interface.
 - (E) To log the event to the console for debugging.
 - *Correct Answer: (B)*
-

19. What is a "Service Worker" and what is its *primary* capability?

- (A) A JavaScript file that runs in the browser and is used to manage and test other scripts.
 - (B) A backend script that "serves" the frontend application.
 - (C) A JavaScript file that runs in the background, separate from the web page, allowing for features like network request interception (caching for offline use) and push notifications.
 - (D) A UI component that shows a "loading" spinner while data is being fetched.
 - (E) A security policy that prevents XSS attacks.
 - *Correct Answer: (C)*
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20. You see "layout thrashing" in your browser's performance profiler. What does this mean?

- (A) The user is "trashing" the UI by clicking too fast.
- (B) The browser is forced to re-calculate the layout of the page multiple times in a single frame due to JavaScript rapidly reading and writing layout properties (e.g., `offsetHeight`, `style.width`).
- (C) The CSS files are "trashing" each other by overriding styles.
- (D) The application is loading too many large image files at once.
- (E) The JavaScript bundle is corrupted and needs to be rebuilt.
- *Correct Answer: (B)*

Backend Development track.

Level 1: Basic Foundations

1. What is the primary role of a "backend" in a web application?

- (A) To define the style, color, and layout of a website.
 - (B) To manage the server, application logic, and database.
 - (C) To create the clickable buttons and forms in a browser.
 - (D) To store files directly on the user's computer.
 - (E) To optimize the website for search engines.
 - *Correct Answer: (B)*
-

2. What does "API" stand for?

- (A) Application Programming Interface

- (B) Advanced Programming Integration
 - (C) Application Protocol Instance
 - (D) Automated Programming Interaction
 - (E) Application Process Information
 - *Correct Answer: (A)*
-

3. What does "HTTP" stand for?

- (A) HyperText Transfer Protocol
 - (B) High-Tech Transfer Program
 - (C) HyperText Transmission Protocol
 - (D) Hyper-Terminal Transfer Policy
 - (E) Home Terminal Transfer Protocol
 - *Correct Answer: (A)*
-

4. What is the main purpose of a "database"?

- (A) To style a web page.
 - (B) To write the business logic of an application.
 - (C) To store, organize, and retrieve data efficiently.
 - (D) To handle user click events.
 - (E) To display images and videos.
 - *Correct Answer: (C)*
-

5. What is the role of a "server" in backend development?

- (A) A computer that runs the backend software and listens for requests.
 - (B) A text editor for writing code.
 - (C) The user's web browser (like Chrome or Firefox).
 - (D) A program that designs the user interface.
 - (E) A file that contains all the HTML code.
 - *Correct Answer: (A)*
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6. What is "JSON"?

- (A) A type of programming language for styling.
- (B) A database management system.
- (C) A lightweight format for storing and transporting data, often used in APIs.

- (D) A backend framework for Python.
 - (E) A type of server.
 - *Correct Answer: (C)*
-

7. What is the basic difference between a "client" and a "server"?

- (A) The client runs on a Mac, and the server runs on Windows.
 - (B) The client is the backend, and the server is the frontend.
 - (C) The client (e.g., a browser) requests data, and the server (backend) provides it.
 - (D) The client stores data, and the server displays data.
 - (E) The client writes code, and the server runs code.
 - *Correct Answer: (C)*
-

8. What does "SQL" stand for?

- (A) Structured Query Language
 - (B) Simple Query Logic
 - (C) Server-Side Query Language
 - (D) System Query Link
 - (E) Structured Question Logic
 - *Correct Answer: (A)*
-

9. What is an "HTTP Request"?

- (A) A message sent *by the server* to the client.
 - (B) A message sent *by the client* (e.g., browser) to the server, asking for data.
 - (C) A type of database.
 - (D) A file containing HTML and CSS.
 - (E) An error in the code.
 - *Correct Answer: (B)*
-

10. What is an "HTTP Response"?

- (A) A message sent *by the server* back to the client, containing the requested data (or an error).
- (B) A message sent *by the client* to the server.
- (C) A JavaScript function.
- (D) A command to restart the server.

- (E) A user's click on a button.
 - *Correct Answer: (A)*
-

11. Which of the following is a popular programming language used for backend development?

- (A) HTML
 - (B) CSS
 - (C) Python
 - (D) React
 - (E) SASS
 - *Correct Answer: (C)*
-

12. What is "localhost"?

- (A) The public, live website on the internet.
 - (B) A standard name that refers to your *own* computer (127.0.0.1).
 - (C) A popular hosting provider.
 - (D) A type of database.
 - (E) A security protocol.
 - *Correct Answer: (B)*
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13. What is a "port number" (e.g., 3000, 8080, 5432)?

- (A) A specific "door" on a server that a software application listens on for requests.
 - (B) The total number of users allowed on a server.
 - (C) The version number of the backend software.
 - (D) A type of error code.
 - (E) The line number where your code is running.
 - *Correct Answer: (A)*
-

14. What is a "database schema"?

- (A) The total amount of data stored.
- (B) The blueprint or structure of a database (e.g., table names, column names, data types).
- (C) A query used to get data.
- (D) The password to access the database.

- (E) A backup file of the database.
 - *Correct Answer: (B)*
-

15. What is an "HTTP status code"?

- (A) A number sent in a response that indicates the status of the request (e.g., success, error, not found).
 - (B) The line number of an error in the code.
 - (C) The total number of users on the server.
 - (D) The version of the HTTP protocol.
 - (E) A secret code to access the API.
 - *Correct Answer: (A)*
-

16. What does the HTTP status code 200 generally mean?

- (A) Not Found
 - (B) Server Error
 - (C) OK (The request was successful)
 - (D) Unauthorized
 - (E) Bad Request
 - *Correct Answer: (C)*
-

17. What does the HTTP status code 404 generally mean?

- (A) OK
 - (B) Server Error
 - (C) Not Found (The requested resource could not be found)
 - (D) Unauthorized
 - (E) Moved Permanently
 - *Correct Answer: (C)*
-

18. What is a "route" or "endpoint" in a backend application?

- (A) The main function that starts the server.
- (B) A specific URL path (e.g., `/api/users`) that performs a specific function.
- (C) A file that contains all the HTML.
- (D) The connection cable to the internet.
- (E) A database table.

- *Correct Answer: (B)*
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19. What does the acronym "CRUD" stand for?

- (A) Create, Read, Update, Delete
 - (B) Connect, Run, Update, Debug
 - (C) Create, Route, Understand, Deploy
 - (D) Call, Read, Use, Destroy
 - (E) Code, Review, Undo, Deploy
 - *Correct Answer: (A)*
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20. What is a "query parameter" in a URL?

- (A) The main domain name (e.g., `https://www.google.com/search?q=google.com`).
 - (B) The HTTP method (e.g., GET, POST).
 - (C) The part *after* the `?` (e.g., `id=123`) used to filter or pass data.
 - (D) The password to access the website.
 - (E) The protocol (e.g., `https://`).
 - *Correct Answer: (C)*
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Level 2: Intermediate Application

1. What is the main difference between an HTTP `GET` and `POST` request?

- (A) `GET` is for sending data, `POST` is for retrieving data.
 - (B) `GET` retrieves data (and passes data in the URL), `POST` submits new data (and passes data in the request body).
 - (C) `GET` is secure, `POST` is not secure.
 - (D) `GET` is for databases, `POST` is for APIs.
 - (E) There is no difference; they are interchangeable.
 - *Correct Answer: (B)*
-

2. Which SQL command is used to retrieve all data from a table named `users`?

- (A) `GET * FROM users;`
- (B) `SELECT * FROM users;`

- (C) `READ * FROM users;`
 - (D) `FETCH users ALL;`
 - (E) `SELECT users ALL;`
 - *Correct Answer: (B)*
-

3. Which SQL command is used to add a new user to a **users** table?

- (A) `ADD INTO users (name) VALUES ('Alice');`
 - (B) `CREATE USER 'Alice' IN users;`
 - (C) `INSERT INTO users (name) VALUES ('Alice');`
 - (D) `UPDATE users SET name = 'Alice';`
 - (E) `NEW USER ('Alice') IN users;`
 - *Correct Answer: (C)*
-

4. In a REST API, which HTTP method is typically used to *update* an existing resource (e.g., change a user's name)?

- (A) `GET`
 - (B) `POST`
 - (C) `UPDATE`
 - (D) `PUT` or `PATCH`
 - (E) `REFRESH`
 - *Correct Answer: (D)*
-

5. In a REST API, which HTTP method is typically used to *delete* a resource?

- (A) `GET`
 - (B) `POST`
 - (C) `DELETE`
 - (D) `REMOVE`
 - (E) `STOP`
 - *Correct Answer: (C)*
-

6. What is the purpose of a `try...catch` block in backend code?

- (A) To write database queries.
 - (B) To define API routes.
 - (C) To handle errors gracefully without crashing the server.
 - (D) To loop through an array of data.
 - (E) To create a new function.
 - *Correct Answer: (C)*
-

7. What is a "foreign key" in a SQL database?

- (A) A key used to encrypt the database.
 - (B) The main ID for a table (that is a Primary Key).
 - (C) A field in one table that uniquely identifies a row of another table, linking them.
 - (D) A key that is not from your country.
 - (E) A type of index.
 - *Correct Answer: (C)*
-

8. What is the purpose of an **index in a database table?**

- (A) To significantly speed up data retrieval (queries) on specific columns.
 - (B) To define the data type of a column.
 - (C) To create a backup of the table.
 - (D) To provide a numbered list of all rows.
 - (E) To delete data from the table.
 - *Correct Answer: (A)*
-

9. How would you select only the **name column from a **users** table where the **age** is 30?**

- (A) `SELECT * FROM users WHERE age = 30;`
 - (B) `SELECT name FROM users WHERE age = 30;`
 - (C) `GET name FROM users WHERE age = 30;`
 - (D) `SELECT name FROM users (age = 30);`
 - (E) `SELECT name WHERE age = 30 FROM users;`
 - *Correct Answer: (B)*
-

10. What is "hashing" in the context of user passwords?

- (A) Encrypting a password so it can be decrypted later.

- (B) Hiding the password file.
 - (C) Storing the password in plain text.
 - (D) Running a one-way algorithm to turn a password into a unique, fixed-length string that cannot be reversed.
 - (E) Deleting the password after the user logs out.
 - *Correct Answer: (D)*
-

11. Why should you *never* store plain text passwords in a database?

- (A) It takes up too much storage space.
 - (B) It is slower to query.
 - (C) If the database is breached, an attacker will have all user passwords.
 - (D) It is not supported by SQL.
 - (E) It makes user login too fast.
 - *Correct Answer: (C)*
-

12. What is the **body of an HTTP **POST** request used for?**

- (A) To specify the URL of the API.
 - (B) To carry the data (e.g., JSON payload) being sent to the server.
 - (C) To store the HTTP status code.
 - (D) To store the authentication token.
 - (E) It is not used for anything.
 - *Correct Answer: (B)*
-

13. What is the **WHERE clause used for in a SQL query?**

- (A) To specify which table to query.
 - (B) To filter the results based on a condition.
 - (C) To sort the results.
 - (D) To join two tables.
 - (E) To specify which columns to return.
 - *Correct Answer: (B)*
-

14. What is the main difference between SQL and NoSQL databases?

- (A) SQL is free; NoSQL is paid.

- (B) SQL stores data in tables (structured); NoSQL is non-relational (e.g., document-based) and more flexible.
 - (C) SQL is for backend; NoSQL is for frontend.
 - (D) SQL is old; NoSQL is new.
 - (E) SQL is fast; NoSQL is slow.
 - *Correct Answer: (B)*
-

15. What is an "environment variable" (e.g., in a `.env` file)?

- (A) A variable that stores the weather.
 - (B) A variable that is hard-coded directly into the application logic.
 - (C) A configuration variable (e.g., API key, database password) stored *outside* the application code.
 - (D) A global variable in the frontend.
 - (E) A type of CSS variable.
 - *Correct Answer: (C)*
-

16. What is a `JOIN` clause used for in SQL?

- (A) To combine rows from two or more tables based on a related column.
 - (B) To join two different databases together.
 - (C) To add a new user to a table.
 - (D) To filter results.
 - (E) To connect to the database.
 - *Correct Answer: (A)*
-

17. How do you parse a JSON string named `jsonString` into a JavaScript object?

- (A) `JSON.parse(jsonString)`
 - (B) `JSON.stringify(jsonString)`
 - (C) `jsonString.toObject()`
 - (D) `new Object(jsonString)`
 - (E) `JSON.load(jsonString)`
 - *Correct Answer: (A)*
-

18. What does the HTTP status code `500` generally mean?

- (A) OK
 - (B) Not Found
 - (C) Internal Server Error (The server encountered an unexpected error).
 - (D) Unauthorized
 - (E) Bad Request
 - *Correct Answer: (C)*
-

19. What is "middleware" in a backend framework (like Express.js)?

- (A) A function that runs *in the middle* of a request-response cycle, often used for logging, authentication, or parsing.
 - (B) Software that is halfway between frontend and backend.
 - (C) The database.
 - (D) A type of testing software.
 - (E) A code editor.
 - *Correct Answer: (A)*
-

20. In the route `/users/123`, what is `123` typically referred to as?

- (A) A query parameter
 - (B) A path parameter (or route parameter)
 - (C) A header
 - (D) The request body
 - (E) A cookie
 - *Correct Answer: (B)*
-

Level 3: Advanced Scenarios

1. A user tries to sign up with an email that already exists in the database. What is the *most appropriate* HTTP status code to return?

- (A) 200 OK
- (B) 404 Not Found
- (C) 409 Conflict
- (D) 500 Internal Server Error
- (E) 301 Moved Permanently
- *Correct Answer: (C)*

2. What is a "JWT" (JSON Web Token)?

- (A) A way to store JSON files in a database.
- (B) A compact, secure way to represent a user's identity (authentication) that is passed between the client and server.
- (C) A JavaScript framework for building APIs.
- (D) A type of NoSQL database.
- (E) A testing tool for web tokens.
- *Correct Answer: (B)*

3. What is the main difference between `async/await` and using `.then()` callbacks for asynchronous operations?

- (A) `async/await` is older and slower; `.then()` is modern and faster.
- (B) `async/await` allows you to write asynchronous code that *looks* synchronous, making it easier to read.
- (C) `async/await` can handle errors, while `.then()` cannot.
- (D) `async/await` is for databases; `.then()` is for APIs.
- (E) There is no functional difference.
- *Correct Answer: (B)*

4. What is an "ORM" (Object-Relational Mapper) like Prisma, Sequelize, or Django ORM?

- (A) A tool that lets you write SQL queries using programming language objects and methods, instead of writing raw SQL.
- (B) A tool for optimizing database memory.
- (C) A type of database.
- (D) A monitoring tool for API traffic.
- (E) A hardware component of a server.
- *Correct Answer: (A)*

5. What is the "N+1 query problem"?

- (A) An error where you write one query, but it needs N+1 tables.
- (B) A performance issue where you execute one query to get N items, and then N *additional* queries to get related data for each item.
- (C) A security vulnerability.

- (D) The process of testing your API N+1 times.
 - (E) A good practice for optimizing databases.
 - *Correct Answer: (B)*
-

6. How do you write a SQL query to get all **posts** and the **name** of the user who wrote them, from **posts** and **users** tables?

- (A) `SELECT * FROM posts; SELECT * FROM users;`
 - (B) `SELECT * FROM posts MERGE users ON posts.user_id = users.id;`
 - (C) `SELECT posts.*, users.name FROM posts JOIN users ON posts.user_id = users.id;`
 - (D) `SELECT posts, users.name FROM posts, users;`
 - (E) `GET posts, users.name WHERE posts.user_id = users.id;`
 - *Correct Answer: (C)*
-

7. What is a "data model" in a NoSQL database like MongoDB?

- (A) The rigid schema of tables and columns.
 - (B) The physical server model.
 - (C) The structure of the data inside a "document" (e.g., a JSON object), defining its fields and types.
 - (D) A mathematical formula for querying data.
 - (E) A 3D model of the database.
 - *Correct Answer: (C)*
-

8. What is a primary advantage of "GraphQL" over "REST"?

- (A) It is much faster because it uses a different protocol than HTTP.
 - (B) It allows the client to request *exactly* the data it needs, in a single request, preventing over-fetching.
 - (C) It is older and more stable than REST.
 - (D) It is more secure by default.
 - (E) It can only be used with React frontends.
 - *Correct Answer: (B)*
-

9. What is the purpose of a **package.json** file in a Node.js project?

- (A) It contains the main application code.
 - (B) It stores all the API secrets and environment variables.
 - (C) It lists project metadata, dependencies (libraries), and scripts (e.g., `npm start`).
 - (D) It is the database schema.
 - (E) It is a log file for all API requests.
 - *Correct Answer: (C)*
-

10. What is a "database transaction"?

- (A) A single SQL query.
 - (B) A sequence of one or more database operations that are treated as a *single unit*: either they all succeed, or they all fail.
 - (C) The process of selling a database.
 - (D) A backup of the database.
 - (E) The financial cost of a database.
 - *Correct Answer: (B)*
-

11. What is "salting" a password, and why is it important in addition to hashing?

- (A) Adding a random string to each password *before* hashing it, to prevent "rainbow table" attacks.
 - (B) Deleting the password after a set time.
 - (C) Encrypting the entire database.
 - (D) Adding the username to the password before hashing.
 - (E) Storing passwords on a different "salty" server.
 - *Correct Answer: (A)*
-

12. What does `app.use(express.json())` do in an Express.js application?

- (A) It starts the server.
 - (B) It defines an API endpoint for JSON data.
 - (C) It is middleware that parses incoming requests with JSON payloads (e.g., from a POST request).
 - (D) It connects the application to a JSON-based database.
 - (E) It converts all outgoing responses to JSON.
 - *Correct Answer: (C)*
-

13. What is "CORS" (Cross-Origin Resource Sharing)?

- (A) A type of database.
 - (B) A backend programming language.
 - (C) A security mechanism in browsers that restricts a frontend in one domain from making requests to a backend in a *different* domain.
 - (D) A method for styling web pages.
 - (E) A testing framework.
 - *Correct Answer: (C)*
-

14. A user tries to log in with a correct email but an incorrect password. What is the *most appropriate* HTTP status code to return?

- (A) **404 Not Found** (This is insecure, as it reveals the email doesn't exist)
 - (B) **500 Internal Server Error**
 - (C) **200 OK** (with an error message in the body)
 - (D) **401 Unauthorized** or **400 Bad Request**
 - (E) **409 Conflict**
 - *Correct Answer: (D)*
-

15. How do you write a SQL query to get the *count* of all users in the **users table?**

- (A) **SELECT ALL users;**
 - (B) **SELECT COUNT(*) FROM users;**
 - (C) **COUNT users;**
 - (D) **SELECT users.length;**
 - (E) **SELECT * FROM users.count;**
 - *Correct Answer: (B)*
-

16. What is a "database migration"?

- (A) A set of scripts that manage and version-control the database schema (e.g., adding a table, changing a column).
 - (B) Moving the database from one server to another.
 - (C) A backup of the database.
 - (D) A tool for optimizing query speed.
 - (E) A type of SQL query.
 - *Correct Answer: (A)*
-

17. What is the main difference between the `PUT` and `PATCH` HTTP methods?

- (A) `PUT` updates data; `PATCH` deletes data.
 - (B) `PUT` is for new resources; `PATCH` is for existing resources.
 - (C) `PUT` replaces the *entire* resource; `PATCH` applies a *partial* update to the resource.
 - (D) `PUT` is secure; `PATCH` is not.
 - (E) There is no difference.
 - *Correct Answer: (C)*
-

18. What is "dependency injection"?

- (A) A security attack where a hacker injects malicious code.
 - (B) A design pattern where a component's dependencies (e.g., a database connection, a service) are "injected" from the outside rather than created internally.
 - (C) The process of installing libraries like `npm install`.
 - (D) A way to make your code dependent on a specific framework.
 - (E) A tool for managing environment variables.
 - *Correct Answer: (B)*
-

19. What is the difference between "authentication" and "authorization"?

- (A) Authentication is for new users; authorization is for existing users.
 - (B) Authentication is proving *who you are* (e.g., login); authorization is verifying *what you are allowed to do* (e.g., admin access).
 - (C) Authentication is for APIs; authorization is for databases.
 - (D) They are the same thing.
 - (E) Authentication is for the frontend; authorization is for the backend.
 - *Correct Answer: (B)*
-

20. What is a "webhook"?

- (A) An API endpoint that *you* call to get data.
- (B) A user's hook for catching fish.
- (C) An automated message (an HTTP callback) sent from one app to another *when an event happens* (e.g., a payment is completed).
- (D) A type of testing framework.
- (E) A security vulnerability.
- *Correct Answer: (C)*

Level 4: Expert & Specialization

1. What is "caching" and when would you use a tool like Redis?

- (A) Caching is storing copies of frequently accessed data (e.g., database results) in a temporary, fast-access location (like Redis) to reduce load and improve speed.
- (B) Caching is a way to log all API requests.
- (C) Caching is a tool for writing unit tests.
- (D) Caching is the process of hashing passwords.
- (E) Caching is a type of database, like SQL.
- *Correct Answer: (A)*

2. What is a "message queue" (e.g., RabbitMQ, Kafka)?

- (A) A database for storing chat messages.
- (B) A service that allows different parts of an application (e.g., microservices) to communicate *asynchronously* by sending messages to a queue.
- (C) A type of API.
- (D) A load balancer.
- (E) A tool for managing API documentation.
- *Correct Answer: (B)*

3. What is "Docker" and what problem does it solve for backend developers?

- (A) A text editor for backend code.
- (B) A tool that "containerizes" an application and its dependencies, ensuring it runs the same way everywhere (e.g., on a developer's laptop and in production).
- (C) A database management tool.
- (D) A code testing framework.
- (E) A brand of server hardware.
- *Correct Answer: (B)*

4. What is a "CI/CD pipeline"?

- (A) A database connection string.
- (B) An automated process (e.g., using Jenkins, GitHub Actions) that builds, tests, and deploys code changes.

- (C) A type of network protocol.
 - (D) A security feature for APIs.
 - (E) A way to manage user permissions.
 - *Correct Answer: (B)*
-

5. What is the main difference between a "monolithic" architecture and a "microservices" architecture?

- (A) A monolith is for frontend; microservices are for backend.
 - (B) A monolith is built in one large codebase; microservices break the application into many small, independent services.
 - (C) A monolith uses SQL; microservices use NoSQL.
 - (D) A monolith is modern; microservices are old.
 - (E) A monolith is secure; microservices are not.
 - *Correct Answer: (B)*
-

6. What is a "load balancer"?

- (A) A tool that checks for errors in your code.
 - (B) A server that distributes incoming network traffic across *multiple* backend servers to improve responsiveness and reliability.
 - (C) A database feature for balancing data.
 - (D) A security tool for blocking hackers.
 - (E) A tool for managing environment variables.
 - *Correct Answer: (B)*
-

7. What is "unit testing" in a backend context?

- (A) Testing the entire application from end to end.
 - (B) Manually testing the API with a tool like Postman.
 - (C) Testing a single, isolated "unit" of code (e.g., one function, one class) to ensure it works correctly.
 - (D) Testing the frontend user interface.
 - (E) A performance test to see how many units it can handle.
 - *Correct Answer: (C)*
-

8. What is "integration testing" in a backend context?

- (A) Testing a single function.
 - (B) Testing how multiple "units" or services work *together* (e.g., does the API route correctly call the database service?).
 - (C) Testing the user interface.
 - (D) A test to see if a new developer can integrate with the team.
 - (E) Manually testing the application.
 - *Correct Answer: (B)*
-

9. What is a "WebSocket" and how does it differ from HTTP?

- (A) It is a more secure version of HTTP.
 - (B) It is a protocol that allows for two-way, persistent communication between a client and server (ideal for real-time apps like chat).
 - (C) It is a tool for styling web pages.
 - (D) It is a database for storing web data.
 - (E) It is another name for a REST API.
 - *Correct Answer: (B)*
-

10. What is a "reverse proxy" (e.g., Nginx)?

- (A) A proxy that hides the *client's* identity.
 - (B) A server that sits *in front* of the main backend server, handling tasks like load balancing, SSL termination, and caching.
 - (C) A type of database.
 - (D) A tool for debugging code in reverse.
 - (E) A security attack.
 - *Correct Answer: (B)*
-

11. What is "rate limiting" and why is it important for a public API?

- (A) A way to make your API run faster.
 - (B) A process of limiting how many requests a user can make in a given time, to prevent abuse and ensure stability.
 - (C) A feature for ranking API users.
 - (D) A tool for testing API speed.
 - (E) A database optimization technique.
 - *Correct Answer: (B)*
-

12. What does "ACID" stand for in the context of SQL database transactions?

- (A) Atomicity, Consistency, Isolation, Durability
 - (B) Authenticate, Commit, Isolate, Deploy
 - (C) Asynchronous, Consistent, Integrated, Deployed
 - (D) A secure, coded, integrated database
 - (E) All, Combined, In, Data
 - *Correct Answer: (A)*
-

13. What is the "CAP Theorem" in distributed systems?

- (A) A theorem stating that a distributed system can only provide two of three guarantees: Consistency, Availability, and Partition tolerance.
 - (B) A security protocol for APIs.
 - (C) A rule for naming database tables.
 - (D) A theorem about CPU processing limits.
 - (E) A method for caching data.
 - *Correct Answer: (A)*
-

14. What is "database sharding"?

- (A) A way to encrypt a database.
 - (B) A scaling technique that splits a large database into smaller, faster, more manageable parts (shards).
 - (C) A security attack on a database.
 - (D) The process of deleting old data.
 - (E) A way to create a backup of a database.
 - *Correct Answer: (B)*
-

15. What is "Infrastructure as Code" (IaC) (e.g., Terraform, CloudFormation)?

- (A) Writing your backend code directly in assembly language.
 - (B) A "dummy" infrastructure used for testing.
 - (C) The process of managing and provisioning infrastructure (e.g., servers, databases, networks) through machine-readable definition files, rather than manual setup.
 - (D) A type of AI that writes code for you.
 - (E) A way to store your entire application in a single file.
 - *Correct Answer: (C)*
-

16. How would you *begin* to debug a memory leak in a Node.js application?

- (A) By restarting the server, as this permanently fixes memory leaks.
 - (B) By using a profiler or heapdump tool to take snapshots of the memory and analyze what objects are not being garbage collected.
 - (C) By deleting all `console.log` statements.
 - (D) By blaming the frontend team.
 - (E) By adding more RAM to the server.
 - *Correct Answer: (B)*
-

17. What is "gRPC"?

- (A) A modern, high-performance API framework (often used in microservices) that uses HTTP/2 and Protocol Buffers.
 - (B) A graphical tool for managing React projects.
 - (C) A type of SQL database.
 - (D) A security protocol for encrypting data.
 - (E) A text editor for backend developers.
 - *Correct Answer: (A)*
-

18. What is "serverless" computing (e.g., AWS Lambda, Google Cloud Functions)?

- (A) A backend with no servers, as all logic runs in the browser.
 - (B) A model where a cloud provider manages the server infrastructure, and you only provide the code (functions) which run on-demand.
 - (C) An old-fashioned way of programming from the 1980s.
 - (D) A server that has no operating system.
 - (E) A type of frontend framework.
 - *Correct Answer: (B)*
-

19. What is "database normalization"?

- (A) The process of making a database "normal" by deleting all strange data.
- (B) The process of structuring a SQL database to reduce data redundancy and improve data integrity.
- (C) A way to make a NoSQL database behave like a SQL database.
- (D) A scaling technique.
- (E) A security feature.
- *Correct Answer: (B)*

20. What is an "idempotent" operation in the context of an API?

- (A) An operation that is very slow and inefficient.
 - (B) An operation that has no effect on the database.
 - (C) An operation that can be called multiple times with the same input, but will only change the state *once*. (e.g., `DELETE /user/123`).
 - (D) An operation that creates a new resource every time it is called.
 - (E) An operation that is not secure.
 - *Correct Answer: (C)*
-

Level 5: Strategic & Architectural

1. Design a simple system to handle a "forgot password" flow. What are the key components and steps?

- (A) 1. User requests reset. 2. Backend generates a *single-use, expiring token*. 3. Token is stored (hashed) in the DB. 4. Send token in an email link. 5. User clicks link. 6. Backend validates token. 7. User provides new password.
 - (B) 1. User requests reset. 2. Backend sends the user's old password in an email.
 - (C) 1. User requests reset. 2. Backend generates a *permanent* token that never expires. 3. User provides new password.
 - (D) 1. User requests reset. 2. Backend temporarily deletes the user's account. 3. User signs up again.
 - (E) 1. User requests reset. 2. Backend asks the user for their new password immediately and saves it.
 - *Correct Answer: (A)*
-

2. Your application's API is experiencing slow read queries. What are the *first three* things you would investigate?

- (A) 1. The frontend CSS. 2. The user's internet speed. 3. The color of the buttons.
- (B) 1. Check for missing database indexes. 2. Check for "N+1 query" problems. 3. Analyze slow queries (e.g., `EXPLAIN ANALYZE`) to see if they can be optimized.
- (C) 1. Restart the server. 2. Restart the database. 3. Restart the user's computer.
- (D) 1. Blame the DevOps team. 2. Blame the network. 3. Blame the hardware.
- (E) 1. Immediately switch from SQL to NoSQL. 2. Rewrite the entire API in a different language. 3. Buy a faster server.
- *Correct Answer: (B)*

3. A startup asks you to build their backend. What factors influence your choice between a monolith and microservices?

- (A) A monolith is always better because it's simpler.
- (B) Microservices are always better because they are "modern."
- (C) The choice depends on team size, project complexity, and scalability needs. A monolith is faster to start; microservices are more flexible for large, complex systems.
- (D) The choice depends on the frontend framework (e.g., React requires microservices).
- (E) The choice depends on the color of the startup's logo.
- *Correct Answer: (C)*

4. What is a "SQL injection" attack, and what is the *best* way to prevent it?

- (A) An attack where a user inputs malicious SQL code into a form field to dump the database; prevented by using prepared statements (parameterized queries).
- (B) An attack where a user "injects" a new server; prevented by a firewall.
- (C) An attack where a user "injects" JavaScript; prevented by hashing passwords.
- (D) An attack where a user "injects" SQL code; prevented by validating user input on the *frontend*.
- (E) An attack where a user steals the SQL database file; prevented by rate limiting.
- *Correct Answer: (A)*

5. Design a simple URL shortener service (like bit.ly). What would the database schema and API endpoints look like?

- (A) A `urls` table with `id`, `original_url`, and `short_code`. Endpoints: `POST /shorten` and `GET /:short_code`.
- (B) A `users` table only. Endpoint: `GET /users`.
- (C) A `urls` table with the full HTML of the original page. Endpoint: `POST /copy_page`.
- (D) A `urls` table with `short_code` only. Endpoints: `GET /all_urls` and `DELETE /all_urls`.
- (E) No database needed; just use a large JSON file.
- *Correct Answer: (A)*

6. What is "horizontal scaling" (scaling out) vs. "vertical scaling" (scaling up)?

- (A) Horizontal is adding more code; vertical is adding more features.

- (B) Horizontal is adding *more machines* (e.g., more servers); vertical is adding *more power* (e.g., more RAM/CPU) to one machine.
 - (C) Horizontal is for SQL; vertical is for NoSQL.
 - (D) Horizontal is for frontend; vertical is for backend.
 - (E) Horizontal is fast; vertical is slow.
 - *Correct Answer: (B)*
-

7. What is "event-driven architecture"?

- (A) An architecture based on user click events.
 - (B) A system where services communicate by producing and consuming "events" (messages about a state change), often via a message queue.
 - (C) A monolith architecture.
 - (D) An architecture that only runs during special events.
 - (E) A type of database.
 - *Correct Answer: (B)*
-

8. You need to process millions of image uploads, resize them into 3 different formats, and store them. How would you design this system to be scalable and responsive?

- (A) Do all the resizing in the API request *synchronously* and make the user wait.
 - (B) 1. The API receives the image and immediately uploads it to a storage bucket (e.g., S3). 2. It sends a "resize" job to a message queue. 3. A separate "worker" service reads from the queue, does the resizing, and saves the new versions.
 - (C) Store all the images directly in the SQL database.
 - (D) Tell the user to resize the images on their computer *before* uploading.
 - (E) Use a serverless function for the API, but do the resizing in the same function.
 - *Correct Answer: (B)*
-

9. What is "database replication" (master-slave) and what is its primary benefit?

- (A) The process of creating multiple copies (replicas) of a database. It improves read performance and provides high availability.
 - (B) A security attack where a hacker replicates your database.
 - (C) A "master" database that controls all the "slave" APIs.
 - (D) A way to delete data from a database.
 - (E) A method for encrypting the database.
 - *Correct Answer: (A)*
-

10. What is a "blue-green deployment" strategy?

- (A) A deployment where you use blue and green servers.
 - (B) A strategy where you have two identical production environments ("blue" and "green"). You deploy to the "green" environment, test it, and then switch traffic from "blue" (old) to "green" (new).
 - (C) A deployment that is very "green" (environmentally friendly).
 - (D) A strategy where you deploy only on Tuesdays.
 - (E) A strategy where you test code in the "blue" environment and deploy in the "green" one.
 - *Correct Answer: (B)*
-

11. What is "OAuth 2.0"?

- (A) A specific username and password for an API.
 - (B) A type of database.
 - (C) An open standard for *delegated authorization*. It allows users to grant a third-party app limited access to their resources (e.g., "Sign in with Google").
 - (D) A new version of HTTP.
 - (E) A security attack.
 - *Correct Answer: (C)*
-

12. How would you design the database tables for a "friends list" feature in a social media app?

- (A) One `users` table with a column called `friends` that stores a (bad) comma-separated list.
 - (B) A `users` table and a `friendships` table. `friendships` has `user_id_1`, `user_id_2`, and `status` (e.g., 'pending', 'accepted').
 - (C) A `users` table and a `friends` table, where `friends` has every possible combination of users.
 - (D) Just one `users` table and no way to link them.
 - (E) A `friends` table with `user_name` and `friend_name` (bad, as names can change).
 - *Correct Answer: (B)*
-

13. What are the key trade-offs of using GraphQL instead of REST for a new, large-scale project?

- (A) REST is always better because it's simpler.

- (B) GraphQL is always better because it's newer.
 - (C) GraphQL gives frontend flexibility (no over-fetching), but adds backend complexity (schema, resolvers). REST is simple and well-understood, but can lead to many endpoints.
 - (D) REST is for frontend, GraphQL is for backend.
 - (E) GraphQL doesn't work with SQL databases.
 - *Correct Answer: (C)*
-

14. What is a "Distributed Denial-of-Service" (DDoS) attack and what are common mitigation strategies?

- (A) An attack where one person steals all the data. Mitigated by hashing passwords.
 - (B) An attack where a service is flooded with traffic from many sources to overwhelm it. Mitigated by services like Cloudflare, rate limiting, and firewalls.
 - (C) An attack where a user "denies" they made a request. Mitigated by logging.
 - (D) An attack where the database is deleted. Mitigated by backups.
 - (E) An attack where the server is physically stolen.
 - *Correct Answer: (B)*
-

15. What is the "eventual consistency" model in distributed systems?

- (A) A model where the system is "eventually" correct, but is wrong most of the time.
 - (B) A model guaranteeing that *if no new updates are made*, all replicas of a database will *eventually* converge to the same value.
 - (C) A model where the system is "consistent" during events.
 - (D) A model where data is *never* consistent.
 - (E) A model where data is *always* 100% consistent, immediately (that is "strong consistency").
 - *Correct Answer: (B)*
-

16. You are designing a new, public-facing REST API. What are the *three* most important security concerns to address from day one?

- (A) 1. Authentication (who is this?), 2. Authorization (what can they do?), and 3. Input Validation (e.g., prevent SQL injection).
- (B) 1. The color scheme, 2. The API documentation, 3. The server's brand name.
- (C) 1. Making it as fast as possible, 2. Making it as cheap as possible, 3. Using a cool name.
- (D) 1. Hashing passwords, 2. Hashing passwords again, 3. Hashing passwords a third time.

- (E) 1. Making all endpoints **GET** requests, 2. Storing all data in one file, 3. Never using a database.
 - *Correct Answer: (A)*
-

17. What are the key performance indicators (KPIs) you would monitor for a high-traffic backend service?

- (A) 1. Error Rate (e.g., 5xx, 4xx), 2. Latency (response time), 3. Throughput (requests per second).
 - (B) 1. Number of code comments, 2. Number of developers, 3. Number of meetings.
 - (C) 1. The frontend's Lighthouse score.
 - (D) 1. How many new users sign up per day.
 - (E) 1. The total cost of the server.
 - *Correct Answer: (A)*
-

18. What is a "canary release"?

- (A) A release where you only deploy the code to servers in the Canary Islands.
 - (B) A strategy where you release the new version to a *small subset* of users (e.g., 5%), monitor for errors, and then gradually roll it out to everyone.
 - (C) A "dead" release (like a canary in a coal mine) that fails.
 - (D) A release that is very "loud" and announced to all users.
 - (E) Another name for a "blue-green" deployment.
 - *Correct Answer: (B)*
-

19. What is a "service-oriented architecture" (SOA)?

- (A) An older architectural pattern (a precursor to microservices) where applications are built from a collection of distinct, loosely-coupled "services."
 - (B) An architecture where you *only* use "serverless" functions.
 - (C) An architecture where every developer is focused on "customer service."
 - (D) A monolithic architecture.
 - (E) A type of hardware.
 - *Correct Answer: (A)*
-

20. Your API is used by a mobile app with unreliable internet. What strategies can you implement on the *backend* to support this?

- (A) Tell the mobile app user to get better internet.
- (B) Make all API endpoints idempotent, support data compression, and design the API to allow for partial or "delta" updates.
- (C) Send all data as large, uncompressed video files.
- (D) Block all users with slow internet connections.
- (E) Re-build the entire backend on the user's phone.
- *Correct Answer: (B)*

Data Science track.

Level 1: Basic Foundations

1. What is Data Science?

- (A) The process of designing and styling websites.
 - (B) An interdisciplinary field that uses scientific methods, processes, algorithms, and systems to extract knowledge and insights from data.
 - (C) The process of setting up and managing computer networks.
 - (D) The process of writing mobile applications.
 - (E) The process of securing data from hackers.
 - *Correct Answer: (B)*
-

2. What is "data"?

- (A) A collection of facts, such as numbers, words, measurements, observations, or just descriptions of things.
 - (B) A computer program.
 - (C) A visual design for a website.
 - (D) A server in the cloud.
 - (E) A type of programming language.
 - *Correct Answer: (A)*
-

3. What is the difference between "structured" and "unstructured" data?

- (A) Structured data is fast; unstructured data is slow.
- (B) Structured data is organized in a defined format (like a table); unstructured data has no predefined format (like text or images).
- (C) Structured data is in English; unstructured data is in other languages.
- (D) Structured data is numbers; unstructured data is text.

- (E) There is no difference.
 - *Correct Answer: (B)*
-

4. What is "Machine Learning"?

- (A) The process of teaching humans how to use computers.
 - (B) A field of AI that gives computers the ability to learn from data without being explicitly programmed.
 - (C) A type of computer hardware.
 - (D) The process of manually inputting data into a spreadsheet.
 - (E) A type of database.
 - *Correct Answer: (B)*
-

5. Which of the following is a popular programming language for Data Science?

- (A) HTML
 - (B) CSS
 - (C) Python
 - (D) SASS
 - (E) Swift
 - *Correct Answer: (C)*
-

6. What is the "mean" of a set of numbers?

- (A) The middle value in the set.
 - (B) The most frequent value in the set.
 - (C) The sum of all values divided by the number of values (the average).
 - (D) The highest value in the set.
 - (E) The lowest value in the set.
 - *Correct Answer: (C)*
-

7. What is the "median" of a set of numbers?

- (A) The middle value when the set is sorted in order.
- (B) The most frequent value in the set.
- (C) The average of the set.
- (D) The highest value in the set.
- (E) The sum of all the values.

- *Correct Answer: (A)*
-

8. What is the "mode" of a set of numbers?

- (A) The middle value in the set.
 - (B) The value that appears most frequently in the set.
 - (C) The average of the set.
 - (D) The highest value in the set.
 - (E) The difference between the highest and lowest values.
 - *Correct Answer: (B)*
-

9. What is a "feature" in a dataset?

- (A) An entire row of data.
 - (B) The name of the dataset.
 - (C) An individual measurable property or characteristic of a phenomenon being observed (e.g., a column in a table).
 - (D) A bug in the data.
 - (E) The final prediction.
 - *Correct Answer: (C)*
-

10. What is a "target variable" (or "label")?

- (A) The dataset's title.
 - (B) A feature that is not important.
 - (C) The outcome or value that you are trying to predict with a machine learning model.
 - (D) The first column in a dataset.
 - (E) A variable that is deleted.
 - *Correct Answer: (C)*
-

11. What is "data visualization"?

- (A) The process of writing code.
- (B) The graphical representation of data (e.g., charts, graphs, maps).
- (C) The process of storing data.
- (D) The process of cleaning data.
- (E) A type of machine learning model.
- *Correct Answer: (B)*

12. What is a "histogram"?

- (A) A graph that shows the relationship between two variables.
- (B) A graph that shows the distribution of a single numerical variable by grouping numbers into "bins".
- (C) A pie chart.
- (D) A type of machine learning algorithm.
- (E) A graph that shows data over time.
- *Correct Answer: (B)*

13. What is a "scatter plot"?

- (A) A graph used to display values for two different variables, represented as points.
- (B) A graph showing the distribution of one variable.
- (C) A chart showing percentages of a whole.
- (D) A graph that shows data over time.
- (E) A map.
- *Correct Answer: (A)*

14. What is a "database"?

- (A) A visual chart.
- (B) An organized collection of data, generally stored and accessed electronically from a computer system.
- (C) A machine learning model.
- (D) A Python library.
- (E) A single text file.
- *Correct Answer: (B)*

15. What is "SQL"?

- (A) A programming language for building websites.
- (B) A programming language used to manage and query data stored in relational databases.
- (C) A type of data visualization.
- (D) A machine learning algorithm.
- (E) A data science library in Python.
- *Correct Answer: (B)*

16. What does a "bar chart" represent?

- (A) The distribution of a numerical variable.
- (B) The relationship between two numerical variables.
- (C) A comparison of values for different categories (categorical data).
- (D) Data changing over time.
- (E) Percentages of a whole.
- *Correct Answer: (C)*

17. What is "data cleaning"?

- (A) The process of visualizing data.
- (B) The process of building a machine learning model.
- (C) The process of detecting and correcting (or removing) corrupt or inaccurate records from a dataset.
- (D) The process of deleting old data.
- (E) The process of storing data in a database.
- *Correct Answer: (C)*

18. What is a "Jupyter Notebook"?

- (A) A type of machine learning model.
- (B) A database.
- (C) An interactive web tool used by data scientists to write and run code, visualize data, and share results.
- (D) A physical notebook for writing down data.
- (E) A Python library for data cleaning.
- *Correct Answer: (C)*

19. What is "Python"?

- (A) A type of snake.
- (B) A high-level, general-purpose programming language widely used in data science.
- (C) A database.
- (D) A specific type of chart.
- (E) A data visualization tool.
- *Correct Answer: (B)*

20. What is a "CSV" file?

- (A) A type of database.
 - (B) A "Comma-Separated Values" file; a plain text file that stores tabular data.
 - (C) A machine learning model.
 - (D) A Python program.
 - (E) An image file.
 - *Correct Answer: (B)*
-

Level 2: Intermediate Application

1. What is the key difference between "Classification" and "Regression"?

- (A) Classification predicts a continuous value (e.g., price); Regression predicts a category (e.g., yes/no).
 - (B) Classification predicts a category (e.g., "spam" or "not spam"); Regression predicts a continuous value (e.g., "price").
 - (C) Classification is for text; Regression is for numbers.
 - (D) Classification is simple; Regression is complex.
 - (E) There is no difference.
 - *Correct Answer: (B)*
-

2. What is "overfitting" in machine learning?

- (A) When a model performs very well on new, unseen data.
 - (B) When a model is too simple to capture the underlying trend in the data.
 - (C) When a model learns the *noise* and details in the training data so well that it performs poorly on new, unseen data.
 - (D) The process of fitting a model to a dataset.
 - (E) When a model has too many features.
 - *Correct Answer: (C)*
-

3. What is "underfitting" in machine learning?

- (A) When a model performs too well on the training data.
- (B) When a model is too complex and learns the noise.

- (C) When a model is too simple to capture the underlying pattern of the data, and performs poorly on both training and test data.
 - (D) The process of putting a model under a dataset.
 - (E) When a model has too few features.
 - *Correct Answer: (C)*
-

4. What is the purpose of a "train/test split"?

- (A) To make the dataset smaller and easier to use.
 - (B) To have two different datasets for two different models.
 - (C) To train the model on one subset of data ("train set") and evaluate its performance on a separate, unseen subset ("test set").
 - (D) To combine two datasets into one.
 - (E) To clean the data.
 - *Correct Answer: (C)*
-

5. What is a "Decision Tree" algorithm?

- (A) A model that uses a tree-like graph of decisions and their possible consequences.
 - (B) A model that draws a straight line through data points.
 - (C) A model for finding groups (clusters) in data.
 - (D) A model for visualizing data.
 - (E) A type of database.
 - *Correct Answer: (A)*
-

6. What is "Linear Regression"?

- (A) An algorithm for classification.
 - (B) An algorithm that attempts to model the relationship between variables by fitting a linear equation (a straight line) to the observed data.
 - (C) An algorithm for finding hidden groups in data.
 - (D) A data visualization technique.
 - (E) A method for cleaning data.
 - *Correct Answer: (B)*
-

7. What is "feature engineering"?

- (A) The process of building a machine learning model.

- (B) The process of visualizing data.
 - (C) The process of using domain knowledge to create, select, or transform features (variables) to make a model perform better.
 - (D) The process of debugging a model.
 - (E) The process of splitting data.
 - *Correct Answer: (C)*
-

8. What is the "K-Nearest Neighbors" (k-NN) algorithm?

- (A) An algorithm that classifies a new data point based on the majority class of its "k" closest neighbors.
 - (B) An algorithm that fits a straight line to data.
 - (C) An algorithm that builds a tree of decisions.
 - (D) An algorithm for finding clusters.
 - (E) A method for cleaning data.
 - *Correct Answer: (A)*
-

9. What is a common way to handle "missing values" in a dataset?

- (A) Delete the entire dataset and start over.
 - (B) Replace the missing values with the mean, median, or mode of the column.
 - (C) Ignore them; models can handle missing values automatically.
 - (D) Replace them all with the number zero.
 - (E) Both B and D are valid approaches.
 - *Correct Answer: (E)*
-

10. What is "accuracy" as a model evaluation metric?

- (A) The total number of predictions made.
 - (B) The number of correct predictions divided by the total number of predictions.
 - (C) The number of incorrect predictions.
 - (D) How fast the model is.
 - (E) The number of features used.
 - *Correct Answer: (B)*
-

11. What is the "Pandas" library in Python?

- (A) A library for building websites.

- (B) A library for building machine learning models.
 - (C) A library for data manipulation and analysis, providing data structures like the "DataFrame".
 - (D) A library for data visualization.
 - (E) A library for database connections.
 - *Correct Answer: (C)*
-

12. What is the "NumPy" library in Python?

- (A) A library for creating charts.
 - (B) A fundamental library for numerical computing, providing support for large, multi-dimensional arrays and matrices.
 - (C) A library for web development.
 - (D) A library for data cleaning.
 - (E) A library for machine learning.
 - *Correct Answer: (B)*
-

13. What is the "Scikit-learn" (sklearn) library in Python?

- (A) A popular, easy-to-use library featuring a wide range of machine learning algorithms and tools.
 - (B) A library for data visualization.
 - (C) A library for working with text data.
 - (D) A library for database management.
 - (E) A library for data manipulation (like Pandas).
 - *Correct Answer: (A)*
-

14. What is a "correlation coefficient" (e.g., Pearson's r)?

- (A) A measure of how important a feature is.
 - (B) A statistical measure of the strength and direction of a linear relationship between two variables (from -1 to 1).
 - (C) The average of a variable.
 - (D) The accuracy of a model.
 - (E) A metric for cleaning data.
 - *Correct Answer: (B)*
-

15. What does a correlation coefficient of +0.9 suggest?

- (A) A strong negative linear relationship.
 - (B) A strong positive linear relationship.
 - (C) No relationship.
 - (D) A weak positive linear relationship.
 - (E) A weak negative linear relationship.
 - *Correct Answer: (B)*
-

16. What does a correlation coefficient of 0 suggest?

- (A) A strong positive linear relationship.
 - (B) A strong negative linear relationship.
 - (C) A perfect relationship.
 - (D) No *linear* relationship between the variables.
 - (E) The data is perfectly correlated.
 - *Correct Answer: (D)*
-

17. What is "categorical" data?

- (A) Data that is numerical (e.g., age, price).
 - (B) Data that can be divided into distinct groups or categories (e.g., "color", "city").
 - (C) Data that is missing.
 - (D) Data that is in a text document.
 - (E) Data that is an image.
 - *Correct Answer: (B)*
-

18. What is "one-hot encoding"?

- (A) A method for visualizing data.
 - (B) A process of converting categorical string variables into a numerical format (binary vectors) that machine learning models can understand.
 - (C) A method for cleaning data.
 - (D) A type of machine learning model.
 - (E) A way to make models run hotter and faster.
 - *Correct Answer: (B)*
-

19. What is a "Pandas DataFrame"?

- (A) A 3D data structure.

- (B) A 1D array of data.
 - (C) A 2D labeled data structure with columns of potentially different types, similar to a spreadsheet or SQL table.
 - (D) A machine learning model.
 - (E) A plot or chart.
 - *Correct Answer: (C)*
-

20. What is "Matplotlib" in Python?

- (A) A popular library for creating static, animated, and interactive data visualizations.
 - (B) A library for machine learning.
 - (C) A library for data manipulation.
 - (D) A library for web development.
 - (E) A library for database access.
 - *Correct Answer: (A)*
-

Level 3: Advanced Scenarios

1. What is the difference between "Supervised" and "Unsupervised" learning?

- (A) Supervised learning uses labeled data (features and targets); Unsupervised learning uses unlabeled data (features only) to find patterns.
 - (B) Supervised learning is for regression; Unsupervised learning is for classification.
 - (C) Supervised learning is complex; Unsupervised learning is simple.
 - (D) Supervised learning requires a "supervisor" (a human); Unsupervised does not.
 - (E) Supervised learning is for numbers; Unsupervised learning is for text.
 - *Correct Answer: (A)*
-

2. Which of the following is an example of an "Unsupervised" learning task?

- (A) Predicting house prices based on features (Linear Regression).
 - (B) Classifying emails as "spam" or "not spam" (Classification).
 - (C) Grouping customers into different segments based on their purchase behavior (Clustering).
 - (D) Predicting if a customer will default on a loan (Classification).
 - (E) Both A and B.
 - *Correct Answer: (C)*
-

3. What is the "K-Means" algorithm?

- (A) A popular "clustering" algorithm (unsupervised) that groups data into 'K' number of clusters.
 - (B) A "classification" algorithm (supervised) for predicting categories.
 - (C) A "regression" algorithm (supervised) for predicting values.
 - (D) A method for cleaning data.
 - (E) A method for feature engineering.
 - *Correct Answer: (A)*
-

4. What is a "confusion matrix"?

- (A) A table used to evaluate the performance of a classification model, showing True Positives, True Negatives, False Positives, and False Negatives.
 - (B) A matrix that is very confusing to read.
 - (C) A chart for visualizing regression results.
 - (D) A data structure in Pandas.
 - (E) A method for clustering data.
 - *Correct Answer: (A)*
-

5. What is "Precision" in a confusion matrix?

- (A) Of all the *actual* positive cases, how many did the model predict correctly?
 - (B) Of all the cases the model predicted as *positive*, how many were *actually* positive?
 - (C) The overall accuracy of the model.
 - (D) The number of True Positives.
 - (E) The number of False Negatives.
 - *Correct Answer: (B)*
-

6. What is "Recall" in a confusion matrix?

- (A) Of all the *actual* positive cases, how many did the model predict correctly?
 - (B) Of all the cases the model predicted as *positive*, how many were *actually* positive?
 - (C) The overall accuracy of the model.
 - (D) The number of True Negatives.
 - (E) The number of False Positives.
 - *Correct Answer: (A)*
-

7. When would you prioritize "Recall" over "Precision"?

- (A) In an email spam filter (you don't want to miss important emails).
 - (B) In a medical diagnosis for a serious disease (it's more important to find all actual cases, even if you get some false positives).
 - (C) When the cost of a false positive is very high.
 - (D) When the cost of a false negative is very low.
 - (E) When the dataset is perfectly balanced.
 - *Correct Answer: (B)*
-

8. What is the "F1-Score"?

- (A) The average of Precision and Recall.
 - (B) The harmonic mean of Precision and Recall, providing a single score that balances both.
 - (C) The same as Accuracy.
 - (D) The product of Precision and Recall.
 - (E) A measure of how fast a model is.
 - *Correct Answer: (B)*
-

9. What is "K-Fold Cross-Validation"?

- (A) A method for training a model K times.
 - (B) A technique where the training data is split into 'K' folds; the model is trained on K-1 folds and tested on the remaining fold, rotating K times.
 - (C) A method for splitting data into a train and test set just once.
 - (D) A type of clustering algorithm.
 - (E) A method for visualizing data in K dimensions.
 - *Correct Answer: (B)*
-

10. What is "hyperparameter tuning"?

- (A) The process of the model learning its parameters (like weights) from the data.
- (B) The process of finding the optimal "settings" (hyperparameters, like 'K' in k-NN) for a model to improve its performance.
- (C) The process of cleaning the data.
- (D) The process of tuning a server to be "hyper" fast.
- (E) The process of visualizing high-dimensional data.
- *Correct Answer: (B)*

11. What is "Grid Search"?

- (A) A method for searching a grid-like dataset.
- (B) A common hyperparameter tuning technique that exhaustively tries every combination of specified hyperparameter values.
- (C) A data visualization technique.
- (D) A method for cleaning data.
- (E) A type of database query.
- *Correct Answer: (B)*

12. What is a "Random Forest"?

- (A) An ensemble machine learning model that operates by building a multitude of Decision Trees and outputting their combined (e.g., average or mode) prediction.
- (B) A single, very deep Decision Tree.
- (C) A clustering algorithm.
- (D) A linear regression model.
- (E) A visualization of a forest.
- *Correct Answer: (A)*

13. What is "dimensionality reduction"?

- (A) The process of adding more features (dimensions) to a dataset.
- (B) The process of reducing the number of features (dimensions) in a dataset, while trying to retain as much information as possible.
- (C) The process of visualizing data in 2D or 3D.
- (D) A method for data cleaning.
- (E) A way to make a model more complex.
- *Correct Answer: (B)*

14. What is "Principal Component Analysis" (PCA)?

- (A) A popular unsupervised algorithm for dimensionality reduction.
- (B) A popular algorithm for classification.
- (C) A popular algorithm for regression.
- (D) A method for data cleaning.
- (E) A method for hyperparameter tuning.
- *Correct Answer: (A)*

15. In Pandas, what does the `.groupby()` method do?

- (A) It sorts the DataFrame by a specific column.
- (B) It deletes a group of columns.
- (C) It splits the data into groups based on some criteria (e.g., a column value) so you can perform aggregate functions (like mean, sum) on them.
- (D) It creates a new column.
- (E) It joins two DataFrames together.
- *Correct Answer: (C)*

16. In Pandas, what does the `.merge()` method do?

- (A) It calculates the mean of a column.
- (B) It splits the DataFrame into groups.
- (C) It combines two DataFrames based on a common column, similar to a SQL JOIN.
- (D) It deletes rows with missing values.
- (E) It plots the data.
- *Correct Answer: (C)*

17. What is "Natural Language Processing" (NLP)?

- (A) A field of AI that helps computers understand, interpret, and generate human language.
- (B) A method for visualizing text data.
- (C) A type of SQL database.
- (D) A programming language that is "natural" to read.
- (E) A data cleaning technique.
- *Correct Answer: (A)*

18. What is "tokenization" in NLP?

- (A) The process of converting text into a secret code.
 - (B) The process of visualizing text.
 - (C) The process of breaking down a piece of text into smaller units, such as words or sentences (called "tokens").
 - (D) The process of checking for spelling errors.
 - (E) A machine learning model for text.
 - *Correct Answer: (C)*
-

19. What is "TF-IDF"?

- (A) A type of machine learning model.
 - (B) A numerical statistic that reflects how important a word is to a document in a collection (corpus).
 - (C) A data visualization technique.
 - (D) A Python library for NLP.
 - (E) A method for cleaning text data.
 - *Correct Answer: (B)*
-

20. What is "Logistic Regression"?

- (A) A machine learning algorithm used for *classification* problems (it predicts a probability).
 - (B) A machine learning algorithm used for *regression* problems.
 - (C) A machine learning algorithm used for *clustering*.
 - (D) A method for data cleaning.
 - (E) A method for visualizing data.
 - *Correct Answer: (A)*
-

Level 4: Expert & Specialization

1. What is "Gradient Boosting" (e.g., XGBoost, LightGBM)?

- (A) An ensemble learning technique that builds models (typically decision trees) sequentially, where each new model corrects the errors of the previous ones.
 - (B) A type of Linear Regression.
 - (C) A clustering algorithm.
 - (D) A method for dimensionality reduction.
 - (E) A type of neural network.
 - *Correct Answer: (A)*
-

2. What is a "Neural Network"?

- (A) A computing system inspired by the biological neural networks that constitute animal brains, used for complex pattern recognition.
- (B) A data visualization technique.
- (C) A data cleaning method.
- (D) A type of SQL database.

- (E) A data structure in Python.
 - *Correct Answer: (A)*
-

3. What is "Deep Learning"?

- (A) A subfield of machine learning based on artificial neural networks with *multiple* layers (deep architectures).
 - (B) A method for learning about "deep" topics.
 - (C) A type of data visualization.
 - (D) A clustering algorithm.
 - (E) A data cleaning technique.
 - *Correct Answer: (A)*
-

4. What is a "hyperparameter"?

- (A) A parameter *learned by the model* during training (e.g., weights in a linear regression).
 - (B) A parameter *set by the data scientist before* training (e.g., the 'K' in K-Means).
 - (C) A very "hyper" or unstable parameter.
 - (D) The target variable.
 - (E) A feature in the dataset.
 - *Correct Answer: (B)*
-

5. What is the main problem with using "accuracy" as a metric for an *imbalanced dataset*?

- (A) It is too complex to calculate.
 - (B) It can be misleadingly high if the model simply predicts the majority class all the time.
 - (C) It only works for regression, not classification.
 - (D) It cannot be used for deep learning models.
 - (E) It is not a well-known metric.
 - *Correct Answer: (B)*
-

6. What are "SMOTE" (Synthetic Minority Over-sampling Technique)?

- (A) A technique for handling imbalanced datasets by *over-sampling* the minority class (creating synthetic new samples).
- (B) A technique for *under-sampling* the majority class.

- (C) A type of machine learning model.
 - (D) A data visualization technique.
 - (E) A method for feature engineering.
 - *Correct Answer: (A)*
-

7. What is the "curse of dimensionality"?

- (A) A phenomenon where having too many features (dimensions) makes data sparse, models harder to train, and performance worse.
 - (B) The difficulty of visualizing data in more than 3 dimensions.
 - (C) A type of error in a neural network.
 - (D) A problem with clustering algorithms only.
 - (E) A problem with regression models only.
 - *Correct Answer: (A)*
-

8. What is "A/B Testing"?

- (A) A statistical method of comparing two versions (A and B) of something (e.g., a website) to determine which one performs better.
 - (B) A type of machine learning model.
 - (C) A method for cleaning data.
 - (D) A data visualization technique.
 - (E) A method for hyperparameter tuning.
 - *Correct Answer: (A)*
-

9. What is a "p-value" in statistical hypothesis testing?

- (A) The probability that the model is correct.
 - (B) The probability of observing the data (or something more extreme) *if the null hypothesis is true*.
 - (C) The "precision-value" of a model.
 - (D) The impact of a feature.
 - (E) The accuracy of a test.
 - *Correct Answer: (B)*
-

10. What is a "Time Series" dataset?

- (A) A dataset where values are recorded in alphabetical order.

- (B) A dataset where values are recorded at different time intervals (e.g., daily stock prices, monthly sales).
 - (C) A dataset with no order.
 - (D) A dataset used for clustering.
 - (E) A dataset used for NLP.
 - *Correct Answer: (B)*
-

11. What is "feature scaling" (e.g., Standardization, Normalization)?

- (A) The process of adding more features.
 - (B) The process of removing features.
 - (C) The process of transforming numerical features to a common scale, which is important for algorithms like k-NN or SVM.
 - (D) The process of one-hot encoding.
 - (E) The process of visualizing features.
 - *Correct Answer: (C)*
-

12. What is the main difference between "Standardization" and "Normalization"?

- (A) Standardization scales data to a mean of 0 and std dev of 1; Normalization scales data to a range (e.g., 0 to 1).
 - (B) Standardization scales data to a range (e.g., 0 to 1); Normalization scales data to a mean of 0 and std dev of 1.
 - (C) Standardization is for text; Normalization is for numbers.
 - (D) Standardization is for regression; Normalization is for classification.
 - (E) There is no difference.
 - *Correct Answer: (A)*
-

13. What is an "ROC Curve" (Receiver Operating Characteristic)?

- (A) A plot that shows the performance of a classification model by plotting the True Positive Rate against the False Positive Rate at various threshold settings.
 - (B) A plot for regression models.
 - (C) A plot for clustering models.
 - (D) A data visualization for time series.
 - (E) A type of "rock" music for data scientists.
 - *Correct Answer: (A)*
-

14. What does "AUC" (Area Under the Curve) represent for an ROC curve?

- (A) The area of the dataset.
 - (B) A single number (from 0 to 1) that summarizes the model's ability to distinguish between positive and negative classes. A higher AUC is better.
 - (C) The speed of the model.
 - (D) The complexity of the model.
 - (E) The amount of data used.
 - *Correct Answer: (B)*
-

15. What is "TensorFlow"?

- (A) A popular open-source library developed by Google, primarily used for deep learning and neural networks.
- (B) A data manipulation library like Pandas.
- (C) A data visualization library.
- (D) A database.
- (E) A clustering algorithm.
- *Correct Answer: (A)*

16. What is "PyTorch"?

- (A) An open-source machine learning library (developed by Facebook) known for its flexibility and use in research, especially for deep learning.
 - (B) A data manipulation library.
 - (C) A data visualization library.
 - (D) A database.
 - (E) A regression algorithm.
 - *Correct Answer: (A)*
-

17. What is "word embedding" (e.g., Word2Vec, GloVe)?

- (A) A visualization of words.
 - (B) A technique in NLP where words are represented as dense numerical vectors in a multi-dimensional space.
 - (C) A method for counting words.
 - (D) A method for checking spelling.
 - (E) A type of machine learning model.
 - *Correct Answer: (B)*
-

18. What is a "Support Vector Machine" (SVM)?

- (A) A supervised learning model that finds an optimal "hyperplane" to separate data into different classes.
 - (B) A clustering algorithm.
 - (C) A deep learning model.
 - (D) A database.
 - (E) A data visualization tool.
 - *Correct Answer: (A)*
-

19. What is "bias" in the context of a machine learning model?

- (A) The error introduced by approximating a real-world problem with a model that is too simple (leading to underfitting).
 - (B) The model's sensitivity to small fluctuations in the training data (leading to overfitting).
 - (C) The model's preference for one outcome over another.
 - (D) The model's speed.
 - (E) The model's accuracy.
 - *Correct Answer: (A)*
-

20. What is "variance" in the context of a machine learning model?

- (A) The error introduced by a model that is too simple.
 - (B) The error from a model's sensitivity to small fluctuations in the training data (leading to overfitting).
 - (C) The model's preference for one outcome over another.
 - (D) The model's speed.
 - (E) The model's accuracy.
 - *Correct Answer: (B)*
-

Level 5: Strategic & Architectural

1. A business wants to know *which customers are likely to churn (leave)*. What type of machine learning problem is this?

- (A) A binary classification problem (predicting "churn" or "not churn").
- (B) A clustering problem (grouping customers).
- (C) A regression problem (predicting how much they will spend).

- (D) A dimensionality reduction problem.
 - (E) An NLP problem.
 - *Correct Answer: (A)*
-

2. A business wants to *segment its customers* into different groups to create targeted marketing. What *type* of machine learning problem is this?

- (A) A classification problem.
 - (B) A clustering problem (unsupervised).
 - (C) A regression problem.
 - (D) A time-series forecasting problem.
 - (E) An A/B testing problem.
 - *Correct Answer: (B)*
-

3. What is "MLOps" (Machine Learning Operations)?

- (A) A set of practices that aims to deploy and maintain machine learning models in production reliably and efficiently.
 - (B) A specific machine learning algorithm.
 - (C) A data visualization tool.
 - (D) A data cleaning technique.
 - (E) A type of database.
 - *Correct Answer: (A)*
-

4. What is "ethical AI" or "responsible AI"?

- (A) A focus on ensuring AI systems are fair, unbiased, transparent, and accountable.
 - (B) An AI that is cheap to run.
 - (C) An AI that is very fast.
 - (D) An AI that is 100% accurate.
 - (E) An AI that can write its own code.
 - *Correct Answer: (A)*
-

5. What is "algorithmic bias"?

- (A) When an AI model produces results that are systematically prejudiced due to biased assumptions in the algorithm or biased training data.
- (B) The bias-variance trade-off.

- (C) A model that is too simple (underfitting).
 - (D) A model that is too complex (overfitting).
 - (E) A model that is 100% fair.
 - *Correct Answer: (A)*
-

6. How would you explain a complex model's prediction to a *non-technical* business stakeholder?

- (A) By explaining the mathematical formulas in detail.
 - (B) By using explainable AI (XAI) techniques like SHAP values to show *which features* (e.g., "high purchase frequency") were most important for the prediction.
 - (C) By telling them the model is a "black box" and they just have to trust it.
 - (D) By showing them the 10,000-line Python script.
 - (E) By giving them the model's accuracy score and nothing else.
 - *Correct Answer: (B)*
-

7. What is a "data pipeline"?

- (A) A physical pipe that transports data.
 - (B) A series of automated steps that extract, transform, and load (ETL) data from a source to a destination (like a data warehouse).
 - (C) A machine learning model.
 - (D) A data visualization.
 - (E) A type of database.
 - *Correct Answer: (B)*
-

8. What is a "data warehouse"?

- (A) A small database on a laptop.
 - (B) A large, central repository of data that is optimized for analysis and reporting (OLAP), often containing historical data from many sources.
 - (C) A CSV file.
 - (D) A machine learning model.
 - (E) A data visualization tool.
 - *Correct Answer: (B)*
-

9. What is a "data lake"?

- (A) A large, central repository that stores *raw data* in its native format, often used for data exploration and machine learning.
 - (B) A highly structured database like a data warehouse.
 - (C) A small CSV file.
 - (D) A machine learning model.
 - (E) A data visualization.
 - *Correct Answer: (A)*
-

10. You've built a model with 95% accuracy. What is the *first* question you should ask before deploying it?

- (A) "How can I make it 96% accurate?"
 - (B) "What is the *baseline* accuracy?" (e.g., if 94% of customers *don't* churn, a model that *always* predicts "no churn" is 94% accurate).
 - (C) "What programming language is it written in?"
 - (D) "Can I run it on my laptop?"
 - (E) "What is the F1-Score?" (This is a good question, but baseline is more fundamental).
 - *Correct Answer: (B)*
-

11. What is the "bias-variance trade-off"?

- (A) A model with high bias is always better.
 - (B) A model with high variance is always better.
 - (C) A fundamental concept that as a model's complexity increases, its bias decreases but its variance increases (and vice-versa). The goal is to find a balance.
 - (D) The trade-off between a model being "biased" (unfair) and "varied" (random).
 - (E) A trade-off between model speed and accuracy.
 - *Correct Answer: (C)*
-

12. What is "model deployment"?

- (A) The process of training a machine learning model.
 - (B) The process of cleaning data.
 - (C) The process of integrating a trained model into an existing application (e.g., as an API) so it can make predictions on new, real-world data.
 - (D) The process of visualizing model results.
 - (E) The process of deleting a model.
 - *Correct Answer: (C)*
-

13. What is a "Transformer" model (e.g., BERT, GPT)?

- (A) A deep learning architecture (used in NLP) that relies heavily on "self-attention" mechanisms to understand context and relationships in text.
 - (B) A model that transforms data from one format to another.
 - (C) A type of clustering algorithm.
 - (D) A type of regression algorithm.
 - (E) A data visualization tool.
 - *Correct Answer: (A)*
-

14. A business wants to predict *next month's sales*. What *type* of machine learning problem is this?

- (A) A time-series forecasting problem (a type of regression).
 - (B) A classification problem.
 - (C) A clustering problem.
 - (D) An NLP problem.
 - (E) A dimensionality reduction problem.
 - *Correct Answer: (A)*
-

15. What is a common way to evaluate a *clustering* model (like K-Means), since it has no "labels"?

- (A) Using Accuracy and Precision.
 - (B) Using a metric like the "Silhouette Score," which measures how similar an object is to its own cluster compared to other clusters.
 - (C) By manually looking at every cluster.
 - (D) Using the R-squared value.
 - (E) Using the p-value.
 - *Correct Answer: (B)*
-

16. What is "reinforcement learning"?

- (A) A type of supervised learning.
- (B) A type of unsupervised learning.
- (C) An area of machine learning where an "agent" learns to make optimal decisions by taking actions in an environment to maximize a cumulative "reward".
- (D) A method for making a model "stronger" by adding more data.
- (E) A method for cleaning data.
- *Correct Answer: (C)*

17. What is a "feature store"?

- (A) A central repository for documenting, storing, and sharing features for machine learning models across an organization.
- (B) A store that sells new features for your models.
- (C) A database for storing raw data.
- (D) A data visualization tool.
- (E) A machine learning model.
- *Correct Answer: (A)*

18. What is the *business* value of a data science model?

- (A) How complex and mathematically interesting it is.
- (B) Its ability to drive a business outcome, such as increasing revenue, reducing costs, or improving customer satisfaction.
- (C) Its accuracy score (e.g., 99% accuracy).
- (D) The number of features it uses.
- (E) The programming language it is written in.
- *Correct Answer: (B)*

19. What is "model drift"?

- (A) When a model's performance degrades over time because the real-world data it's seeing in production has changed from the data it was trained on.
- (B) When a model "drifts" from one server to another.
- (C) A method for improving model performance.
- (D) A data visualization technique.
- (E) A type of data cleaning.
- *Correct Answer: (A)*

20. You have to choose between a simple, interpretable model (like Logistic Regression) and a complex, "black box" model (like a Deep Neural Network) that is 2% more accurate. How do you decide?

- (A) Always choose the complex model for the best accuracy.
- (B) Always choose the simple model because it's easier.

- (C) It depends on the business problem. For high-stakes decisions (e.g., loan approval, medical), interpretability and fairness (the simple model) may be more important than a small accuracy boost.
- (D) Ask the frontend team what they prefer.
- (E) Flip a coin.

Correct Answer: ©

Data Analytics track.

This set is distinct from Data Science, focusing more on SQL, Business Intelligence (BI) tools, and interpreting past data, rather than building predictive machine learning models.

Level 1: Basic Foundations

1. What is the primary goal of Data Analytics?

- (A) To build artificial intelligence and predictive models.
 - (B) To inspect, clean, transform, and model data to find useful information and support decision-making.
 - (C) To design and build databases from scratch.
 - (D) To write the code for websites and mobile apps.
 - (E) To secure a company's network from hackers.
 - *Correct Answer: (B)*
-

2. What does "SQL" stand for?

- (A) Simple Query Language
 - (B) Structured Query Language
 - (C) Server Quality Logic
 - (D) System Query Link
 - (E) Standard Question Language
 - *Correct Answer: (B)*
-

3. What is a "KPI"?

- (A) Key Performance Indicator: A measurable value that shows how effectively a company is achieving key business objectives.
- (B) Key Python Integration: A library for coding.

- (C) Key Performance Index: A ranking of all employees.
 - (D) Key Programming Interface: Another name for an API.
 - (E) Known Problem Indicator: A list of bugs.
 - *Correct Answer: (A)*
-

4. What is a "dashboard"?

- (A) A code editor for writing SQL.
 - (B) A visual display of the most important information, consolidated on a single screen, to monitor performance at a glance.
 - (C) The main page of a Microsoft Excel file.
 - (D) A server that stores data.
 - (E) A type of database.
 - *Correct Answer: (B)*
-

5. Which of the following is a "data visualization" tool?

- (A) Microsoft Word
 - (B) Tableau or Power BI
 - (C) Node.js
 - (D) GitHub
 - (E) VS Code
 - *Correct Answer: (B)*
-

6. What is a "CSV" file?

- (A) A "Comma-Separated Values" file, a plain text file for storing tabular data.
 - (B) A "Computer Security Version" file, used for antivirus.
 - (C) A "Calculated Sheet Version" file, specific to Excel.
 - (D) A "Customer Service validation" file.
 - (E) A type of image file.
 - *Correct Answer: (A)*
-

7. What is the "mean" of a set of numbers?

- (A) The middle value.
- (B) The most common value.
- (C) The average value (sum of values divided by the count).

- (D) The highest value.
 - (E) The lowest value.
 - *Correct Answer: (C)*
-

8. What is the "median" of a set of numbers?

- (A) The middle value *after* the numbers have been sorted.
 - (B) The most common value.
 - (C) The average value.
 - (D) The difference between the highest and lowest value.
 - (E) The sum of all values.
 - *Correct Answer: (A)*
-

9. In a database, what is a "table"?

- (A) A visual chart or graph.
 - (B) A collection of related data organized in a grid of rows and columns.
 - (C) A single piece of data, like a name.
 - (D) A query used to get data.
 - (E) The entire database itself.
 - *Correct Answer: (B)*
-

10. What is a "row" in a database table?

- (A) A single, complete record or entry (e.g., all information for one customer).
 - (B) A single characteristic or attribute (e.g., the "Email" column).
 - (C) The name of the table.
 - (D) The title of a chart.
 - (E) A calculation.
 - *Correct Answer: (A)*
-

11. What is a "column" in a database table?

- (A) A single, complete record (e.g., all info for one customer).
- (B) A set of data values for a specific attribute (e.g., "Age" for all customers).
- (C) The entire table.
- (D) A filter on a dashboard.
- (E) A security setting.

- *Correct Answer: (B)*
-

12. What is "data cleaning"?

- (A) The process of visualizing data.
 - (B) The process of deleting all old data.
 - (C) The process of finding and fixing errors, inconsistencies, and missing values in a dataset.
 - (D) The process of writing SQL queries.
 - (E) The process of designing a dashboard.
 - *Correct Answer: (C)*
-

13. Which chart type is best for showing trends over time?

- (A) A pie chart
 - (B) A line chart
 - (C) A scatter plot
 - (D) A bar chart
 - (E) A histogram
 - *Correct Answer: (B)*
-

14. Which chart type is best for comparing values across different categories?

- (A) A bar chart
 - (B) A line chart
 - (C) A scatter plot
 - (D) A pie chart
 - (E) A treemap
 - *Correct Answer: (A)*
-

15. Which chart type is best for showing the *relationship* between two numerical variables?

- (A) A pie chart
- (B) A bar chart
- (C) A scatter plot
- (D) A line chart
- (E) A KPI card

- *Correct Answer: (C)*
-

16. What is the "mode" of a set of data?

- (A) The average value.
 - (B) The middle value.
 - (C) The value that appears most frequently.
 - (D) The highest value.
 - (E) The range of values.
 - *Correct Answer: (C)*
-

17. What is "Microsoft Excel"?

- (A) A database management system.
 - (B) A programming language.
 - (C) A spreadsheet program used for organizing, analyzing, and visualizing data.
 - (D) A cloud computing service.
 - (E) A tool for writing SQL code.
 - *Correct Answer: (C)*
-

18. What is a "data type"?

- (A) The name of a dashboard.
 - (B) A classification that specifies which type of value a variable has (e.g., String, Integer, Date).
 - (C) A SQL query.
 - (D) A type of chart.
 - (E) A folder for storing data.
 - *Correct Answer: (B)*
-

19. What does "ETL" stand for in data analytics?

- (A) Extract, Transform, Load
- (B) Error, Test, Launch
- (C) Export, Translate, Link
- (D) Estimate, Test, Learn
- (E) E-commerce, Tools, Logic
- *Correct Answer: (A)*

20. What is "data integrity"?

- (A) The speed of the database.
 - (B) The visualization of the data.
 - (C) The maintenance, accuracy, consistency, and completeness of data.
 - (D) The process of deleting data.
 - (E) The file format of the data.
 - *Correct Answer: (C)*
-

Level 2: Intermediate Application

1. Which SQL command is used to retrieve data from a database?

- (A) GET
 - (B) READ
 - (C) SELECT
 - (D) FETCH
 - (E) OPEN
 - *Correct Answer: (C)*
-

2. Which SQL clause is used to filter the results of a query?

- (A) WHERE
 - (B) FILTER
 - (C) GROUP BY
 - (D) SELECT
 - (E) SORT
 - *Correct Answer: (A)*
-

3. What is a "Pivot Table" in Excel used for?

- (A) To write SQL code.
- (B) To summarize, group, and analyze large amounts of data in a table.
- (C) To create a visual line chart.
- (D) To format the color of cells.
- (E) To check for spelling errors.

- *Correct Answer: (B)*
-

4. What is the purpose of the **VLOOKUP function in Excel?**

- (A) To calculate the average of a column.
 - (B) To sort data alphabetically.
 - (C) To find a value in one column of a table and return a corresponding value from another column.
 - (D) To create a chart.
 - (E) To filter rows based on a condition.
 - *Correct Answer: (C)*
-

5. Which SQL clause is used to combine rows from two or more tables?

- (A) **MERGE**
 - (B) **COMBINE**
 - (C) **GROUP**
 - (D) **JOIN**
 - (E) **APPEND**
 - *Correct Answer: (D)*
-

6. Which SQL aggregate function returns the total number of rows?

- (A) **SUM()**
 - (B) **TOTAL()**
 - (C) **COUNT()**
 - (D) **AVG()**
 - (E) **MAX()**
 - *Correct Answer: (C)*
-

7. Which SQL clause is used to sort the results of a query?

- (A) **SORT BY**
- (B) **ALIGN BY**
- (C) **ORDER BY**
- (D) **GROUP BY**

- (E) **ARRANGE**
 - *Correct Answer: (C)*
-

8. In a BI tool, what is a "Dimension"?

- (A) A numerical, measurable value (e.g., "Sales", "Quantity").
 - (B) A qualitative, descriptive value (e.g., "Region", "Product Name").
 - (C) The name of the dashboard.
 - (D) A filter.
 - (E) A type of chart.
 - *Correct Answer: (B)*
-

9. In a BI tool, what is a "Measure"?

- (A) A numerical, quantitative value that can be aggregated (e.g., "Sales", "Profit").
 - (B) A descriptive, categorical value (e.g., "Customer Name").
 - (C) The title of a chart.
 - (D) A comment or note.
 - (E) A date or time.
 - *Correct Answer: (A)*
-

10. What does the SQL **GROUP BY clause do?**

- (A) It sorts the results in a group.
 - (B) It groups rows that have the same values in specified columns, so aggregate functions can be applied.
 - (C) It combines two tables together.
 - (D) It filters the results.
 - (E) It updates the data in a table.
 - *Correct Answer: (B)*
-

11. What is a "primary key" in a database table?

- (A) A column that contains non-essential data.
- (B) A column (or set of columns) that uniquely identifies each row in a table.
- (C) The first column in every table.
- (D) A key used to unlock the database.
- (E) A link to another table.

- *Correct Answer: (B)*
-

12. What is a "foreign key" in a database table?

- (A) A column that is not from your country.
 - (B) A column that uniquely identifies each row (this is a primary key).
 - (C) A column (or set of columns) that links to the primary key of another table.
 - (D) A password to access the table.
 - (E) The last column in every table.
 - *Correct Answer: (C)*
-

13. Which SQL aggregate function calculates the average of a column?

- (A) `MEAN()`
 - (B) `MEDIAN()`
 - (C) `AVERAGE()`
 - (D) `AVG()`
 - (E) `TOTAL() / COUNT()`
 - *Correct Answer: (D)*
-

14. Which SQL command is used to add a new row of data to a table?

- (A) `ADD ROW`
 - (B) `CREATE ROW`
 - (C) `INSERT INTO`
 - (D) `UPDATE`
 - (E) `ALTER TABLE`
 - *Correct Answer: (C)*
-

15. Which SQL command is used to modify existing data in a table?

- (A) `MODIFY`
- (B) `CHANGE`
- (C) `INSERT INTO`
- (D) `UPDATE`
- (E) `ALTER TABLE`

- *Correct Answer: (D)*
-

16. What is a common way to handle *missing data* (null values) during analysis?

- (A) Delete the entire dataset.
 - (B) Report the data as-is; the errors don't matter.
 - (C) Remove the rows, or replace the missing value with a default (e.g., 0, mean, median).
 - (D) Close the program and restart it.
 - (E) Replace all missing values with the word "missing".
 - *Correct Answer: (C)*
-

17. What is the purpose of the **SUMIF function in Excel?**

- (A) To sum all numbers in a range.
 - (B) To sum numbers in a range *only if* a specific condition is met.
 - (C) To check if a sum is correct.
 - (D) To find the sum of two different tables.
 - (E) To create a chart.
 - *Correct Answer: (B)*
-

18. What does the **LIKE operator do in a SQL **WHERE** clause?**

- (A) It checks for an exact match.
 - (B) It is used for pattern matching (e.g., **WHERE name LIKE 'A%'**).
 - (C) It checks if a value is similar, but not identical.
 - (D) It joins two tables based on similar values.
 - (E) It calculates the average.
 - *Correct Answer: (B)*
-

19. What is a "calculated field" in Excel or a BI tool?

- (A) Any field that contains a number.
- (B) A new field (column) that you create using a formula to transform existing data.
- (C) A field that is hidden from view.
- (D) A field that is used as a filter.
- (E) A field containing text data.
- *Correct Answer: (B)*

20. What is "data validation"?

- (A) The process of checking the accuracy and quality of data.
 - (B) The process of visualizing data.
 - (C) The process of writing a SQL query.
 - (D) The process of creating a dashboard.
 - (E) The process of deleting data.
 - *Correct Answer: (A)*
-

Level 3: Advanced Scenarios

1. What is the difference between an **INNER JOIN and a **LEFT JOIN** in SQL?**

- (A) **INNER JOIN** is faster; **LEFT JOIN** is slower.
 - (B) **INNER JOIN** returns *only* matching rows; **LEFT JOIN** returns *all* rows from the left table and matching rows from the right.
 - (C) **INNER JOIN** returns all rows; **LEFT JOIN** returns only matching rows.
 - (D) **INNER JOIN** is for numbers; **LEFT JOIN** is for text.
 - (E) There is no difference.
 - *Correct Answer: (B)*
-

2. What is the fundamental problem with the "Correlation does not imply causation" concept?

- (A) Just because two variables move together (correlation) doesn't mean one *caused* the other.
 - (B) Correlation is a more powerful finding than causation.
 - (C) If two variables are correlated, one must have caused the other.
 - (D) This concept only applies to pie charts.
 - (E) This concept is outdated and no longer relevant.
 - *Correct Answer: (A)*
-

3. A manager asks for a report on "Sales Performance." What is the *first* question you should ask?

- (A) "What color do you want the dashboard to be?"
- (B) "How soon do you need this?"

- (C) "What business question are you trying to answer?" or "How do you define 'performance'?"
 - (D) "What software should I use?"
 - (E) "Who should I blame if sales are bad?"
 - *Correct Answer: (C)*
-

4. What is "data storytelling"?

- (A) The process of making up stories about data.
 - (B) The process of writing a SQL query.
 - (C) The skill of communicating data insights effectively using a narrative and visuals.
 - (D) A feature in Microsoft Excel.
 - (E) A type of database.
 - *Correct Answer: (C)*
-

5. What is the difference between the SQL **WHERE** clause and the **HAVING** clause?

- (A) **WHERE** filters rows *before* **GROUP BY**; **HAVING** filters groups *after* **GROUP BY**.
 - (B) **WHERE** filters groups; **HAVING** filters rows.
 - (C) **WHERE** is for numbers; **HAVING** is for text.
 - (D) **WHERE** is for **SELECT**; **HAVING** is for **UPDATE**.
 - (E) They are identical and can be used interchangeably.
 - *Correct Answer: (A)*
-

6. Which chart is best for showing how different parts make up a *whole* (e.g., market share)?

- (A) A line chart
 - (B) A scatter plot
 - (C) A pie chart or a stacked bar chart
 - (D) A histogram
 - (E) A map
 - *Correct Answer: (C)*
-

7. How would you calculate "Year-over-Year Growth" (YoY) for sales?

- (A) $(\text{Current Year Sales} - \text{Last Year Sales}) / \text{Last Year Sales}$

- (B) $(\text{Current Year Sales} / \text{Last Year Sales}) - 1$
 - (C) $(\text{Current Year Sales} - \text{Last Year Sales})$
 - (D) $\text{Current Year Sales} / \text{Last Year Sales}$
 - (E) Both (A) and (B) are correct ways to calculate it.
 - *Correct Answer: (E)*
-

8. What is a "data dictionary"?

- (A) A central file or document that defines all the tables, columns, data types, and business definitions for a database.
 - (B) A tool that checks spelling in your SQL queries.
 - (C) A list of all data analysts in a company.
 - (D) A dashboard that shows all company data.
 - (E) A type of SQL join.
 - *Correct Answer: (A)*
-

9. What is the purpose of a **CASE** statement in SQL?

- (A) To join two tables.
 - (B) To filter results like a **WHERE** clause.
 - (C) To create conditional "if-then-else" logic within a query (e.g., to create new categories).
 - (D) To sort results.
 - (E) To create a new table.
 - *Correct Answer: (C)*
-

10. What is a "funnel analysis"?

- (A) An analysis of fluid dynamics.
 - (B) A way to visualize the user's journey through a series of steps (e.g., homepage -> product page -> cart -> purchase).
 - (C) A type of SQL query.
 - (D) A method for cleaning data.
 - (E) A way to build a dashboard.
 - *Correct Answer: (B)*
-

11. What is a "cohort analysis"?

- (A) An analysis of military groups.
 - (B) An analysis that tracks a group of users (a "cohort") who share a common characteristic (e.g., "signed up in January") over time.
 - (C) A way to join tables in SQL.
 - (D) A type of pie chart.
 - (E) A method for cleaning data.
 - *Correct Answer: (B)*
-

12. What is an "outlier"?

- (A) A data point that is a "lie" and must be deleted.
 - (B) The most common data point.
 - (C) A data point that is significantly different from other data points in a dataset.
 - (D) The average value.
 - (E) A data point with a missing value.
 - *Correct Answer: (C)*
-

13. What is a potential problem with a chart's Y-axis not starting at zero?

- (A) It can exaggerate the differences between values, making them look more significant than they are.
 - (B) It makes the chart load slower.
 - (C) It is not possible; all charts must start at zero.
 - (D) It can hide the most important data.
 - (E) It makes the chart more accurate.
 - *Correct Answer: (A)*
-

14. What is a "subquery" in SQL?

- (A) A query that is not as good as a normal query.
 - (B) A query that is embedded (nested) inside another SQL query.
 - (C) A query that updates data.
 - (D) A query that deletes a table.
 - (E) A query that only runs on a subset of data.
 - *Correct Answer: (B)*
-

15. What is "DAX"?

- (A) A brand of soap.
 - (B) A programming language used in Microsoft Power BI for creating formulas and calculations.
 - (C) A type of SQL database.
 - (D) A data visualization chart.
 - (E) A Python library.
 - *Correct Answer: (B)*
-

16. What is a "star schema" in data modeling?

- (A) A complex model with tables linking to many other tables in a web.
 - (B) A model with one central "fact" table (e.g., Sales) connected to multiple "dimension" tables (e.g., Time, Product, Region).
 - (C) A database model that is new and popular.
 - (D) A visual chart that looks like a star.
 - (E) A way to rate your data.
 - *Correct Answer: (B)*
-

17. How would you find the "top 10" selling products in SQL?

- (A) `SELECT product_name, SUM(sales) FROM sales GROUP BY product_name ORDER BY SUM(sales) DESC LIMIT 10;`
 - (B) `SELECT product_name FROM sales WHERE sales > 10;`
 - (C) `SELECT product_name, SUM(sales) FROM sales;`
 - (D) `SELECT TOP 10 product_name FROM sales;`
 - (E) `SELECT product_name FROM sales LIMIT 10;`
 - *Correct Answer: (A)*
-

18. What is a "calculated field" in Tableau?

- (A) A field that is hidden.
 - (B) A new field created using a formula, similar to in Excel.
 - (C) The name of a dashboard.
 - (D) A filter.
 - (E) A connection to a database.
 - *Correct Answer: (B)*
-

19. Why is it important to understand the *business context* of your data?

- (A) It is not important; the numbers speak for themselves.
 - (B) To ensure the analysis is relevant, the right questions are asked, and the insights are actionable.
 - (C) To know who to blame if the data is wrong.
 - (D) To help you write better SQL queries.
 - (E) To make your charts look better.
 - *Correct Answer: (B)*
-

20. What is "data quality"?

- (A) A measure of how "good" the data is for its intended purpose (e.g., accurate, complete, timely).
 - (B) The total amount of data you have.
 - (C) The speed at which you can query data.
 - (D) The type of database you use.
 - (E) The number of charts on your dashboard.
 - *Correct Answer: (A)*
-

Level 4: Expert & Specialization

1. What is a "window function" in SQL (e.g., `ROW_NUMBER()`, `RANK()`)?

- (A) A function that opens a new window with the results.
 - (B) A function that performs a calculation across a *set of rows* related to the current row, without collapsing them.
 - (C) A function that filters data.
 - (D) A function that joins tables.
 - (E) A function that updates data.
 - *Correct Answer: (B)*
-

2. What is a "CTE" (Common Table Expression) in SQL?

- (A) A temporary, named result set that you can reference within a `SELECT`, `INSERT`, `UPDATE`, or `DELETE` statement.
- (B) A permanent table stored in the database.
- (C) A function for cleaning data.
- (D) A "Critical Table Error."

- (E) A type of index.
 - *Correct Answer: (A)*
-

3. What is the difference between "OLTP" and "OLAP"?

- (A) OLTP is for analytics; OLAP is for transactions.
 - (B) OLTP (Online Transaction Processing) is for fast, day-to-day operations; OLAP (Online Analytical Processing) is for complex analysis and reporting.
 - (C) OLTP is old; OLAP is new.
 - (D) OLTP is for data visualization; OLAP is for data storage.
 - (E) There is no difference.
 - *Correct Answer: (B)*
-

4. What is a "data warehouse"?

- (A) A small database on a laptop.
 - (B) A large, central repository of data, optimized for analysis and reporting (OLAP), that integrates data from many sources.
 - (C) A single Excel file.
 - (D) A tool for data visualization.
 - (E) A server for hosting a website.
 - *Correct Answer: (B)*
-

5. What is the purpose of the **CALCULATE** function in DAX (Power BI)?

- (A) It is a simple sum function.
 - (B) It is a super-function that modifies the *filter context* in which an expression is evaluated.
 - (C) It formats the text.
 - (D) It joins two tables.
 - (E) It creates a new chart.
 - *Correct Answer: (B)*
-

6. What is a "Level of Detail" (LOD) expression in Tableau?

- (A) A setting that makes the dashboard graphics low-quality.
- (B) A way to filter the data.

- (C) A syntax that allows you to compute aggregations at a different level of detail than the visualization itself.
 - (D) A way to connect to a database.
 - (E) A type of chart.
 - *Correct Answer: (C)*
-

7. Why might you use "Python" (with libraries like Pandas) for data analytics?

- (A) It is the only way to make a dashboard.
 - (B) To perform complex data cleaning, manipulation, and automation tasks that are too large or difficult for Excel or SQL alone.
 - (C) To build websites.
 - (D) To design the database schema.
 - (E) You would not; Python is only for Data Science.
 - *Correct Answer: (B)*
-

8. What is "A/B testing"?

- (A) A test to see if your code is "A" quality or "B" quality.
 - (B) A method of comparing two versions (A and B) of something to see which performs better.
 - (C) A way to join two tables, "A" and "B".
 - (D) A data cleaning technique.
 - (E) A type of SQL query.
 - *Correct Answer: (B)*
-

9. How do you *begin* to optimize a very slow SQL query?

- (A) By adding more data to the table.
 - (B) By checking if the tables have "indexes" on the columns used in **JOIN** or **WHERE** clauses.
 - (C) By deleting the table and starting over.
 - (D) By rewriting the query in Python.
 - (E) By telling the user to be more patient.
 - *Correct Answer: (B)*
-

10. What is a "data model" in the context of Power BI or Tableau?

- (A) The collection of tables and the *relationships* (joins) defined between them.
 - (B) The visual dashboard.
 - (C) A single SQL query.
 - (D) A machine learning model.
 - (E) The data cleaning steps.
 - *Correct Answer: (A)*
-

11. What is the difference between a "live connection" and an "extract" in Tableau?

- (A) A live connection is fast; an extract is slow.
 - (B) A live connection queries the database directly; an extract is a snapshot of the data stored in memory.
 - (C) A live connection is for charts; an extract is for tables.
 - (D) A live connection is for big data; an extract is for small data.
 - (E) There is no difference.
 - *Correct Answer: (B)*
-

12. What is "data governance"?

- (A) The process of governing a country using data.
 - (B) The overall management of the availability, usability, integrity, and security of data in an enterprise.
 - (C) A type of database.
 - (D) A dashboard for executives.
 - (E) A SQL command.
 - *Correct Answer: (B)*
-

13. What is a "rolling average" (or moving average)?

- (A) An average that is wrong.
 - (B) A calculation used to analyze data points by creating a series of averages of different subsets of the full data set (e.g., a 7-day rolling average).
 - (C) The average of an entire dataset.
 - (D) A type of chart.
 - (E) A database function.
 - *Correct Answer: (B)*
-

14. What is "statistical significance" (e.g., p-value)?

- (A) A measure of how *important* or *large* an effect is.
 - (B) A measure of the probability that an observed difference is due to *random chance* rather than a real effect.
 - (C) The accuracy of a model.
 - (D) The R-squared value.
 - (E) The title of a chart.
 - *Correct Answer: (B)*
-

15. What is "R"?

- (A) A letter grade for a bad analysis.
 - (B) A programming language and environment widely used for statistical analysis and visualization.
 - (C) A special type of SQL query.
 - (D) A "Regression" model.
 - (E) A setting in Microsoft Excel.
 - *Correct Answer: (B)*
-

16. What does `pd.read_csv()` do in the Python Pandas library?

- (A) It creates a new CSV file.
 - (B) It reads a CSV file into a Pandas DataFrame.
 - (C) It calculates the sum of a CSV file.
 - (D) It deletes a CSV file.
 - (E) It sends a CSV file in an email.
 - *Correct Answer: (B)*
-

17. What does the `.groupby()` method in Pandas allow you to do?

- (A) Split the data into groups, apply a function (like `sum` or `mean`), and combine the results.
 - (B) Sort the DataFrame.
 - (C) Join two DataFrames.
 - (D) Create a chart.
 - (E) Read a CSV file.
 - *Correct Answer: (A)*
-

18. What is a "dashboard action" in Tableau?

- (A) A button that deletes the dashboard.
 - (B) An interactive element that allows a user to click on one part of a dashboard (e.g., a bar) and have it filter or highlight other parts.
 - (C) The "Save" button.
 - (D) A calculation like `SUM()`.
 - (E) A connection to a database.
 - *Correct Answer: (B)*
-

19. What is "sentiment analysis"?

- (A) An analysis of how a person is feeling.
 - (B) The use of NLP to determine if a piece of text (like a review) is positive, negative, or neutral.
 - (C) A chart that shows emotions.
 - (D) A type of SQL query.
 - (E) A filter in Excel.
 - *Correct Answer: (B)*
-

20. What is a "data pipeline"?

- (A) A physical pipe that moves data.
 - (B) A set of automated processes that moves and transforms data from a source to a destination.
 - (C) A type of dashboard.
 - (D) A SQL query.
 - (E) A filter in Tableau.
 - *Correct Answer: (B)*
-

Level 5: Strategic & Architectural

1. What is the difference between "Descriptive" and "Diagnostic" analytics?

- (A) Descriptive is "What happened?"; Diagnostic is "Why did it happen?".
- (B) Descriptive is "Why did it happen?"; Diagnostic is "What happened?".
- (C) Descriptive is "What will happen?"; Diagnostic is "What should we do?".
- (D) Descriptive is for charts; Diagnostic is for tables.
- (E) They are the same thing.
- *Correct Answer: (A)*

2. What is the difference between "Predictive" and "Prescriptive" analytics?

- (A) Predictive is "What should we do?"; Prescriptive is "What will happen?".
- (B) Predictive is "What happened?"; Prescriptive is "Why did it happen?".
- (C) Predictive is "What *will* happen?"; Prescriptive is "What *should we do*?".
- (D) Predictive is for Data Science; Prescriptive is for Data Analytics.
- (E) They are the same thing.
- *Correct Answer: (C)*

3. What is a "data-driven culture"?

- (A) A culture where all decisions are made by computers.
- (B) A culture where employees are "driven" hard to produce data.
- (C) A culture where data and evidence are used to inform and make business decisions at all levels.
- (D) A culture that only uses SQL.
- (E) A culture that has a lot of dashboards.
- *Correct Answer: (C)*

4. What is a "Single Source of Truth" (SSOT)?

- (A) A single, central repository (like a data warehouse) that is trusted as the one correct source for all company data.
- (B) The most senior data analyst in the company.
- (C) A dashboard that is never wrong.
- (D) A legal document.
- (E) A Python script.
- *Correct Answer: (A)*

5. How do you measure the "ROI" (Return on Investment) of a data analytics project?

- (A) By how beautiful the dashboard is.
- (B) By how many people use the dashboard.
- (C) By measuring the business value it generated (e.g., increased revenue, saved costs) minus the cost of the project.
- (D) By how long it took to build.
- (E) You cannot measure the ROI of analytics.
- *Correct Answer: (C)*

6. An executive asks for a dashboard. What is the *first* thing you should do?

- (A) Start building it immediately in Tableau.
- (B) Schedule a meeting to understand the *business questions* they need to answer and the *KPIs* they need to track.
- (C) Ask them what colors they like.
- (D) Tell them it will take six months.
- (E) Send them a raw CSV file.
- *Correct Answer: (B)*

7. How do you handle a stakeholder who *disagrees* with your data-driven finding?

- (A) Tell them they are wrong and that the data is always right.
- (B) Change your finding to match their opinion.
- (C) Thank them for the feedback, re-check your data and methodology, and then (if still correct) try to understand *why* they disagree.
- (D) Delete the dashboard and pretend it never happened.
- (E) Show them the raw SQL query.
- *Correct Answer: (C)*

8. What is "data democratization"?

- (A) The idea that all data should be public on the internet.
- (B) The practice of making data accessible to more people within an organization (not just analysts).
- (C) A vote on which data is correct.
- (D) A type of database.
- (E) A security risk.
- *Correct Answer: (B)*

9. What is a "self-service BI" model?

- (A) A model where analysts serve their own data.
- (B) An approach where business users can access and analyze data themselves using BI tools, without needing an analyst for every request.
- (C) A BI tool that is free.
- (D) A BI tool that only works on your phone.
- (E) A model where the BI tool builds its own dashboards.

- *Correct Answer: (B)*
-

10. How would you design a KPI dashboard for an e-commerce executive? What 3 KPIs would you *definitely* include?

- (A) 1. Number of SQL queries run, 2. Dashboard load time, 3. Number of charts.
 - (B) 1. Revenue (by day/week/month), 2. Conversion Rate, 3. Average Order Value (AOV).
 - (C) 1. Employee birthdays, 2. Number of support tickets, 3. Office locations.
 - (D) 1. Website color, 2. Website font, 3. Number of pages.
 - (E) 1. Number of Python scripts, 2. Database size, 3. Server uptime.
 - *Correct Answer: (B)*
-

11. What is "data privacy" (e.g., GDPR, CCPA) and how does it affect an analyst?

- (A) It has no effect; analysts can see all data.
 - (B) It is a set of laws that govern how "Personally Identifiable Information" (PII) must be collected, stored, and used.
 - (C) It is a suggestion to keep data private.
 - (D) It is a type of data cleaning.
 - (E) It is a feature in Excel.
 - *Correct Answer: (B)*
-

12. How do you prioritize data requests when you have *too many* from different departments?

- (A) First-come, first-served.
 - (B) By who shouts the loudest.
 - (C) By prioritizing tasks based on their potential *business impact* and urgency.
 - (D) By doing the easiest ones first.
 - (E) By doing the ones for your boss first.
 - *Correct Answer: (C)*
-

13. What is the *most important* skill for a data analyst?

- (A) Knowing how to write complex SQL.
- (B) Being an expert in Tableau.

- (C) Business acumen and curiosity: The ability to ask the right questions and understand the "why".
 - (D) Being very fast at Excel.
 - (E) Knowing Python.
 - *Correct Answer: (C)*
-

14. What are the ethical implications of "algorithmic bias" in analytics?

- (A) The algorithm might be too slow.
 - (B) If the data is biased (e.g., based on past discrimination), the analytical model will learn and *amplify* that bias.
 - (C) The algorithm might be too expensive to run.
 - (D) The algorithm might be hard to understand.
 - (E) There are no ethical implications.
 - *Correct Answer: (B)*
-

15. A manager sees a *correlation* between employee happiness and company profit. What should be your *first* step?

- (A) Tell the manager to give everyone a raise to increase profits.
 - (B) Investigate for *causation*. Does happiness *cause* profit, does profit *cause* happiness, or is there a *third factor*?
 - (C) Build a dashboard to show the correlation.
 - (D) Dismiss the finding as a coincidence.
 - (E) Run a complex machine learning model.
 - *Correct Answer: (B)*
-

16. How would you set up an A/B test for a new "Buy Now" button color on a website?

- (A) Change the button for *all* users and see if sales go up.
 - (B) Show the *old* button (Control, A) to 50% of users and the *new* button (Variant, B) to the other 50%, then measure the conversion rate for each group.
 - (C) Ask 10 users which button they like better.
 - (D) Change the button, but only on the weekends.
 - (E) Put both buttons on the page and see which one users click.
 - *Correct Answer: (B)*
-

17. What is "data-driven attribution" in marketing?

- (A) Attributing all sales to the last ad a customer clicked.
 - (B) The process of using data to assign credit for a conversion (e.g., a sale) to different touchpoints in a customer's journey.
 - (C) A dashboard for the marketing team.
 - (D) A way to clean marketing data.
 - (E) A/B testing a marketing email.
 - *Correct Answer: (B)*
-

18. What is a major *disadvantage* of a "self-service BI" model?

- (A) It is too fast and efficient.
 - (B) It can lead to "data chaos" if users are untrained, creating conflicting reports and KPIs.
 - (C) It is too expensive.
 - (D) It only works for small data.
 - (E) It makes the data analysts' jobs obsolete.
 - *Correct Answer: (B)*
-

19. Why is "data cleaning" often the most time-consuming part of an analysis?

- (A) Because real-world data is often messy, incomplete, inconsistent, and in the wrong format.
 - (B) Because the cleaning tools are very slow.
 - (C) Because analysts enjoy cleaning data.
 - (D) Because managers require it to take a long time.
 - (E) It is not; it is the fastest part.
 - *Correct Answer: (A)*
-

20. What is the *final* step in any data analysis project?

- (A) Finishing the SQL query.
- (B) Creating the dashboard.
- (C) Communicating the findings and recommendations to the stakeholders.
- (D) Saving the file.
- (E) Deleting the raw data.
- *Correct Answer: (C)*

Cybersecurity track.

Level 1: Basic Foundations

1. What is "Cybersecurity"?

- (A) The practice of repairing computers and servers.
 - (B) The practice of protecting systems, networks, and data from digital attacks.
 - (C) The practice of writing code for websites.
 - (D) The practice of analyzing business data.
 - (E) The practice of creating computer networks.
 - *Correct Answer: (B)*
-

2. What is "malware"?

- (A) A general term for any malicious software designed to cause harm or gain unauthorized access.
 - (B) A specific type of computer hardware.
 - (C) A secure piece of code that protects a computer.
 - (D) A computer network.
 - (E) A user's password.
 - *Correct Answer: (A)*
-

3. What is a "virus"?

- (A) A piece of code that protects a file.
 - (B) A malicious program that attaches itself to legitimate files and spreads when they are opened.
 - (C) A piece of hardware that speeds up a computer.
 - (D) A secure login screen.
 - (E) A network connection.
 - *Correct Answer: (B)*
-

4. What is a "firewall"?

- (A) A program that finds and deletes viruses.
- (B) A network security device that monitors and filters incoming and outgoing network traffic.
- (C) A type of malicious software.
- (D) A physical wall in a server room to prevent fires.
- (E) A tool for writing code.
- *Correct Answer: (B)*

5. What is "phishing"?

- (A) A method of securing an email account.
- (B) A fraudulent attempt to obtain sensitive information (like passwords or credit card numbers) by disguising as a trustworthy entity in an email or message.
- (C) A type of computer virus.
- (D) A secure way to back up files.
- (E) A tool for catching network traffic.
- *Correct Answer: (B)*

6. What does the "C" in the "CIA Triad" stand for?

- (A) Cybersecurity
- (B) Confidentiality
- (C) Compliance
- (D) Computer
- (E) Code
- *Correct Answer: (B)*

7. What does the "I" in the "CIA Triad" stand for?

- (A) Integrity
- (B) Internet
- (C) Information
- (D) Intrusion
- (E) Identity
- *Correct Answer: (A)*

8. What does the "A" in the "CIA Triad" stand for?

- (A) Attack
 - (B) Authentication
 - (C) Availability
 - (D) Access
 - (E) Antivirus
 - *Correct Answer: (C)*
-

9. What is "authentication"?

- (A) The process of protecting a network with a firewall.
 - (B) The process of verifying the identity of a user (proving you are who you say you are).
 - (C) The process of giving a user permission to access a file.
 - (D) The process of encrypting data.
 - (E) The process of attacking a system.
 - *Correct Answer: (B)*
-

10. What is "authorization"?

- (A) The process of verifying a user's identity.
 - (B) The process of determining what a user is *allowed* to do or access *after* they have been authenticated.
 - (C) The process of scanning for viruses.
 - (D) The process of writing a security policy.
 - (E) The process of backing up data.
 - *Correct Answer: (B)*
-

11. What is a "strong password"?

- (A) A short, easy-to-remember password like "12345".
 - (B) A long, complex password with a mix of upper/lowercase letters, numbers, and symbols.
 - (C) A password that is written down on a sticky note.
 - (D) A password that is the same as your username.
 - (E) A password that is a single, common dictionary word.
 - *Correct Answer: (B)*
-

12. What is "antivirus" software?

- (A) A program designed to detect, prevent, and remove malicious software.
 - (B) A type of malware.
 - (C) A network firewall.
 - (D) A tool for sending phishing emails.
 - (E) A hardware device.
 - *Correct Answer: (A)*
-

13. What is "encryption"?

- (A) The process of deleting data permanently.
 - (B) The process of converting data into a secret code to prevent unauthorized access.
 - (C) The process of stealing data.
 - (D) The process of finding viruses.
 - (E) The process of creating a backup.
 - *Correct Answer: (B)*
-

14. What is a "patch"?

- (A) A type of computer virus.
 - (B) A software update released by a vendor to fix a bug or security vulnerability.
 - (C) A piece of hardware.
 - (D) A fraudulent email.
 - (E) A weak password.
 - *Correct Answer: (B)*
-

15. What is a "vulnerability"?

- (A) A strong security setting.
 - (B) A piece of antivirus software.
 - (C) A weakness in a system that an attacker could exploit.
 - (D) A user's password.
 - (E) A backup of your data.
 - *Correct Answer: (C)*
-

16. What is "social engineering"?

- (A) A security hardware device.
 - (B) The act of psychologically manipulating people into performing actions or divulging confidential information.
 - (C) The process of engineering a secure network.
 - (D) A type of encryption.
 - (E) A tool for scanning for viruses.
 - *Correct Answer: (B)*
-

17. What is a "backup"?

- (A) A copy of data that can be used to restore the original in case of data loss.
 - (B) A type of malware.
 - (C) A firewall.
 - (D) A security vulnerability.
 - (E) A user's login name.
 - *Correct Answer: (A)*
-

18. What is a "username"?

- (A) A secret code used to encrypt files.
 - (B) A type of computer virus.
 - (C) A name that uniquely identifies someone on a computer system.
 - (D) A piece of malicious software.
 - (E) A secure network.
 - *Correct Answer: (C)*
-

19. What is a "hacker"?

- (A) Someone who sells computer hardware.
 - (B) An individual who uses computer, networking, or other skills to overcome a technical problem. (Can be ethical or malicious).
 - (C) A software program that deletes files.
 - (D) A customer support agent.
 - (E) A type of computer server.
 - *Correct Answer: (B)*
-

20. What is "data"?

- (A) A physical computer.
 - (B) Information, such as facts or numbers, that can be stored and used.
 - (C) A type of antivirus software.
 - (D) A network firewall.
 - (E) A security attack.
 - *Correct Answer: (B)*
-

Level 2: Intermediate Application

1. What is the main difference between a "virus" and a "worm"?

- (A) A virus is malicious, but a worm is not.
 - (B) A virus needs a host file to spread; a worm can self-replicate and spread across a network independently.
 - (C) A virus spreads through email; a worm spreads through websites.
 - (D) A virus is hardware; a worm is software.
 - (E) There is no difference.
 - *Correct Answer: (B)*
-

2. What is "ransomware"?

- (A) A type of malware that encrypts a victim's files and demands a ransom payment to decrypt them.
 - (B) A security tool that "ransoms" a hacker's computer.
 - (C) A hardware device for encrypting files.
 - (D) A type of antivirus software.
 - (E) A free software update.
 - *Correct Answer: (A)*
-

3. What is a "Trojan" in cybersecurity?

- (A) A security program from Troy.
 - (B) Malware that disguises itself as legitimate software (like a game or utility) to trick a user into running it.
 - (C) A type of computer hardware.
 - (D) A network firewall rule.
 - (E) A secure backup file.
 - *Correct Answer: (B)*
-

4. What is a "VPN" (Virtual Private Network)?

- (A) A type of computer virus.
- (B) A secure, encrypted connection over a public network, used to protect privacy and secure data.
- (C) A public, open Wi-Fi network.
- (D) A tool for designing websites.
- (E) A type of malicious software.
- *Correct Answer: (B)*

5. What is "Two-Factor Authentication" (2FA)?

- (A) Using two different computers to log in.
- (B) A security process that requires two different methods of authentication (e.g., a password and a code from your phone).
- (C) Using two different passwords for the same account.
- (D) A type of firewall.
- (E) A type of virus.
- *Correct Answer: (B)*

6. What is the main difference between "hashing" and "encryption"?

- (A) Hashing is reversible; encryption is not.
- (B) Hashing is for text; encryption is for files.
- (C) Hashing is a one-way function (cannot be reversed); encryption is a two-way function (can be decrypted with a key).
- (D) Hashing is insecure; encryption is secure.
- (E) They are the same thing.
- *Correct Answer: (C)*

7. Why is it important to "patch" software?

- (A) To change the color and design.
- (B) To fix security vulnerabilities before attackers can exploit them.
- (C) To make the software run slower.
- (D) To delete old files.
- (E) It is not important.
- *Correct Answer: (B)*

8. What is "HTTPS"?

- (A) A faster, "hyper" version of HTTP.
- (B) The secure version of HTTP; it encrypts the data sent between your browser and the website.
- (C) A type of malware.
- (D) A standard for writing HTML code.
- (E) A type of firewall.
- *Correct Answer: (B)*

9. What is a "DDoS" attack?

- (A) A "Distributed Denial-of-Service" attack, where an attacker floods a server with traffic from many sources to overwhelm it.
- (B) A "Data Deletion of Service" attack, where an attacker deletes all files.
- (C) A security tool for "Detecting and Denying" attacks.
- (D) A type of antivirus software.
- (E) A method for backing up a server.
- *Correct Answer: (A)*

10. What is a "botnet"?

- (A) A network of secure computers used for research.
- (B) A network of private computers infected with malicious software and controlled as a group by an attacker.
- (C) A type of fishing net for robots.
- (D) A security company's internal network.
- (E) An antivirus program.
- *Correct Answer: (B)*

11. What is "spyware"?

- (A) Software that "spies" on your computer, secretly monitoring your activity and collecting personal information.
- (B) A tool used by security professionals to spy on hackers.
- (C) A type of hardware device.
- (D) A secure web browser.
- (E) A password manager.
- *Correct Answer: (A)*

12. What does a "password manager" do?

- (A) It generates weak, easy-to-guess passwords.
- (B) It securely stores all your unique, complex passwords in an encrypted vault.
- (C) It is a type of malware that steals passwords.
- (D) It is a spreadsheet for typing passwords into.
- (E) It is a hardware device for locking your computer.
- *Correct Answer: (B)*

13. What is the "Principle of Least Privilege"?

- (A) The idea that important people ("princes") get the most access.
- (B) The idea that users and programs should *only* be given the minimum permissions necessary to perform their job.
- (C) The idea that all users should have admin access.
- (D) A law that requires all software to be free.
- (E) A type of security attack.
- *Correct Answer: (B)*

14. What is a "security policy"?

- (A) A type of firewall.
- (B) A document that outlines an organization's rules and procedures for information security.
- (C) A piece of malicious software.
- (D) A tool for hacking.
- (E) A computer's hardware.
- *Correct Answer: (B)*

15. What is the main difference between "symmetric" and "asymmetric" encryption?

- (A) Symmetric is fast; asymmetric is slow.
- (B) Symmetric uses one *single key* for both encryption and decryption; asymmetric uses *two keys* (a public and a private key).
- (C) Symmetric is secure; asymmetric is not.
- (D) Symmetric is new; asymmetric is old.
- (E) Symmetric is for files; asymmetric is for text.
- *Correct Answer: (B)*

16. What is a "digital certificate"?

- (A) A type of malware.
- (B) A digital file used to prove the identity of a website or individual (e.g., used in HTTPS).
- (C) A certificate you get for completing a cybersecurity course.
- (D) A password.
- (E) A firewall.

- *Correct Answer: (B)*
-

17. What is "BYOD"?

- (A) "Bring Your Own Device" - A policy allowing employees to use their personal devices for work.
 - (B) "Buy Your Own Data" - A policy for internet usage.
 - (C) "Backup Your Own Data" - A backup policy.
 - (D) "Block Your Own Device" - A security attack.
 - (E) A type of antivirus software.
 - *Correct Answer: (A)*
-

18. What is a "security risk" of public Wi-Fi?

- (A) It is usually too slow.
 - (B) It costs too much money.
 - (C) Attackers can "sniff" the traffic, intercepting unencrypted data like passwords or credit card numbers.
 - (D) It is not compatible with most new devices.
 - (E) There is no risk; all public Wi-Fi is secure.
 - *Correct Answer: (C)*
-

19. What is "adware"?

- (A) Software that securely blocks all advertisements.
 - (B) Software that automatically downloads and installs security "add-ons".
 - (C) Software that displays unwanted pop-up advertisements.
 - (D) A hardware device for advertising.
 - (E) A type of secure web browser.
 - *Correct Answer: (C)*
-

20. What is "data breach"?

- (A) A successful security incident in which sensitive, confidential, or protected data is stolen or accessed by an unauthorized person.
- (B) The process of backing up data.
- (C) The process of encrypting data.
- (D) A type of firewall.

- (E) A software patch.
 - *Correct Answer: (A)*
-

Level 3: Advanced Scenarios

1. What is a "SQL Injection" (SQLi) attack?

- (A) An attack where a user "injects" JavaScript into a website.
 - (B) An attack where an attacker inserts malicious SQL code into a web form to manipulate the backend database.
 - (C) A method for "injecting" more speed into a database.
 - (D) A type of secure database query.
 - (E) A tool for cleaning a database.
 - *Correct Answer: (B)*
-

2. How do you *prevent* a basic SQL Injection attack?

- (A) By using a strong firewall.
 - (B) By using prepared statements (parameterized queries) and input validation.
 - (C) By encrypting the entire database.
 - (D) By using HTTPS.
 - (E) By hiding the web form from users.
 - *Correct Answer: (B)*
-

3. What is a "Cross-Site Scripting" (XSS) attack?

- (A) An attack where malicious code is injected into a trusted website, which then runs in a victim's browser.
 - (B) An attack where an attacker "crosses" two different websites.
 - (C) An attack that steals a website's CSS files.
 - (D) An attack that targets the server's database.
 - (E) A method for securing a website.
 - *Correct Answer: (A)*
-

4. What is the difference between "Stored XSS" and "Reflected XSS"?

- (A) Stored XSS is permanent; Reflected XSS is temporary.

- (B) Stored XSS is saved on the server (e.g., in a comment); Reflected XSS is sent in a request (e.g., a URL) and reflected back.
 - (C) Stored XSS is harmless; Reflected XSS is dangerous.
 - (D) Stored XSS is for databases; Reflected XSS is for websites.
 - (E) There is no difference.
 - *Correct Answer: (B)*
-

5. What is a "Man-in-the-Middle" (MITM) attack?

- (A) An attack where a hacker is "in the middle" of two other hackers.
 - (B) An attack where an attacker secretly intercepts and relays communication between two parties who believe they are communicating directly.
 - (C) A security tool for monitoring network traffic.
 - (D) A type of firewall.
 - (E) A type of server.
 - *Correct Answer: (B)*
-

6. What is a "zero-day" vulnerability?

- (A) A vulnerability that has zero impact.
 - (B) A vulnerability that is "day zero" of a new software.
 - (C) A software vulnerability that is known to the vendor, but they have "zero days" to fix it.
 - (D) A vulnerability that is discovered and exploited by hackers *before* the vendor is aware of it or can release a patch.
 - (E) A vulnerability that only exists for one day.
 - *Correct Answer: (D)*
-

7. What is "penetration testing" (or "pen testing")?

- (A) The act of illegally hacking into a system.
 - (B) An *authorized* simulated cyberattack on a computer system to evaluate its security.
 - (C) A test to see if a pen will write on a computer.
 - (D) A type of antivirus software.
 - (E) A method for backing up a system.
 - *Correct Answer: (B)*
-

8. What is the main difference between "Red Team" and "Blue Team"?

- (A) Red Team attacks a system; Blue Team defends it.
 - (B) Red Team is for hardware; Blue Team is for software.
 - (C) Red Team is for Windows; Blue Team is for Linux.
 - (D) Red Team is the "bad guys"; Blue Team is the "good guys". (Note: Both are "good guys" in a test).
 - (E) Red Team is for the network; Blue Team is for the computers.
 - *Correct Answer: (A)*
-

9. What is an "IDS" (Intrusion Detection System)?

- (A) A system that *only* monitors network traffic and alerts an admin of suspicious activity.
 - (B) A system that *actively blocks* suspicious activity.
 - (C) A system for "Installing, Deploying, and Securing" software.
 - (D) A type of firewall.
 - (E) A type of antivirus software.
 - *Correct Answer: (A)*
-

10. What is an "IPS" (Intrusion Prevention System)?

- (A) A system that *only* monitors and alerts.
 - (B) A system that monitors traffic and *actively blocks* detected threats.
 - (C) A system for "Installing, Protecting, and Securing" software.
 - (D) A type of antivirus software.
 - (E) A password manager.
 - *Correct Answer: (B)*
-

11. What is "spear phishing"?

- (A) A phishing attack that targets a *specific individual, group, or organization*.
 - (B) A phishing attack that uses a "spear" in the email.
 - (C) A general, non-targeted phishing email.
 - (D) A type of fishing.
 - (E) A type of secure email.
 - *Correct Answer: (A)*
-

12. What is "whaling" in cybersecurity?

- (A) A spear phishing attack that specifically targets high-profile individuals like executives (CEOs, CFOs).
 - (B) An attack that targets fishing boats.
 - (C) A type of DDoS attack.
 - (D) A security tool for "watching" network traffic.
 - (E) A type of firewall.
 - *Correct Answer: (A)*
-

13. What is a "honeypot"?

- (A) A decoy computer system or server, set up to attract and trap attackers.
 - (B) A type of malware.
 - (C) A secure database for storing passwords.
 - (D) A social engineering technique.
 - (E) A type of firewall.
 - *Correct Answer: (A)*
-

14. What is "encryption key management"?

- (A) The process of managing the *lifecycle* of cryptographic keys (generation, storage, use, and deletion).
 - (B) The process of managing employee passwords.
 - (C) A type of software for typing.
 - (D) A hardware device for storing files.
 - (E) The process of encrypting and decrypting data.
 - *Correct Answer: (A)*
-

15. What is "Cross-Site Request Forgery" (CSRF)?

- (A) An attack where a user is tricked into submitting a malicious request to a website *they are already logged into*.
 - (B) An attack where an attacker "forges" a new website.
 - (C) An attack where an attacker injects a script (this is XSS).
 - (D) An attack that steals a database.
 - (E) A security tool for forging certificates.
 - *Correct Answer: (A)*
-

16. What is a "rootkit"?

- (A) A type of malware designed to gain "root" (admin) access to a computer and remain hidden.
 - (B) A tool for securing the "root" directory.
 - (C) A type of antivirus software.
 - (D) A hardware device.
 - (E) A tool for penetration testers.
 - *Correct Answer: (A)*
-

17. What is "hardening" a system?

- (A) The process of making a computer physically harder.
 - (B) The process of making a system more difficult to use.
 - (C) The process of securing a system by reducing its attack surface (e.g., disabling unused services, applying patches).
 - (D) The process of backing up a system.
 - (E) The process of installing a virus.
 - *Correct Answer: (C)*
-

18. What is a "VPN Concentrator"?

- (A) A device that *terminates* multiple VPN tunnels, typically used at a corporate headquarters.
 - (B) A piece of software on a user's laptop.
 - (C) A type of attack against VPNs.
 - (D) A protocol for securing VPNs.
 - (E) A tool for "concentrating" on network traffic.
 - *Correct Answer: (A)*
-

19. What is "data loss prevention" (DLP)?

- (A) The process of backing up data.
 - (B) A set of tools and policies designed to prevent sensitive data from *leaving* an organization's network.
 - (C) A type of encryption.
 - (D) A type of firewall.
 - (E) A method for recovering lost data.
 - *Correct Answer: (B)*
-

20. What is "Access Control"?

- (A) The selective restriction of access to a resource; the implementation of "authorization".
 - (B) A type of password.
 - (C) A hardware device for opening a door.
 - (D) A type of malware.
 - (E) The process of logging in.
 - *Correct Answer: (A)*
-

Level 4: Expert & Specialization

1. What is a "SIEM" (Security Information and Event Management) system?

- (A) A tool that collects, aggregates, and analyzes security log data from across an organization to detect threats.
 - (B) A tool for performing penetration tests.
 - (C) A type of firewall.
 - (D) A system for managing employee passwords.
 - (E) A hardware-based encryption device.
 - *Correct Answer: (A)*
-

2. What is "Threat Intelligence"?

- (A) Information about an organization's internal security threats.
 - (B) Evidence-based knowledge (e.g., context, mechanisms, indicators) about existing or emerging threats, which can be used to inform security decisions.
 - (C) A type of AI that is a "threat".
 - (D) A list of all company vulnerabilities.
 - (E) A tool for performing DDoS attacks.
 - *Correct Answer: (B)*
-

3. What is the "Cyber Kill Chain"?

- (A) A model (by Lockheed Martin) that outlines the stages of a cyberattack, from reconnaissance to post-exploitation.
- (B) A piece of malware that "kills" a network.
- (C) A tool for "killing" viruses.
- (D) A security policy.

- (E) A type of encryption.
 - *Correct Answer: (A)*
-

4. What is "sandboxing"?

- (A) The practice of testing code on a beach.
 - (B) A security mechanism for running suspicious programs in a separate, isolated environment to see what they do.
 - (C) A type of network firewall.
 - (D) A method for storing passwords.
 - (E) A type of penetration test.
 - *Correct Answer: (B)*
-

5. What is "digital forensics"?

- (A) The process of building secure digital systems.
 - (B) The practice of collecting, analyzing, and preserving digital evidence in a way that is legally admissible.
 - (C) A type of antivirus software.
 - (D) A method for performing DDoS attacks.
 - (E) A type of encryption.
 - *Correct Answer: (B)*
-

6. What is the "MITRE ATT&CK" framework?

- (A) A globally accessible knowledge base of *adversary tactics and techniques* based on real-world observations.
 - (B) A specific piece of attack software.
 - (C) A security compliance standard, like PCI.
 - (D) A type of firewall.
 - (E) A tool for penetration testing.
 - *Correct Answer: (A)*
-

7. What is "Incident Response" (IR)?

- (A) The process of responding to all emails and support tickets.
- (B) An organization's formal plan and methodology for handling a security breach or cyberattack.

- (C) A tool that blocks attacks.
 - (D) The process of patching software.
 - (E) The process of backing up data.
 - *Correct Answer: (B)*
-

8. What is a "Web Application Firewall" (WAF)?

- (A) A firewall that protects a user's web browser.
 - (B) A firewall that filters, monitors, and blocks HTTP traffic to and from a web application (e.g., to block SQLi and XSS).
 - (C) A firewall built into the Windows operating system.
 - (D) A type of antivirus software.
 - (E) A physical hardware device.
 - *Correct Answer: (B)*
-

9. What is "container security"?

- (A) The practice of securing physical shipping containers.
 - (B) The practice of securing software "containers" (e.g., Docker, Kubernetes) and their lifecycle.
 - (C) A type of firewall.
 - (D) A method for securing backups.
 - (E) A type of physical security.
 - *Correct Answer: (B)*
-

10. What is a "Security Operations Center" (SOC)?

- (A) A command center facility where a team of security professionals monitors and analyzes an organization's security posture.
 - (B) A piece of software for securing a PC.
 - (C) A type of security attack.
 - (D) A secure data backup location.
 - (E) A type of encryption.
 - *Correct Answer: (A)*
-

11. What is "OAuth 2.0"?

- (A) A protocol for *authentication* (proving who you are).

- (B) An open standard for *delegated authorization*, allowing apps to access resources on behalf of a user (e.g., "Sign in with Google").
 - (C) A type of encryption.
 - (D) A specific type of security attack.
 - (E) A version of a firewall.
 - *Correct Answer: (B)*
-

12. What is "Server-Side Request Forgery" (SSRF)?

- (A) An attack where an attacker forges a request *from* a client.
 - (B) A vulnerability where an attacker can induce a server-side application to make requests to an unintended location.
 - (C) A type of XSS attack.
 - (D) A type of SQL Injection attack.
 - (E) A tool for testing servers.
 - *Correct Answer: (B)*
-

13. What is a "vulnerability scan"?

- (A) An *automated* process that scans a system for known vulnerabilities (e.g., missing patches, misconfigurations).
 - (B) A *manual* process of trying to exploit vulnerabilities (this is a pen test).
 - (C) A scan for viruses.
 - (D) A scan for hardware errors.
 - (E) A scan for user passwords.
 - *Correct Answer: (A)*
-

14. What is "Network Access Control" (NAC)?

- (A) A policy that denies all access to the network.
 - (B) A security approach that attempts to unify endpoint security (e.g., antivirus, patches) with network security and access control.
 - (C) A type of VPN.
 - (D) A type of firewall.
 - (E) A tool for monitoring network speed.
 - *Correct Answer: (B)*
-

15. What is a "password-spraying" attack?

- (A) An attack that tries *many different passwords* for a *single username*.
 - (B) An attack that tries a *single, common password* against *many different usernames*.
 - (C) An attack where a user "sprays" their password on a wall.
 - (D) A tool for generating strong passwords.
 - (E) A type of ransomware.
 - *Correct Answer: (B)*
-

16. What is "credential stuffing"?

- (A) The act of "stuffing" credentials into a secure vault.
- (B) An attack where stolen usernames and passwords from one data breach are used to try and log in to other services.
- (C) A type of password-spraying attack.
- (D) A security feature.
- (E) A tool for penetration testers.
- *Correct Answer: (B)*

17. What is "threat modeling"?

- (A) A process for building a model of a computer network.
 - (B) A structured process for identifying, quantifying, and addressing potential threats and vulnerabilities in a system *during its design phase*.
 - (C) A tool for creating "honeypots".
 - (D) A method for responding to an incident.
 - (E) A type of security compliance.
 - *Correct Answer: (B)*
-

18. What is "SOAR" (Security Orchestration, Automation, and Response)?

- (A) A tool that *replaces* a SIEM.
 - (B) A platform that helps security teams manage and respond to alarms by *automating* and *orchestrating* security workflows.
 - (C) A type of security attack.
 - (D) A physical security device.
 - (E) A compliance standard.
 - *Correct Answer: (B)*
-

19. What is "fuzzing"?

- (A) An automated testing technique that provides invalid, unexpected, or random data as input to a program to find bugs.
 - (B) A manual testing technique.
 - (C) A type of social engineering attack.
 - (D) A method for encrypting data.
 - (E) A type of network scan.
 - *Correct Answer: (A)*
-

20. What is "reverse engineering" in a malware context?

- (A) The process of building new malware.
 - (B) The process of decompiling and analyzing a piece of malware to understand what it does and how it works.
 - (C) A type of security attack.
 - (D) A method for patching software.
 - (E) The process of designing a secure system in reverse.
 - *Correct Answer: (B)*
-

Level 5: Strategic & Architectural

1. What is "Governance, Risk, and Compliance" (GRC)?

- (A) A specific type of security software.
 - (B) A structured approach to aligning an organization's IT with its business goals, while managing risks and meeting compliance requirements.
 - (C) A team of hackers (Red Team).
 - (D) A model for a cyberattack.
 - (E) A type of firewall.
 - *Correct Answer: (B)*
-

2. What is a "Business Continuity Plan" (BCP)?

- (A) A plan for how a business will *continue to operate* during and after a disaster.
- (B) A plan for how to stop a cyberattack.
- (C) A plan for how to back up data.
- (D) A plan for how to patch servers.
- (E) A plan for the company's "continuity" party.
- *Correct Answer: (A)*

3. What is a "Disaster Recovery" (DR) plan?

- (A) A plan focused *specifically* on restoring the IT infrastructure and data after a disaster. (It is a *part* of the BCP).
- (B) A plan for firing employees after a disaster.
- (C) A plan for preventing disasters.
- (D) A plan for managing security risks.
- (E) A plan for a company's financial recovery.
- *Correct Answer: (A)*

4. What is a "security risk assessment"?

- (A) The process of *identifying, analyzing, and evaluating* risks to an organization's assets.
- (B) The process of buying insurance for a security risk.
- (C) A list of all company assets.
- (D) A tool that blocks all risks.
- (E) A penetration test.
- *Correct Answer: (A)*

5. What is the difference between "quantitative" and "qualitative" risk analysis?

- (A) Quantitative uses *numerical values* (e.g., money); Qualitative uses *descriptive categories* (e.g., high, medium, low).
- (B) Quantitative is for "quantity" of risks; Qualitative is for "quality" of risks.
- (C) Quantitative is fast; Qualitative is slow.
- (D) Quantitative is for hardware; Qualitative is for software.
- (E) There is no difference.
- *Correct Answer: (A)*

6. What is "defense in depth"?

- (A) A security strategy that relies on a *single, very strong* defense (like a firewall).
- (B) A security strategy that uses *multiple, layered* security controls, so if one fails, another may stop the attack.
- (C) A type of military strategy.
- (D) A strategy for "deeply" analyzing a threat.
- (E) A type of data backup.

- *Correct Answer: (B)*
-

7. What is "ISO 27001"?

- (A) A specific type of firewall.
 - (B) An *international standard* that specifies the requirements for an Information Security Management System (ISMS).
 - (C) A US government law.
 - (D) A type of encryption.
 - (E) A piece of hacking software.
 - *Correct Answer: (B)*
-

8. What is "PCI DSS"?

- (A) A "Payment Card Industry Data Security Standard," a set of requirements for any organization that handles credit card data.
 - (B) A "Personal Computer Information" standard.
 - (C) A government law for data privacy.
 - (D) A type of antivirus software.
 - (E) A standard for building secure PCs.
 - *Correct Answer: (A)*
-

9. What is "HIPAA"?

- (A) A US law (Health Insurance Portability and Accountability Act) that includes rules for protecting the privacy and security of patient health information.
 - (B) A security standard for banks.
 - (C) A type of encryption.
 - (D) A type of "hippopotamus" virus.
 - (E) A European data privacy law.
 - *Correct Answer: (A)*
-

10. What is "GDPR"?

- (A) A US data privacy law.
- (B) A "General Data Protection Regulation" - a European Union law on data protection and privacy.
- (C) A "General Data Risk" policy.

- (D) A standard for building secure networks.
 - (E) A type of security attack.
 - *Correct Answer: (B)*
-

11. What is a "Chief Information Security Officer" (CISO)?

- (A) The senior-level executive responsible for an organization's entire information security program.
 - (B) The lead developer on the security team.
 - (C) A consultant who performs penetration tests.
 - (D) A government agent in charge of cybersecurity.
 - (E) The person who manages the firewall.
 - *Correct Answer: (A)*
-

12. What is "security architecture"?

- (A) The physical architecture of a secure building.
 - (B) The *design* of a secure IT system, based on security principles, to protect assets and manage risk.
 - (C) A diagram of a network.
 - (D) A list of security rules.
 - (E) A type of firewall.
 - *Correct Answer: (B)*
-

13. What is a "Zero Trust" security model?

- (A) A model where you "trust no one" on your team.
 - (B) A model where you "trust" all devices inside the network, but "zero" outside.
 - (C) A security model based on the principle of "never trust, always verify," treating all users and devices as untrusted.
 - (D) A model with "zero" firewalls.
 - (E) A model that has "zero" cost.
 - *Correct Answer: (C)*
-

14. What is a "supply chain" attack in cybersecurity?

- (A) An attack on a company's physical shipping and logistics.

- (B) An attack where an attacker compromises a trusted *third-party vendor* (e.g., a software supplier) to gain access to the real target.
 - (C) An attack on a grocery store's supply chain.
 - (D) A type of phishing attack.
 - (E) A type of DDoS attack.
 - *Correct Answer: (B)*
-

15. What are the key phases of an "Incident Response" plan?

- (A) 1. Preparation, 2. Identification, 3. Containment, 4. Eradication, 5. Recovery, 6. Lessons Learned.
 - (B) 1. Attack, 2. Defend, 3. Win.
 - (C) 1. Blame, 2. Fire, 3. Hire.
 - (D) 1. Detect, 2. Delete, 3. Restore.
 - (E) 1. Call the CEO, 2. Call the police, 3. Call the news.
 - *Correct Answer: (A)*
-

16. What is "threat hunting"?

- (A) The *proactive* and *iterative* process of searching through networks and datasets to detect and isolate advanced threats that evade existing security tools.
 - (B) The *reactive* process of responding to a security alert.
 - (C) A type of penetration test.
 - (D) A type of social engineering.
 - (E) A tool for "hunting" for files.
 - *Correct Answer: (A)*
-

17. What is "Cloud Security Posture Management" (CSPM)?

- (A) A tool for managing employee posture (sitting) in the cloud.
 - (B) A class of tools designed to identify and remediate misconfigurations and compliance risks in cloud environments.
 - (C) A type of cloud firewall.
 - (D) A cloud-based antivirus.
 - (E) A cloud provider's physical security.
 - *Correct Answer: (B)*
-

18. What is "Business Email Compromise" (BEC)?

- (A) A type of malware that compromises an email server.
 - (B) A type of phishing attack.
 - (C) An attack where an attacker gains access to a business email account and impersonates the owner to defraud the company.
 - (D) A security tool for scanning emails.
 - (E) A compliance standard for emails.
 - *Correct Answer: (C)*
-

19. What is a "purple team"?

- (A) A team that is a mix of managers and engineers.
 - (B) A team that works to foster communication and collaboration between the Red Team (attack) and Blue Team (defense).
 - (C) A team of external consultants.
 - (D) A compliance and audit team.
 - (E) A team that develops security software.
 - *Correct Answer: (B)*
-

20. What is "cyber resilience"?

- (A) The ability of an organization to *resist* all cyberattacks so they never fail.
- (B) The ability of an organization to *prepare for, respond to, and recover from* a cyberattack, while continuing to operate.
- (C) A type of antivirus software.
- (D) A type of insurance policy.
- (E) A government agency.
- *Correct Answer: (B)*

Database Administration track.

Level 1: Basic Foundations

1. What does "DBA" stand for?

- (A) Data Backup and Access
- (B) Database Administrator
- (C) Database Application
- (D) Data by Algorithm
- (E) Database Allocation

- *Correct Answer: (B)*
-

2. What is a "database"?

- (A) A computer program for browsing the internet.
 - (B) A physical server.
 - (C) An organized collection of structured data, stored electronically.
 - (D) A single spreadsheet file.
 - (E) A type of programming language.
 - *Correct Answer: (C)*
-

3. What does "RDBMS" stand for?

- (A) Relational Database Management System
 - (B) Real-time Data Management Service
 - (C) Remote Database Mainframe System
 - (D) Relational Data Backup System
 - (E) Real-time Database Module
 - *Correct Answer: (A)*
-

4. What is "SQL"?

- (A) A type of server hardware.
 - (B) A standard language for communicating with and managing relational databases.
 - (C) A security protocol.
 - (D) A data visualization tool.
 - (E) A type of database backup.
 - *Correct Answer: (B)*
-

5. What is a "table" in a database?

- (A) A visual report or dashboard.
- (B) A single piece of data, like a name.
- (C) A data structure that organizes information into rows and columns.
- (D) A user's password.
- (E) A connection to the database.
- *Correct Answer: (C)*

6. What is a "column" in a database table?

- (A) A single, complete record (e.g., all data for one user).
- (B) A single attribute or field of data (e.g., "Email" or "Age").
- (C) The entire database.
- (D) A security permission.
- (E) A backup file.
- *Correct Answer: (B)*

7. What is a "row" in a database table?

- (A) A single attribute or field (e.g., the "Name" column).
- (B) A single, complete record or entry (e.g., all data for one user).
- (C) The name of the table.
- (D) A type of data (e.g., text).
- (E) A SQL command.
- *Correct Answer: (B)*

8. What is a "primary key"?

- (A) A column that is used for sorting data.
- (B) A column (or columns) that uniquely identifies each row in a table.
- (C) The main password to the database.
- (D) The first column in every table.
- (E) A key that links to another table.
- *Correct Answer: (B)*

9. What is a "foreign key"?

- (A) A key that is not from your country.
 - (B) A column that uniquely identifies each row (this is a primary key).
 - (C) A column in one table that links to the primary key of another table.
 - (D) A password to access the table.
 - (E) The last column in every table.
 - *Correct Answer: (C)*
-

10. What is a "data type"?

- (A) A type of chart (e.g., bar, pie).
 - (B) A classification that specifies the type of data a column can hold (e.g., INT, VARCHAR, DATE).
 - (C) The name of the database.
 - (D) A SQL command.
 - (E) A user's role.
 - *Correct Answer: (B)*
-

11. What is a "query"?

- (A) A request sent to the database to retrieve, add, or modify data.
 - (B) A table of data.
 - (C) A database backup.
 - (D) A security vulnerability.
 - (E) A user's permission.
 - *Correct Answer: (A)*
-

12. What does "DDL" stand for in SQL?

- (A) Data Deletion Language
 - (B) Data Definition Language (e.g., CREATE, ALTER, DROP)
 - (C) Data Duplication Language
 - (D) Database Download Logic
 - (E) Data Manipulation Language
 - *Correct Answer: (B)*
-

13. What does "DML" stand for in SQL?

- (A) Data Model Language
 - (B) Database Management Logic
 - (C) Data Manipulation Language (e.g., SELECT, INSERT, UPDATE, DELETE)
 - (D) Data Migration Language
 - (E) Data Definition Language
 - *Correct Answer: (C)*
-

14. What does "NULL" represent in a database?

- (A) The number zero (0).
 - (B) An empty string ("").
 - (C) The absence of a value, or an unknown value.
 - (D) A "false" boolean value.
 - (E) An error in the data.
 - *Correct Answer: (C)*
-

15. What is the basic purpose of an "index"?

- (A) To secure the database from hackers.
 - (B) To back up the data.
 - (C) To create a table.
 - (D) To speed up the retrieval of data (queries) from a table.
 - (E) To delete data.
 - *Correct Answer: (D)*
-

16. What is "data integrity"?

- (A) The speed of the database.
 - (B) The overall accuracy, completeness, and consistency of data.
 - (C) The process of backing up data.
 - (D) The physical size of the database.
 - (E) The programming language used.
 - *Correct Answer: (B)*
-

17. What is a "database backup"?

- (A) A tool for speeding up queries.
 - (B) A copy of the database, used to restore it in case of data loss.
 - (C) A user's password.
 - (D) A firewall.
 - (E) A table of data.
 - *Correct Answer: (B)*
-

18. What is a "schema"?

- (A) A single SQL query.
- (B) The logical structure or "blueprint" of a database (e.g., tables, columns, keys).

- (C) A user account.
 - (D) A data visualization.
 - (E) A backup file.
 - *Correct Answer: (B)*
-

19. What does "DCL" stand for in SQL?

- (A) Data Cleaning Language
 - (B) Data Connection Language
 - (C) Data Control Language (e.g., GRANT, REVOKE)
 - (D) Data Column Language
 - (E) Data Creation Language
 - *Correct Answer: (C)*
-

20. What is a "NoSQL" database?

- (A) A database that has no security.
 - (B) A database that does not use a relational (table-based) model.
 - (C) A database that cannot be queried.
 - (D) An old, outdated type of database.
 - (E) A database that only stores numbers.
 - *Correct Answer: (B)*
-

Level 2: Intermediate Application

1. Which SQL command is used to retrieve all columns from a table named **Employees?**

- (A) `GET * FROM Employees;`
 - (B) `SELECT Employees;`
 - (C) `READ * FROM Employees;`
 - (D) `SELECT * FROM Employees;`
 - (E) `SELECT ALL FROM Employees;`
 - *Correct Answer: (D)*
-

2. Which SQL clause is used to filter the results of a **SELECT statement?**

- (A) `FILTER BY`
 - (B) `WHERE`
 - (C) `GROUP BY`
 - (D) `IF`
 - (E) `LIMIT`
 - *Correct Answer: (B)*
-

3. Which SQL command is used to add a new row of data to a table?

- (A) `ADD ROW`
 - (B) `CREATE ROW`
 - (C) `INSERT INTO`
 - (D) `UPDATE`
 - (E) `NEW ROW`
 - *Correct Answer: (C)*
-

4. Which SQL command is used to modify existing data in a table?

- (A) `MODIFY`
 - (B) `CHANGE`
 - (C) `INSERT INTO`
 - (D) `UPDATE`
 - (E) `ALTER TABLE`
 - *Correct Answer: (D)*
-

5. Which SQL command is used to delete a row from a table?

- (A) `REMOVE ROW FROM ... WHERE ...;`
 - (B) `DELETE FROM ... WHERE ...;`
 - (C) `DROP ROW FROM ... WHERE ...;`
 - (D) `TRUNCATE ... WHERE ...;`
 - (E) `CLEAR ROW FROM ... WHERE ...;`
 - *Correct Answer: (B)*
-

6. Which SQL command is used to create a new table?

- (A) NEW TABLE ...
 - (B) BUILD TABLE ...
 - (C) INSERT TABLE ...
 - (D) ALTER TABLE ...
 - (E) CREATE TABLE ...
 - *Correct Answer: (E)*
-

7. Which SQL command is used to add a new column to an existing table?

- (A) ALTER TABLE ... ADD COLUMN ...
 - (B) UPDATE TABLE ... ADD COLUMN ...
 - (C) CREATE COLUMN ... ON ...
 - (D) MODIFY TABLE ... ADD ...
 - (E) INSERT COLUMN ... INTO ...
 - *Correct Answer: (A)*
-

8. Which SQL command completely removes a table (and all its data) from the database?

- (A) DELETE TABLE ...
 - (B) REMOVE TABLE ...
 - (C) TRUNCATE TABLE ...
 - (D) DROP TABLE ...
 - (E) CLEAR TABLE ...
 - *Correct Answer: (D)*
-

9. Which SQL clause is used to sort the result set?

- (A) SORT BY
 - (B) ORDER BY
 - (C) GROUP BY
 - (D) ALIGN BY
 - (E) WHERE
 - *Correct Answer: (B)*
-

10. Which SQL aggregate function returns the total number of rows?

- (A) SUM()
 - (B) TOTAL()
 - (C) COUNT(*)
 - (D) ROWS()
 - (E) MAX()
 - *Correct Answer: (C)*
-

11. Which SQL aggregate function calculates the average of a numerical column?

- (A) MEAN()
 - (B) MEDIAN()
 - (C) AVG()
 - (D) AVERAGE()
 - (E) CALCULATE()
 - *Correct Answer: (C)*
-

12. Which SQL clause is used with aggregate functions to group rows with the same values?

- (A) ORDER BY
 - (B) GROUP BY
 - (C) WHERE
 - (D) JOIN
 - (E) CLUSTER BY
 - *Correct Answer: (B)*
-

13. Which SQL command is used to combine rows from two tables based on a related column?

- (A) MERGE
 - (B) COMBINE
 - (C) JOIN
 - (D) CONNECT
 - (E) RELATE
 - *Correct Answer: (C)*
-

14. What is the difference between DELETE and TRUNCATE?

- (A) DELETE removes rows one by one (and can be filtered); TRUNCATE removes all rows at once (and is faster).
 - (B) DELETE removes a table; TRUNCATE removes a row.
 - (C) DELETE is permanent; TRUNCATE can be undone.
 - (D) DELETE removes all rows; TRUNCATE removes only some.
 - (E) There is no difference.
 - *Correct Answer: (A)*
-

15. Which SQL command is used to give a user permission to SELECT data?

- (A) ALLOW SELECT ON ... TO ...;
 - (B) GRANT SELECT ON ... TO ...;
 - (C) CREATE USER ... WITH SELECT;
 - (D) SET PERMISSION SELECT ON ... TO ...;
 - (E) AUTHORIZE SELECT ON ... TO ...;
 - *Correct Answer: (B)*
-

16. Which SQL command is used to remove a user's permission?

- (A) REMOVE
 - (B) DELETE
 - (C) DENY
 - (D) UNGRANT
 - (E) REVOKE
 - *Correct Answer: (E)*
-

17. Which SQL command is used to create an index?

- (A) CREATE INDEX ... ON ... (...);
- (B) ADD INDEX ... ON ... (...);
- (C) CREATE NEW INDEX ...;
- (D) ALTER TABLE ... ADD INDEX ...;
- (E) INDEX TABLE ... (...);
- *Correct Answer: (A)*

18. What is a **LEFT JOIN**?

- (A) It returns *only* the rows that match in both tables.
- (B) It returns *all* rows from the left table, and the matched rows from the right table.
- (C) It returns *all* rows from the right table, and the matched rows from the left table.
- (D) It returns *all* rows from both tables.
- (E) It returns only the rows on the "left" side of the database.
- *Correct Answer: (B)*

19. Which operator is used for pattern matching in a **WHERE** clause?

- (A) **MATCH**
- (B) **LIKE**
- (C) **SIMILAR**
- (D) **REGEX**
- (E) **CONTAINS**
- *Correct Answer: (B)*

20. What does the **DISTINCT** keyword do in a **SELECT** statement?

- (A) It sorts the results.
- (B) It returns only unique (non-duplicate) values for a column.
- (C) It counts the number of rows.
- (D) It joins two tables.
- (E) It makes the query run faster.
- *Correct Answer: (B)*

Level 3: Advanced Scenarios

1. What is the difference between the **WHERE** clause and the **HAVING** clause?

- (A) **WHERE** filters rows *before* grouping; **HAVING** filters groups *after* grouping.
- (B) **WHERE** filters groups; **HAVING** filters rows.
- (C) **WHERE** is for **SELECT**; **HAVING** is for **UPDATE**.
- (D) **WHERE** is for numbers; **HAVING** is for text.

- (E) They are identical.
 - *Correct Answer: (A)*
-

2. What is a "transaction" in a database?

- (A) A single SQL query.
 - (B) A sequence of one or more database operations (e.g., **INSERT**, **UPDATE**) that are treated as a single, atomic unit.
 - (C) A backup of the database.
 - (D) A user's permission.
 - (E) The financial cost of a query.
 - *Correct Answer: (B)*
-

3. What does "ACID" stand for?

- (A) Access, Control, Integrity, Data
 - (B) Atomicity, Consistency, Isolation, Durability
 - (C) Authorize, Commit, Isolate, Delete
 - (D) A secure, coded, integrated database
 - (E) Asynchronous, Concurrent, Independent, Data
 - *Correct Answer: (B)*
-

4. What is "database normalization"?

- (A) The process of making a database "normal" by deleting all old data.
 - (B) The process of organizing data in a database to reduce redundancy and improve data integrity.
 - (C) The process of adding more indexes to a database.
 - (D) The process of backing up a database.
 - (E) The process of encrypting a database.
 - *Correct Answer: (B)*
-

5. What is a "view" in SQL?

- (A) A physical copy of a table.
- (B) A virtual table based on the result-set of a stored SQL query.
- (C) A user's permission.
- (D) A visual dashboard.

- (E) A database log file.
 - *Correct Answer: (B)*
-

6. What is a "stored procedure"?

- (A) A backup file that is "stored" safely.
 - (B) A set of pre-compiled SQL statements that are stored in the database and can be executed as a single unit.
 - (C) A table that stores procedures.
 - (D) A type of database user.
 - (E) A log of all past queries.
 - *Correct Answer: (B)*
-

7. What is a "trigger"?

- (A) A type of security attack.
 - (B) A stored procedure that automatically executes (fires) in response to a specific event (e.g., **INSERT**, **UPDATE**).
 - (C) A button on a dashboard.
 - (D) A hardware device for a server.
 - (E) A manual backup process.
 - *Correct Answer: (B)*
-

8. What is a "subquery"?

- (A) A query that is not as good as a normal query.
 - (B) A query that is embedded (nested) inside another SQL query.
 - (C) A query that updates data.
 - (D) A query that only runs on a subset of data.
 - (E) A query that is saved as a view.
 - *Correct Answer: (B)*
-

9. What is a "full backup"?

- (A) A backup of only the new or changed files.
- (B) A complete copy of the entire database.
- (C) A backup of only the database schema.
- (D) A backup of only the transaction log.

- (E) A backup that is "full" and has no more space.
 - *Correct Answer: (B)*
-

10. What is a "differential backup"?

- (A) A backup of only the data that has changed *since the last full backup*.
 - (B) A backup of only the data that has changed *since the last backup of any kind*.
 - (C) A complete copy of the database.
 - (D) A backup of only the transaction log.
 - (E) A backup that is "different" every time.
 - *Correct Answer: (A)*
-

11. What is the purpose of the **EXPLAIN or **EXPLAIN PLAN** command?**

- (A) To "explain" the data in a table.
 - (B) To get help on a SQL command.
 - (C) To show the execution plan (how the database will run the query), used for performance tuning.
 - (D) To describe the table schema.
 - (E) To create a database diagram.
 - *Correct Answer: (C)*
-

12. Why can too many indexes be bad for performance?

- (A) They slow down **SELECT** queries.
 - (B) They take up too much disk space and slow down write operations (e.g., **INSERT**, **UPDATE**).
 - (C) They cause security vulnerabilities.
 - (D) They corrupt the data.
 - (E) They cannot be bad; more indexes are always better.
 - *Correct Answer: (B)*
-

13. What is the difference between a "clustered" and "non-clustered" index?

- (A) A clustered index is for a cluster of servers; a non-clustered index is for one server.
- (B) A clustered index defines the *physical order* of the data; a non-clustered index is a separate structure that points to the data.
- (C) A clustered index is slow; a non-clustered index is fast.

- (D) A clustered index is for text; a non-clustered index is for numbers.
 - (E) There is no difference.
 - *Correct Answer: (B)*
-

14. What is the purpose of a "transaction log"?

- (A) A log of all users who have logged in.
 - (B) A log of all slow queries.
 - (C) A serial record of all changes made to the database, used for recovery.
 - (D) A file that stores the database schema.
 - (E) A backup file.
 - *Correct Answer: (C)*
-

15. What is "denormalization"?

- (A) The process of making a database "abnormal" or corrupt.
 - (B) The process of intentionally adding redundant data to a normalized database to improve query performance.
 - (C) The process of deleting data.
 - (D) Another name for normalization.
 - (E) A security attack.
 - *Correct Answer: (B)*
-

16. What is the difference between **UNION and **UNION ALL**?**

- (A) **UNION** selects all rows; **UNION ALL** selects only some.
 - (B) **UNION** removes duplicate rows; **UNION ALL** includes all rows, including duplicates.
 - (C) **UNION** is for joining tables; **UNION ALL** is for joining databases.
 - (D) **UNION** is fast; **UNION ALL** is slow.
 - (E) There is no difference.
 - *Correct Answer: (B)*
-

17. What is a "composite key"?

- (A) A key made of two or more columns, used together to uniquely identify a row.
- (B) A key that is stored in a separate file.
- (C) A key that is encrypted.
- (D) Another name for a foreign key.

- (E) A key that is not reliable.
 - *Correct Answer: (A)*
-

18. What is a "CHECK constraint"?

- (A) A constraint that "checks" if a user has permission.
 - (B) A rule that limits the values that can be placed in a column (e.g., `Age > 18`).
 - (C) A constraint that checks if a primary key is unique.
 - (D) A constraint that checks for viruses.
 - (E) A constraint that links to a foreign key.
 - *Correct Answer: (B)*
-

19. What is a "CTE" (Common Table Expression)?

- (A) A "Critical Table Error."
 - (B) A temporary, named result set that you can reference within a `SELECT`, `INSERT`, `UPDATE`, or `DELETE` statement.
 - (C) A permanent table.
 - (D) A tool for database cleaning.
 - (E) A type of index.
 - *Correct Answer: (B)*
-

20. What is an "incremental backup"?

- (A) A backup of only the data that has changed *since the last backup* (full, differential, or incremental).
 - (B) A backup of only the data that has changed *since the last full backup*.
 - (C) A complete copy of the database.
 - (D) A backup that slowly "increments" in size.
 - (E) A backup of only the transaction log.
 - *Correct Answer: (A)*
-

Level 4: Expert & Specialization

1. What is a "database lock"?

- (A) A physical lock on the server room.

- (B) A mechanism to prevent two users from modifying the same piece of data at the same time (concurrency control).
 - (C) A security feature that "locks" a user out.
 - (D) A type of encryption key.
 - (E) A backup file.
 - *Correct Answer: (B)*
-

2. What is a "deadlock"?

- (A) A situation where a database server has crashed.
 - (B) A situation where two or more transactions are "stuck," each waiting for the other to release a lock.
 - (C) A security vulnerability.
 - (D) A corrupt backup file.
 - (E) A lock on a user account.
 - *Correct Answer: (B)*
-

3. What is "SQL injection" and what is the *best* way to prevent it?

- (A) An attack where malicious SQL is "injected" into a query; prevented by using prepared statements (parameterized queries).
 - (B) An attack where a user "injects" JavaScript; prevented by a firewall.
 - (C) An attack where a server is flooded with traffic; prevented by rate limiting.
 - (D) An attack where a user steals a backup; prevented by encryption.
 - (E) An attack where a database is deleted; prevented by **GRANT/REVOKE**.
 - *Correct Answer: (A)*
-

4. What is "database replication"?

- (A) The process of creating multiple copies of a database (replicas), often on different servers, for high availability and read scalability.
 - (B) A security attack where a hacker replicates your database.
 - (C) A "master" database that controls all the "slave" APIs.
 - (B) The process of creating a single backup.
 - (E) A method for encrypting the database.
 - *Correct Answer: (A)*
-

5. What is "database clustering"?

- (A) A group of databases that are all "clustered" in one location.
 - (B) A group of servers (nodes) that work together as a single system to provide high availability and load balancing.
 - (C) A type of index.
 - (D) A security feature.
 - (E) The process of sharding a database.
 - *Correct Answer: (B)*
-

6. What is "database sharding"?

- (A) A way to encrypt a database.
 - (B) A "horizontal partitioning" technique that splits a large database into smaller, faster parts (shards).
 - (C) A security attack on a database.
 - (D) The process of deleting old data.
 - (E) A way to create a backup of a database.
 - *Correct Answer: (B)*
-

7. What is the difference between "OLTP" and "OLAP"?

- (A) OLTP is for analytics; OLAP is for transactions.
 - (B) OLTP (Online Transaction Processing) is for fast, day-to-day operations; OLAP (Online Analytical Processing) is for complex analysis.
 - (C) OLTP is old; OLAP is new.
 - (D) OLTP is for backups; OLAP is for recovery.
 - (E) There is no difference.
 - *Correct Answer: (B)*
-

8. What is a "data warehouse"?

- (A) A small database on a laptop.
 - (B) A large, central repository of data, optimized for analysis and reporting (OLAP), that integrates data from many sources.
 - (C) A single Excel file.
 - (D) A tool for data visualization.
 - (E) A server for hosting a website.
 - *Correct Answer: (B)*
-

9. What is "Point-in-Time Recovery" (PITR)?

- (A) The ability to recover a database *exactly* as it was at a specific time (e.g., 9:15 AM), usually by using backups and transaction logs.
 - (B) The ability to recover only one "point" of data.
 - (C) A backup that is taken at a specific "point in time."
 - (D) A type of security attack.
 - (E) A tool for query optimization.
 - *Correct Answer: (A)*
-

10. What is a "query optimizer"?

- (A) A tool that rewrites your SQL queries to be more efficient.
 - (B) A hardware device that speeds up a server.
 - (C) A component of the database that determines the most efficient way to execute a query.
 - (D) A DBA who specializes in writing fast queries.
 - (E) A type of index.
 - *Correct Answer: (C)*
-

11. What is a "connection pool"?

- (A) A "pool" of water used to cool a server.
 - (B) A cache of database connections that are kept open and reused, reducing the overhead of opening/closing connections.
 - (C) A group of users who are allowed to connect.
 - (D) A type of network security group.
 - (E) A table that lists all database connections.
 - *Correct Answer: (B)*
-

12. What is "Transparent Data Encryption" (TDE)?

- (A) Encryption that is "transparent" and easy to break.
- (B) A feature that encrypts the *entire* database at rest (on the disk) without the application needing to be aware of it.
- (C) Encryption of data as it travels over the network (e.g., SSL/TLS).
- (D) A tool for encrypting passwords.
- (E) A type of firewall.
- *Correct Answer: (B)*

13. What is "auditing" in a database context?

- (A) The process of checking the database for financial errors.
- (B) The process of tracking and logging specific database events (e.g., who accessed what data at what time).
- (C) The process of optimizing a query.
- (D) The process of backing up a database.
- (E) The process of installing a database.
- *Correct Answer: (B)*

14. What is a "column-store" index/database?

- (A) A database that only stores one column.
- (B) A database that stores data *by column* rather than *by row*, which is highly efficient for analytical (OLAP) queries.
- (C) A database that is stored in an Excel column.
- (D) A type of encrypted index.
- (E) A standard row-based (row-store) database.
- *Correct Answer: (B)*

15. What is "Multi-Version Concurrency Control" (MVCC)?

- (A) A "concurrency control" method where readers do not block writers, by showing them a "snapshot" of the data.
- (B) A tool for managing multiple "versions" of a database.
- (C) A security protocol.
- (D) A backup strategy.
- (E) A type of database clustering.
- *Correct Answer: (A)*

16. What is a "Disaster Recovery" (DR) plan?

- (A) A plan for how to prevent all disasters.
- (B) A plan for how to restore IT infrastructure (including databases) at a *secondary site* after a major disaster.
- (C) A plan for firing employees after a disaster.
- (D) A plan for backing up a single file.
- (E) A plan for managing a security breach.

- *Correct Answer: (B)*
-

17. What is a "hot standby" vs. a "cold standby"?

- (A) A hot standby is a server that is overheating.
 - (B) A hot standby is a replica server that is on and ready to take over; a cold standby is a server that is off and must be started.
 - (C) A hot standby is for summer; a cold standby is for winter.
 - (D) A hot standby is a backup; a cold standby is a live server.
 - (E) A hot standby is insecure; a cold standby is secure.
 - *Correct Answer: (B)*
-

18. What is "capacity planning"?

- (A) The process of planning how many users can access the database.
 - (B) The process of monitoring and forecasting resource usage (e.g., disk, CPU, RAM) to ensure the database can meet future demand.
 - (C) The process of cleaning up old data to free "capacity."
 - (D) The process of designing a new database.
 - (E) The process of planning a backup.
 - *Correct Answer: (B)*
-

19. What is a "maintenance plan"?

- (A) A plan for physically cleaning the server room.
 - (B) A set of automated tasks (e.g., backups, index rebuilds, integrity checks) that run on a schedule to keep the database healthy.
 - (C) A plan for updating the database software.
 - (D) A plan for training DBAs.
 - (E) A user's guide to the database.
 - *Correct Answer: (B)*
-

20. What is "Role-Based Access Control" (RBAC)?

- (A) A method of managing permissions by assigning users to "roles" (e.g., "Analyst," "Admin") and then granting permissions to those roles.
- (B) A method where every user gets their own specific permissions.
- (C) A security attack.

- (D) A type of database.
 - (E) A hardware device for access control.
 - *Correct Answer: (A)*
-

Level 5: Strategic & Architectural

1. What are the key factors when choosing a SQL vs. a NoSQL database for a new project?

- (A) SQL is for old projects; NoSQL is for new projects.
 - (B) SQL is for simple data; NoSQL is for complex data.
 - (C) The choice depends on data structure (schema-on-write vs. schema-on-read), scalability needs, and consistency (ACID) requirements.
 - (D) SQL is for small data; NoSQL is for "big data."
 - (E) The choice depends on the programming language being used.
 - *Correct Answer: (C)*
-

2. You need to design a database schema for a simple blog (with users, posts, and comments). What are the key tables and relationships?

- (A) One big table for users, posts, and comments.
 - (B) **Users** (PK: user_id), **Posts** (PK: post_id, FK: user_id), **Comments** (PK: comment_id, FK: post_id, FK: user_id).
 - (C) **Posts** (PK: post_id) and **Comments** (PK: comment_id).
 - (D) **Users** (PK: user_id) and **Posts_and_Comments** (PK: id).
 - (E) **Blog** (PK: id, Text: all_data).
 - *Correct Answer: (B)*
-

3. A developer reports that a specific query is running very slowly. What are your *first five steps* to troubleshoot it?

- (A) 1. Restart the server. 2. Blame the network. 3. Blame the developer. 4. Add more RAM. 5. Delete old data.
- (B) 1. Get the query. 2. Run an **EXPLAIN PLAN**. 3. Check for missing/unused indexes. 4. Check for table scans. 5. Check for out-of-date statistics.
- (C) 1. Ask the user to run it at night. 2. Tell the developer to rewrite the app. 3. Switch to NoSQL. 4. Buy a new server. 5. Defragment the disk.

- (D) 1. Check the firewall. 2. Check the user's password. 3. Check the transaction log. 4. Run a full backup. 5. Update the server OS.
 - (E) 1. Add indexes to every column. 2. Denormalize the tables. 3. Drop all constraints. 4. Give the user admin rights. 5. Restart the query.
 - *Correct Answer: (B)*
-

4. What is the business case for "denormalization"? When would you strategically use it?

- (A) Never; it is a bad practice.
 - (B) When you want to save disk space.
 - (C) In a read-heavy (OLAP) system where query performance is more important than data redundancy, and **JOINS** are too costly.
 - (D) In a write-heavy (OLTP) system to make **INSERTs** faster.
 - (E) When you are first designing the database and don't know the schema.
 - *Correct Answer: (C)*
-

5. What is a "High Availability" (HA) strategy, and how does it differ from "Disaster Recovery" (DR)?

- (A) HA is about surviving a *disaster*; DR is about surviving a *component failure*.
 - (B) HA is about *automatic* recovery from component failure (e.g., clustering); DR is about *manual* recovery from a site-wide disaster (e.g., restoring backups).
 - (C) They are the same thing.
 - (D) HA is a backup plan; DR is a replication plan.
 - (E) HA is for software; DR is for hardware.
 - *Correct Answer: (B)*
-

6. What is the "CAP Theorem," and why is it important for distributed databases?

- (A) A theorem stating a distributed system can only provide two of three guarantees: Consistency, Availability, and Partition Tolerance.
 - (B) A theorem about CPU processing (Capacity, Availability, Performance).
 - (C) A rule for naming database tables.
 - (D) A security protocol for APIs.
 - (E) A method for caching data.
 - *Correct Answer: (A)*
-

7. What are the pros and cons of using a "Database-as-a-Service" (DBaaS) like AWS RDS vs. a self-managed server?

- (A) DBaaS is more expensive and complex; self-managed is cheaper and easier.
 - (B) DBaaS is easier (handles patches, backups); self-managed gives you *full control* but requires *more admin overhead*.
 - (C) DBaaS is less secure; self-managed is more secure.
 - (D) DBaaS is for small data; self-managed is for big data.
 - (E) There are no pros to using DBaaS.
 - *Correct Answer: (B)*
-

8. What is a "database migration" strategy, and how would you perform one with minimal downtime?

- (A) A strategy for moving data; best way is to shut down the app for a weekend.
 - (B) A strategy for changing database vendors; best way is to use a "blue-green" deployment or "log-shipping" replication.
 - (C) A strategy for deleting old data.
 - (D) A strategy for backing up data.
 - (E) A strategy for hiring a new DBA.
 - *Correct Answer: (B)*
-

9. What is a "data retention policy," and what is the DBA's role in enforcing it?

- (A) A policy for how long to "retain" employees; the DBA has no role.
 - (B) A business/legal rule for how long data must be kept; the DBA implements it using automated scripts to archive or delete old data.
 - (C) A policy for how much data a user can "retain" in their head.
 - (D) A policy about database backups.
 - (E) A policy for how many tables a database can "retain."
 - *Correct Answer: (B)*
-

10. What is "data governance"?

- (A) The process of governing a country using data.
- (B) The overall management of the availability, usability, integrity, and security of data in an enterprise.
- (C) A type of database.
- (D) A dashboard for executives.
- (E) A SQL command.

- *Correct Answer: (B)*
-

11. How would you design a backup strategy for a critical 24/7 e-commerce database?

- (A) One full backup every night.
 - (B) A daily full backup, combined with differential backups every hour, and transaction log backups every 10 minutes.
 - (C) One full backup per month.
 - (D) Just use replication; no backups are needed.
 - (E) A weekly full backup.
 - *Correct Answer: (B)*
-

12. What is a "database fire drill" and why is it crucial?

- (A) A drill for how to evacuate the server room during a fire.
 - (B) A scheduled test of your *entire* disaster recovery plan, including restoring backups to a test server, to *prove* you can recover.
 - (C) A security test to see if your firewall works.
 - (D) A performance test.
 - (E) A meeting to discuss what to do in a disaster.
 - *Correct Answer: (B)*
-

13. How would you *architecturally* secure a new database server?

- (A) Put it on the public internet, but with a very strong password.
 - (B) Place it in a private subnet, use a firewall/security groups, enable TDE, enforce SSL, and use RBAC.
 - (C) Just install antivirus software on it.
 - (D) Give the 'sa' (admin) password to all developers.
 - (E) Turn on auditing and nothing else.
 - *Correct Answer: (B)*
-

14. What is the strategic trade-off of using "read replicas"?

- (A) Pro: Improves read performance. Con: Costs more money.
- (B) Pro: Improves *read* performance. Con: Can have "replication lag" (stale data) and adds complexity.
- (C) Pro: Improves *write* performance. Con: Slows down reads.

- (D) Pro: More secure. Con: Slower performance.
 - (E) There are no trade-offs; they are always better.
 - *Correct Answer: (B)*
-

15. What is the DBA's main responsibility in a "root cause analysis" (RCA) after a database outage?

- (A) To find which developer or user to blame for the outage.
 - (B) To provide all relevant logs and metrics to help identify the *technical* cause of the failure.
 - (C) To write the final report for the executives.
 - (D) To immediately buy a new server.
 - (E) To restore the database from backup (this is part of recovery, not RCA).
 - *Correct Answer: (B)*
-

16. How would you *strategically* implement a "least privilege" security model?

- (A) Give everyone "read-only" access to everything.
 - (B) Give no one access to anything.
 - (C) Start with *no permissions*, create "roles" for each job function, grant *only* the minimum permissions needed to those roles, and assign users to roles.
 - (D) Give everyone "admin" rights, but tell them not to do anything bad.
 - (E) Use a single "app" user for all developers and applications.
 - *Correct Answer: (C)*
-

17. What is the DBA's role in a "CI/CD pipeline"?

- (A) The DBA has no role; it is a DevOps process.
 - (B) To manage and automate database schema changes (migrations) and test them to ensure they don't break the application.
 - (C) To manually approve every single deployment.
 - (D) To run the application's unit tests.
 - (E) To build the application code.
 - *Correct Answer: (B)*
-

18. What is the difference between "horizontal scaling" and "vertical scaling" for a database?

- (A) Horizontal is adding more code; vertical is adding more features.
 - (B) Horizontal is adding *more servers* (e.g., sharding/clustering); vertical is adding *more power* (e.g., CPU/RAM) to one server.
 - (C) Horizontal is for SQL; vertical is for NoSQL.
 - (D) Horizontal is for frontend; vertical is for backend.
 - (E) Horizontal is fast; vertical is slow.
 - *Correct Answer: (B)*
-

19. What is a "fill factor" in an index, and what is the strategic trade-off of setting it?

- (A) How "full" the database is; set it to 100% to save space.
 - (B) The percentage of space on an index page to fill; 100% is good for reads, but a *lower* value (e.g., 80%) is better for *write-heavy* tables to reduce page splits.
 - (C) A setting for how full the transaction log can get.
 - (D) A setting for how much RAM the database can use.
 - (E) There is no trade-off; it should always be 100%.
 - *Correct Answer: (B)*
-

20. What is a database "isolation level" and what is the trade-off of using "Serializable"?

- (A) A setting for how "isolated" a server is; Serializable is the most secure.
- (B) A setting that controls "concurrency"; Serializable is the *strongest* (prevents all issues), but it *kills performance* by using many locks.
- (C) A setting for backups; Serializable means one at a time.
- (D) A setting for users; Serializable means only one user can connect.
- (E) There is no trade-off; Serializable is always the best.
- *Correct Answer: (B)*

Cloud Engineering track.

Level 1: Basic Foundations

1. What is "cloud computing"?

- (A) A service that stores all data on a single, local hard drive.
 - (B) A network of physical servers located in a private home or office.
 - (C) The on-demand delivery of IT resources (e.g., servers, storage) over the internet.
 - (D) A type of software used exclusively for advanced data analysis.
 - (E) A programming framework for building desktop applications.
 - *Correct Answer: (C)*
-

2. What does "IaaS" stand for?

- (A) Internet as a Service
 - (B) Infrastructure as a Service
 - (C) Integration as a Service
 - (D) Information as a Service
 - (E) Identity as a Service
 - *Correct Answer: (B)*
-

3. Which option is an example of "SaaS" (Software as a Service)?

- (A) A service providing raw virtual machines for you to manage.
 - (B) A service providing a platform to build and deploy your own apps.
 - (C) A fully managed application delivered over the web, like Gmail or Salesforce.
 - (D) A type of secure network connection to a data center.
 - (E) A local database installed on your personal computer.
 - *Correct Answer: (C)*
-

4. What is "PaaS" (Platform as a Service)?

- (A) A service providing raw virtual machines and configurable networking.
 - (B) A managed platform (e.g., database, OS) to build and deploy apps.
 - (C) A fully managed application like a word processor or email.
 - (D) A physical server that you purchase and keep in an office.
 - (E) A type of security software that scans for viruses.
 - *Correct Answer: (B)*
-

5. Which of these is a major cloud provider?

- (A) Linux
 - (B) Intel
 - (C) Oracle
 - (D) Python
 - (E) Amazon Web Services (AWS)
 - *Correct Answer: (E)*
-

6. What is a "virtual machine" (VM)?

- (A) A physical, high-performance server.
 - (B) A software-based emulation of a physical computer.
 - (C) A type of network firewall.
 - (D) A storage device for backups.
 - (E) A tool for writing application code.
 - *Correct Answer: (B)*
-

7. What is a "Region" in a cloud context?

- (A) A specific, physical data center.
 - (B) A geographic area in the world (e.g., "US East") containing data centers.
 - (C) A user's local neighborhood.
 - (D) A section of a hard drive used for storage.
 - (E) A group of users with the same permissions.
 - *Correct Answer: (B)*
-

8. What is an "Availability Zone" (AZ)?

- (A) A single, isolated data center (or group of data centers) within a Region.
 - (B) The entire geographical area, such as "North America".
 - (C) A setting that determines if a service is available.
 - (D) A type of network connection.
 - (E) A specific virtual machine.
 - *Correct Answer: (A)*
-

9. What is "object storage" (e.g., AWS S3, Azure Blob)?

- (A) A service for storing data in a relational database.
 - (B) A service for storing files and data as "objects" in a flat structure.
 - (C) A service for running application code.
 - (D) A service for managing user identities.
 - (E) A physical hard drive attached to a server.
 - *Correct Answer: (B)*
-

10. What is "block storage" (e.g., AWS EBS, Azure Disk)?

- (A) A virtual hard drive that attaches to a single VM, used for its OS and files.
- (B) A service for storing files that can be accessed from the internet.

- (C) A service for managing user passwords.
 - (D) A database for storing data in blocks.
 - (E) A type of network firewall.
 - *Correct Answer: (A)*
-

11. What is a "public cloud"?

- (A) A cloud environment built and operated by a company for its own private use.
 - (B) A cloud environment where IT resources are owned and operated by a third-party provider and delivered over the internet.
 - (C) A mix of both public and private cloud environments.
 - (D) A cloud that is not secure and is open to the public.
 - (E) A small-scale cloud built in a local office.
 - *Correct Answer: (B)*
-

12. What is a "private cloud"?

- (A) A cloud environment built and operated by a single company for its own use.
 - (B) A cloud environment that is open to the public for anyone to use.
 - (C) A mix of public and on-premises environments.
 - (D) A small, personal laptop.
 - (E) A cloud provider like AWS or Google Cloud.
 - *Correct Answer: (A)*
-

13. What is a "hybrid cloud"?

- (A) A cloud environment that only uses virtual machines.
 - (B) A cloud environment that only uses SaaS products.
 - (C) An architecture that combines a private cloud with one or more public clouds.
 - (D) A cloud provider that has not been fully launched.
 - (E) A cloud that is powered by hybrid (gas and electric) generators.
 - *Correct Answer: (C)*
-

14. What does "scalability" mean in the cloud?

- (A) The ability to increase or decrease IT resources (e.g., servers) as needed.
- (B) The ability to access resources from anywhere in the world.
- (C) The ability to secure data using encryption.

- (D) The ability to run code.
 - (E) The ability to create user accounts.
 - *Correct Answer: (A)*
-

15. What is "elasticity" in a cloud context?

- (A) The physical flexibility of the server hardware.
 - (B) The ability of the cloud to *automatically* scale resources up and down.
 - (C) The process of stretching a network cable.
 - (D) A type of subscription model for cloud services.
 - (E) A security feature for data.
 - *Correct Answer: (B)*
-

16. What is the "pay-as-you-go" pricing model?

- (A) A model where you pay a large, fixed fee once per year.
 - (B) A model where you only pay for the specific IT resources you consume.
 - (C) A model where you pay for resources, but only if your application is successful.
 - (D) A free model where you do not have to pay.
 - (E) A model where you must pay for all resources in a region.
 - *Correct Answer: (B)*
-

17. What is a "VPC" (Virtual Private Cloud)?

- (A) A physical, private network in your office.
 - (B) A logically isolated section of the public cloud, creating a private network.
 - (C) A type of virtual machine.
 - (D) A storage device for files.
 - (E) A user's personal computer.
 - *Correct Answer: (B)*
-

18. What is "Azure"?

- (A) The cloud computing platform from Google.
- (B) The cloud computing platform from Microsoft.
- (C) The cloud computing platform from Amazon.
- (D) A type of open-source database.
- (E) A programming language for cloud.

- *Correct Answer: (B)*
-

19. What is "GCP"?

- (A) The cloud computing platform from Google.
 - (B) The cloud computing platform from Microsoft.
 - (C) The cloud computing platform from Amazon.
 - (D) A type of security protocol.
 - (E) A hardware manufacturer.
 - *Correct Answer: (A)*
-

20. What is "IAM" (Identity and Access Management)?

- (A) A service for managing users, groups, and permissions.
 - (B) A service for monitoring application performance.
 - (C) A service for creating virtual machines.
 - (D) A service for storing files and objects.
 - (E) A service for creating virtual networks.
 - *Correct Answer: (A)*
-

Level 2: Intermediate Application

1. What is the AWS service for running virtual machines?

- (A) S3 (Simple Storage Service)
 - (B) RDS (Relational Database Service)
 - (C) EC2 (Elastic Compute Cloud)
 - (D) VPC (Virtual Private Cloud)
 - (E) Lambda
 - *Correct Answer: (C)*
-

2. What is the Azure equivalent of AWS EC2?

- (A) Azure Blob Storage
- (B) Azure SQL Database
- (C) Azure Virtual Network
- (D) Azure Virtual Machines

- (E) Azure Functions
 - *Correct Answer: (D)*
-

3. What is an "AMI" (Amazon Machine Image)?

- (A) A type of virtual network configuration.
 - (B) A template containing the software (OS, apps) to create a VM.
 - (C) A managed database service instance.
 - (D) A storage device used for backups.
 - (E) A user's security credential file.
 - *Correct Answer: (B)*
-

4. What is a "Security Group" in AWS?

- (A) A group of users who all have the same security clearance.
 - (B) A virtual firewall at the *instance level* that controls inbound/outbound traffic.
 - (C) A physical security device in a data center.
 - (D) A type of IAM policy.
 - (E) A tool for scanning for viruses.
 - *Correct Answer: (B)*
-

5. What is the Azure equivalent of an AWS "Security Group"?

- (A) A "Resource Group"
 - (B) An "Availability Set"
 - (C) A "Network Security Group" (NSG)
 - (D) A "Virtual Network" (VNet)
 - (E) An "Azure Policy"
 - *Correct Answer: (C)*
-

6. What is a "subnet" in a VPC?

- (A) A sub-section of a larger network (VPC), often used to isolate resources.
- (B) A type of storage device.
- (C) A template for creating a virtual machine.
- (D) A group of users.
- (E) A container for storing files.
- *Correct Answer: (A)*

7. What is an "IAM Role"?

- (A) A set of permissions that you *attach* to a user or group.
- (B) An identity with permissions that a service (like an EC2 instance) can *assume*.
- (C) A password policy for an account.
- (D) A type of virtual machine.
- (E) A physical security key.
- *Correct Answer: (B)*

8. What is the "CLI"?

- (A) "Cloud Login Interface," a graphical tool for logging in.
- (B) "Command Line Interface," a text-based tool for interacting with cloud services.
- (C) "Cloud Logic Inspector," a tool for debugging code.
- (D) "Client License Information," a file for software activation.
- (E) "Container Launch Initiative," a tool for Docker.
- *Correct Answer: (B)*

9. What is an "SDK" (Software Development Kit)?

- (A) A set of libraries that let you interact with cloud services from your code.
- (B) A virtual server for development.
- (C) A security key for accessing an account.
- (D) A text-based command-line tool.
- (E) A pre-built virtual machine image.
- *Correct Answer: (A)*

10. What is a "load balancer"?

- (A) A service that distributes incoming network traffic across multiple servers.
 - (B) A service that "balances" the cost of your servers.
 - (C) A service that "balances" the storage on your hard drives.
 - (D) A security tool that blocks all traffic.
 - (E) A database for managing user loads.
 - *Correct Answer: (A)*
-

11. What is "AWS S3" used for?

- (A) Running virtual machines.
 - (B) Storing files (objects) in a highly scalable way.
 - (C) Managing relational databases.
 - (D) Creating private networks.
 - (E) Running serverless code.
 - *Correct Answer: (B)*
-

12. What is "AWS RDS"?

- (A) A service for running virtual machines.
 - (B) A service for storing files as objects.
 - (C) A managed service for running relational databases (e.g., MySQL, Postgres).
 - (D) A service for managing user permissions.
 - (E) A service for monitoring application logs.
 - *Correct Answer: (C)*
-

13. What is the *primary* difference between an "IAM User" and an "IAM Role"?

- (A) A User is for a person; a Role is for a service. Both provide permissions.
 - (B) A User has a password; a Role does not.
 - (C) A User is permanent; a Role is temporary.
 - (D) A User has more permissions than a Role.
 - (E) A User is for AWS; a Role is for Azure.
 - *Correct Answer: (A)*
-

14. What is a "snapshot"?

- (A) A point-in-time backup of a block storage volume (like an EBS volume).
 - (B) A screenshot of your virtual machine's desktop.
 - (C) A copy of a file from an S3 bucket.
 - (D) A type of security scan.
 - (E) A log file from a server.
 - *Correct Answer: (A)*
-

15. What is a "public subnet"?

- (A) A subnet that is accessible to all users within your organization.
 - (B) A subnet whose traffic is routed to an "Internet Gateway," making it reachable from the internet.
 - (C) A subnet that is not secure and should not be used.
 - (D) A subnet that contains only storage devices.
 - (E) A subnet that is shared between multiple AWS accounts.
 - *Correct Answer: (B)*
-

16. What is a "private subnet"?

- (A) A subnet that cannot access any other services.
 - (B) A subnet that is *not* directly reachable from the public internet.
 - (C) A subnet that is encrypted.
 - (D) A subnet that is more expensive.
 - (E) A subnet that can only be used by one person.
 - *Correct Answer: (B)*
-

17. What is a "key pair" used for when launching an AWS EC2 instance?

- (A) To encrypt the instance's hard drive.
 - (B) To authenticate (log in) to a Linux instance via SSH.
 - (C) To define the username and password for the instance.
 - (D) To unlock the instance from a "locked" state.
 - (E) To provide the "key" and "value" for a tag.
 - *Correct Answer: (B)*
-

18. What is the GCP service for object storage?

- (A) Google Kubernetes Engine
 - (B) Google Compute Engine
 - (C) Google Cloud Storage
 - (D) Google Cloud SQL
 - (E) Google BigQuery
 - *Correct Answer: (C)*
-

19. What is the GCP service for running virtual machines?

- (A) Google Kubernetes Engine

- (B) Google Compute Engine
 - (C) Google Cloud Storage
 - (D) Google Cloud SQL
 - (E) Google App Engine
 - *Correct Answer: (B)*
-

20. What is an "S3 Bucket"?

- (A) A container (folder) for storing objects (files) in AWS S3.
 - (B) A physical hard drive used for S3.
 - (C) A type of virtual machine.
 - (D) A security policy for S3.
 - (E) A user account for S3.
 - *Correct Answer: (A)*
-

Level 3: Advanced Scenarios

1. What is an "Auto Scaling Group"?

- (A) A tool that automatically encrypts all data on a virtual machine.
 - (B) A service that automatically adjusts the number of VMs based on traffic or a schedule.
 - (C) A service that automatically backs up a database.
 - (D) A tool that automatically assigns IP addresses to new VMs.
 - (E) A service that automatically patches the operating system.
 - *Correct Answer: (B)*
-

2. What is a "NAT Gateway"?

- (A) A service that allows instances in a *private subnet* to initiate outbound traffic to the internet.
 - (B) A service that allows *inbound* traffic from the internet to a private subnet.
 - (C) A service that translates domain names into IP addresses.
 - (D) A type of load balancer.
 - (E) A security service that blocks all internet traffic.
 - *Correct Answer: (A)*
-

3. What is a "read replica" for a database?

- (A) A read-only copy of a database, used to offload read queries and improve performance.
 - (B) A full backup of a database.
 - (C) A security tool that "reads" database logs for threats.
 - (D) A type of database that can only be read, not written to.
 - (E) A user who only has read-only permissions.
 - *Correct Answer: (A)*
-

4. What is "Multi-AZ" for a database (e.g., AWS RDS)?

- (A) A feature that creates a *standby replica* in a different Availability Zone for high availability.
 - (B) A feature that creates multiple read replicas in different regions.
 - (C) A feature that allows a database to be accessed from multiple zones.
 - (D) A security feature that encrypts the database in multiple zones.
 - (E) A feature that scales the database across multiple zones.
 - *Correct Answer: (A)*
-

5. What is the difference between an "Application Load Balancer" (ALB) and a "Network Load Balancer" (NLB)?

- (A) ALB operates at Layer 7 (HTTP); NLB operates at Layer 4 (TCP).
 - (B) ALB operates at Layer 4 (TCP); NLB operates at Layer 7 (HTTP).
 - (C) ALB is for internal traffic; NLB is for external traffic.
 - (D) ALB is less secure than an NLB.
 - (E) ALB is an older, deprecated service.
 - *Correct Answer: (A)*
-

6. What is "AWS CloudWatch"?

- (A) A service for managing user permissions.
 - (B) A service for monitoring resources, collecting logs, and setting alarms.
 - (C) A service for storing files.
 - (D) A service for creating virtual networks.
 - (E) A service for building websites.
 - *Correct Answer: (B)*
-

7. What is "Azure Monitor"?

- (A) A service for managing virtual machines.
 - (B) A service for monitoring resources, collecting logs, and setting alerts.
 - (C) A service for managing databases.
 - (D) A service for user authentication.
 - (E) A service for storing files.
 - *Correct Answer: (B)*
-

8. What is an "S3 Lifecycle Policy"?

- (A) A policy that automatically moves objects to cheaper storage tiers or deletes them after a set time.
 - (B) A security policy that defines who can access an S3 bucket.
 - (C) A policy that defines the maximum size of an S3 bucket.
 - (D) A backup policy for S3.
 - (E) A policy that tracks the "life" of an S3 bucket.
 - *Correct Answer: (A)*
-

9. What does an "IAM Policy" do?

- (A) It defines a user's password requirements.
 - (B) It defines the *permissions* (e.g., "Allow S3:GetObject") for a user, group, or role.
 - (C) It creates a new IAM user.
 - (D) It monitors a user's activity.
 - (E) It is a physical security policy.
 - *Correct Answer: (B)*
-

10. What is "AWS Route 53"?

- (A) A scalable Domain Name System (DNS) web service.
 - (B) A service for creating private networks.
 - (C) A service for monitoring network traffic.
 - (D) A security service for blocking attacks.
 - (E) A service for creating virtual machines.
 - *Correct Answer: (A)*
-

11. What is "Azure DNS"?

- (A) A managed service for running relational databases.
 - (B) A hosting service for DNS domains, providing name resolution.
 - (C) A service for managing virtual networks.
 - (D) A security service for monitoring traffic.
 - (E) A service for creating virtual machines.
 - *Correct Answer: (B)*
-

12. What is a "VPC Peering" connection?

- (A) A connection that links a VPC to the public internet.
 - (B) A connection that links a VPC to an on-premises data center.
 - (C) A private network connection between two separate VPCs.
 - (D) A connection that links a user's laptop to a VPC.
 - (E) A connection that links a VM to a storage bucket.
 - *Correct Answer: (C)*
-

13. What is a "security best practice" for an AWS account?

- (A) Use the "root" user for all daily tasks.
 - (B) Enable Multi-Factor Authentication (MFA) on the "root" user.
 - (C) Store access keys directly in your application code.
 - (D) Open all ports on your security groups.
 - (E) Never update your IAM policies.
 - *Correct Answer: (B)*
-

14. What is a "Resource Group" in Azure?

- (A) A logical container that holds related Azure resources (e.g., VMs, storage, VNets).
 - (B) A group of users with the same permissions.
 - (C) A set of security rules.
 - (D) A specific data center location.
 - (E) A type of virtual machine.
 - *Correct Answer: (A)*
-

15. What is the difference between "AWS EBS" and "AWS S3"?

- (A) EBS is object storage; S3 is block storage.
- (B) EBS is block storage (a VM hard drive); S3 is object storage (for files).

- (C) EBS is a database; S3 is a virtual machine.
 - (D) EBS is free; S3 is expensive.
 - (E) There is no difference; they are the same service.
 - *Correct Answer: (B)*
-

16. What is a "CDN" (Content Delivery Network) like AWS CloudFront?

- (A) A service that "caches" content (e.g., images, videos) in locations closer to users.
 - (B) A service for creating virtual networks.
 - (C) A service for managing databases.
 - (D) A service for monitoring security.
 - (E) A service for running application code.
 - *Correct Answer: (A)*
-

17. What is an "API Gateway"?

- (A) A managed service that acts as a "front door" for APIs to handle traffic, security, and monitoring.
 - (B) A physical hardware device for network routing.
 - (C) A security service that only blocks traffic.
 - (D) A tool for writing application code.
 - (E) A database for storing API keys.
 - *Correct Answer: (A)*
-

18. What is the "Shared Responsibility Model"?

- (A) The idea that you and the cloud provider share the *cost* of all services.
 - (B) The idea that the cloud provider is *fully* responsible for all security.
 - (C) The idea that *you* are *fully* responsible for all security.
 - (D) The idea that the provider is responsible for security *of* the cloud, and you are responsible for security *in* the cloud.
 - (E) A legal document that transfers all responsibility to you.
 - *Correct Answer: (D)*
-

19. What is "AWS CloudFormation"?

- (A) A service for monitoring application performance.

- (B) An "Infrastructure as Code" (IaC) service for provisioning AWS resources with templates.
 - (C) A service for storing and managing files.
 - (D) A managed database service.
 - (E) A tool for designing application architecture.
 - *Correct Answer: (B)*
-

20. What is "Azure Resource Manager" (ARM)?

- (A) The underlying "Infrastructure as Code" (IaC) service for provisioning Azure resources.
 - (B) A tool for managing employee access to resources.
 - (C) A monitoring and logging service.
 - (D) A physical manager for Azure data centers.
 - (E) A service for building virtual machines.
 - *Correct Answer: (A)*
-

Level 4: Expert & Specialization

1. What is "Infrastructure as Code" (IaC)?

- (A) The process of managing and provisioning infrastructure through code (e.g., Terraform, CloudFormation).
 - (B) A hardware device that writes infrastructure code for you.
 - (C) The practice of storing all application code inside the server's OS.
 - (D) A type of networking protocol for cloud.
 - (E) A monitoring tool that checks your code.
 - *Correct Answer: (A)*
-

2. What is "Terraform"?

- (A) An AWS-native service for IaC.
 - (B) A cloud-agnostic (multi-cloud) "Infrastructure as Code" (IaC) tool.
 - (C) A service for monitoring application performance.
 - (D) A container orchestration service.
 - (E) A serverless computing platform.
 - *Correct Answer: (B)*
-

3. What is "Docker"?

- (A) A tool for provisioning and managing virtual machines.
 - (B) A platform for packaging applications and their dependencies into a "container."
 - (C) A serverless computing service.
 - (D) A tool for managing infrastructure as code.
 - (E) A managed database service.
 - *Correct Answer: (B)*
-

4. What is "Kubernetes" (e.g., EKS, AKS, GKE)?

- (A) A service for automating the deployment, scaling, and management of containerized applications.
 - (B) A service for building and storing container images.
 - (C) A service for running single, event-driven functions.
 - (D) A service for managing virtual networks.
 - (E) A managed NoSQL database service.
 - *Correct Answer: (A)*
-

5. What is "serverless" computing (e.g., AWS Lambda, Azure Functions)?

- (A) A model where the cloud provider manages the servers, and you run code in response to events.
 - (B) A model where all code runs directly on the user's browser.
 - (C) A type of container service (like Kubernetes).
 - (D) A physical server that has no operating system.
 - (E) An old model of computing using mainframes.
 - *Correct Answer: (A)*
-

6. What is a "CI/CD pipeline"?

- (A) An automated workflow for building, testing, and deploying application code.
 - (B) A physical "pipeline" for network traffic.
 - (C) A security protocol for data.
 - (D) A tool for managing databases.
 - (E) A monitoring and logging service.
 - *Correct Answer: (A)*
-

7. What is "AWS CodePipeline"?

- (A) A managed "Continuous Integration/Continuous Delivery" (CI/CD) service.
 - (B) A code editor that runs in the cloud.
 - (C) A service for storing and versioning code (like GitHub).
 - (D) A service for running virtual machines.
 - (E) A service for monitoring code performance.
 - *Correct Answer: (A)*
-

8. What is "Azure DevOps"?

- (A) A service for managing virtual networks.
 - (B) A suite of services that includes "Pipelines" (CI/CD), "Repos" (Git), and "Boards" (planning).
 - (C) A managed database service.
 - (D) A service for monitoring application logs.
 - (E) A serverless computing platform.
 - *Correct Answer: (B)*
-

9. What is "AWS Direct Connect" or "Azure ExpressRoute"?

- (A) A dedicated, private network connection from your on-premises data center to the cloud.
 - (B) A standard internet-based VPN connection.
 - (C) A type of load balancer.
 - (D) A service for managing DNS.
 - (E) A high-speed connection to the public internet.
 - *Correct Answer: (A)*
-

10. What is a "Web Application Firewall" (WAF)?

- (A) A firewall that protects a user's web browser.
 - (B) A firewall that filters Layer 7 (HTTP) traffic to protect web apps from attacks like XSS or SQLi.
 - (C) A firewall that protects a virtual machine (this is a Security Group / NSG).
 - (D) A type of antivirus software.
 - (E) A physical hardware device in a data center.
 - *Correct Answer: (B)*
-

11. What is an "AWS Key Management Service" (KMS) or "Azure Key Vault"?

- (A) A service for securely storing and managing encryption keys and other secrets.
 - (B) A service for storing and managing SSH key pairs.
 - (C) A service for storing user passwords.
 - (D) A hardware security device.
 - (E) A service for managing API keys.
 - *Correct Answer: (A)*
-

12. What is "AWS DynamoDB"?

- (A) A managed relational (SQL) database service.
 - (B) A managed NoSQL (key-value and document) database service.
 - (C) A service for data warehousing.
 - (D) A service for monitoring database performance.
 - (E) A service for running virtual machines.
 - *Correct Answer: (B)*
-

13. What is "Azure Cosmos DB"?

- (A) A globally distributed, multi-model NoSQL database service.
 - (B) A managed relational (SQL) database service.
 - (C) A data warehousing service.
 - (D) A service for creating virtual networks.
 - (E) A service for managing user identities.
 - *Correct Answer: (A)*
-

14. What is the difference between "AWS CloudFormation" and "Terraform"?

- (A) CloudFormation is for AWS; Terraform is for Azure only.
 - (B) CloudFormation is cloud-agnostic; Terraform is AWS-only.
 - (C) CloudFormation is AWS-native; Terraform is a cloud-agnostic (multi-cloud) tool.
 - (D) CloudFormation uses HCL; Terraform uses JSON/YAML.
 - (E) CloudFormation is for servers; Terraform is for networks.
 - *Correct Answer: (C)*
-

15. What is a "container registry" (e.g., ECR, ACR)?

- (A) A service for running containers (this is Kubernetes).
 - (B) A service for storing, managing, and deploying container images.
 - (C) A tool for building container images.
 - (D) A security scanner for containers.
 - (E) A virtual machine optimized for containers.
 - *Correct Answer: (B)*
-

16. What is a "service mesh" (e.g., Istio, Linkerd)?

- (A) A dedicated infrastructure layer for managing service-to-service communication in a microservices architecture.
- (B) A physical network mesh in a data center.
- (C) A tool for monitoring network traffic.
- (D) A type of firewall.
- (E) A service for building virtual machines.
- *Correct Answer: (A)*

17. What is a "VPC Endpoint"?

- (A) A connection that allows you to privately access AWS services without using the public internet.
 - (B) A public IP address for a VPC.
 - (C) A VPN connection to a VPC.
 - (D) A tool for monitoring VPC traffic.
 - (E) A user's login "endpoint" for a VPC.
 - *Correct Answer: (A)*
-

18. What is "AWS Shield"?

- (A) A managed "Distributed Denial of Service" (DDoS) protection service.
 - (B) A service for managing encryption keys.
 - (C) A service for managing user permissions.
 - (D) A hardware security device.
 - (E) A Web Application Firewall (WAF).
 - *Correct Answer: (A)*
-

19. What is "Azure Sentinel"?

- (A) A "Security Information and Event Management" (SIEM) and "SOAR" tool.
- (B) A "Distributed Denial of Service" (DDoS) protection service.

- (C) A service for managing encryption keys.
 - (D) A Web Application Firewall (WAF).
 - (E) A service for managing virtual networks.
 - *Correct Answer: (A)*
-

20. What is "AWS EKS"?

- (A) A managed "Kubernetes" service.
 - (B) A managed "Elasticsearch" service.
 - (C) A managed "Kafka" service.
 - (D) A managed "container registry" service.
 - (E) A managed "serverless" platform.
 - *Correct Answer: (A)*
-

Level 5: Strategic & Architectural

1. What is "High Availability" (HA) in a cloud architecture?

- (A) An architecture designed to be very fast and performant.
 - (B) An architecture designed to be very low-cost.
 - (C) An architecture designed to remain operational by eliminating single points of failure (e.g., using multiple AZs).
 - (D) An architecture that is deployed in a single data center.
 - (E) An architecture that is managed by a third-party vendor.
 - *Correct Answer: (C)*
-

2. How does a "Disaster Recovery" (DR) strategy differ from "High Availability" (HA)?

- (A) HA is about surviving a component failure (e.g., one server); DR is about surviving a site-wide disaster (e.g., an earthquake).
 - (B) HA is about backups; DR is about live replication.
 - (C) HA is more expensive than DR and is considered a replacement for it.
 - (D) HA is for data; DR is for applications.
 - (E) They are identical strategies with different names.
 - *Correct Answer: (A)*
-

3. What is a "lift-and-shift" (rehosting) cloud migration?

- (A) Re-architecting an application to be fully cloud-native.
 - (B) Moving an application to the cloud with minimal or no code changes.
 - (C) Decommissioning an old application and replacing it with a new SaaS product.
 - (D) Moving an application from one cloud provider to another.
 - (E) Throwing away the old server ("lift") and writing a new one ("shift").
 - *Correct Answer: (B)*
-

4. What is a "re-platform" (lift-and-reshape) cloud migration?

- (A) Moving an application to the cloud with no changes.
 - (B) Moving an application to the cloud while making minor optimizations (e.g., moving to a managed database).
 - (C) Fully rewriting the application to be cloud-native.
 - (D) Decommissioning the application.
 - (E) Moving the application to a new physical data center.
 - *Correct Answer: (B)*
-

5. What is "RPO" (Recovery Point Objective)?

- (A) The maximum *amount of data* (measured in time) that a business is willing to lose in a disaster (e.g., "1 hour of data").
 - (B) The maximum *amount of time* it should take to get a system back online after a disaster (e.g., "4 hours").
 - (C) The "Real Performance Objective" of a system.
 - (D) The "Resource Provisioning Objective" for a new project.
 - (E) The "Role-based Permission Objective" for a user.
 - *Correct Answer: (A)*
-

6. What is "RTO" (Recovery Time Objective)?

- (A) The maximum *amount of data* (measured in time) that a business is willing to lose.
 - (B) The maximum *amount of time* it should take to get a system back online after a disaster.
 - (C) The "Real Time Objective" for a system's performance.
 - (D) The "Resource Timing Objective" for a project.
 - (E) The "Role-based Time-out Objective" for a user.
 - *Correct Answer: (B)*
-

7. What is a strategic reason to use "Reserved Instances" (RIs) or "Savings Plans"?

- (A) To gain the ability to automatically scale your instances.
 - (B) To receive a significant discount in exchange for a 1- or 3-year commitment.
 - (C) To get instances with more powerful CPUs.
 - (D) To get instances with better security.
 - (E) To allow your instances to be moved between regions.
 - *Correct Answer: (B)*
-

8. What is a "Spot Instance"?

- (A) An instance with a fixed, long-term price.
 - (B) An instance that can be purchased for a very large discount, but can be *terminated* by the cloud provider at any time.
 - (C) An instance used specifically for security "spot checks."
 - (D) An instance that is reserved for your account.
 - (E) A type of serverless function.
 - *Correct Answer: (B)*
-

9. What is a "multi-cloud" strategy?

- (A) The practice of using *multiple public clouds* (e.g., AWS and Azure) to avoid vendor lock-in.
 - (B) The practice of using a *mix* of public and private cloud.
 - (C) The practice of deploying an application across multiple regions in a *single* cloud.
 - (D) The practice of using a CDN.
 - (E) A strategy that involves multiple layers of cloud security.
 - *Correct Answer: (A)*
-

10. What is a "hybrid cloud" strategy?

- (A) The practice of using *multiple public clouds* (e.g., AWS and Azure).
 - (B) The practice of using a *mix* of public cloud and on-premises infrastructure.
 - (C) The practice of deploying an application across multiple regions.
 - (D) A strategy that uses "hybrid" virtual machines.
 - (E) A strategy for migrating applications.
 - *Correct Answer: (B)*
-

11. What is the "Cloud Adoption Framework" (CAF)?

- (A) A specific piece of software for migrating to the cloud.
 - (B) A set of best practices and guidance from a cloud provider to help plan and execute a migration.
 - (C) A security-only framework for locking down cloud accounts.
 - (D) A sales and marketing framework for selling cloud services.
 - (E) A legal document required to open a cloud account.
 - *Correct Answer: (B)*
-

12. What is "cloud-native" architecture?

- (A) An architecture that avoids using any cloud provider-specific services.
 - (B) An architecture that is designed *specifically* to leverage cloud services, like serverless, containers, and microservices.
 - (C) An architecture that was "born" in the cloud (i.e., a new company).
 - (D) An architecture that is lifted-and-shifted from on-premises.
 - (E) An architecture that runs on a private cloud.
 - *Correct Answer: (B)*
-

13. What is a "Zero Trust" security model?

- (A) A model where you "trust no one" on your team.
 - (B) A model where you "trust" all devices inside the network, but "zero" outside.
 - (C) A security model based on the principle of "never trust, always verify," treating all users and devices as untrusted.
 - (D) A model with "zero" firewalls.
 - (E) A model that has "zero" cost.
 - *Correct Answer: (C)*
-

14. What is "cloud governance"?

- (A) The process of "governing" a country using cloud data.
- (B) The set of policies, rules, and controls to manage costs, security, and compliance in the cloud.
- (C) A specific cloud service for monitoring.
- (D) A team of people who approve cloud accounts.
- (E) A legal agreement with a cloud provider.
- *Correct Answer: (B)*

15. What is a "FinOps" (Cloud Financial Operations)?

- (A) A "finance" team that operates in the cloud.
- (B) A cultural practice that brings financial accountability to the variable spending model of the cloud.
- (C) A specific tool for tracking cloud costs.
- (D) A type of security audit.
- (E) A model for "financing" a cloud migration.
- *Correct Answer: (B)*

16. What is a "Data Lake"?

- (A) A central repository that stores *raw, unstructured* and *structured* data at any scale.
- (B) A central repository that stores *only structured, cleaned* data for analysis (this is a Data Warehouse).
- (C) A type of NoSQL database.
- (D) A service for monitoring data.
- (E) A physical lake used to cool a data center.
- *Correct Answer: (A)*

17. What is a "serverless" architecture (e.g., using Lambda, API Gateway, DynamoDB)?

- (A) An architecture where all the code runs on the client's browser.
- (B) An architecture where you *only* use virtual machines.
- (C) An architecture where you *only* use containers.
- (D) An architecture composed of event-driven services where you don't manage any servers.
- (E) An architecture that is not "production-ready."
- *Correct Answer: (D)*

18. What is the strategic trade-off of using a *managed service* (e.g., AWS RDS)?

- (A) Pro: Full control over the OS. Con: You must manage patching.
- (B) Pro: Reduced operational overhead. Con: Less control and potential vendor lock-in.
- (C) Pro: It is always cheaper. Con: It is less secure.
- (D) Pro: It is more secure. Con: It is always slower.
- (E) There are no trade-offs; they are always better.
- *Correct Answer: (B)*

19. What is a "blue-green" deployment strategy?

- (A) A strategy where you deploy to "blue" (test) and "green" (prod) servers at the same time.
- (B) A strategy where you have two identical environments (blue/green), deploy to the inactive one, and then switch traffic.
- (C) A strategy that uses "blue" (fast) and "green" (eco-friendly) servers.
- (D) A strategy for slowly rolling out a new feature to users.
- (E) A strategy for database backups.
- *Correct Answer: (B)*

20. What is a "canary release" strategy?

- (A) A strategy where you deploy a new version to *all* users at once.
- (B) A strategy where you deploy a new version to a *small subset* of users, monitor it, and then roll it out to everyone.
- (C) A strategy where you "warn" users (like a canary in a coal mine) before deploying.
- (D) A strategy where you deploy two versions (A and B) at the same time.
- (E) A strategy for database disaster recovery.
- *Correct Answer: (B)*

UX/UI Design track.

Level 1: Basic Foundations

1. What is the primary focus of "UX Design" (User Experience)?

- (A) It focuses on the visual look, such as colors and fonts, of an application.
- (B) It focuses on the overall experience a user has with a product, including usability.
- (C) It focuses on coding the frontend of the website using HTML and CSS.
- (D) It focuses on writing the marketing copy and text for the application.
- (E) It focuses on managing the server and database for the application.
- *Correct Answer: (B)*

2. What is the primary focus of "UI Design" (User Interface)?

- (A) It focuses on the user's emotional journey and psychological experience.
- (B) It focuses on the visual layout and graphical elements, like buttons and menus.
- (C) It focuses on the backend server logic and database connections.
- (D) It focuses on conducting user interviews and analyzing research data.
- (E) It focuses on the information architecture and content hierarchy.

- *Correct Answer: (B)*
-

3. What is a "wireframe"?

- (A) A high-fidelity, full-color visual design of the final product.
 - (B) A basic, low-fidelity visual guide that represents the skeletal framework of a website.
 - (C) A piece of code that makes the website interactive for users.
 - (D) A final, clickable simulation of the application used for testing.
 - (E) A document containing all the text and copy for the application.
 - *Correct Answer: (B)*
-

4. What is a "prototype"?

- (A) A document that lists all the features for a future product.
 - (B) A final, coded version of the application that is ready to be sold.
 - (C) A simulation or sample version of a product used to test concepts or user flows.
 - (D) A low-fidelity sketch of the basic layout, often done on paper.
 - (E) A marketing plan for how the product will be launched.
 - *Correct Answer: (C)*
-

5. What is "usability"?

- (A) The measure of how visually attractive and appealing a design is.
 - (B) The measure of how easy a product is to use for its intended purpose.
 - (C) The number of features a product has.
 - (D) The total number of users who access the product each day.
 - (E) The process of coding the final application.
 - *Correct Answer: (B)*
-

6. What is a "User Flow"?

- (A) A diagram showing the path a user takes through an application to complete a task.
- (B) A type of animation that shows how a user "flows" from one screen to the next.
- (C) A database table that lists all of the users of an application.
- (D) A research method where a user describes their experience.
- (E) A list of all the features a user has requested.
- *Correct Answer: (A)*

7. What is a "mockup"?

- (A) A low-fidelity, basic layout sketch of a product.
- (B) A high-fidelity, static, full-color design that represents the final product's look.
- (C) A clickable, interactive simulation used for user testing.
- (D) A piece of code that mocks the behavior of a backend server.
- (E) A document containing user research and interview notes.
- *Correct Answer: (B)*

8. What is a "font"?

- (A) A set of printable or displayable text characters in a specific style (e.g., Arial, Times New Roman).
- (B) The main title or heading on a webpage.
- (C) A small, graphical icon used in a user interface.
- (D) The color of the text in an application.
- (E) A type of layout grid used in web design.
- *Correct Answer: (A)*

9. What is a "color palette"?

- (A) The collection of all images and icons used in an application.
- (B) The specific set of colors chosen for use in a brand or product's design.
- (C) A design principle related to the spacing of elements.
- (D) A document that describes the user's emotional state.
- (E) The physical tool artists use to mix paints.
- *Correct Answer: (B)*

10. What is a "button" in UI design?

- (A) An interactive element that a user clicks to perform an action.
 - (B) A static piece of text used for a title or heading.
 - (C) An image or photograph displayed on the page.
 - (D) A link that takes the user to a completely different website.
 - (E) A wireframe that shows the layout of a page.
 - *CorrectAnswer: (A)*
-

11. What is "Figma"?

- (A) A popular, collaborative interface design tool that runs in a browser.
 - (B) A programming language used to code websites.
 - (C) A type of database for storing user data.
 - (D) A project management tool for tracking tasks.
 - (E) A service for hosting live websites.
 - *Correct Answer: (A)*
-

12. What is "Sketch"?

- (A) A database management system for designers.
 - (B) A project management tool for tracking tasks.
 - (C) A popular vector graphics editor for UI design, used primarily on macOS.
 - (D) A service for conducting user research and interviews.
 - (E) A framework for coding interactive prototypes.
 - *Correct Answer: (C)*
-

13. What is "Adobe XD"?

- (A) A tool for writing and editing website code.
 - (B) A vector-based design tool for web and mobile apps, part of the Adobe suite.
 - (C) A service for hosting and managing web servers.
 - (D) A tool for editing videos and adding special effects.
 - (E) A spreadsheet program for organizing design components.
 - *Correct Answer: (B)*
-

14. What is a "grid system" in UI design?

- (A) A set of invisible lines used to structure content and create a consistent layout.
 - (B) A security feature that locks users out of the system.
 - (C) A chart or graph used to display data.
 - (D) A list of all the pages in a website.
 - (E) A tool for testing the speed of an application.
 - *Correct Answer: (A)*
-

15. What is "white space" (or negative space)?

- (A) The parts of a design that are colored white.
 - (B) The empty, unmarked space between elements in a design, used to create breathing room.
 - (C) A type of error message that appears when a page is blank.
 - (D) A document used for writing down ideas.
 - (E) A specific font that is difficult to read.
 - *Correct Answer: (B)*
-

16. What is "responsive design"?

- (A) A design that responds to user clicks with fast animations.
 - (B) A design that learns and adapts to a user's personal preferences.
 - (C) A design approach that allows a layout to adapt to different screen sizes (e.g., desktop, mobile).
 - (D) A design that includes a chatbot to respond to user questions.
 - (E) A design that is visually very bright and colorful.
 - *Correct Answer: (C)*
-

17. What is an "icon"?

- (A) A large, high-resolution photograph used as a background.
 - (B) A block of text that explains a feature.
 - (C) A small, simple graphic symbol used to represent an object, action, or idea.
 - (D) The main logo of a company.
 - (E) A clickable, interactive prototype of an application.
 - *Correct Answer: (C)*
-

18. What is "typography"?

- (A) The art and technique of arranging type (text) to make it legible and appealing.
 - (B) The process of creating a high-fidelity mockup.
 - (C) The study of how users type on a keyboard.
 - (D) A type of user research method.
 - (E) A list of all the colors used in a design.
 - *Correct Answer: (A)*
-

19. What is "contrast" in UI design?

- (A) The difference between visual elements (e.g., light vs. dark) used to create emphasis.
 - (B) A design that is intentionally confusing and difficult to use.
 - (C) The process of comparing two different design mockups.
 - (D) A specific font style that is very thin.
 - (E) The total number of colors used in a design palette.
 - *Correct Answer: (A)*
-

20. What is "alignment"?

- (A) The process of making a design "agree" with the company's brand.
 - (B) The placement of visual elements so they line up along a common edge or center.
 - (C) A user's personal preference for a specific layout.
 - (D) A type of grid system used for mobile devices only.
 - (E) The process of getting all stakeholders to agree on a design.
 - *Correct Answer: (B)*
-

Level 2: Intermediate Application

1. What is the *primary* goal of a usability test?

- (A) To determine if the visual design is aesthetically pleasing to users.
 - (B) To identify potential problems in a design by observing real users.
 - (C) To ask users what new features they would like to have in the product.
 - (D) To test if the website's backend server can handle a high load of traffic.
 - (E) To see if the user likes the brand's logo and color scheme.
 - *Correct Answer: (B)*
-

2. What is "visual hierarchy"?

- (A) The arrangement of elements to imply importance, guiding the user's eye.
 - (B) The list of all pages on a website, organized by section.
 - (C) The company's organizational chart for the design team.
 - (D) The specific font and color choices used in the application.
 - (E) The process of making a design look symmetrical and balanced.
 - *Correct Answer: (A)*
-

3. What is the main difference between a wireframe and a mockup?

- (A) A wireframe is interactive; a mockup is a static image.
 - (B) A wireframe focuses on layout (low-fidelity); a mockup focuses on visuals (high-fidelity).
 - (C) A wireframe is for mobile; a mockup is for desktop.
 - (D) A wireframe is built with code; a mockup is built in Figma.
 - (E) A wireframe is a final design; a mockup is just a basic sketch.
 - *Correct Answer: (B)*
-

4. What is a "user story"?

- (A) A short, simple description of a feature from the perspective of the user.
 - (B) A long, detailed biography of a target user (this is a persona).
 - (C) A review or testimonial from a real customer.
 - (D) A diagram showing the user's path (this is a user flow).
 - (E) A list of all the steps in a usability test.
 - *Correct Answer: (A)*
-

5. What is "Fitt's Law"?

- (A) A design principle stating that colors should be bright and appealing.
 - (B) A design principle stating that a target (e.g., a button) should be large enough for users to click.
 - (C) A design principle stating that users prefer simple designs.
 - (D) A design principle stating that a user's "fit" with a product determines satisfaction.
 - (E) A design principle related to website loading speed.
 - *Correct Answer: (B)*
-

6. What is "Hick's Law"?

- (A) A principle stating that the time it takes to make a decision increases with the number of choices.
 - (B) A principle stating that users prefer designs they have seen before.
 - (C) A principle stating that a user's "hiccups" (errors) should be minimized.
 - (D) A principle stating that text should always be left-aligned.
 - (E) A principle stating that icons should always have text labels.
 - *Correct Answer: (A)*
-

7. What is the primary purpose of "user research"?

- (A) To create aesthetically pleasing visual designs.
 - (B) To understand user behaviors, needs, and motivations to inform design decisions.
 - (C) To test the website's code for bugs and errors.
 - (D) To build the final, high-fidelity mockups of an application.
 - (E) To decide which features the stakeholders want in the product.
 - *Correct Answer: (B)*
-

8. What is a "heuristic evaluation"?

- (A) A method where a user tries to complete tasks on a prototype.
 - (B) A method where an expert inspects a design against a list of usability principles.
 - (C) A method where designers brainstorm new feature ideas.
 - (D) A final check of the code before a product is launched.
 - (E) A survey sent to users to gauge their overall satisfaction.
 - *Correct Answer: (B)*
-

9. What is "accessibility" (a11y) in design?

- (A) The practice of making a design available on both mobile and desktop.
 - (B) The practice of making a design usable by people with disabilities.
 - (C) The practice of making a design "accessible" to a global audience with translation.
 - (D) The practice of ensuring a design is affordable for all users.
 - (E) The practice of making sure a website is easy to find on Google.
 - *Correct Answer: (B)*
-

10. What is a "call to action" (CTA)?

- (A) A prominent button or link designed to prompt the user to take a specific action.
 - (B) A phone number for the user to call for customer support.
 - (C) A notification that alerts the user to a new message.
 - (D) The main heading or title of a webpage.
 - (E) A user story that describes a specific action.
 - *Correct Answer: (A)*
-

11. What is "information architecture" (IA)?

- (A) The practice of designing the server and database architecture.
 - (B) The structural design of shared information, organizing content so users can find it.
 - (C) The visual design of the user interface.
 - (D) The process of writing the code for a website.
 - (E) A document that lists all the information a user needs.
 - *Correct Answer: (B)*
-

12. What is "kerning"?

- (A) The adjustment of space between *individual characters* in a piece of text.
 - (B) The adjustment of space between *lines* of text (this is leading).
 - (C) A specific type of font that is very bold.
 - (D) The process of choosing a color palette.
 - (E) A usability principle related to error messages.
 - *Correct Answer: (A)*
-

13. What is "leading" (or line-height)?

- (A) The adjustment of space between individual characters.
 - (B) The main, "leading" piece of text on a page.
 - (C) The adjustment of space between lines of text.
 - (D) The line that "leads" a user's eye, created by alignment.
 - (E) The first item in a navigation menu.
 - *Correct Answer: (C)*
-

14. What is a "sitemap"?

- (A) A diagram showing the hierarchical structure of all pages in a website.
 - (B) A map showing the physical location of the company's data centers.
 - (C) A low-fidelity wireframe of the homepage.
 - (D) A visual design of a map interface, like Google Maps.
 - (E) A user flow that shows a user's path to a specific page.
 - *Correct Answer: (A)*
-

15. What is a "style guide"?

- (A) A document that outlines all the user personas and their stories.
- (B) A document that defines the visual design standards (colors, fonts, etc.) for a brand.

- (C) A clickable, high-fidelity prototype of an application.
 - (D) A user manual that guides a person on how to use the product.
 - (E) A list of all the pages in a sitemap.
 - *Correct Answer: (B)*
-

16. What is a "breadcrumb"?

- (A) A navigation aid that shows the user's current location in the site's hierarchy.
 - (B) A type of notification that appears briefly at the bottom of the screen.
 - (C) A user research method for tracking a user's "trail."
 - (D) A small, unimportant feature in an application.
 - (E) A usability heuristic related to finding food-related content.
 - *Correct Answer: (A)*
-

17. What is a "modal" (or modal dialog)?

- (A) A full-screen page that replaces the current page.
 - (B) A UI element that appears *on top* of the main content, disabling it until the user interacts.
 - (C) A type of navigation menu that slides in from the side.
 - (D) A notification that appears at the bottom of the screen.
 - (E) A wireframe that shows the "mode" of the user.
 - *Correct Answer: (B)*
-

18. What is "affinity mapping"?

- (A) A research method for grouping and finding patterns in qualitative data (e.g., from interviews).
 - (B) A design technique for creating visually "affiliated" or similar elements.
 - (C) A UI layout where elements are arranged based on their "affinity" to each other.
 - (D) A way to map out a user flow based on their emotions.
 - (E) A test to see how much "affinity" a user has for a brand.
 - *Correct Answer: (A)*
-

19. What is a "card" in UI design?

- (A) A UI component that groups related information in a flexible, box-like container.
- (B) A modal dialog that appears on top of the content.

- (C) A research method (this is "card sorting").
 - (D) A type of payment form for a credit card.
 - (E) A wireframe of a single screen.
 - *Correct Answer: (A)*
-

20. What is "empathy" in UX design?

- (A) The ability to understand and share the feelings of the user.
 - (B) A specific visual style that is soft and inviting.
 - (C) A set of usability heuristics.
 - (D) The process of agreeing with all of a user's feature requests.
 - (E) A type of user interface animation.
 - *Correct Answer: (A)*
-

Level 3: Advanced Scenarios

1. What is a "user-centered design" (UCD) process?

- (A) A design process that focuses on the business goals and stakeholder opinions.
 - (B) A design process that focuses on the visual aesthetics and brand identity.
 - (C) A design process that focuses on the user's needs and feedback at every stage.
 - (D) A design process that is led by the engineering team's technical constraints.
 - (E) A design process that copies the features of competitors.
 - *Correct Answer: (C)*
-

2. What is a "user persona"?

- (A) A real-time log of a user's clicks and interactions on a live website.
 - (B) A fictional, composite character created to represent a key user type.
 - (C) A short biography of a stakeholder or member of the design team.
 - (D) A list of all the features a user has requested for the product.
 - (E) The final, high-fidelity mockup of the user's profile page.
 - *Correct Answer: (B)*
-

3. What is the main difference between "qualitative" and "quantitative" research?

- (A) Qualitative is numbers (e.g., surveys); Quantitative is "why" (e.g., interviews).

- (B) Qualitative is "why" (e.g., interviews); Quantitative is numbers (e.g., analytics).
 - (C) Qualitative is for testing prototypes; Quantitative is for testing wireframes.
 - (D) Qualitative is fast and cheap; Quantitative is slow and expensive.
 - (E) Qualitative is for UI; Quantitative is for UX.
 - *Correct Answer: (B)*
-

4. What are "Jakob's Ten Usability Heuristics"?

- (A) A set of strict, inflexible rules that must be followed for all UI design.
 - (B) A list of ten popular design trends that are updated each year.
 - (C) A checklist of ten key features that every successful application must have.
 - (D) A set of broad, guiding principles for usable interface design.
 - (E) A list of ten common user types (personas) for software.
 - *Correct Answer: (D)*
-

5. What is "WCAG"?

- (A) "Web Content Accessibility Guidelines," a global standard for web accessibility.
 - (B) "Website Color and Graphics," a set of rules for visual design.
 - (C) "World Class Animation Group," a community for designers.
 - (D) "Wireframe and Component Assembly," a design methodology.
 - (E) "Web Coding and Grids," a framework for developers.
 - *Correct Answer: (A)*
-

6. What is an example of a "WCAG" guideline?

- (A) All text must be a specific font, like Arial.
 - (B) All websites must have a dark mode.
 - (C) All images must have alternative text ("alt text").
 - (D) All websites must load in under 3 seconds.
 - (E) All buttons must be circular.
 - *Correct Answer: (C)*
-

7. What is a "design system"?

- (A) A single Sketch or Figma file that contains all of the design mockups.
- (B) A collection of reusable components, guided by standards, to build products.
- (C) A software program used for creating wireframes and prototypes.

- (D) The process of testing a design with users (usability testing).
 - (E) A document outlining the information architecture of a website.
 - *Correct Answer: (B)*
-

8. What is "Atomic Design"?

- (A) A methodology for creating design systems by breaking UIs into "atoms," "molecules," "organisms," etc.
 - (B) A design style that uses very small, "atomic" text and icons.
 - (C) A type of user research that breaks down a user's needs.
 - (D) A coding framework for building websites.
 - (E) A way to design for nuclear power plant interfaces.
 - *Correct Answer: (A)*
-

9. What is a "user journey map"?

- (A) A diagram showing the path a user takes to complete a task (this is a user flow).
 - (B) A visualization of a user's steps, feelings, and pain points throughout their interaction with a product.
 - (C) A sitemap that is personalized for a specific user.
 - (D) A high-fidelity mockup of the "travel" or "maps" section of an app.
 - (E) A list of all the features a user "journeys" through.
 - *Correct Answer: (B)*
-

10. What is "card sorting"?

- (A) A research method where users organize topics into groups to help create an information architecture.
 - (B) A design technique for laying out content in "cards" on a UI.
 - (C) A brainstorming method where designers write ideas on cards.
 - (D) A usability test where users are asked to find a "card" in the design.
 - (E) A project management tool for organizing design tasks.
 - *Correct Answer: (A)*
-

11. What is the difference between "Open" and "Closed" card sorting?

- (A) "Open" is done online; "Closed" is done in person.

- (B) "Open" lets users create their own category names; "Closed" uses predefined category names.
 - (C) "Open" is for stakeholders; "Closed" is for real users.
 - (D) "Open" is for websites; "Closed" is for mobile apps.
 - (E) "Open" has no limit on cards; "Closed" has a strict limit.
 - *Correct Answer: (B)*
-

12. What is a "heuristic" (as in "heuristic evaluation")?

- (A) A strict, unbreakable rule.
 - (B) A broad, experience-based "rule of thumb" or guideline.
 - (C) A type of error message.
 - (D) A new feature request.
 - (E) A specific visual design style.
 - *Correct Answer: (B)*
-

13. What is the "Gestalt" principle of "Proximity"?

- (A) Objects that are close together are perceived as a group.
 - (B) Objects that look similar are perceived as a group.
 - (C) The human eye tends to follow the simplest path.
 - (D) The mind tends to see incomplete shapes as complete.
 - (E) An object in the foreground is seen as more important.
 - *Correct Answer: (A)*
-

14. What is the "Gestalt" principle of "Similarity"?

- (A) Objects that are close together are perceived as a group.
 - (B) Objects that look similar (e.g., in color or shape) are perceived as a group.
 - (C) The human eye prefers "similar" or symmetrical designs.
 - (D) The mind tends to see incomplete shapes as complete.
 - (E) An object's importance is similar to its size.
 - *Correct Answer: (B)*
-

15. What is "gamification"?

- (A) The process of designing a video game.
- (B) The process of testing a design to see if it is "game" for launch.

- (C) The application of game-like elements (e.g., points, badges) to a non-game product.
 - (D) A usability test where users "play a game" with the prototype.
 - (E) A design style that looks like a video game.
 - *Correct Answer: (C)*
-

16. What is a "dark pattern"?

- (A) A UI design that uses a "dark mode" or dark color scheme.
 - (B) A UI pattern that is intentionally deceptive and tricks a user into an unintended action.
 - (C) A design that has not been finished or is still in the "dark."
 - (D) A wireframe that has not been approved by stakeholders.
 - (E) A design that is hidden from most users.
 - *Correct Answer: (B)*
-

17. What is "A/B testing"?

- (A) To ask users which design (A or B) they find more visually appealing.
 - (B) To test two versions of a design to see which one performs better on a specific metric.
 - (C) To test the "accessibility" and "bugs" (A and B) in a design.
 - (D) To allow users to choose between two different layouts (A or B) in their settings.
 - (E) To see if the "A" team of designers is better than the "B" team.
 - *Correct Answer: (B)*
-

18. What is "interaction design" (IxD)?

- (A) The practice of designing the social "interactions" between users.
 - (B) The practice of designing the *interactive* elements and behaviors of a product (e.g., animations, transitions).
 - (C) The practice of designing the database and server.
 - (D) The practice of designing the company's "interaction" with its customers.
 - (E) The practice of interviewing users.
 - *Correct Answer: (B)*
-

19. What is "mobile-first" design?

- (A) A strategy of designing the *mobile* version of a product *before* the desktop version.

- (B) A strategy of designing *only* for mobile phones and not for desktops.
 - (C) A design style that looks like a mobile app, even on desktop.
 - (D) A business strategy to only sell mobile phones.
 - (E) A coding framework for building mobile apps.
 - *Correct Answer: (A)*
-

20. What is a "hamburger menu"?

- (A) A navigation menu used for restaurant or food-delivery apps.
 - (B) A UI component that shows a list of different types of hamburgers.
 - (C) An icon (three stacked lines) that reveals a navigation menu when clicked.
 - (D) A "dark pattern" that tricks a user into ordering food.
 - (E) A full-screen menu that is always visible on the page.
 - *Correct Answer: (C)*
-

Level 4: Expert & Specialization

1. What is the "System Usability Scale" (SUS)?

- (A) A tool for measuring the physical "scale" or size of a design system.
 - (B) A quick, 10-question survey used to measure the perceived usability of a product.
 - (C) A set of guidelines for making a system scalable.
 - (D) A type of design system.
 - (E) A hardware device for testing system performance.
 - *Correct Answer: (B)*
-

2. What is "design thinking"?

- (A) A specific visual style that is very minimalist and thoughtful.
 - (B) A five-stage, human-centered process for creative problem-solving (Empathize, Define, Ideate, Prototype, Test).
 - (C) The process of thinking about a design's colors and fonts.
 - (D) A software for managing design files.
 - (E) A set of usability heuristics.
 - *Correct Answer: (B)*
-

3. What is the "ideate" stage of design thinking?

- (A) The stage where you test a prototype with real users.
 - (B) The stage where you build a high-fidelity mockup.
 - (C) The stage focused on brainstorming and generating a wide range of potential solutions.
 - (D) The stage where you research and empathize with your users.
 - (E) The stage where you define the core problem to be solved.
 - *Correct Answer: (C)*
-

4. What is a "design sprint"?

- (A) A long-term, multi-month project to build a new design system.
 - (B) A time-constrained (usually 5-day) process for answering critical business questions through design.
 - (C) The final, fast-paced coding "sprint" to launch a product.
 - (D) A type of font that looks like it's moving fast.
 - (E) A quick meeting to review and critique a design.
 - *Correct Answer: (B)*
-

5. What is the main difference between "divergent" and "convergent" thinking?

- (A) Divergent is generating many ideas; convergent is narrowing them down.
 - (B) Divergent is narrowing ideas; convergent is generating them.
 - (C) Divergent is for UI; convergent is for UX.
 - (D) Divergent is for designers; convergent is for developers.
 - (E) Divergent is logical; convergent is creative.
 - *Correct Answer: (A)*
-

6. What is a "mental model"?

- (A) A visualization of a user's brain activity during a usability test.
 - (B) A user's internal belief about how a system works, based on their past experiences.
 - (C) A diagram of the designer's thought process.
 - (D) A type of user persona that focuses on psychology.
 - (E) A security model that prevents users from thinking of bad ideas.
 - *Correct Answer: (B)*
-

7. What is the main difference between a "user flow" and a "user journey map"?

- (A) A flow is a simple path; a journey map includes emotions and pain points.
 - (B) A flow includes emotions; a journey map is a simple path.
 - (C) A flow is for desktop; a journey map is for mobile.
 - (D) A flow is a prototype; a journey map is a wireframe.
 - (E) A flow is a sitemap; a journey map is a persona.
 - *Correct Answer: (A)*
-

8. What is the purpose of "stakeholder interviews"?

- (A) To understand the needs and goals of the target users.
 - (B) To test the usability of a prototype with internal employees.
 - (C) To understand the business goals, constraints, and success criteria from key decision-makers.
 - (D) To sell the final design to the company's investors.
 - (E) To create a list of all stakeholders in a "company map."
 - *Correct Answer: (C)*
-

9. What is "Guerilla" usability testing?

- (A) A highly structured, formal usability test conducted in a lab.
 - (B) A quick, informal method of testing with random users in a public place.
 - (C) A test that is "at war" with the current design.
 - (D) A test that uses animal-themed (e.g., gorilla) icons.
 - (E) A test that is conducted by the military.
 - *Correct Answer: (B)*
-

10. What is "eye-tracking" in user research?

- (A) A technology that measures and records where a user is looking on a screen.
 - (B) A design principle that ensures all elements are "easy on the eyes."
 - (C) A test where an observer manually "tracks" a user's eyes.
 - (D) A UI component that follows the user's cursor.
 - (E) A security feature that scans a user's retina.
 - *Correct Answer: (A)*
-

11. What is "microcopy"?

- (A) The small, instructional text in a UI (e.g., button labels, error messages).

- (B) A very small, difficult-to-read font.
 - (C) A "copy and paste" function for small elements.
 - (D) A document containing all the legal text for an app.
 - (E) A type of user research note.
 - *Correct Answer: (A)*
-

12. What is an "empathy map"?

- (A) A map that shows where users live.
 - (B) A collaborative tool for visualizing what a user "says, thinks, feels, and does."
 - (C) A high-fidelity mockup of a user's profile.
 - (D) A chart that tracks a user's emotions over time.
 - (E) A list of all the features a user feels are "empathetic."
 - *Correct Answer: (B)*
-

13. What is a "component" in a design system?

- (A) A reusable UI element, like a button or a form field.
 - (B) A single, static mockup of one screen.
 - (C) A user persona.
 - (D) A document of usability test results.
 - (E) A project management task.
 - *Correct Answer: (A)*
-

14. What is a "design token"?

- (A) A "token" of appreciation for a good design.
 - (B) A password or "token" to access a design file.
 - (C) A core, indivisible design value (e.g., a specific color `#FF0000`, a font size `16px`).
 - (D) A UI component, like a button or a card.
 - (E) A security feature for a design system.
 - *Correct Answer: (C)*
-

15. What is the difference between "prototyping" and "wireframing"?

- (A) Wireframing is about *structure*; prototyping is about *interaction*.
- (B) Wireframing is *interactive*; prototyping is *static*.
- (C) Wireframing is *high-fidelity*; prototyping is *low-fidelity*.

- (D) Wireframing is done with code; prototyping is done in Figma.
 - (E) They are the same thing.
 - *Correct Answer: (A)*
-

16. What is a "diary study"?

- (A) A research method where users self-report their activities and experiences over a period of time.
 - (B) A study of a designer's personal diary.
 - (C) A usability test that is performed every day.
 - (D) A document containing all the notes from a design sprint.
 - (E) A UI for a "diary" or "journal" application.
 - *Correct Answer: (A)*
-

17. What is "ethnographic research"?

- (A) The study of ethnic and cultural backgrounds.
 - (B) The study of users in their *natural environment* to understand their behaviors.
 - (C) A usability test conducted in a formal lab.
 - (D) A survey sent to thousands of users.
 - (E) The process of designing for different cultures.
 - *Correct Answer: (B)*
-

18. What is the "Kano Model"?

- (A) A model for prioritizing features based on their ability to satisfy users (e.g., Basic, Performance, Delighter).
 - (B) A model for designing information architecture.
 - (C) A model for creating a color palette.
 - (D) A specific type of design system.
 - (E) A usability testing method.
 - *Correct Answer: (A)*
-

19. What is a "contextual inquiry"?

- (A) An interview with a user *while* they are performing a task in their normal environment.
- (B) An interview with a stakeholder to understand business "context."
- (C) A usability test in a lab.

- (D) A survey about a user's background.
 - (E) A method for analyzing competitors.
 - *Correct Answer: (A)*
-

20. What is a "design critique"?

- (A) A meeting where designers present their work and receive constructive feedback.
 - (B) A negative review of a product from a user.
 - (C) A heuristic evaluation.
 - (D) A usability test.
 - (E) A stakeholder interview.
 - *Correct Answer: (A)*
-

Level 5: Strategic & Architectural

1. What is a "key performance indicator" (KPI) for evaluating UX design?

- (A) The number of lines of code written by the development team.
 - (B) The personal opinion of the CEO about the color scheme.
 - (C) A measurable value, like "Task Success Rate" or "System Usability Scale" (SUS) score.
 - (D) The total number of hours the design team worked on the project.
 - (E) The specific design software (e.g., Figma, Sketch) used by the team.
 - *Correct Answer: (C)*
-

2. What is the "HEART" framework from Google?

- (A) A framework for measuring the user experience (Happiness, Engagement, Adoption, Retention, Task Success).
 - (B) A design principle for creating "emotional" and "heartfelt" designs.
 - (C) A coding framework for building user interfaces.
 - (D) A usability heuristic related to error messages.
 - (E) A method for stakeholder interviews.
 - *Correct Answer: (A)*
-

3. What is the strategic *disadvantage* of a "dark pattern"?

- (A) It can erode user trust and damage the brand's long-term reputation.
 - (B) It is difficult to code and implement for developers.
 - (C) It only works on users who are not paying attention.
 - (D) It is not visually appealing.
 - (E) It does not work on mobile devices.
 - *Correct Answer: (A)*
-

4. What is the strategic role of a "UX Designer" on an Agile/Scrum team?

- (A) To provide final, polished mockups *before* the first sprint begins.
 - (B) To work *outside* the development team and hand off designs.
 - (C) To work *iteratively* with the team, providing just-in-time research, flows, and designs.
 - (D) To act as the "Scrum Master" and manage the project timeline.
 - (E) To test the code after the developers have finished building it.
 - *Correct Answer: (C)*
-

5. What is the "Double Diamond" model of design?

- (A) A model that describes a design process of *diverging* and *converging* twice (Discover, Define, Develop, Deliver).
 - (B) A visual design style that uses two diamond shapes in the logo.
 - (C) A usability test that is "twice as hard" as a normal test.
 - (D) A type of design system.
 - (E) A pricing model for design software.
 - *Correct Answer: (A)*
-

6. What is "ethical design"?

- (A) The practice of designing products that meet all legal requirements.
 - (B) The practice of designing products that are visually appealing and trendy.
 - (C) The practice of considering the user's well-being and avoiding harm (e.g., addiction, bias).
 - (D) The practice of designing products that are free to use.
 - (E) The practice of donating a portion of profits to charity.
 - *Correct Answer: (C)*
-

7. What is "inclusive design"?

- (A) A design that includes as many features as possible.
 - (B) A design that is intended for *everyone* and considers the full range of human diversity.
 - (C) A design that is "accessible" (this is a component, but not the full definition).
 - (D) A design that is "all-inclusive" and costs one single price.
 - (E) A design process that "includes" all stakeholders in every meeting.
 - *Correct Answer: (B)*
-

8. What is a "jobs-to-be-done" (JTBD) framework?

- (A) A framework that focuses on the "job" (the underlying goal) a user is trying to accomplish.
 - (B) A project management tool for assigning "jobs" to designers.
 - (C) A list of all the job openings on the design team.
 - (D) A framework for designing the "Careers" page of a website.
 - (E) A usability test to see if a user can find a "job" in the app.
 - *Correct Answer: (A)*
-

9. How does a "design system" provide *business value*?

- (A) It makes the product look more beautiful and trendy.
 - (B) It increases *efficiency* and *consistency*, allowing teams to build and ship products faster.
 - (C) It replaces the need for user research.
 - (D) It allows a company to fire all of its junior designers.
 - (E) It guarantees that a product will be successful.
 - *Correct Answer: (B)*
-

10. What is "design governance"?

- (A) The process of "governing" a country using design principles.
 - (B) The process of managing and standardizing a design system to ensure it is used correctly.
 - (C) A legal team that "governs" what a designer can and cannot do.
 - (D) A type of design thinking.
 - (E) A security policy for design files.
 - *Correct Answer: (B)*
-

11. What is the main strategic difference between a "product designer" and a "UX designer"?

- (A) A product designer is more senior than a UX designer.
 - (B) A product designer is often more involved in the *business strategy* and product lifecycle.
 - (C) A product designer only works on physical "products."
 - (D) A UX designer only does research, while a product designer only does visuals.
 - (E) There is no difference; the titles are completely interchangeable.
 - *Correct Answer: (B)*
-

12. What is "UX writing"?

- (A) The practice of writing the code for a website.
 - (B) The practice of writing the marketing "copy" for advertisements.
 - (C) The practice of crafting the UI "microcopy" to guide the user (this is related, but not the full definition).
 - (D) The practice of writing user manuals and documentation.
 - (E) The strategic practice of crafting all text a user encounters in a product.
 - *Correct Answer: (E)*
-

13. What is the "Aha!" moment in UX?

- (A) The moment a designer "Aha!" thinks of a new idea.
 - (B) The point in a usability test where the user says "Aha!"
 - (C) The moment a user first understands the *core value* of a product.
 - (D) The moment a stakeholder (e.g., the CEO) finally approves a design.
 - (E) An error message that appears when a user makes a mistake.
 - *Correct Answer: (C)*
-

14. What is a "UX metric"?

- (A) A quantitative score (e.g., SUS, NPS) used to track and measure the user experience.
- (B) A qualitative quote from a user interview.
- (C) A design principle related to the "metric" system.
- (D) The number of designers on a team.
- (E) The specific font size (metric) used in a design.
- *Correct Answer: (A)*

15. What is the "Net Promoter Score" (NPS)?

- (A) A metric that measures a user's *satisfaction* with a specific task.
- (B) A metric that measures a user's *willingness to recommend* a product to others.
- (C) A metric that measures how *fast* a user can complete a task.
- (D) A metric that measures a website's "net" profit.
- (E) A metric that measures how "new" a user is.
- *Correct Answer: (B)*

16. What is a "UX strategy"?

- (A) A document that outlines all the design principles (e.g., Gestalt).
- (B) A detailed plan for a single, two-week design sprint.
- (C) A long-term plan that connects the user experience to the overall business goals.
- (D) The "strategy" a user employs to complete a task in a usability test.
- (E) A plan for hiring and managing the design team.
- *Correct Answer: (C)*

17. What is the "Task Success Rate" (TSR)?

- (A) The percentage of users who "successfully" rate the app 5 stars.
- (B) The percentage of users who correctly and completely achieve a specific goal in a usability test.
- (C) The "success" of the design team in meeting their project deadline.
- (D) The "rate" at which a developer can code a new feature.
- (E) The percentage of users who find the design "successful."
- *Correct Answer: (B)*

18. Why is it dangerous to only listen to "vocal users"?

- (A) Because they are "vocal" and their voices are too loud.
- (B) Because they represent the "silent majority" of users.
- (C) Because they often represent the *extremes* (lovers or haters) and not the average user.
- (D) Because they are usually competitors trying to steal your ideas.
- (E) Because they only give positive feedback.
- *Correct Answer: (C)*

19. What is a "UX roadmap"?

- (A) A visualization of a user's "road" or journey (this is a journey map).
- (B) A strategic plan that outlines the timeline and priorities for UX projects and improvements.
- (C) A sitemap that uses a "road" metaphor.
- (D) A document that lists all the "stops" (pages) a user visits.
- (E) A plan for the design team's company "road trip."
- *Correct Answer: (B)*

20. What is "Service Design"?

- (A) The design of a single digital product, like a mobile app.
- (B) The design of the *entire experience* of a service, including all digital touchpoints, physical locations, and staff interactions.
- (C) The design of the "customer service" department's scripts.
- (D) The design of the "Terms of Service" legal document.
- (E) The design of a server's architecture.
- *Correct Answer: (B)*

iOS Developer track.

Level 1: Basic Foundations

1. What is the primary programming language used for modern iOS development?

- (A) Java
- (B) Swift
- (C) Objective-C
- (D) Python
- (E) C#
- *Correct Answer: (B)*

2. What is "Xcode"?

- (A) The official "Integrated Development Environment" (IDE) used to build iOS apps.
- (B) A specific programming language created by Apple for UI design.

- (C) The hardware simulator used for testing apps on a Mac.
 - (D) A cloud service for storing and managing iOS app data.
 - (E) The online store where users download iOS applications.
 - *Correct Answer: (A)*
-

3. What is the difference between **let** and **var** in Swift?

- (A) **let** is used for integers; **var** is used for strings.
 - (B) **let** is used for variables; **var** is used for constants.
 - (C) **let** is used for constants (unchangeable); **var** is used for variables (changeable).
 - (D) **let** is for public properties; **var** is for private properties.
 - (E) **let** is used in classes; **var** is used in structs.
 - *Correct Answer: (C)*
-

4. What is a "Simulator" in Xcode?

- (A) A tool that simulates network conditions for testing.
 - (B) A program on your Mac that emulates an iPhone or iPad for testing.
 - (C) A tool for designing user interface layouts.
 - (D) A security feature for encrypting app data.
 - (E) A code-checking tool that finds bugs.
 - *Correct Answer: (B)*
-

5. What is a **UIViewController**?

- (A) A UI element that displays text, like a label.
 - (B) An object that manages a single screen (a "view") of your app's content.
 - (C) A UI element that a user can tap, like a button.
 - (D) A configuration file that controls the app's settings.
 - (E) A piece of code that controls the app's internet connection.
 - *Correct Answer: (B)*
-

6. What is "Interface Builder"?

- (A) A programming language for building apps.
- (B) A visual, drag-and-drop editor inside Xcode for designing your UI.
- (C) A tool for managing and storing app assets like images.

- (D) A server that "builds" the final app file.
 - (E) A text editor for writing Swift code.
 - *Correct Answer: (B)*
-

7. What is a "Storyboard"?

- (A) A Swift code file for writing application logic.
 - (B) A visual file in Interface Builder that shows multiple app screens and their connections.
 - (C) A configuration file for managing app permissions, like camera access.
 - (D) A document for planning the app's features with the design team.
 - (E) A type of animation for screen transitions.
 - *Correct Answer: (B)*
-

8. What UI element is used to display non-editable text?

- (A) `UIButton`
 - (B) `UITextField`
 - (C) `UILabel`
 - (D) `UISlider`
 - (E) `UIImageView`
 - *Correct Answer: (C)*
-

9. What UI element is used to trigger an action when tapped?

- (A) `UILabel`
 - (B) `UIButton`
 - (C) `UITextField`
 - (D) `UIView`
 - (E) `UIImageView`
 - *Correct Answer: (B)*
-

10. What UI element is used to display an image?

- (A) `UILabel`
- (B) `UIImage`

- (C) `UIButton`
 - (D) `UIImageView`
 - (E) `UITextField`
 - *Correct Answer: (D)*
-

11. What is an `IBOutlet`?

- (A) A special keyword that marks a function to be triggered by a UI event.
 - (B) A property in your code that creates a *connection* to a UI element in a Storyboard.
 - (C) A setting that defines the layout and constraints of a view.
 - (D) A variable that stores the app's name and version number.
 - (E) A type of animation for a button.
 - *Correct Answer: (B)*
-

12. What is an `IBAction`?

- (A) A special keyword that marks a *function* to be triggered by a UI event, like a button tap.
 - (B) A property in your code that connects to a UI element in a Storyboard.
 - (C) A variable that stores the current "action" of the user.
 - (D) A setting for a view's background color.
 - (E) A type of network request.
 - *Correct Answer: (A)*
-

13. What is the basic data type for a whole number in Swift?

- (A) `String`
 - (B) `Double`
 - (C) `Bool`
 - (D) `Int`
 - (E) `Array`
 - *Correct Answer: (D)*
-

14. What is the basic data type for a "true" or "false" value in Swift?

- (A) `String`
- (B) `Double`

- (C) `Bool`
 - (D) `Int`
 - (E) `Boolean`
 - *Correct Answer: (C)*
-

15. What is the basic data type for a piece of text in Swift?

- (A) `String`
 - (B) `Text`
 - (C) `Label`
 - (D) `Int`
 - (E) `Character`
 - *Correct Answer: (A)*
-

16. What is an "Asset Catalog"?

- (A) A file that lists all the assets a company owns.
 - (B) A place in Xcode for managing your app's media assets, like images and app icons.
 - (C) A type of database for storing app data.
 - (D) A third-party library for downloading images.
 - (E) A list of all the code files in your project.
 - *Correct Answer: (B)*
-

17. What is the `print()` function used for in Swift?

- (A) To print a document to a physical printer.
 - (B) To display a message to the user in the app's UI.
 - (C) To output a message to the debug console in Xcode.
 - (D) To save a piece of text to a file.
 - (E) To define a new variable.
 - *Correct Answer: (C)*
-

18. What is a `//` (double slash) used for in Swift code?

- (A) To divide two numbers.
- (B) To create a "comment," a note in the code that is ignored by the compiler.
- (C) To check if two values are equal.

- (D) To define a new function.
 - (E) To import a library.
 - *Correct Answer: (B)*
-

19. What is the "App Store"?

- (A) Apple's official platform for distributing and downloading iOS applications.
 - (B) A place in Xcode where you "store" your app's files.
 - (C) A type of database for storing app data.
 - (D) A cloud service for backing up your code.
 - (E) A tool for designing an app's UI.
 - *Correct Answer: (A)*
-

20. What is a "Project" in Xcode?

- (A) The final, compiled application file.
 - (B) A file that holds all the files, settings, and targets to build an app.
 - (C) A single Swift code file.
 - (D) A user interface design file.
 - (E) A plan for what features to build.
 - *Correct Answer: (B)*
-

Level 2: Intermediate Application

1. What is an "optional" in Swift (e.g., `String?`)?

- (A) A value that is "optional" for the user to fill in.
 - (B) A variable that can hold either a value or `nil` (no value).
 - (C) A type of `Bool` that can be true, false, or "optional".
 - (D) A setting in Xcode that is not required.
 - (E) A function that may or may not return a value.
 - *Correct Answer: (B)*
-

2. How do you safely "unwrap" an optional value?

- (A) By using `if let` or `guard let`.
- (B) By using the `!` (force-unwrap) operator.

- (C) By using the `?` (optional chaining) operator.
 - (D) By using the `??` (nil-coalescing) operator.
 - (E) By converting it to a `String`.
 - *Correct Answer: (A)*
-

3. What is the `!` (force-unwrap) operator, and why can it be dangerous?

- (A) It safely unwraps an optional; it is not dangerous.
 - (B) It unwraps an optional, but *crashes the app* if the value is `nil`.
 - (C) It converts an optional to a non-optional type.
 - (D) It marks an optional as "important".
 - (E) It is a logical "NOT" operator, used in `if` statements.
 - *Correct Answer: (B)*
-

4. What is a `UITableView`?

- (A) A UI element for displaying a scrolling, single-column list of items.
 - (B) A UI element for displaying a grid of items.
 - (C) A static view that holds text and images.
 - (D) A type of database for storing data.
 - (E) A navigation controller for managing screens.
 - *Correct Answer: (A)*
-

5. What are the two *required* protocols for a `UITableView`?

- (A) `UITableViewDataSource` and `UITableViewDelegate`
 - (B) `UINavigationController` and `UIView`
 - (C) `Codable` and `Equatable`
 - (D) `UITableViewDataSource` and `UITableViewCell`
 - (E) `UITableViewDelegate` and `UINavigationController`
 - *Correct Answer: (A)*
-

6. What is the `UITableViewDataSource` protocol responsible for?

- (A) Handling cell taps and row selection.
- (B) Defining the visual style and height of the rows.

- (C) Providing the data: how many rows to show and what cell to use for each row.
 - (D) Managing the app's internet connection.
 - (E) Animating the table view when rows are added.
 - *Correct Answer: (C)*
-

7. What is the `UITableViewDelegate` protocol responsible for?

- (A) Providing the data and number of rows for the table.
 - (B) Handling user interactions, like row taps, and customizing row appearance.
 - (C) Loading the data from a remote server.
 - (D) Defining the `UIViewController` that contains the table.
 - (E) Creating the database to store the table's data.
 - *Correct Answer: (B)*
-

8. What is a "struct" in Swift?

- (A) A "reference type" used for building complex objects.
 - (B) A "value type" used for grouping related data, like a `User`'s name and age.
 - (C) A type of function that returns a value.
 - (D) A visual UI element.
 - (E) A file for storing app configuration.
 - *Correct Answer: (B)*
-

9. What is a "class" in Swift?

- (A) A "value type" used for simple data.
 - (B) A "reference type" used as a blueprint for creating complex objects.
 - (C) A type of array for storing a collection of items.
 - (D) A file that defines the app's structure.
 - (E) A special type of `UIViewController`.
 - *Correct Answer: (B)*
-

10. What is the main difference between a `struct` and a `class`?

- (A) `structs` are value types (copied); `classes` are reference types (shared).
- (B) `structs` are reference types; `classes` are value types.
- (C) `structs` can have functions; `classes` cannot.

- (D) **structs** are for data; **classes** are for UI.
 - (E) **structs** are new in Swift; **classes** are from Objective-C.
 - *Correct Answer: (A)*
-

11. What is a **UINavigationController**?

- (A) A UI element that shows a stack of view controllers, with a navigation bar at the top.
 - (B) A UI element that shows a row of tabs at the bottom.
 - (C) A view controller that plays video.
 - (D) A view controller that shows a scrolling list of items.
 - (E) A non-visual object for managing network requests.
 - *Correct Answer: (A)*
-

12. What is a **UITabBarController**?

- (A) A UI element that shows a stack of view controllers.
 - (B) A UI element that manages a set of view controllers, displayed via a tab bar at the bottom.
 - (C) A view controller that shows a grid of items.
 - (D) A view controller for text input.
 - (E) A non-visual object for managing app settings.
 - *Correct Answer: (B)*
-

13. What is a "Segue"?

- (A) A type of animation for a button.
 - (B) A transition (connection) between two view controllers in a Storyboard.
 - (C) A piece of data passed from one screen to another.
 - (D) A file for storing layout constraints.
 - (E) A function that runs when the app launches.
 - *Correct Answer: (B)*
-

14. What is the purpose of the **prepare(for:sender:)** function?

- (A) To prepare the app for launch.
- (B) To prepare a network request before it is sent.
- (C) To pass data to a new view controller *before* a segue is performed.

- (D) To prepare a `UITableView` cell for display.
 - (E) To set up the app's initial state.
 - *Correct Answer: (C)*
-

15. What is an "Array" in Swift?

- (A) An ordered collection of values of the same type.
 - (B) An unordered collection of unique values.
 - (C) A collection of key-value pairs.
 - (D) A single variable that holds one value.
 - (E) A type of UI layout.
 - *Correct Answer: (A)*
-

16. What is a "Dictionary" in Swift?

- (A) An ordered collection of values.
 - (B) An unordered collection of key-value pairs.
 - (C) A built-in tool for spell-checking.
 - (D) A file for storing text translations.
 - (E) A UI element for displaying definitions.
 - *Correct Answer: (B)*
-

17. What is "Auto Layout"?

- (A) A system for automatically writing Swift code.
 - (B) A system for automatically designing app icons.
 - (C) A constraint-based system for defining adaptive UI layouts.
 - (D) A tool for automatically fixing bugs in your code.
 - (E) A file that defines the app's colors.
 - *Correct Answer: (C)*
-

18. What is a "closure" in Swift?

- (A) A type of `class` that cannot be subclassed.
- (B) A self-contained block of functionality (a function) that can be passed around.
- (C) The process of "closing" an app.
- (D) A security feature that "closes" network ports.
- (E) A UI element that can be opened and closed.

- *Correct Answer: (B)*
-

19. What is a "protocol" in Swift?

- (A) A blueprint of methods and properties for a specific task (e.g., `UITableViewDataSource`).
 - (B) A type of `class` or `struct`.
 - (C) A network setting for `HTTP` or `HTTPS`.
 - (D) A strict rule for naming your variables.
 - (E) A file that defines the app's permissions.
 - *Correct Answer: (A)*
-

20. What is the `??` (nil-coalescing) operator used for?

- (A) To force-unwrap an optional, crashing if `nil`.
 - (B) To safely unwrap an optional using `if let`.
 - (C) To provide a *default value* for an optional that is `nil`.
 - (D) To check if two optional values are equal.
 - (E) To mark a variable as an optional.
 - *Correct Answer: (C)*
-

Level 3: Advanced Scenarios

1. What is the "Codable" protocol used for?

- (A) To make your app's code more readable.
 - (B) To easily encode and decode data (like JSON) into Swift types.
 - (C) To handle user taps and gestures.
 - (D) To create and manage layout constraints.
 - (E) To manage navigation between view controllers.
 - *Correct Answer: (B)*
-

2. What is `URLSession`?

- (A) An API (set of tools) for performing network requests, like fetching data from a server.

- (B) A UI element for displaying a web page.
 - (C) A security feature for managing web session cookies.
 - (D) A tool for "sessioning" a user during a usability test.
 - (E) A file for storing user login session data.
 - *Correct Answer: (A)*
-

3. What is "Grand Central Dispatch" (GCD)?

- (A) A UI element for laying out views.
 - (B) A low-level API for managing concurrent operations (e.g., running tasks on a background thread).
 - (C) A tool for navigating between view controllers.
 - (D) A service for managing app "dispatch" to the App Store.
 - (E) A network framework for sending messages.
 - *Correct Answer: (B)*
-

4. Why would you run a network request on a *background* thread?

- (A) To make the network request run faster.
 - (B) To prevent the UI from freezing while waiting for the request to complete.
 - (C) To make the request more secure.
 - (D) You should not; all code must run on the main thread.
 - (E) To automatically retry the request if it fails.
 - *Correct Answer: (B)*
-

5. After a network request finishes, on which thread must you *a/ways* update the UI?

- (A) On any background thread.
 - (B) On the "network" thread.
 - (C) On the "main" thread.
 - (D) On a new thread created just for the UI.
 - (E) On the same thread the request was made on.
 - *Correct Answer: (C)*
-

6. What is **UserDefaults**?

- (A) A secure, high-performance database for storing large amounts of data.
- (B) A simple key-value store for persisting *small* pieces of data, like user settings.

- (C) A file that defines the app's default UI layout.
 - (D) A list of all the "default" users for testing.
 - (E) A network service for storing user profiles.
 - *Correct Answer: (B)*
-

7. What is **Core Data**?

- (A) A lightweight key-value store for simple settings.
 - (B) A powerful, built-in framework for managing and persisting a large object graph (database).
 - (C) A framework for fetching data from the internet.
 - (D) A framework for designing UI elements.
 - (E) A tool for analyzing "core" app performance.
 - *Correct Answer: (B)*
-

8. What is the **UIViewController** "lifecycle"?

- (A) A set of functions (e.g., **viewDidLoad**, **viewDidAppear**) that are called at different stages.
 - (B) A diagram showing how all view controllers are connected.
 - (C) The process of creating and destroying a **UIViewController**.
 - (D) A setting that determines how long a view is "alive".
 - (E) A tool for testing view controller performance.
 - *Correct Answer: (A)*
-

9. When is **viewDidLoad()** called?

- (A) Every time the view appears on screen.
 - (B) Only *once*, after the view controller's view has been loaded into memory.
 - (C) Right before the view is about to appear on screen.
 - (D) When the user taps on the view.
 - (E) When the app is about to close.
 - *Correct Answer: (B)*
-

10. What is the difference between **viewDidLoad()** and **viewDidAppear()**?

- (A) **viewDidLoad** is for UI setup; **viewDidAppear** is for logic.

- (B) `viewDidLoad` is called once; `viewDidAppear` is called *every time* the view becomes visible.
 - (C) `viewDidAppear` is called once; `viewDidLoad` is called every time.
 - (D) `viewDidLoad` runs on a background thread; `viewDidAppear` runs on the main thread.
 - (E) There is no difference; they can be used interchangeably.
 - *Correct Answer: (B)*
-

11. What is a `UICollectionView`?

- (A) A UI element for displaying a scrolling list of items (this is `UITableView`).
 - (B) A UI element for displaying items in a customizable grid or layout.
 - (C) A UI element for "collecting" user input, like a form.
 - (D) A non-visual object for "collecting" data.
 - (E) A view controller for managing tabs.
 - *Correct Answer: (B)*
-

12. What is the difference between "Frame" and "Bounds" of a `UIView`?

- (A) "Frame" is the view's location *relative to its parent*; "Bounds" is the view's location *relative to itself*.
 - (B) "Frame" is the view's location *relative to itself*; "Bounds" is the view's location *relative to its parent*.
 - (C) "Frame" defines the view's color; "Bounds" defines its size.
 - (D) "Frame" is for `classes`; "Bounds" is for `structs`.
 - (E) There is no difference; they are the same.
 - *Correct Answer: (A)*
-

13. What is a "protocol extension"?

- (A) A way to add new functions to an *existing* protocol.
 - (B) A way to provide a *default implementation* for methods in a protocol.
 - (C) A `class` that "extends" a protocol.
 - (D) A file that lists all the protocols in your app.
 - (E) A network protocol like `HTTP` or `HTTPS`.
 - *Correct Answer: (B)*
-

14. What is the "delegate" pattern?

- (A) A design pattern where one object (the delegate) acts *on behalf of*, or provides data to, another object.
 - (B) A pattern for "delegating" tasks to a background thread.
 - (C) A pattern for creating and managing `UITableViewController`s.
 - (D) A pattern for managing user authentication.
 - (E) A security pattern for "delegating" permissions.
 - *Correct Answer: (A)*
-

15. What is "JSON"?

- (A) A "JavaScript Object Notation," a lightweight format for sending data over a network.
 - (B) A Swift-specific data format for storing objects.
 - (C) A type of `Core Data` database.
 - (D) A tool for designing UI layouts.
 - (E) A framework for building animations.
 - *Correct Answer: (A)*
-

16. What is the purpose of a "completion handler" (e.g., in a network request)?

- (A) A closure that is *called* when an asynchronous task (like a network request) *finishes*.
 - (B) A function that "completes" the app and shuts it down.
 - (C) A UI element that shows a "completed" state.
 - (D) A variable that is true when a task is done.
 - (E) A tool for handling app "completions" in the App Store.
 - *Correct Answer: (A)*
-

17. What is an "enum" (enumeration) in Swift?

- (A) A type that defines a common group of *related values* (e.g., `Compass.north`, `Compass.south`).
 - (B) A type of `class` that can be "enumerated" (counted).
 - (C) A loop that "enumerates" over an array.
 - (D) A UI element for a numbered list.
 - (E) A variable that holds a single number.
 - *Correct Answer: (A)*
-

18. What is a "computed property"?

- (A) A property that *stores* a value.
 - (B) A property that *calculates* a value when it is accessed, rather than storing it.
 - (C) A property that is shared by all instances of a class.
 - (D) A property that is defined in a protocol.
 - (E) A property that is related to "computing" (e.g., CPU usage).
 - *Correct Answer: (B)*
-

19. What is "optional chaining" (e.g., `user?.address?.street`)?

- (A) A safe way to call properties or methods on an optional that might be `nil`.
 - (B) A "chain" of `if let` statements.
 - (C) A "chain" of force-unwrapped optionals.
 - (D) A UI element that links, or "chains," views together.
 - (E) A network request that "chains" multiple calls.
 - *Correct Answer: (A)*
-

20. What is a "guard let" statement?

- (A) A statement that runs code *only if* an optional is `nil`.
 - (B) A statement that safely unwraps an optional, or *exits* the current scope (e.g., `return`) if it is `nil`.
 - (C) A statement that "guards" a variable from being changed.
 - (D) A statement that is the same as an `if let` statement.
 - (E) A security feature for "guarding" user data.
 - *Correct Answer: (B)*
-

Level 4: Expert & Frameworks

1. What is "SwiftUI"?

- (A) An older, imperative framework for building UIs.
- (B) A modern, *declarative* framework for building UIs across all Apple platforms.
- (C) A framework for managing network requests.
- (D) A database for storing UI layouts.
- (E) A programming language that is faster than Swift.
- *Correct Answer: (B)*

2. What is a **@State** property wrapper in SwiftUI?

- (A) A property that stores a *constant* value.
- (B) A property wrapper that marks a value as a *source of truth*, causing the view to re-render when it changes.
- (C) A property that is shared across the entire app.
- (D) A property that defines the "state" of the network (e.g., "connected").
- (E) A property that stores data in **Core Data**.
- *Correct Answer: (B)*

3. What is a **@Binding** property wrapper in SwiftUI?

- (A) A property that creates a *two-way connection* to a **@State** variable in a parent view.
- (B) A property that "binds" a view to a network request.
- (C) A property that stores a value.
- (D) A property that defines a view's layout constraints.
- (E) A property that is "bound" to the app's lifecycle.
- *Correct Answer: (A)*

4. What is the difference between "UIKit" and "SwiftUI"?

- (A) UIKit is declarative; SwiftUI is imperative.
- (B) UIKit is imperative; SwiftUI is declarative.
- (C) UIKit is for iOS; SwiftUI is for macOS.
- (D) UIKit is a programming language; SwiftUI is a framework.
- (E) There is no difference; they are two names for the same thing.
- *Correct Answer: (B)*

5. What is "Combine"?

- (A) A framework for "combining" multiple Storyboards.
- (B) A *declarative* Swift API for processing values over time (e.g., handling asynchronous events).
- (C) A UI framework for building layouts.
- (D) A tool for "combining" Swift and Objective-C code.
- (E) A database for "combining" data.
- *Correct Answer: (B)*

6. What is an `@ObservableObject` in SwiftUI/Combine?

- (A) A protocol for an object that can "publish" its changes, allowing views to subscribe.
- (B) A UI element that is "observable" by the user.
- (C) A `@State` variable for objects.
- (D) A property for "observing" network traffic.
- (E) A security feature for "observing" user behavior.
- *Correct Answer: (A)*

7. What is "Automatic Reference Counting" (ARC)?

- (A) Swift's system for *manually* managing memory.
- (B) Swift's system for *automatically* managing memory by tracking references to class instances.
- (C) A tool for "counting" how many times a user "references" a feature.
- (D) A framework for building augmented reality (AR) apps.
- (E) A security feature for "counting" login attempts.
- *Correct Answer: (B)*

8. What is a "retain cycle" (or strong reference cycle)?

- (A) A situation where two class instances hold *weak* references to each other.
- (B) A situation where two class instances hold *strong* references to each other, preventing ARC from deallocating them.
- (C) A "cycle" of testing an app and "retaining" the build.
- (D) A good practice for keeping objects in memory.
- (E) A UI layout that is circular.
- *Correct Answer: (B)*

9. How do you *break* a retain cycle, for example in a closure?

- (A) By using `[weak self]` or `[unowned self]` in the closure's capture list.
- (B) By using `[strong self]`.
- (C) By setting the object to `nil` inside the closure.
- (D) By running the closure on the main thread.
- (E) You cannot break them; they are a permanent part of Swift.
- *Correct Answer: (A)*

10. What is the main difference between `weak` and `unowned`?

- (A) `weak` must be an optional; `unowned` does not need to be.
- (B) `weak` is for `classes`; `unowned` is for `structs`.
- (C) `weak` is a strong reference; `unowned` is a weak reference.
- (D) `weak` can only be used in closures.
- (E) There is no difference.
- *Correct Answer: (A)*

11. What is an "NSPredicate"?

- (A) A "predicate" (a filter condition) used to fetch or filter data, most commonly with `Core Data`.
- (B) A type of network request.
- (C) A UI layout constraint.
- (D) A security rule for user authentication.
- (E) A tool for predicting app performance.
- *Correct Answer: (A)*

12. What is `XCTest`?

- (A) A framework for "testing" network speed.
- (B) A framework for building and running unit tests, UI tests, and performance tests.
- (C) A tool for "testing" different UI color palettes.
- (D) A third-party library for analytics.
- (E) A framework for "testing" beta versions of iOS.
- *Correct Answer: (B)*

13. What is a "unit test"?

- (A) A test that checks a small, isolated "unit" of code, like a single function.
- (B) A test that checks the *entire* application flow from start to finish.
- (C) A test that checks the *user interface* by simulating taps.
- (D) A "unit" of measurement for app performance.
- (E) A manual test performed by a QA tester.
- *Correct Answer: (A)*

14. What is a "UI test"?

- (A) A test that checks a small, isolated function.
- (B) A test that launches the app and simulates user interactions (taps, swipes) on the UI.
- (C) A test to see if users "like" the UI.
- (D) A "unit" of measurement for UI performance.
- (E) A test that checks the app's networking logic.
- *Correct Answer: (B)*

15. What is "Test-Driven Development" (TDD)?

- (A) A process where you write the *tests before* you write the code.
- (B) A process where you write code "driven" by manual QA testers.
- (C) A process where you "drive" the UI with test scripts.
- (D) A process where you *only* write tests and no application code.
- (E) A process where you write tests *after* the code is finished.
- *Correct Answer: (A)*

16. What is the "App Transport Security" (ATS)?

- (A) A security feature that enforces secure (e.g., [HTTPS](#)) network connections.
- (B) A tool for "transporting" an app to the App Store.
- (C) A type of encryption for [Core Data](#).
- (D) A service for managing user authentication.
- (E) A hardware "transport" for a physical device.
- *Correct Answer: (A)*

17. What is the [Info.plist](#) file?

- (A) A Swift code file for app logic.
 - (B) A configuration file that stores an app's metadata, like its name, icon, and permissions.
 - (C) A file for storing UI layouts.
 - (D) A file for writing unit tests.
 - (E) A text file for user "info".
 - *Correct Answer: (B)*
-

18. What is a "CocoaPod"?

- (A) A first-party dependency manager built by Apple.
 - (B) A third-party dependency manager for Swift and Objective-C projects.
 - (C) A type of UI element for displaying "pods" of content.
 - (D) A "pod" of servers for cloud hosting.
 - (E) A security feature for "podding" an app.
 - *Correct Answer: (B)*
-

19. What is "Swift Package Manager" (SPM)?

- (A) A third-party dependency manager.
 - (B) Apple's official, integrated dependency manager for Swift projects.
 - (C) A tool for "managing" and "packaging" your app for the App Store.
 - (D) A file for storing Swift "packages".
 - (E) A service for encrypting Swift code.
 - *Correct Answer: (B)*
-

20. What is a "generic" in Swift (e.g., `Array<Element>`)?

- (A) A feature that allows you to write flexible, reusable code (e.g., a function or type) that can work with *any* type.
 - (B) A "generic" or "common" type, like `String` or `Int`.
 - (C) A variable that has no specific type.
 - (D) A "generic" or "default" UI layout.
 - (E) A non-specific error message.
 - *Correct Answer: (A)*
-

Level 5: Strategic & Architectural

1. What is "MVC" (Model-View-Controller)?

- (A) A "Model-View-Combine" architecture for SwiftUI.
- (B) A common Apple design pattern that separates objects into three roles: Model (data), View (UI), and Controller (logic).
- (C) A "Massive-View-Controller," a bad practice where one view controller does everything.
- (D) A "Model-View-Connection" pattern for networking.
- (E) A unit testing framework.

- *Correct Answer: (B)*
-

2. What is a "Massive View Controller" problem?

- (A) A `UIViewController` that is physically too large on the screen.
 - (B) A common anti-pattern where a `UIViewController` becomes bloated with too much responsibility (e.g., logic, network calls).
 - (C) A good architectural pattern for simple apps.
 - (D) An architecture that uses multiple `UITableViewController`s.
 - (E) A type of memory leak.
 - *Correct Answer: (B)*
-

3. What is "MVVM" (Model-View-ViewModel)?

- (A) An architectural pattern that "moves" UI logic from the Controller into a "ViewModel."
 - (B) A "Model-View-View-Model" pattern with two views.
 - (C) Apple's required architecture for all SwiftUI apps.
 - (D) A testing framework.
 - (E) A pattern for managing memory.
 - *Correct Answer: (A)*
-

4. What is a "coordinator" pattern?

- (A) A pattern where a "Coordinator" object manages *navigation logic*, taking it out of the `UIViewController`.
 - (B) A pattern for "coordinating" network requests.
 - (C) A pattern for "coordinating" animations.
 - (D) A pattern for managing `Core Data`.
 - (E) A built-in framework from Apple.
 - *Correct Answer: (A)*
-

5. What is "Instruments"?

- (A) A tool for *profiling* an app to find performance bottlenecks, memory leaks, and high CPU usage.
- (B) A tool for playing musical "instruments" in an app.
- (C) A framework for unit testing.
- (D) A framework for UI testing.

- (E) A tool for "instrumenting" an app with analytics.
 - *Correct Answer: (A)*
-

6. What is a "memory leak"?

- (A) A situation where an app uses too much memory and is terminated.
 - (B) A situation where an object is no longer needed but is *not* deallocated from memory (e.g., a retain cycle).
 - (C) A situation where sensitive data (like a password) is "leaked" to a log.
 - (D) A bug where data is "leaked" from one user to another.
 - (E) A hardware fault in the phone's RAM.
 - *Correct Answer: (B)*
-

7. What is "XCTest" used for?

- (A) To "test" different UI color palettes.
 - (B) To "test" network connectivity.
 - (C) To create and run "unit tests" and "UI tests".
 - (D) To submit an app for "testing" on TestFlight.
 - (E) To "test" the app's name.
 - *Correct Answer: (C)*
-

8. What is "TestFlight"?

- (A) An Apple platform for distributing *beta* versions of an app to testers *before* releasing it on the App Store.
 - (B) A tool for simulating a "flight" in an app.
 - (C) A framework for running unit tests.
 - (D) A third-party tool for testing.
 - (E) The final "flight check" before an app is approved.
 - *Correct Answer: (A)*
-

9. What is "App Store Connect"?

- (A) A tool for "connecting" your app to the internet.
- (B) The web portal used to manage your app (e.g., submit for review, set pricing, view analytics).
- (C) A framework for "connecting" to the App Store.

- (D) A social network for iOS developers.
 - (E) A tool for managing database connections.
 - *Correct Answer: (B)*
-

10. What is a "provisioning profile"?

- (A) A file that "provisions" (sets up) a new `UIViewController`.
 - (B) A file that authorizes an app to be installed and run on *specific devices* and use *specific services*.
 - (C) A "profile" of the app's performance.
 - (D) A "profile" of the app's target user.
 - (E) A file that "provisions" a new database.
 - *Correct Answer: (B)*
-

11. What is a "Certificate" (e.g., Development, Distribution)?

- (A) A "certificate" of completion for an app.
 - (B) A cryptographic file that *identifies you* (the developer) to Apple and proves you are trusted.
 - (C) A file that lists the app's features.
 - (D) A file that manages app permissions.
 - (E) A type of `Info.plist` file.
 - *Correct Answer: (B)*
-

12. What are the "Human Interface Guidelines" (HIG)?

- (A) A strict set of *rules* that your app *must* follow to be accepted.
 - (B) A set of *recommendations* and *guidelines* from Apple for creating a great, platform-consistent user experience.
 - (C) A "guideline" for how to write Swift code.
 - (D) A "guideline" for managing memory.
 - (E) A legal document.
 - *Correct Answer: (B)*
-

13. What is a "dependency manager" (e.g., SPM, CocoaPods)?

- (A) A tool for managing "dependencies" (third-party code) in your project.
- (B) A tool for managing your "tasks" and "dependencies" in a project.

- (C) A pattern for managing "dependent" view controllers.
 - (D) A tool for managing "dependencies" between users.
 - (E) A tool for managing network "dependencies".
 - *Correct Answer: (A)*
-

14. What is a "sandbox" in iOS?

- (A) A UI element for "drawing" in a sandbox.
 - (B) A security mechanism that restricts an app to its *own* container, limiting its access to the file system and other apps.
 - (C) A "sandbox" project for testing new ideas.
 - (D) A version of an app for beta testers.
 - (E) A server for testing network code.
 - *Correct Answer: (B)*
-

15. What is "on-device" processing (e.g., for Core ML)?

- (A) The practice of processing data *on a remote server*.
 - (B) The practice of processing data *directly on the user's device*, which is faster and more private.
 - (C) A tool for testing on a physical "device".
 - (D) A type of memory storage.
 - (E) A security feature for "on-device" authentication.
 - *Correct Answer: (B)*
-

16. What is the "main thread"?

- (A) The *only* thread where it is safe to perform *UI updates*.
 - (B) The *only* thread where you should perform *network requests*.
 - (C) The "main" code file in your project.
 - (D) The "main" `UIViewController` of your app.
 - (E) The "main" flow of a user journey.
 - *Correct Answer: (A)*
-

17. What is "app thinning"?

- (A) The process of "thinning" out your code to make it simpler.

- (B) An App Store process that delivers a *tailored* app bundle (e.g., only the assets for that specific device) to save space.
 - (C) A UI design trend that uses "thin" fonts.
 - (D) A security feature that "thins" out an app's permissions.
 - (E) A type of memory leak.
 - *Correct Answer: (B)*
-

18. What is the strategic trade-off of "cross-platform" (e.g., React Native) vs. "native" (e.g., Swift)?

- (A) Native is faster to build but has worse performance.
 - (B) Cross-platform is faster (one codebase); Native provides the *best* performance and platform integration.
 - (C) Native is cheaper; cross-platform is more expensive.
 - (D) Cross-platform is more secure than native.
 - (E) There is no trade-off; cross-platform is always better.
 - *Correct Answer: (B)*
-

19. What is a "protocol-oriented programming"?

- (A) A style of programming (favored in Swift) that emphasizes designing your app's architecture using *protocols* and *extensions*.
 - (B) A style of programming that focuses on *network protocols*.
 - (C) A style of programming that uses *only* **classes**.
 - (D) A style of programming that is "protocol-driven" (i.e., very formal).
 - (E) An old style of programming from Objective-C.
 - *Correct Answer: (A)*
-

20. What is "Core ML"?

- (A) A framework for "managing" the "core" of the app.
- (B) A framework for "machine learning" models.
- (C) A framework for integrating *pre-trained machine learning models* into your app.
- (D) A framework for "core" animations.
- (E) A framework for "core" data storage.
- *Correct Answer: (C)*

Android Developer track.

Level 1: Basic Foundations

1. What is the primary, modern programming language officially recommended for Android development?

- (A) Java
 - (B) Kotlin
 - (C) Swift
 - (D) C#
 - (E) JavaScript
 - *Correct Answer: (B)*
-

2. What is "Android Studio"?

- (A) The official "Integrated Development Environment" (IDE) used to build, test, and debug Android apps.
 - (B) The online store where users download Android applications.
 - (C) A specific programming language created by Google for UI design.
 - (D) A cloud service for storing and managing Android app data.
 - (E) A virtual device that emulates an Android phone on your computer.
 - *Correct Answer: (A)*
-

3. What is an "Activity" in Android?

- (A) A component that runs a background task, like downloading a file.
 - (B) A component that provides a single screen with a user interface.
 - (C) A configuration file that defines the app's permissions.
 - (D) A visual element, like a button or a text box.
 - (E) A message that is sent between different app components.
 - *Correct Answer: (B)*
-

4. What is an "XML layout" file used for?

- (A) To define the user interface (the visual structure) of a screen.
 - (B) To write the main application logic in the Kotlin language.
 - (C) To store user data, like login information.
 - (D) To define the app's name, icon, and permissions.
 - (E) To store image assets and app icons.
 - *Correct Answer: (A)*
-

5. What is the `AndroidManifest.xml` file?

- (A) A file that defines the visual layout of your main screen.
 - (B) A core file that declares the app's components, permissions, and hardware features.
 - (C) A script used to build and compile the application.
 - (D) A file that stores all the text (strings) for the application.
 - (E) A file used for writing automated tests.
 - *Correct Answer: (B)*
-

6. Which UI component is used to display non-editable text?

- (A) `Button`
 - (B) `EditText`
 - (C) `TextView`
 - (D) `ImageView`
 - (E) `CheckBox`
 - *Correct Answer: (C)*
-

7. Which UI component is used to get text input from a user?

- (A) `TextView`
 - (B) `Button`
 - (C) `Layout`
 - (D) `EditText`
 - (E) `Spinner`
 - *Correct Answer: (D)*
-

8. Which UI component is used to display a single image?

- (A) `BitmapView`
 - (B) `Drawable`
 - (C) `PhotoView`
 - (D) `ImageHolder`
 - (E) `ImageView`
 - *Correct Answer: (E)*
-

9. What is the "Google Play Store"?

- (A) The official IDE for building Android apps.
 - (B) A cloud database for storing app data.
 - (C) The official digital distribution service (app store) for Android.
 - (D) A virtual device for testing apps.
 - (E) A library for showing Google Maps in an app.
 - *Correct Answer: (C)*
-

10. What is an "Emulator" (or Android Virtual Device - AVD)?

- (A) A physical Android phone used for development.
 - (B) A software program that simulates an Android device on your computer.
 - (C) A tool for compiling your Kotlin code into an app.
 - (D) A file that defines your app's permissions.
 - (E) A security tool that scans your app for viruses.
 - *Correct Answer: (B)*
-

11. What is an "APK" (Android Package)?

- (A) The file format used to distribute and install Android applications.
 - (B) A text file containing your app's source code.
 - (C) A settings file in Android Studio.
 - (D) A file that defines the app's UI layout.
 - (E) A type of network request.
 - *Correct Answer: (A)*
-

12. What is `Log.d()` used for in Android development?

- (A) To display a "dialog" box to the user.
 - (B) To "delete" a file from the user's storage.
 - (C) To "draw" a view on the screen.
 - (D) To write a "debug" message to the Logcat (console).
 - (E) To "define" a new variable.
 - *Correct Answer: (D)*
-

13. In Kotlin, what keyword is used to declare a variable that *can* be changed?

- (A) `val`
 - (B) `var`
 - (C) `let`
 - (D) `const`
 - (E) `static`
 - *Correct Answer: (B)*
-

14. In Kotlin, what keyword is used to declare a constant (a variable that *cannot* be changed)?

- (A) `val`
 - (B) `var`
 - (C) `let`
 - (D) `const`
 - (E) `final`
 - *Correct Answer: (A)*
-

15. What is the traditional (legacy) programming language for Android, now secondary to Kotlin?

- (A) Java
 - (B) C++
 - (C) Swift
 - (D) Python
 - (E) Dart
 - *Correct Answer: (A)*
-

16. What is the `R` file or `R` object in Android?

- (A) A "resource" file that automatically generated IDs for all your app's assets (layouts, strings, images).
 - (B) A "runtime" file that executes your main application code.
 - (C) A "release" file that holds your app's production settings.
 - (D) A "read-only" file for storing secure data.
 - (E) A "reference" file that lists all your app's functions.
 - *Correct Answer: (A)*
-

17. What is the **setOnClickListener** method used for?

- (A) To define what happens when a **TextView** is edited.
 - (B) To define what happens when a **Button** (or other View) is tapped.
 - (C) To set the text color of a View.
 - (D) To set the visibility of a View.
 - (E) To define what happens when the screen is "clicked."
 - *Correct Answer: (B)*
-

18. What is a **LinearLayout**?

- (A) A layout that arranges its child views in a single row (horizontal) or a single column (vertical).
 - (B) A layout that arranges views based on their relationship to each other and to the parent.
 - (C) A layout that lets you place views anywhere using X/Y coordinates.
 - (D) A layout that displays a scrolling list of items.
 - (E) A layout that shows a grid of items.
 - *Correct Answer: (A)*
-

19. What is a **ConstraintLayout**?

- (A) A layout that arranges views in a single row or column.
 - (B) A layout that arranges views based on constraints (relationships) to each other and the parent.
 - (C) A layout that is "constrained" and cannot be changed.
 - (D) An old, legacy layout that is no longer used.
 - (E) A layout for displaying text with "constraints."
 - *Correct Answer: (B)*
-

20. What data type is used to store text, like "Hello World"?

- (A) **Int**
- (B) **Double**
- (C) **Char**
- (D) **String**
- (E) **Text**
- *Correct Answer: (D)*

Level 2: Intermediate Application

1. What is an **Intent**?

- (A) A component that runs a long-running background task.
- (B) A messaging object used to request an action, such as starting another **Activity**.
- (C) A file that defines the app's permissions.
- (D) A UI element that shows the user's "intention."
- (E) A variable that stores a user's text input.
- *Correct Answer: (B)*

2. What is the difference between an "explicit" and an "implicit" **Intent**?

- (A) An explicit **Intent** specifies the *exact component* to start (e.g., **MainActivity.class**); an implicit **Intent** specifies an *action* (e.g., "open a map").
- (B) An explicit **Intent** is for starting **Activities**; an implicit **Intent** is for starting **Services**.
- (C) An explicit **Intent** carries data; an implicit **Intent** does not.
- (D) An explicit **Intent** is secure; an implicit **Intent** is not.
- (E) An explicit **Intent** is written in Kotlin; an implicit **Intent** is written in XML.
- *Correct Answer: (A)*

3. What is the **onCreate()** method in an **Activity**?

- (A) It is called every time the **Activity** becomes visible to the user.
- (B) It is called only *once* when the **Activity** is first created, and is used for setup.
- (C) It is called when the **Activity** is about to be destroyed.
- (D) It is called when the app is "created" for the first time.
- (E) It is called when the user "creates" a new file.
- *Correct Answer: (B)*

4. What is a **RecyclerView**?

- (A) A UI component for displaying a simple, static list of items.
- (B) A UI component for displaying a large, *efficiently scrolling* list of items.

- (C) A UI component for displaying items in a grid.
 - (D) A database for "recycling" old data.
 - (E) A tool for "recycling" memory in an app.
 - *Correct Answer: (B)*
-

5. What is the purpose of a **RecyclerView.Adapter**?

- (A) It "adapts" the **RecyclerView** to fit different screen sizes.
 - (B) It creates and "binds" the data from your app (e.g., an **Array**) to the **Views** in the list.
 - (C) It handles click events on the rows.
 - (D) It defines the layout of the **RecyclerView** (e.g., linear vs. grid).
 - (E) It manages the network connection to "adapt" the data.
 - *Correct Answer: (B)*
-

6. What is the purpose of the **ViewHolder** pattern?

- (A) It "holds" the data for the entire list.
 - (B) It "holds" references to a single row's views, avoiding repeated **findViewById** calls.
 - (C) It "holds" the user's place in the list when they scroll.
 - (D) It "holds" the **Activity** that contains the list.
 - (E) It "holds" the database connection.
 - *Correct Answer: (B)*
-

7. How do you pass data from one **Activity** to another?

- (A) By using a global "static" variable.
 - (B) By saving the data to a file and reading it in the new **Activity**.
 - (C) By putting the data as "extras" in the **Intent**.
 - (D) By using **SharedPreferences**.
 - (E) You cannot pass data between **Activities**.
 - *Correct Answer: (C)*
-

8. What is a **Fragment**?

- (A) A "fragment" of an app that has been broken.

- (B) A reusable portion of a user interface that is hosted within an **Activity**.
 - (C) A configuration file.
 - (D) A small piece of data.
 - (E) A type of background task.
 - *Correct Answer: (B)*
-

9. What is the primary purpose of a **ViewModel?**

- (A) To store and manage UI-related data in a way that *survives* configuration changes (like screen rotation).
 - (B) To define the "model" of the UI (the XML layout).
 - (C) To "view" the data from the database.
 - (D) To run network requests and other background tasks.
 - (E) To replace the **Activity** and **Fragment** classes.
 - *Correct Answer: (A)*
-

10. What is **SharedPreferences used for?**

- (A) Storing large, complex, relational data.
 - (B) Storing and retrieving small amounts of simple, key-value data (like user settings).
 - (C) "Sharing" data between your app and other apps.
 - (D) "Sharing" your app with other users.
 - (E) Storing image and media files.
 - *Correct Answer: (B)*
-

11. Why is it best practice to store user-facing text (e.g., "Login") in the **strings.xml file?**

- (A) It makes the text more secure.
 - (B) It makes the text run faster.
 - (C) It allows for easy "localization" (translating the app into other languages).
 - (D) It is the only place **TextView** can read text from.
 - (E) It allows the text to be changed by the user.
 - *Correct Answer: (C)*
-

12. What is the modern, recommended way to get references to Views in your code?

- (A) Using `findViewById()` for every single View.
 - (B) Using "View Binding," which generates a binding class for each layout file.
 - (C) Storing all Views in a global "static" list.
 - (D) Giving all Views a number and accessing them by index.
 - (E) Using reflection to find Views by their name.
 - *Correct Answer: (B)*
-

13. What is a "nullable" type in Kotlin (e.g., `String?`)?

- (A) A variable that is "null" and cannot be used.
 - (B) A variable that can hold either a value (e.g., a `String`) or `null`.
 - (C) A type of `Boolean` that is neither true nor false.
 - (D) A variable that has been deleted.
 - (E) A variable that is private and cannot be accessed.
 - *Correct Answer: (B)*
-

14. What is the `?.` (safe call) operator in Kotlin used for?

- (A) To "call" a function in a safer, encrypted way.
 - (B) To check if a variable is `null`, and if not, call a method on it.
 - (C) To "force" a `null` variable to have a value.
 - (D) To mark a function as "safe" to call.
 - (E) To check if two variables are equal.
 - *Correct Answer: (B)*
-

15. What is `Gradle`?

- (A) A "build automation" tool used to manage dependencies and compile your app.
 - (B) A UI design tool for creating layouts.
 - (C) A testing framework for writing unit tests.
 - (D) A database for storing app data.
 - (E) A server for hosting your app's backend.
 - *Correct Answer: (A)*
-

16. What is the purpose of the `onPause()` method in an `Activity`?

- (A) It is called when the `Activity` is first created.

- (B) It is called when the **Activity** is no longer visible (e.g., user hits "Home").
 - (C) It is called when the **Activity** is partially obscured (e.g., by a dialog).
 - (D) It is called when the app crashes.
 - (E) It is called to "pause" a media player.
 - *Correct Answer: (C)*
-

17. What is the purpose of the `onDestroy()` method in an **Activity?**

- (A) It is called just before the **Activity** is "destroyed" and removed from memory.
 - (B) It is called to "destroy" or delete a file.
 - (C) It is called when the app is installed.
 - (D) It is called when the user "destroys" the UI by tapping on it.
 - (E) It is called every time the **Activity** is hidden.
 - *Correct Answer: (A)*
-

18. What is the "Context" in Android?

- (A) An object that provides access to global app resources and services (e.g., strings, assets).
 - (B) The "context" or text inside a **TextView**.
 - (C) A file that defines the app's layout.
 - (D) The "context" of a user's session (e.g., logged in).
 - (E) A background thread for running tasks.
 - *Correct Answer: (A)*
-

19. What is a "style" in an XML layout?

- (A) A file that defines the app's permissions.
 - (B) A collection of attributes (e.g., color, size) that can be reused across multiple **Views**.
 - (C) A Kotlin file for writing UI logic.
 - (D) A component for "styling" images with filters.
 - (E) A **style** of programming, like "object-oriented."
 - *Correct Answer: (B)*
-

20. What is a "drawable"?

- (A) A "drawable" UI element that the user can move.

- (A) A general term for a graphic that can be drawn to the screen (e.g., an image, a shape).
 - (C) A "drawable"-type pen for a stylus.
 - (D) A background task.
 - (E) A Kotlin file.
 - *Correct Answer: (B)*
-

Level 3: Advanced Scenarios

1. A user rotates their phone, and the text they typed into an **EditText** disappears. Why does this happen?

- (A) The **Activity** is *destroyed and recreated* on rotation, and the **ViewModel** was not used to save the text.
 - (B) The **EditText** view is not marked as "rotatable" in the XML.
 - (C) The internet connection is lost during the rotation.
 - (D) The text is saved, but the color is set to transparent.
 - (E) The font does not support landscape mode.
 - *Correct Answer: (A)*
-

2. What is **LiveData**?

- (A) A class that provides a "live" video stream from the camera.
 - (B) A lifecycle-aware, observable data holder class.
 - (C) A "live" database connection that is always open.
 - (D) A "live" test that runs on a real device.
 - (E) A **TextView** that updates "live" as the user types.
 - *Correct Answer: (B)*
-

3. What is the primary advantage of using **LiveData**?

- (A) It automatically updates the UI *only* when the **Activity/Fragment** is in an active state, preventing crashes.
- (B) It automatically encrypts all data.
- (C) It makes network requests run much faster.
- (D) It automatically creates UI layouts.
- (E) It is the only way to store data in a **ViewModel**.
- *Correct Answer: (A)*

4. What is **Room**?

- (A) A UI layout for designing a "room" in an app.
- (B) A "networking" library for REST APIs.
- (C) An "Object-Relational Mapping" (ORM) library that provides an abstraction layer over **SQLite**.
- (D) A "room" where developers can chat.
- (E) A "dependency injection" framework.
- *Correct Answer: (C)*

5. What is a "DAO" (Data Access Object) in the context of **Room**?

- (A) The main database object.
- (B) The XML file that defines the database schema.
- (C) An interface where you define your database queries (e.g., **@Query**, **@Insert**).
- (D) A background task for accessing data.
- (E) A "Data Access Owner" (the user).
- *Correct Answer: (C)*

6. What is **Retrofit**?

- (A) A type-safe "networking" library for Android and Java, used for consuming REST APIs.
- (B) A "database" library for storing data.
- (C) A "UI" library for creating "retro" style interfaces.
- (D) A "build" tool for "retrofitting" old code.
- (D) An "analytics" library for tracking users.
- *Correct Answer: (A)*

7. What are "Coroutines" in Kotlin?

- (A) A framework for managing and simplifying asynchronous or non-blocking code (like network calls).
- (B) A "routine" or "function" that must be run on the UI thread.
- (C) A "UI" component for showing a list.
- (D) A "security" feature for encrypting data.
- (E) A "testing" library.

- *Correct Answer: (A)*
-

8. Why can you *not* run a network request or database query on the "main thread"?

- (A) It is *insecure* and can leak data.
 - (B) It *blocks* the UI, causing the app to freeze (an "ANR").
 - (C) It *wastes* battery by running too fast.
 - (D) It is *not possible* due to compiler errors.
 - (E) It *corrupts* the data.
 - *Correct Answer: (B)*
-

9. How do you switch to the "main thread" from a background coroutine?

- (A) By using `withContext(Dispatchers.Main) { ... }`
 - (B) By using `Thread.sleep()`
 - (C) By creating a new `Activity`.
 - (D) By calling `runOnMain()`.
 - (E) You cannot; all coroutines run on the main thread.
 - *Correct Answer: (A)*
-

10. What is the "Repository" pattern?

- (A) A pattern where the `Activity` directly talks to the network and database.
 - (B) A "repository" for storing your app's code (like GitHub).
 - (C) A "mediator" class that handles data operations, abstracting the data sources (e.g., network, database) from the `ViewModel`.
 - (D) A `ViewModel` that is "shared" between multiple `Fragments`.
 - (E) A pattern for storing all your app's `Strings`.
 - *Correct Answer: (C)*
-

11. What is "Dependency Injection"?

- (A) A "security attack" where a hacker "injects" bad code.
- (B) A "design pattern" where a class's dependencies (e.g., a `Repository`) are *provided* to it, rather than it creating them itself.
- (C) A "build" feature for "injecting" `Gradle` dependencies.
- (D) A "networking" feature for "injecting" data.

- (E) A "UI" pattern for "injecting" **Views**.
 - *Correct Answer: (B)*
-

12. What is "Hilt"?

- (A) A "networking" library for REST APIs.
 - (B) A "database" library.
 - (C) A "dependency injection" library for Android, built on top of Dagger.
 - (D) A "UI testing" framework.
 - (E) A "build" tool.
 - *Correct Answer: (C)*
-

13. What is a **Service** in Android?

- (A) A component for running long-running operations in the *background* (e.g., playing music) without a UI.
 - (B) A "customer service" SDK for adding a chat bot.
 - (C) An **Activity** that provides a "service" to the user.
 - (D) A **ViewModel** that "serves" data.
 - (E) A "network" class that "serves" data from an API.
 - *Correct Answer: (A)*
-

14. What is a **BroadcastReceiver**?

- (A) A "networking" component for "receiving" data from a "broadcast".
 - (B) A "UI" component for "receiving" user input.
 - (C) A "component" that listens for and "receives" system-wide "broadcast" messages (e.g., "battery low", "boot completed").
 - (D) A "media" component for "receiving" a radio "broadcast".
 - (E) A "security" component for "receiving" threats.
 - *Correct Answer: (C)*
-

15. What is a "data class" in Kotlin?

- (A) A **class** that is "abstract" and must be subclassed.
- (B) A **class** that can *only* hold **Ints** and **Strings**.

- (C) A `class` designed to *hold data*, which automatically generates `equals()`, `hashCode()`, `toString()`, etc.
 - (D) A `class` that connects to a "database".
 - (E) A `class` that is *private* to the file.
 - *Correct Answer: (C)*
-

16. What does **ProGuard** or **R8** do?

- (A) It "guards" your app from viruses.
 - (B) It "shrinks" (removes unused code) and "obfuscates" (renames) your code to make the APK smaller and harder to reverse-engineer.
 - (C) It "upgrades" your code to the latest version of Kotlin.
 - (D) It "tests" your code for bugs.
 - (E) It "installs" your app on a device.
 - *Correct Answer: (B)*
-

17. How do you request a "dangerous" permission (e.g., **CAMERA**)?

- (A) You just add it to the `AndroidManifest.xml` file.
 - (B) You add it to the `AndroidManifest.xml` file *and* explicitly "request" it from the user at runtime.
 - (C) You "request" it at runtime; the `AndroidManifest.xml` is not needed.
 - (D) You "email" Google to get permission.
 - (E) You "cannot" request dangerous permissions.
 - *Correct Answer: (B)*
-

18. What is the "Navigation Component"?

- (A) A "hardware" component for GPS.
 - (B) A "UI" component for a "bottom navigation" bar.
 - (C) A "Jetpack library" for managing `Fragment` transactions and "navigation" within your app.
 - (D) A "networking" component for "navigating" websites.
 - (E) A "Google Maps" API.
 - *Correct Answer: (C)*
-

19. What is the difference between `==` and `===` in Kotlin?

- (A) `==` checks for "structural" equality (do they have the same content?); `===` checks for "referential" equality (do they point to the same object?).
 - (B) `==` checks for "referential" equality; `===` checks for "structural" equality.
 - (C) `==` is for `Ints`; `===` is for `Strings`.
 - (D) `==` is an "assignment" (like `=`); `===` is a "comparison".
 - (E) There is no `===` in Kotlin.
 - *Correct Answer: (A)*
-

20. What is an "ANR" (Application Not Responding)?

- (A) A "crash" that happens when your app uses too much memory.
 - (B) A "dialog" shown by the OS when the "main thread" is blocked for too long (e.g., by a network call).
 - (C) A "network error" when the server is "not responding".
 - (D) A "security error" when an "application is not registered".
 - (E) A "build error".
 - *Correct Answer: (B)*
-

Level 4: Expert & Frameworks

1. What is "Jetpack Compose"?

- (A) A "networking" library for "composing" API requests.
 - (B) A "build" tool for "composing" your app's `Gradle` files.
 - (C) A "modern, declarative" UI toolkit for building native Android UI.
 - (D) A "database" library for "composing" SQL queries.
 - (E) A "testing" library for "composing" UI tests.
 - *Correct Answer: (C)*
-

2. What is a `@Composable` function?

- (A) A "function" that "composes" (builds) a piece of UI.
- (B) A "function" that "composes" (combines) multiple data sources.
- (C) A "function" that "composes" (writes) music.
- (D) A "function" that "composes" (encrypts) data.
- (E) A "function" that "composes" (logs) errors.
- *Correct Answer: (A)*

3. What is "recomposition" in Jetpack Compose?

- (A) The "process" of rebuilding your entire app.
- (B) The "process" of "re-running" your Composable functions to update the UI when their state changes.
- (C) The "process" of "decomposing" your UI into smaller parts.
- (D) The "process" of "re-downloading" data.
- (E) The "process" of "re-installing" your app.
- *Correct Answer: (B)*

4. What is the purpose of **remember** in Jetpack Compose?

- (A) To "remember" (store) an object in the `ViewModel`.
- (B) To "remember" (store) an object in memory *across* recompositions.
- (C) To "remember" (store) an object in `SharedPreferences`.
- (D) To "remember" (store) an object in the `Room` database.
- (E) To "remember" (store) an object in the `AndroidManifest.xml`.
- *Correct Answer: (B)*

5. What is "State Hoisting" in Compose?

- (A) A "pattern" of moving "state" *down* into the Composable that uses it.
- (B) A "pattern" of moving "state" *up* to a common ancestor to make a Composable stateless.
- (C) A "pattern" of "hoisting" (lifting) the entire app into the cloud.
- (D) A "pattern" of "hoisting" (copying) state into a global variable.
- (E) A "pattern" of "hoisting" (deleting) old state.
- *Correct Answer: (B)*

6. What is "Kotlin Flow"?

- (A) A "UI" component for "flowing" text.
- (B) An "asynchronous data stream" (like RxJava) that emits values sequentially.
- (C) A "networking" library.
- (D) A "layout" that lets `Views` "flow" to the next line.
- (E) A "dependency injection" framework.
- *Correct Answer: (B)*

7. What is the difference between a "hot" and "cold" Flow?

- (A) A "hot" Flow is faster; a "cold" Flow is slower.
 - (B) A "hot" Flow is for "networking"; a "cold" Flow is for "databases".
 - (C) A "hot" Flow (e.g., SharedFlow) is *active* and emits values even with no collectors; a "cold" Flow *starts* emitting only when collected.
 - (D) A "hot" Flow can "error"; a "cold" Flow cannot.
 - (E) A "hot" Flow is for the "UI thread"; a "cold" Flow is for "background threads".
 - *Correct Answer: (C)*
-

8. What is StateFlow?

- (A) A "hot" Flow that holds a single, observable "state" value.
 - (B) A "cold" Flow that emits a "state" value once.
 - (C) A LiveData object.
 - (D) A ViewModel state.
 - (E) A "UI" component for showing "state".
 - *Correct Answer: (A)*
-

9. What is a Room "database migration"?

- (A) The "process" of "migrating" (moving) your Room database to a new server.
 - (B) The "process" of "migrating" (updating) your database "schema" (e.g., adding a table) *without* losing user data.
 - (C) The "process" of "migrating" (converting) your Room database to Firebase.
 - (D) The "process" of "migrating" (deleting) old data.
 - (E) The "process" of "migrating" (copying) data from one user to another.
 - *Correct Answer: (B)*
-

10. What is WorkManager?

- (A) A "library" for managing "work" on the *UI thread*.
- (B) A "library" for "deferrable, guaranteed, and constrained" *background* work (e.g., "sync data every 24 hours when charging").
- (C) A "project management" tool.
- (D) A "UI" component for showing a "work" "progress" bar.

- (E) A "networking" library.
 - *Correct Answer: (B)*
-

11. What is a "Content Provider"?

- (A) A "component" that *shares* your app's data securely with *other* apps.
 - (B) A "component" that "provides" **Views** to your **Activity**.
 - (C) A "component" that "provides" (downloads) "content" from the internet.
 - (D) A "component" that "provides" (manages) "dependencies".
 - (E) A "component" that "provides" (shows) "ads".
 - *Correct Answer: (A)*
-

12. How do you create a "Custom View"?

- (A) By "subclassing" **View** or **ViewGroup** and "overriding" **onDraw** and **onMeasure**.
 - (B) By "creating" a new XML "layout" file.
 - (C) By "creating" a new **Activity**.
 - (D) By "creating" a new **ViewModel**.
 - (E) By "subclassing" **RoomDatabase**.
 - *Correct Answer: (A)*
-

13. What is the difference between **launch** and **async** in coroutines?

- (A) **launch** is "slow"; **async** is "fast".
 - (B) **launch** is for the "UI thread"; **async** is for "background threads".
 - (C) **launch** is "fire and forget" (returns a **Job**); **async** is for getting a "result" (returns a **Deferred**).
 - (D) **launch** is "old"; **async** is "new".
 - (E) There is no difference.
 - *Correct Answer: (C)*
-

14. What is an "instrumented test" (e.g., using **Espresso**)?

- (A) A "test" that runs on the *JVM* to check "business logic".
- (B) A "test" that runs on a *device or emulator* to "test" the *UI* and *Android APIs*.
- (C) A "test" for musical "instruments".
- (D) A "test" for the "networking" layer.

- (E) A "manual" test.
 - *Correct Answer: (B)*
-

15. What is a "unit test"?

- (A) A "test" that runs on a *device* to check the *UI*.
 - (B) A "test" that runs on the *JVM* to check a "single, isolated" "unit" (e.g., a function).
 - (C) A "test" that checks the "entire" "app flow".
 - (D) A "test" that "measures" "performance".
 - (E) A "test" that "checks" "security".
 - *Correct Answer: (B)*
-

16. What is **TDD** (Test-Driven Development)?

- (A) A "process" where you write "tests" *after* you write the "code".
 - (B) A "process" where you write "tests" *before* you write the "code".
 - (C) A "process" where you *only* write "tests".
 - (D) A "process" where "testers" "drive" the "development".
 - (E) A "process" for "driving" "UI tests".
 - *Correct Answer: (B)*
-

17. What is a "sealed class" in Kotlin?

- (A) A **class** that is "encrypted" or "sealed".
 - (B) A **class** that "restricts" the class "hierarchy", often used for "representing states" (e.g., **Loading**, **Success**, **Error**).
 - (C) A **class** that "cannot" be "instantiated".
 - (D) A **class** that is "final" and "cannot" be "subclassed".
 - (E) A "data class" that is "sealed".
 - *Correct Answer: (B)*
-

18. What is a **Foreground Service**?

- (A) A **Service** that runs on the "UI thread".
- (B) A **Service** that shows a "UI" component.
- (C) A **Service** that is "less" likely to be "killed" by the OS because it shows a "persistent notification".

- (D) A "networking" **Service**.
 - (E) A "database" **Service**.
 - *Correct Answer: (C)*
-

19. What is **ProGuard**?

- (A) A "security" tool for "guarding" against "professionals".
 - (B) A "build" tool for "shrinking" and "obfuscating" code (now largely replaced by **R8**).
 - (C) A "testing" tool.
 - (D) A "dependency injection" framework.
 - (E) A "networking" library.
 - *Correct Answer: (B)*
-

20. What is **R8**?

- (A) A "networking" library.
 - (B) A "database" library.
 - (C) The "default" "build" tool that "shrinks", "obfuscates", and "desugars" code.
 - (D) A "UI testing" framework.
 - (E) A "unit testing" framework.
 - *Correct Answer: (C)*
-

Level 5: Strategic & Architectural

1. A user's **RecyclerView** is "janky" (stutters) while scrolling. What is the *most likely* cause?

- (A) The **RecyclerView** has too many items in the list.
 - (B) The device's network connection is slow.
 - (C) Complex work (e.g., image decoding, date formatting, database lookups) is being done in **onBindViewHolder** on the main thread.
 - (D) The **ViewModel** is holding too much data, causing a memory leak.
 - (E) The **Activity** is not using **Fragments**.
 - *Correct Answer: (C)*
-

2. What is the key strategic difference between **MVVM** (Model-View-ViewModel) and **MVI** (Model-View-Intent)?

- (A) **MVVM** is for **XML**; **MVI** is for **Compose**.
 - (B) **MVVM** uses **LiveData**; **MVI** uses **Kotlin Flow**.
 - (C) **MVI** enforces a *strict, unidirectional data flow* (State flows down, Intents/Events flow up), which can simplify debugging complex states.
 - (D) **MVVM** is for **Kotlin**; **MVI** is for **Java**.
 - (E) **MVVM** is for simple apps; **MVI** is for databases.
 - *Correct Answer: (C)*
-

3. You've rotated an **Activity** 20 times, and the memory profiler shows 20 instances of it. What is the *most likely* cause?

- (A) This is normal behavior for Android.
 - (B) The **ViewModel** is not correctly configured.
 - (C) A background thread, closure, or static reference is holding a strong reference to the **Activity Context**, preventing garbage collection.
 - (D) The **Room** database is open, which locks the **Activity** in memory.
 - (E) The **onPause()** method was not implemented, so the **Activity** never stopped.
 - *Correct Answer: (C)*
-

4. What is the primary *business value* of adopting a "multi-module architecture"?

- (A) It makes the app *run* faster on the user's device.
 - (B) It *reduces APK size* by splitting the app into multiple **APKs**.
 - (C) It *improves build times* (via caching) and *enforces separation of concerns*, allowing teams to work in parallel.
 - (D) It makes the app *more secure* by default.
 - (E) It is *required* by Google to publish on the Play Store.
 - *Correct Answer: (C)*
-

5. What is the *best practice* for storing a secret API key?

- (A) Hard-code it as a **String** constant in your **Repository** class.
- (B) Place it in **strings.xml** so it's easy to access.
- (C) Store it in **local.properties** (which is in **.gitignore**) and expose it to your app via the **build.gradle** file.

- (D) Ask the user to type in the API key when they first launch the app.
 - (E) Store it in `SharedPreferences` on the first run.
 - *Correct Answer: (C)*
-

6. A startup wants an app for iOS and Android. What is the key *technical trade-off* between "Kotlin Multiplatform (KMP)" and "Flutter"?

- (A) `KMP` shares *logic* but requires *native UI* code (SwiftUI/Compose); `Flutter` shares *both* logic and UI but is a *non-native* ecosystem.
 - (B) `KMP` is "easier" to learn; `Flutter` is "harder".
 - (C) `KMP` is "owned" by Google; `Flutter` is "owned" by Facebook.
 - (D) `KMP` only works for "simple" apps; `Flutter` is for "complex" apps.
 - (E) `KMP` has "better" performance; `Flutter` has "worse" performance.
 - *Correct Answer: (A)*
-

7. Why is "Dependency Injection" (e.g., using `Hilt`) considered essential for a modern, testable architecture?

- (A) It "decouples" components (e.g., a `ViewModel` doesn't know *how* to create a `Repository`), which makes *testing* (by providing "fakes" or "mocks") significantly easier.
 - (B) It "combines" all your dependencies into a single file.
 - (C) It "makes" your app "run faster" by pre-loading all your classes.
 - (D) It "encrypts" your dependencies.
 - (E) It "automatically" "writes" your unit tests.
 - *Correct Answer: (A)*
-

8. Why is it a severe architectural error to fetch data from a network *inside* a Composable function?

- (A) Composable functions *must* be `suspend` functions, and network calls are not.
- (B) Composable functions can "recompose" (re-run) many times per second, which would trigger an excessive number of network calls.
- (C) Network calls are *not* "UI" elements, so the compiler will "reject" them.
- (D) This is "not" an error; it is the "recommended" way to fetch data.
- (E) This will "block" the "background thread" and cause a "crash".
- *Correct Answer: (B)*

9. What is "SSL Pinning," and when would you use it?

- (A) It's an "attack" where a hacker "pins" your SSL certificate.
- (B) It's a "security" "technique" where the app *only* trusts a "specific, hardcoded" certificate, "preventing" "Man-in-the-Middle" (MITM) "attacks".
- (C) It's a "performance" "technique" for "pinning" an SSL "connection" "open".
- (D) It's a "UI" "technique" for "pinning" a "secure" "view" to the "top" of the "screen".
- (E) It's a "build" "setting" that "pins" your "SDK" "version".
- *Correct Answer: (B)*

10. What is the strategic difference between Room and Firebase Firestore?

- (A) Room is an "old" "SQL" "database"; Firestore is a "new" "SQL" "database".
- (B) Room is a "local, on-device" "SQL" "database"; Firestore is a "cloud-hosted, real-time" "NoSQL" "database".
- (C) Room is "insecure"; Firestore is "secure".
- (D) Room is for "Kotlin"; Firestore is for "Java".
- (D) Room is "complex"; Firestore is "simple".
- *Correct Answer: (B)*

11. What is the difference between a "memory leak" and "memory churn"?

- (A) A "leak" is "memory" "lost" (not garbage collected); "churn" is "rapidly" "allocating" and "deallocating" "memory", "causing" "stutters".
- (B) A "leak" is "rapidly" "allocating" "memory"; "churn" is "memory" "lost".
- (C) A "leak" is "harmless"; "churn" is "dangerous".
- (D) A "leak" "happens" on the "UI thread"; "churn" "happens" on a "background thread".
- (E) They are "two" "names" for the "same" "problem".
- *Correct Answer: (A)*

12. Why is it a *bad* idea to store sensitive user data (like a token) in SharedPreferences?

- (A) SharedPreferences is "temporary" and "deleted" when the "app" "closes".
- (B) SharedPreferences is "slow" and "blocks" the "UI thread".
- (C) SharedPreferences is "stored" as "plain text" (XML), "making" it "easy" to "steal" on a "rooted" "device".

- (D) `SharedPreferences` "cannot" "store" `Strings`.
 - (E) `SharedPreferences` is "public" and "can" be "read" by "other" "apps".
 - *Correct Answer: (C)*
-

13. What is "Baseline Profiling"?

- (A) A "profile" of your "app's" "baseline" "security" "vulnerabilities".
 - (B) A "profile" of "code" (generated by a test) that is "used" by the "Android Runtime" to "AOT-compile" "critical" "paths", "improving" "startup" "speed".
 - (C) A "profile" of your "app's" "baseline" "memory" "usage".
 - (D) A "profile" "used" by "developers" to "compare" "performance".
 - (E) A "profile" "used" by "ProGuard" to "determine" "which" "code" to "shrink".
 - *Correct Answer: (B)*
-

14. What is "unidirectional data flow" (UDF)?

- (A) A "pattern" where "data" *only* "flows" from the "network" to the "UI".
 - (B) A "pattern" (used in `MVI/Compose`) where "State" "flows" *down* (from `ViewModel` to `UI`) and "Events" "flow" *up* (from `UI` to `ViewModel`).
 - (C) A "pattern" where "data" "flows" *between* `Activitys` in *one* "direction".
 - (D) A "pattern" where "data" is *not* "cached" and "must" be "fetched" "every" "time".
 - (E) A "pattern" for "networking".
 - *Correct Answer: (B)*
-

15. What is an "ABI" (Application Binary Interface)?

- (A) An "API" for "binary" "code".
 - (B) The "interface" between your "app" and the "CPU" (e.g., `arm64-v8a`, `x86_64`).
 - (C) A "security" "interface".
 - (D) A "build" "script".
 - (E) A "testing" "interface".
 - *Correct Answer: (B)*
-

16. Why would you strategically choose to use `AlarmManager` instead of `WorkManager`?

- (A) You need to "run" a "task" *immediately* on a "background thread".
- (B) You need to "run" a "deferrable" "task" "sometime" in the "next" "hour".

- (C) You need to "run" a "task" at an *exact* "time" (e.g., 9:00 AM) and "wake" the "phone" "up" to "do" "it".
 - (D) You "need" to "run" a "task" that "requires" a "network" "connection".
 - (E) You "should" "never" "use" `AlarmManager`; "always" "use" `WorkManager`.
 - *Correct Answer: (C)*
-

17. What is the strategic difference between "unit tests" and "instrumented tests"?

- (A) "Unit tests" "check" "logic" on the "JVM" (fast); "instrumented tests" "check" "UI" and "Android APIs" on a "device" (slow).
 - (B) "Unit tests" are "slow"; "instrumented tests" are "fast".
 - (C) "Unit tests" are "manual"; "instrumented tests" are "automated".
 - (D) "Unit tests" are for "Compose"; "instrumented tests" are for "XML".
 - (E) "Unit tests" are "written" by "juniors"; "instrumented tests" are "written" by "seniors".
 - *Correct Answer: (A)*
-

18. Your app's build time is very slow. What are the *first three* things you would investigate to fix it?

- (A) 1. "Buy" a "faster" "computer". 2. "Delete" "half" the "code". 3. "Re-install" "Android Studio".
 - (B) 1. "Enable" the "Gradle cache". 2. "Check" for "non-incremental" "annotation processors" (like `kapt`). 3. "Parallelize" the "build" (if using "modules").
 - (C) 1. "Update" "all" "dependencies" to the "latest" "version". 2. "Clean" the "project". 3. "Invalidate" "caches" and "restart".
 - (D) 1. "Change" "all" `vars` to `vals`. 2. "Delete" "all" "comments". 3. "Reduce" the "number" of "layouts".
 - (E) 1. "Move" "all" "code" to "one" "single" "file". 2. "Disable" `R8`. 3. "Disable" "View Binding".
 - *Correct Answer: (B)*
-

19. What is a "cold start" vs. a "warm start" for an app?

- (A) A "cold start" is "launching" the "app" for the "first" "time" "ever"; a "warm start" is "launching" it "later".
- (B) A "cold start" is "launching" the "app" when its "process" is "not" in "memory"; a "warm start" is "launching" it when it "is" "already" in "memory".
- (C) A "cold start" is "launching" the "app" in the "winter"; a "warm start" is "launching" it in the "summer".

- (D) A "cold start" is "slow"; a "warm start" is "also" "slow".
 - (E) A "cold start" is "from" a "notification"; a "warm start" is "from" the "launcher".
 - *Correct Answer: (B)*
-

20. A junior dev asks why the app needs both **Hilt and **Retrofit**. What is the strategic distinction?**

- (A) **Retrofit** makes the network call; **Hilt** provides the configured **Retrofit** "instance" (the "how") to the "class" that "needs" "it" (the "who").
- (B) **Retrofit** is for "downloading" "data"; **Hilt** is for "uploading" "data".
- (C) **Retrod** is for "REST" "APIs"; **Hilt** is for "SOAP" "APIs".
- (D) **Retrofit** is "old"; **Hilt** is "new".
- (E) **Retrofit** is for "Kotlin"; **Hilt** is for "Java".
- *Correct Answer: (A)*

DevOps Engineering track

Level 1: Basic Foundations

1. What is the primary goal of "DevOps"?

- (A) A set of tools for deploying applications.
 - (B) A cultural shift that combines software development (Dev) and IT operations (Ops) to shorten the development life cycle.
 - (C) A new programming language for cloud servers.
 - (D) A job title for a person who only manages servers.
 - (E) A type of software for monitoring website traffic.
 - *Correct Answer: (B)*
-

2. What does "CI/CD" stand for?

- (A) Continuous Integration and Continuous Delivery/Deployment.
 - (B) Cloud Infrastructure and Cloud Deployment.
 - (C) Code Integration and Code Delivery.
 - (D) Continuous Infrastructure and Continuous Development.
 - (E) Cloud Integration and Cloud Delivery.
 - *Correct Answer: (A)*
-

3. What is "Git"?

- (A) A popular "distributed version control system" (VCS) for tracking changes in source code.
 - (B) A tool for building and compiling code into a runnable application.
 - (C) A server for hosting live websites.
 - (D) A "Graphical Interface Tool" for designing UIs.
 - (E) A type of database for storing code.
 - *Correct Answer: (A)*
-

4. What is a "server"?

- (A) A computer program that browses the internet (this is a client).
 - (B) A computer (hardware or software) that provides services or resources to other computers.
 - (C) A "Source Code Management" tool.
 - (D) A file that contains all the application's code.
 - (E) A type of network firewall.
 - *Correct Answer: (B)*
-

5. What is a "container" (e.g., Docker)?

- (A) A type of virtual machine that includes a full guest operating system.
 - (B) A lightweight, standalone package of software that includes everything needed to run it.
 - (C) A physical box for shipping servers.
 - (D) A tool for managing cloud billing.
 - (E) A file that "contains" all your network passwords.
 - *Correct Answer: (B)*
-

6. What is "Infrastructure as a Service" (IaaS)?

- (A) A cloud model where you rent raw computing resources like VMs and storage.
 - (B) A cloud model where you use a fully managed application (like email).
 - (C) A cloud model where you rent a platform to build your app on.
 - (D) A service for writing infrastructure code.
 - (E) A service for monitoring your infrastructure.
 - *Correct Answer: (A)*
-

7. What is an "IP Address"?

- (A) A "permission" that "grants" a user "access" to a system.
 - (B) A "physical" "port" on a "server" for a "network" "cable".
 - (C) A "programming" "language" for "networking".
 - (D) A "unique" "address" that "identifies" a "device" on a "network".
 - (E) A "protocol" for "securing" "internet" "traffic".
 - *Correct Answer: (D)*
-

8. What is a "firewall"?

- (A) A "network security" "device" that "monitors" and "filters" "traffic" based on "rules".
 - (B) A "physical" "wall" in a "data center" to "prevent" "fires".
 - (C) A "software" "tool" for "debugging" "code".
 - (D) A "hardware" "device" for "cooling" a "server".
 - (E) A "type" of "malware" that "burns" "files".
 - *Correct Answer: (A)*
-

9. What is "YAML"?

- (A) A "programming" "language" used for "writing" "application" "logic".
 - (B) A "data-serialization" "format" often "used" for "configuration" "files".
 - (C) A "database" for "storing" "user" "data".
 - (D) A "version control" "system" "similar" to "Git".
 - (E) A "server" "monitoring" "tool".
 - *Correct Answer: (B)*
-

10. What is "SSH" (Secure Shell)?

- (A) A "protocol" for "securely" "accessing" and "managing" a "remote" "server".
 - (B) A "type" of "encryption" for "files" "at rest".
 - (C) A "firewall" "rule" for "blocking" "traffic".
 - (D) A "software" "library" for "building" "web" "applications".
 - (E) A "physical" "key" for "unlocking" a "server" "rack".
 - *Correct Answer: (A)*
-

11. What is a "build" in the context of CI/CD?

- (A) The "process" of "planning" "new" "features".
- (B) The "process" of "compiling" "source code" into a "runnable" "artifact".

- (C) The "process" of "deploying" an "application" to a "server".
 - (D) The "process" of "monitoring" a "live" "application".
 - (E) The "process" of "writing" "unit" "tests".
 - *Correct Answer: (B)*
-

12. What is an "artifact"?

- (A) The "source code" "written" by "developers".
 - (B) The "file" (e.g., a `.jar`, `.zip`, or container image) "produced" by a "build".
 - (C) A "log" "file" from a "server".
 - (D) An "old", "outdated" "server".
 - (E) A "security" "vulnerability".
 - *Correct Answer: (B)*
-

13. What is "AWS"?

- (A) "Amazon Web Services", a major "cloud computing" "provider".
 - (B) "A Web Server", a "type" of "software" for "hosting" "websites".
 - (C) "Advanced Web Security", a "protocol" for "encryption".
 - (D) "Automated Web Services", a "tool" for "testing".
 - (E) "A Windows Server", a "type" of "operating" "system".
 - *Correct Answer: (A)*
-

14. What is "Azure"?

- (A) "Microsoft's" "cloud computing" "platform".
 - (B) "Google's" "cloud computing" "platform".
 - (C) "A" "type" of "database" "management" "system".
 - (D) "A" "version control" "system".
 - (E) "A" "programming" "language" for "automation".
 - *Correct Answer: (A)*
-

15. What is "GCP"?

- (A) "General Cloud Principles", a "set" of "best" "practices".
- (B) "Google Cloud Platform", a major "cloud computing" "provider".
- (C) "Git Control Panel", a "tool" for "managing" "repositories".
- (D) "Global Compute Protocol", a "networking" "standard".

- (E) "Gateway Control Protocol", a "security" "tool".
 - *Correct Answer: (B)*
-

16. What is a "log file"?

- (A) A "file" that "contains" "application" "source code".
 - (B) A "file" that "records" "events", "errors", or "messages" "from" an "application" or "server".
 - (C) A "file" that "stores" "user" "passwords".
 - (D) A "file" that "defines" the "network" "configuration".
 - (E) A "file" that "contains" the "build" "artifact".
 - *Correct Answer: (B)*
-

17. What is "automation"?

- (A) The "process" of "manually" "configuring" a "server".
 - (B) The "practice" of "using" "scripts" and "tools" to "perform" "tasks" "without" "human" "intervention".
 - (C) A "type" of "car" that "drives" "itself".
 - (D) A "job" "title" for a "junior" "developer".
 - (E) The "process" of "writing" "documentation".
 - *Correct Answer: (B)*
-

18. What is "monitoring"?

- (A) The "process" of "collecting" and "analyzing" "data" to "check" the "health" and "performance" of a "system".
 - (B) The "process" of "deploying" a "new" "application".
 - (C) The "process" of "writing" "security" "rules".
 - (D) The "process" of "backing up" a "database".
 - (E) The "process" of "writing" "code".
 - *Correct Answer: (A)*
-

19. What is "source code"?

- (A) The "compiled" "application" "file" (the "artifact").
- (B) The "human-readable" "instructions" (e.g., in Kotlin, Python, Go) "written" by "developers".

- (C) The "log" "files" from a "server".
 - (D) The "configuration" "files" for a "pipeline".
 - (E) The "hardware" "source" for a "computer".
 - *Correct Answer: (B)*
-

20. What is "Docker"?

- (A) A "platform" for "building" and "running" "applications" in "containers".
 - (B) A "cloud" "provider" "like" "AWS" or "Azure".
 - (C) A "version control" "system" "like" "Git".
 - (D) A "monitoring" "tool" for "visualizing" "metrics".
 - (E) A "programming" "language" for "automation".
 - *Correct Answer: (A)*
-

Level 2: Intermediate Application

1. What is a **git commit** command used for?

- (A) To "download" "changes" from a "remote" "repository".
 - (B) To "upload" "local" "changes" to the "remote" "repository".
 - (C) To "create" a "new" "branch" in the "repository".
 - (D) To "save" a "snapshot" of the "current" "changes" to the "local" "repository".
 - (E) To "undo" the "last" "change" "made" to a "file".
 - *Correct Answer: (D)*
-

2. What is a **git pull** command used for?

- (A) To "fetch" "changes" from a "remote" "repository" and "merge" them "into" the "current" "branch".
 - (B) To "upload" "local" "commits" to the "remote" "repository".
 - (C) To "save" a "snapshot" of "changes" "locally".
 - (D) To "create" a "new" "branch".
 - (E) To "discard" "all" "local" "changes".
 - *Correct Answer: (A)*
-

3. What is a **git push** command used for?

- (A) To "fetch" "changes" from a "remote" "repository".
 - (B) To "upload" "local" "commits" to the "remote" "repository".
 - (C) To "save" a "snapshot" of "changes" "locally".
 - (D) To "create" a "new" "branch".
 - (E) To "discard" "all" "local" "changes".
 - *Correct Answer: (B)*
-

4. What is a "Dockerfile"?

- (A) A "text" "file" containing "instructions" to "build" a "Docker" "container" "image".
 - (B) A "running" "instance" of a "Docker" "image".
 - (C) A "server" for "storing" "Docker" "images".
 - (D) A "graphical" "tool" for "managing" "containers".
 - (E) A "script" for "automating" "Git" "commands".
 - *Correct Answer: (A)*
-

5. What is a "Docker Image"?

- (A) A "running" "instance" of a "container".
 - (B) A "read-only" "template" (a "snapshot") "that" "contains" the "application" and "its" "dependencies".
 - (C) A "file" that "defines" "how" to "build" an "image".
 - (D) A "virtual" "machine" that "runs" on "Docker".
 - (E) A "network" for "connecting" "containers".
 - *Correct Answer: (B)*
-

6. What is a "Docker Container"?

- (A) A "read-only" "template" for "an" "application".
 - (B) A "running" "instance" of a "Docker" "image".
 - (C) A "file" that "defines" "how" to "build" an "image".
 - (D) A "physical" "server" for "running" "Docker".
 - (E) A "storage" "volume" for "persisting" "data".
 - *Correct Answer: (B)*
-

7. What is "Ansible"?

- (A) An "automation" "tool" "used" for "configuration management" and "application" "deployment".
 - (B) A "container" "orchestration" "platform".
 - (C) A "monitoring" "and" "alerting" "system".
 - (D) A "version control" "system".
 - (E) A "cloud" "provider".
 - *Correct Answer: (A)*
-

8. What is "Jenkins"?

- (A) A "popular" "open-source" "automation" "server" "used" for "CI/CD" "pipelines".
 - (B) A "cloud" "provider" "specializing" in "storage".
 - (C) A "text" "editor" for "writing" "YAML" "files".
 - (D) A "container" "runtime" "similar" to "Docker".
 - (E) A "monitoring" "tool" for "visualizing" "metrics".
 - *Correct Answer: (A)*
-

9. What is a "pipeline" in CI/CD?

- (A) A "series" of "automated" "steps" (e.g., "build", "test", "deploy") "that" "code" "goes" "through".
 - (B) A "network" "connection" "between" the "developer" and the "server".
 - (C) A "physical" "server" "that" "runs" "the" "application".
 - (D) A "security" "rule" "that" "blocks" "bad" "traffic".
 - (E) A "file" "that" "contains" "all" "the" "source" "code".
 - *Correct Answer: (A)*
-

10. What is a "virtual machine" (VM)?

- (A) A "lightweight" "package" that "shares" the "host" "OS" "kernel".
 - (B) A "software" "emulation" of a "physical" "computer" "that" "runs" "its" "own" "OS".
 - (C) A "physical" "server" "in" a "data" "center".
 - (D) A "software" "application" for "browsing" the "internet".
 - (E) A "network" "routing" "device".
 - *Correct Answer: (B)*
-

11. What is the main difference between a "container" and a "VM"?

- (A) "Containers" "virtualize" the "OS"; "VMs" "virtualize" the "hardware".
 - (B) "Containers" "are" "heavy"; "VMs" "are" "lightweight".
 - (C) "Containers" "run" "their" "own" "full" "OS"; "VMs" "share" the "host" "kernel".
 - (D) "Containers" "are" "for" "development"; "VMs" "are" "for" "production".
 - (E) "Containers" "are" "less" "secure" "than" "VMs".
 - *Correct Answer: (A)*
-

12. What is **ssh-keygen** used for?

- (A) To "generate" a "new" "public/private" "key" "pair" for "SSH" "authentication".
 - (B) To "start" a "secure" "shell" "session" "with" a "remote" "server".
 - (C) To "configure" the "SSH" "server" "settings".
 - (D) To "encrypt" a "file" "using" the "SSH" "protocol".
 - (E) To "scan" the "network" for "open" "SSH" "ports".
 - *Correct Answer: (A)*
-

13. What is a "package manager" (e.g., **apt**, **yum**, **npm**, **pip**)?

- (A) A "tool" for "managing" "project" "timelines".
 - (B) A "tool" for "automating" the "installation", "update", and "removal" of "software" "packages".
 - (C) A "service" for "shipping" "physical" "packages".
 - (D) A "tool" for "packaging" "an" "application" "into" a "container".
 - (E) A "tool" for "managing" "network" "packets".
 - *Correct Answer: (B)*
-

14. What is **curl**?

- (A) A "command-line" "tool" for "transferring" "data" "with" "URLs", "often" "used" to "test" "APIs".
 - (B) A "graphical" "web" "browser".
 - (C) A "programming" "language" for "networking".
 - (D) A "text" "editor" for "configuration" "files".
 - (E) A "tool" for "encrypting" "network" "traffic".
 - *Correct Answer: (A)*
-

15. What is a **git merge** command used for?

- (A) To "delete" a "branch".
 - (B) To "combine" the "changes" "from" "one" "branch" "into" "another".
 - (C) To "rename" a "branch".
 - (D) To "upload" "changes" to the "remote" "repository".
 - (E) To "undo" a "commit".
 - *Correct Answer: (B)*
-

16. What is a `git branch` command used for?

- (A) To "list", "create", or "delete" "branches".
 - (B) To "switch" "between" "different" "branches".
 - (C) To "merge" "two" "branches" "together".
 - (D) To "upload" a "branch" to the "remote".
 - (E) To "delete" a "commit" "from" a "branch".
 - *Correct Answer: (A)*
-

17. What is "GitHub"?

- (A) A "command-line" "tool" for "managing" "Git" "repositories".
 - (B) A "web-based" "hosting" "service" for "Git" "repositories".
 - (C) A "type" of "server" "operating" "system".
 - (D) A "monitoring" "tool".
 - (E) A "cloud" "provider".
 - *Correct Answer: (B)*
-

18. What is a "pull request" (or "merge request")?

- (A) A "command" to "force" "download" "code".
 - (B) A "request" "to" "deploy" "code" "to" "production".
 - (C) A "request" "to" "merge" "your" "changes" (from "one" "branch") "into" "another", "allowing" for "code" "review".
 - (D) A "request" "for" "help" "from" a "senior" "developer".
 - (E) A "request" "to" "delete" a "repository".
 - *Correct Answer: (C)*
-

19. What is a "webhook"?

- (A) An "automated" "HTTP" "callback" "sent" from "one" "app" to "another" "when" an "event" "occurs".
 - (B) A "security" "tool" for "catching" "phishing" "attacks".
 - (C) A "command" "to" "deploy" an "application".
 - (D) A "type" of "network" "cable".
 - (E) A "graphical" "tool" for "managing" "Git".
 - *Correct Answer: (A)*
-

20. What is **chmod**?

- (A) A "command" to "change" the "mode" (file "permissions") of a "file" or "directory".
 - (B) A "command" to "change" the "owner" of a "file".
 - (C) A "command" to "create" a "new" "file".
 - (D) A "command" to "monitor" "changes" to a "file".
 - (E) A "command" to "compress" a "file".
 - *Correct Answer: (A)*
-

Level 3: Advanced Scenarios

1. What is the *main* difference between **git merge** and **git rebase**?

- (A) **merge** "combines" "histories"; **rebase** "deletes" the "history".
 - (B) **merge** "creates" a "new" "merge commit"; **rebase** "re-writes" "history" to "create" a "linear" "sequence".
 - (C) **merge** "is" for "pulling" "changes"; **rebase** "is" for "pushing" "changes".
 - (D) **merge** "is" "safe"; **rebase** "is" "dangerous" and "should" "never" "be" "used".
 - (E) **merge** "is" "done" "locally"; **rebase** "is" "done" on the "remote" "server".
 - *Correct Answer: (B)*
-

2. What is "Infrastructure as Code" (IaC)?

- (A) The "practice" of "managing" and "provisioning" "infrastructure" "using" "code" and "automation".
- (B) "Hardware" that "has" "code" "embedded" "in" "it".
- (C) "Source code" that "is" "stored" "on" the "infrastructure".
- (D) A "type" of "networking" "protocol".
- (E) A "monitoring" "tool" "that" "reads" "code".
- *Correct Answer: (A)*

3. What is "Terraform"?

- (A) A "CI/CD" "pipeline" "tool" "from" "HashiCorp".
- (B) An "Infrastructure as Code" (IaC) "tool" for "building" and "managing" "cloud" "resources".
- (C) A "container" "orchestration" "platform" "similar" to "Kubernetes".
- (D) A "monitoring" "and" "observability" "platform".
- (E) A "service mesh" "tool".
- *Correct Answer: (B)*

4. What is the purpose of a "Terraform state file"?

- (A) To "store" the "source code" for "your" "infrastructure".
- (B) To "store" "secret" "keys" and "passwords".
- (C) To "keep" a "record" of the "current" "state" of "your" "managed" "infrastructure".
- (D) To "define" the "variables" "used" in "your" "Terraform" "code".
- (E) To "log" "all" the "changes" "made" "by" "Terraform".
- *Correct Answer: (C)*

5. What is a "Terraform provider"?

- (A) A "plugin" "that" "allows" "Terraform" "to" "interact" "with" a "specific" "API" (e.g., "AWS", "Azure").
- (B) The "company" "that" "provides" "your" "cloud" "services".
- (C) A "file" "that" "defines" "your" "infrastructure".
- (D) A "server" "that" "runs" "your" "Terraform" "code".
- (E) A "user" "who" "is" "allowed" "to" "run" "Terraform".
- *Correct Answer: (A)*

6. What is a "reverse proxy" (e.g., Nginx)?

- (A) A "proxy" "that" "hides" the "client's" "IP" "address".
- (B) A "server" "that" "sits" "in" "front" of "web" "servers" "and" "forwards" "client" "requests".
- (C) A "database" "that" "stores" "proxy" "settings".
- (D) A "security" "tool" "that" "runs" "attacks" "in" "reverse".
- (E) A "DNS" "server".
- *Correct Answer: (B)*

7. What is a "load balancer"?

- (A) A "server" "that" "distributes" "incoming" "network" "traffic" "across" "multiple" "servers".
- (B) A "tool" "that" "balances" the "load" (CPU/memory) "on" a "single" "server".
- (C) A "database" "that" "balances" "data" "storage".
- (D) A "security" "tool" "that" "blocks" "high-load" "traffic".
- (E) A "build" "tool" "that" "balances" "pipeline" "jobs".
- *Correct Answer: (A)*

8. What is the main difference between a "load balancer" and a "reverse proxy"?

- (A) "They" "are" "the" "same" "thing".
- (B) A "load balancer" "distributes" "traffic"; a "reverse proxy" "can" "do" "that" *and* "other" "tasks" (e.g., "caching", "SSL").
- (C) A "load balancer" "is" "for" "internal" "traffic"; a "reverse proxy" "is" "for" "external" "traffic".
- (D) A "load balancer" "is" "hardware"; a "reverse proxy" "is" "software".
- (E) A "load balancer" "is" "for" "VMs"; a "reverse proxy" "is" "for" "containers".
- *Correct Answer: (B)*

9. What is "Prometheus"?

- (A) An "open-source" "monitoring" and "alerting" "system" that "pulls" "metrics" "from" "services".
- (B) An "open-source" "dashboard" "tool" for "visualizing" "data".
- (C) An "open-source" "container" "orchestrator".
- (D) An "open-source" "CI/CD" "pipeline" "tool".
- (E) An "open-source" "Infrastructure as Code" "tool".
- *Correct Answer: (A)*

10. What is "Grafana"?

- (A) An "open-source" "monitoring" "system" "that" "collects" "metrics".
- (B) An "open-source" "dashboard" and "visualization" "tool", "often" "used" "with" "Prometheus".
- (C) An "open-source" "database" for "storing" "logs".
- (D) An "open-source" "security" "scanner".

- (E) An "open-source" "build" "automation" "tool".
 - *Correct Answer: (B)*
-

11. What is a "GitHub Actions workflow"?

- (A) A "YAML" "file" "in" a `.github/workflows` "directory" "that" "defines" "an" "automated" "CI/CD" "process".
 - (B) A "graphical" "tool" "in" "GitHub" "for" "visualizing" "code".
 - (C) A "text" "file" "in" your "code" "that" "defines" "permissions".
 - (D) A "script" "for" "automating" "pull" "requests".
 - (E) A "security" "policy" "for" a "repository".
 - *Correct Answer: (A)*
-

12. What is "configuration management" (e.g., Ansible, Puppet, Chef)?

- (A) The "practice" of "managing" "application" "code".
 - (B) The "practice" of "automating" the "configuration", "patching", and "management" of "servers".
 - (C) The "practice" of "managing" "network" "firewall" "rules".
 - (D) The "practice" of "managing" "billing" and "costs".
 - (E) The "practice" of "managing" "container" "images".
 - *Correct Answer: (B)*
-

13. What is the difference between "mutable" and "immutable" infrastructure?

- (A) "Mutable" "servers" "are" "changed" "in-place"; "immutable" "servers" "are" "replaced" "with" "new" "ones".
 - (B) "Mutable" "servers" "are" "physical"; "immutable" "servers" "are" "virtual".
 - (C) "Mutable" "servers" "are" "fast"; "immutable" "servers" "are" "slow".
 - (D) "Mutable" "servers" "are" "for" "development"; "immutable" "servers" "are" "for" "production".
 - (E) "Mutable" "servers" "are" "insecure"; "immutable" "servers" "are" "secure".
 - *Correct Answer: (A)*
-

14. What is "GitFlow"?

- (A) A "branching" "model" "that" "uses" "long-lived" `develop` "and" `main` "branches" "with" "feature" "branches".

- (B) A "CI/CD" "tool" "for" "Git".
 - (C) A "network" "protocol" "for" "Git".
 - (D) A "graphical" "client" "for" "Git".
 - (E) A "command" "to" "visualize" the "flow" of "commits".
 - *Correct Answer: (A)*
-

15. What is "Trunk-Based Development"?

- (A) A "branching" "model" "where" "all" "developers" "commit" "directly" "to" the "main" "branch" ("trunk").
 - (B) A "branching" "model" "with" "many" "long-lived" "feature" "branches".
 - (C) A "development" "style" "where" "no" "version" "control" "is" "used".
 - (D) A "way" "to" "develop" "on" "a" "single", "shared" "server".
 - (E) A "CI/CD" "pipeline" "strategy".
 - *Correct Answer: (A)*
-

16. What is "observability"?

- (A) The "same" "as" "monitoring" (just "collecting" "metrics" and "logs").
 - (B) A "superset" of "monitoring" "that" "includes" "metrics", "logs", and "traces" "to" "understand" a "system's" "state".
 - (C) A "security" "practice" of "observing" "hackers".
 - (D) A "development" "practice" of "observing" "users".
 - (E) A "tool" "for" "visualizing" "data".
 - *Correct Answer: (B)*
-

17. What is "distributed tracing"?

- (A) "Tracing" a "request" "as" "it" "moves" "through" "a" "single" "application".
 - (B) "Tracing" a "request" "as" "it" "moves" "across" "multiple" "microservices".
 - (C) "Tracing" "network" "packets" "on" a "server".
 - (D) "Tracing" "file" "changes" "in" a "Git" "repository".
 - (E) "Tracing" "user" "clicks" "on" a "website".
 - *Correct Answer: (B)*
-

18. What is a "service mesh" (e.g., Istio, Linkerd)?

- (A) A "dedicated" "infrastructure" "layer" "for" "managing" "service-to-service" "communication" in "microservices".
 - (B) A "physical" "network" "mesh" "in" a "data" "center".
 - (C) A "tool" "for" "monitoring" "network" "traffic".
 - (D) A "type" of "firewall".
 - (E) A "service" "for" "building" "virtual" "machines".
 - *Correct Answer: (A)*
-

19. What is "blue-green deployment"?

- (A) A "strategy" "where" "you" "deploy" a "new" "version" to a "small" "subset" of "users".
 - (B) A "strategy" "where" "you" "have" "two" "identical" "environments" ("blue"/"green") "and" "switch" "traffic" "over".
 - (C) A "strategy" "that" "uses" "blue" and "green" "servers".
 - (D) A "strategy" "for" "slowly" "rolling" "out" a "new" "feature" "to" "users".
 - (E) A "strategy" "for" "database" "backups".
 - *Correct Answer: (B)*
-

20. What is "canary release"?

- (A) A "strategy" "where" "you" "deploy" a "new" "version" "to" a "small" "subset" of "users" "and" "monitor" "it".
 - (B) A "strategy" "where" "you" "have" "two" "identical" "production" "environments".
 - (C) A "strategy" "where" "you" "warn" "users" (like a "canary") "before" "deploying".
 - (D) A "strategy" "where" "you" "deploy" "two" "versions" (A and B) "at" "the" "same" "time".
 - (E) A "strategy" "for" "database" "disaster" "recovery".
 - *Correct Answer: (A)*
-

Level 4: Expert & Specialization

1. What is "Kubernetes" (K8s)?

- (A) A "container" "runtime" for "running" "a" "single" "container".
- (B) An "open-source" "platform" for "automating" "the" "deployment", "scaling", and "management" of "containers".
- (C) A "cloud" "provider" "from" "Google".
- (D) An "Infrastructure as Code" "tool".
- (E) A "monitoring" "and" "logging" "system".

- *Correct Answer: (B)*
-

2. What is a "Pod" in Kubernetes?

- (A) The "smallest" "deployable" "unit", "which" "can" "hold" "one" "or" "more" "containers".
 - (B) A "physical" "server" (a "node") "in" the "cluster".
 - (C) A "set" of "rules" for "networking".
 - (D) A "load" "balancer" "for" "distributing" "traffic".
 - (E) A "storage" "unit" "for" "persisting" "data".
 - *Correct Answer: (A)*
-

3. What is a "Deployment" in Kubernetes?

- (A) A "resource" "that" "provides" "declarative" "updates" "for" "Pods" "and" "ReplicaSets".
 - (B) A "physical" "server" (a "node").
 - (C) A "network" "service" "for" "exposing" "a" "Pod".
 - (D) A "storage" "volume".
 - (E) A "configuration" "file" "for" "storing" "secrets".
 - *Correct Answer: (A)*
-

4. What is a "Service" in Kubernetes?

- (A) A "running" "instance" of "an" "application" (a "Pod").
 - (B) An "abstract" "way" "to" "expose" "an" "application" (a "set" of "Pods") "as" a "network" "service".
 - (C) A "worker" "machine" "in" the "cluster".
 - (D) A "storage" "unit".
 - (E) A "CI/CD" "pipeline".
 - *Correct Answer: (B)*
-

5. What is an "Ingress" in Kubernetes?

- (A) An "API" "object" "that" "manages" "external" "access" (HTTP/HTTPS) "to" "services" "in" a "cluster".
- (B) A "security" "rule" "that" "blocks" "all" "incoming" "traffic".
- (C) A "container" "that" "runs" "inside" a "Pod".

- (D) A "network" "connection" "between" "two" "Pods".
 - (E) A "log" "file" "for" "all" "incoming" "requests".
 - *Correct Answer: (A)*
-

6. What is a **ConfigMap**?

- (A) A "Kubernetes" "object" "for" "storing" "non-confidential" "configuration" "data" "in" "key-value" "pairs".
 - (B) A "Kubernetes" "object" "for" "storing" "sensitive" "data" (like "passwords").
 - (C) A "map" of "the" "network" "configuration".
 - (D) A "map" of "all" "the" "nodes" "in" "a" "cluster".
 - (E) A "Terraform" "file".
 - *Correct Answer: (A)*
-

7. What is a "Secret"?

- (A) A "Kubernetes" "object" "for" "storing" "non-confidential" "data".
 - (B) A "Kubernetes" "object" "for" "storing" "sensitive" "data" (like "passwords" or "API" "keys").
 - (C) A "secret" "project" "that" "is" "not" "public".
 - (D) A "file" "used" "by" "SSH".
 - (E) A "log" "file" "that" "is" "encrypted".
 - *Correct Answer: (B)*
-

8. What is "Helm"?

- (A) A "package manager" "for" "Kubernetes", "used" to "manage" "and" "deploy" "applications".
 - (B) A "security" "scanner" "for" "Kubernetes".
 - (C) A "monitoring" "tool" "for" "Kubernetes".
 - (D) A "container" "runtime".
 - (E) A "cloud-native" "database".
 - *Correct Answer: (A)*
-

9. What is a **PersistentVolume (PV)**?

- (A) A "piece" of "storage" "in" the "cluster" "that" "survives" "even" "if" "a" "Pod" "is" "destroyed".

- (B) A "type" of "Pod" "that" "is" "persistent".
 - (C) A "network" "connection" "that" "is" "persistent".
 - (D) A "configuration" "file" "that" "is" "persistent".
 - (E) A "log" "file" "that" "is" "never" "deleted".
 - *Correct Answer: (A)*
-

10. What is "Terraform remote state"?

- (A) The "practice" of "storing" the "Terraform" "state" "file" "in" a "remote", "shared" "location" (like "S3").
 - (B) The "practice" of "running" "Terraform" "on" a "remote" "server".
 - (C) The "state" of a "remote" "server" "managed" "by" "Terraform".
 - (D) A "Terraform" "file" "that" "is" "not" "stored" "locally".
 - (E) A "security" "vulnerability" "in" "Terraform".
 - *Correct Answer: (A)*
-

11. What is a "Terraform module"?

- (A) A "container" for "a" "set" of "reusable" "Terraform" "resources".
 - (B) A "plugin" "for" "a" "specific" "cloud" "provider".
 - (C) The "Terraform" "state" "file".
 - (D) A "command" "to" "run" "Terraform".
 - (E) A "variable" "in" "a" "Terraform" "file".
 - *Correct Answer: (A)*
-

12. What is "GitLab"?

- (A) A "command-line" "tool" "for" "Git".
 - (B) A "complete" "DevOps" "platform" (SCM, "CI/CD", "registry") "delivered" "as" a "single" "application".
 - (C) A "monitoring" "tool" "for" "Git".
 - (D) A "competitor" "to" "GitHub" "that" "is" "only" "for" "lab" "projects".
 - (E) A "text" "editor" "for" "Git".
 - *Correct Answer: (B)*
-

13. What is "DevSecOps"?

- (A) The "practice" of "integrating" "security" "testing" "and" "practices" "into" "every" "stage" of the "DevOps" "pipeline".
 - (B) A "team" of "developers" "who" "are" "also" "security" "experts".
 - (C) A "set" of "tools" "for" "securing" "DevOps" "pipelines".
 - (D) The "practice" of "separating" "security" "from" "the" "DevOps" "process".
 - (E) A "job" "title" "for" a "security" "manager".
 - *Correct Answer: (A)*
-

14. What is a "linter"?

- (A) A "tool" "that" "analyzes" "source code" "to" "flag" "programming" "errors", "bugs", and "stylistic" "errors".
 - (B) A "tool" "that" "compiles" "code" "into" an "artifact".
 - (C) A "tool" "that" "deploys" "an" "application".
 - (D) A "tool" "that" "scans" "for" "security" "vulnerabilities".
 - (E) A "tool" "for" "removing" "lint" "from" "a" "server".
 - *Correct Answer: (A)*
-

15. What is "static code analysis" (SAST)?

- (A) "Analyzing" "an" "application" "by" "running" "it" "and" "observing" "its" "behavior".
 - (B) "Analyzing" "an" "application's" "source code" "without" "actually" "running" "it".
 - (C) "Analyzing" "only" "code" "that" "is" "static" (never "changes").
 - (D) "Analyzing" "the" "network" "traffic" "of" "an" "application".
 - (E) "Analyzing" "the" "logs" "from" "an" "application".
 - *Correct Answer: (B)*
-

16. What is "dynamic code analysis" (DAST)?

- (A) "Analyzing" "an" "application" "by" "running" "it" "and" "testing" "it" "for" "vulnerabilities".
 - (B) "Analyzing" "an" "application's" "source code" "without" "running" "it".
 - (C) "Analyzing" "code" "that" "is" "dynamic" (changes "often").
 - (D) "Analyzing" "the" "dependencies" "of" "an" "application".
 - (E) "Analyzing" "the" "build" "script" "of" "an" "application".
 - *Correct Answer: (A)*
-

17. What is **kubect1**?

- (A) The "main" "control" "server" (control "plane") "in" a "Kubernetes" "cluster".
- (B) A "command-line" "tool" "for" "interacting" "with" "a" "Kubernetes" "cluster".
- (C) A "graphical" "dashboard" "for" "Kubernetes".
- (D) A "security" "scanner" "for" "Kubernetes".
- (E) A "package" "manager" "for" "Kubernetes".
- *Correct Answer: (B)*

18. What is a "sidecar" container?

- (A) A "container" "that" "runs" *before* "the" "main" "application" "container".
 - (B) A "container" "that" "runs" *alongside* "the" "main" "application" "container" "in" "a" "Pod" (e.g., "for" "logging").
 - (C) A "container" "that" "is" "not" "as" "important" "as" "the" "main" "container".
 - (D) A "backup" "container" "that" "runs" "if" "the" "main" "one" "fails".
 - (E) A "container" "that" "runs" "on" a "different" "node".
 - *Correct Answer: (B)*
-

19. What is "chaos engineering"?

- (A) The "practice" of "intentionally" "introducing" "failures" "into" a "system" "to" "test" "its" "resilience".
 - (B) A "development" "practice" "with" "no" "rules" "or" "planning".
 - (C) A "security" "attack" "that" "causes" "chaos".
 - (D) A "monitoring" "tool" "that" "finds" "chaos" "in" "logs".
 - (E) A "management" "style" "that" "is" "chaotic".
 - *Correct Answer: (A)*
-

20. What is "GitOps"?

- (A) A "practice" "of" "managing" "IT" "operations" "using" "only" "Git" "commands".
 - (B) A "practice" "of" "using" "a" "Git" "repository" "as" the "single" "source" of "truth" "for" "declarative" "infrastructure" "and" "applications".
 - (C) A "new" "version" of "Git".
 - (D) A "CI/CD" "tool" "from" "GitHub".
 - (E) A "security" "policy" "for" "Git".
 - *Correct Answer: (B)*
-

Level 5: Strategic & Architectural

1. A large team is experiencing frequent, painful `git merge` conflicts and slow releases. What is the *most likely* architectural cause?

- (A) The "team" "is" "using" "`git rebase`" "instead" of "`git merge`".
 - (B) The "team" "is" "using" "long-lived" "feature" "branches" "instead" of "Trunk-Based" "Development".
 - (C) The "team" "is" "not" "using" "enough" "comments" "in" "their" "code".
 - (D) The "team's" "CI" "pipeline" "is" "too" "fast".
 - (E) The "team" "is" "using" "GitHub" "instead" of "GitLab".
 - *Correct Answer: (B)*
-

2. A startup is deploying its first app. What is the key *strategic trade-off* of using Serverless (e.g., Lambda) vs. Kubernetes?

- (A) "Serverless" "is" "more" "expensive" "for" "all" "workloads"; "Kubernetes" "is" "always" "cheaper".
 - (B) "Serverless" "has" "low" "overhead" "but" "vendor" "lock-in"; "Kubernetes" "is" "portable" "but" "has" "high" "operational" "complexity".
 - (C) "Serverless" "is" "for" "monoliths"; "Kubernetes" "is" "for" "microservices".
 - (D) "Serverless" "is" "insecure"; "Kubernetes" "is" "secure".
 - (E) "Serverless" "is" "for" "databases"; "Kubernetes" "is" "for" "web" "servers".
 - *Correct Answer: (B)*
-

3. Why would an organization choose to use "Terraform" *instead of* a provider-native IaC tool (e.g., AWS CloudFormation)?

- (A) "Terraform" "is" "the" "only" "tool" "that" "can" "manage" "AWS" "resources".
 - (B) "Terraform" "is" "cloud-agnostic", "allowing" "them" "to" "manage" "a" "multi-cloud" "environment" "with" "one" "tool".
 - (C) "Terraform" "is" "less" "complex" "and" "does" "not" "use" "state" "files".
 - (D) "Terraform" "is" "graphical"; "CloudFormation" "is" "code-only".
 - (E) "Terraform" "is" "significantly" "faster" "at" "provisioning" "resources".
 - *Correct Answer: (B)*
-

4. A service is experiencing high latency. You check its metrics: CPU is low, memory is low, but network I/O is very high. What is the *most likely* bottleneck?

- (A) The "service" "is" "CPU-bound" "and" "needs" "a" "faster" "processor".
- (B) The "service" "is" "memory-bound" "and" "is" "leaking" "memory".

- (C) The "service" "is" "waiting" "on" "a" "slow" "downstream" "dependency", "like" "a" "database" "or" "API".
 - (D) The "service's" "local" "disk" "is" "full".
 - (E) The "service" "needs" "to" "be" "rewritten" "in" "a" "different" "language".
 - *Correct Answer: (C)*
-

5. What is the primary *architectural* benefit of using a "service mesh" in a large microservices deployment?

- (A) It "combines" "all" "microservices" "into" "a" "single" "monolith" "for" "easier" "deployment".
 - (B) It "offloads" "complex" "network" "logic" (e.g., "retries", "mTLS", "tracing") "from" the "application" "code" "to" the "infrastructure".
 - (C) It "replaces" "the" "need" "for" "Kubernetes".
 - (D) It "acts" "as" "a" "database" "for" "all" "microservices".
 - (E) It "is" "a" "CI/CD" "tool" "for" "deploying" "microservices".
 - *Correct Answer: (B)*
-

6. A team wants to store their Terraform state file in a Git repository. Why is this a *bad* architectural practice?

- (A) "Git" "cannot" "handle" "text" "files" "like" "the" "state" "file".
 - (B) The "state" "file" "often" "contains" "sensitive" "secrets" "in" "plain" "text".
 - (C) The "state" "file" "will" "corrupt" "the" "Git" "repository".
 - (D) "Terraform" "is" "not" "compatible" "with" "Git".
 - (E) This "is" "a" "good" "practice" "for" "easy" "versioning".
 - *Correct Answer: (B)*
-

7. An application fails intermittently. What is the strategic difference between "monitoring" and "observability" in debugging this?

- (A) "Monitoring" "tells" "you" *if* "it's" "broken"; "observability" (with "traces") "helps" "you" "ask" *why* "it's" "broken".
 - (B) "Monitoring" "is" "for" "logs"; "observability" "is" "for" "metrics".
 - (C) "Monitoring" "is" "for" "production"; "observability" "is" "for" "development".
 - (D) "Monitoring" "is" "old"; "observability" "is" "new".
 - (E) "Monitoring" "is" "reactive"; "observability" "is" "also" "reactive".
 - *Correct Answer: (A)*
-

8. What is the strategic trade-off of using a "push" (e.g., StatsD) vs. a "pull" (e.g., Prometheus) model for metrics?

- (A) "Push" "is" "simpler" "for" "firewalls" "and" "ephemeral" "jobs"; "Pull" "is" "simpler" "for" "service" "discovery" "and" "health" "checking".
 - (B) "Push" "is" "always" "more" "reliable" "than" "Pull".
 - (C) "Push" "is" "for" "Kubernetes"; "Pull" "is" "for" "VMs".
 - (D) "Push" "is" "for" "metrics"; "Pull" "is" "for" "logs".
 - (E) "Push" "is" "old"; "Pull" "is" "new" "and" "better".
 - *Correct Answer: (A)*
-

9. A **Deployment** in Kubernetes is failing to update. The new Pods are stuck in a **CrashLoopBackOff** state. What is the *first* command you should run?

- (A) `kubectl delete deployment ...` (to "delete" "it" "and" "start" "over").
 - (B) `kubectl apply -f ...` (to "re-apply" "the" "configuration").
 - (C) `kubectl describe pod ...` (to "check" "for" "events", "like" "image" "pull" "errors").
 - (D) `kubectl logs ...` (to "check" "the" "application's" "logs" "for" "a" "crash").
 - (E) Both (C) and (D) are "essential" "first" "steps".
 - *Correct Answer: (E)*
-

10. What is the "Immutable Infrastructure" pattern, and what is its *main security benefit*?

- (A) "Servers" "are" "never" "changed"; "this" "is" "bad" "for" "security" "as" "they" "get" "old".
 - (B) "Servers" "are" "replaced" "with" "new", "patched" "images"; "this" "eliminates" "configuration" "drift" "and" "makes" "auditing" "easier".
 - (C) "Infrastructure" "is" "managed" "by" "code", "which" "is" "insecure".
 - (D) "Infrastructure" "is" "managed" "manually", "which" "is" "more" "secure".
 - (E) "Infrastructure" "is" "stored" "on" "immutable" "hard" "drives".
 - *Correct Answer: (B)*
-

11. Why is it a bad practice to use **latest** as your Docker image tag in production?

- (A) The **latest** "tag" "is" "slower" "to" "pull" "than" "a" "versioned" "tag".
- (B) The **latest** "tag" "is" "mutable"; "it's" "not" "deterministic" "and" "can" "cause" "unexpected" "rollbacks" "or" "updates".

- (C) The **latest** "tag" "is" "less" "secure" "by" "default".
 - (D) The **latest** "tag" "is" "not" "supported" "by" "Kubernetes".
 - (E) The **latest** "tag" "takes" "up" "more" "storage" "space".
 - *Correct Answer: (B)*
-

12. An organization is choosing between "GitFlow" and "Trunk-Based Development". What is the key *cultural* prerequisite for "Trunk-Based"?

- (A) A "large", "distributed" "team" "that" "needs" "many" "long-lived" "branches".
 - (B) A "mature", "high-trust" "culture" "with" "a" "very" "fast", "reliable" "CI/CD" "pipeline" "and" "strong" "tests".
 - (C) A "team" "that" "prefers" "to" "do" "large", "quarterly" "releases".
 - (D) A "team" "that" "does" "not" "like" "to" "use" "automated" "testing".
 - (E) A "team" "that" "does" "not" "use" "pull" "requests".
 - *Correct Answer: (B)*
-

13. A Kubernetes **Pod cannot start due to an **ImagePullBackOff** error. What is the *most likely* cause?**

- (A) The "Pod" "does" "not" "have" "enough" "CPU" "or" "memory".
 - (B) The "cluster" "cannot" "find" or "authenticate" "with" "the" "container" "registry".
 - (C) The "application" "inside" "the" "container" "is" "crashing".
 - (D) The "network" "firewall" "is" "blocking" "the" "Pod".
 - (E) The "PersistentVolume" "is" "not" "available".
 - *Correct Answer: (B)*
-

14. What is the *architectural trade-off* of using a "monolith" vs. "microservices"?

- (A) "Monoliths" "are" "simple" "to" "deploy" "but" "hard" "to" "scale"; "microservices" "are" "easy" "to" "scale" "but" "complex" "to" "deploy".
 - (B) "Monoliths" "are" "for" "new" "projects"; "microservices" "are" "for" "old" "projects".
 - (C) "Monoliths" "are" "always" "fast"; "microservices" "are" "always" "slow" "due" "to" "network" "calls".
 - (D) "Monoliths" "are" "for" "Kubernetes"; "microservices" "are" "for" "VMs".
 - (E) "Monoliths" "are" "insecure"; "microservices" "are" "secure".
 - *Correct Answer: (A)*
-

15. What is "RTO" (Recovery Time Objective) and "RPO" (Recovery Point Objective)?

- (A) "RTO" "is" "how" "much" "data" "you" "can" "lose"; "RPO" "is" "how" "fast" "you" "can" "recover".
 - (B) "RTO" "is" "how" "fast" "you" "must" "recover"; "RPO" "is" "how" "much" "data" "you" "can" "lose".
 - (C) "RTO" "is" "the" "cost" "of" "recovery"; "RPO" "is" "the" "point" "of" "failure".
 - (D) "RTO" "is" "the" "test" "time"; "RPO" "is" "the" "test" "point".
 - (E) "They" "are" "both" "metrics" "for" "CI/CD" "pipeline" "speed".
 - *Correct Answer: (B)*
-

16. What is the primary purpose of a "Disaster Recovery" (DR) plan?

- (A) To "prevent" "a" "disaster" "from" "ever" "happening".
 - (B) To "define" "the" "process" "of" "restoring" "critical" "services" "after" "a" "major" "failure".
 - (C) To "define" "the" "security" "rules" "to" "stop" "hackers".
 - (D) To "define" "the" "CI/CD" "pipeline" "for" "deployments".
 - (E) To "define" "the" "monitoring" "and" "alerting" "rules".
 - *Correct Answer: (B)*
-

17. Why is "cost management" (FinOps) a critical DevOps concern?

- (A) "DevOps" "is" "responsible" "for" "paying" "the" "cloud" "bill".
 - (B) "The" "automation" "and" "scalability" "of" "DevOps" "can" "lead" "to" "massive", "unexpected" "costs" "if" "not" "governed".
 - (C) "FinOps" "is" "a" "tool" "for" "running" "CI/CD" "pipelines".
 - (D) "It" "is" "not" "a" "DevOps" "concern"; "it" "is" "a" "finance" "concern" "only".
 - (E) "DevOps" "tools" "are" "the" "most" "expensive" "part" "of" "a" "Two" "cloud" "bill".
 - *Correct Answer: (B)*
-

18. What is "Site Reliability Engineering" (SRE)?

- (A) A "practice" (originating "at" "Google") "that" "applies" "software" "engineering" "principles" "to" "infrastructure" "and" "operations".
 - (B) A "different" "name" "for" "the" "traditional" "IT" "SysAdmin" "role".
 - (C) A "job" "focused" "only" "on" "the" "reliability" "of" "the" "company" "website".
 - (D) A "job" "focused" "only" "on" "CI/CD" "pipelines".
 - (E) A "job" "focused" "only" "on" "security".
 - *Correct Answer: (A)*
-

19. What is an "SLO" (Service Level Objective)?

- (A) A "legal" "contract" "with" "a" "customer" (this "is" "an" "SLA").
 - (B) A "specific", "measurable" "target" (e.g., "99.9%", "uptime") "for" "a" "service's" "reliability".
 - (C) A "general" "goal" "like" "make" "the" "website" "fast".
 - (D) A "metric" "that" "shows" "the" "current" "performance" (this "is" "an" "SLI").
 - (E) A "log" "file" "for" a "service".
 - *Correct Answer: (B)*
-

20. A Pod is in a Pending state. The cluster has capacity. What is the *most likely* cause?

- (A) The "application" "inside" "the" "container" "is" "crashing".
- (B) The "image" "cannot" "be" "pulled".
- (C) The "Pod" "is" "waiting" "for" "a" PersistentVolume "that" "is" "not" "available".
- (D) The "network" "firewall" "is" "blocking" "the" "Pod".
- (E) The Service "is" "not" "configured" "correctly".
- *Correct Answer: (C)*

QA Automation Engineer / SET (Software Engineer in Test) track.

Level 1: Basic Foundations

1. What is the primary purpose of Software Quality Assurance (QA)?

- (A) To write the source code for new features and applications.
 - (B) To design the user interface and user experience of an application.
 - (C) To identify defects (bugs) and ensure the software meets requirements.
 - (D) To deploy the application to production servers for users to access.
 - (E) To manage the project schedule and budget.
 - *Correct Answer: (C)*
-

2. What is a "bug" or "defect" in software?

- (A) An error, flaw, or fault in a computer program that causes it to produce an incorrect result.
- (B) A new feature request that has not been implemented yet.
- (C) A piece of hardware that is not working correctly.
- (D) A log file that is generated by the application.
- (E) A task for the developer to complete.
- *Correct Answer: (A)*

3. What is "Black-Box Testing"?

- (A) Testing the internal code structure and logic of the software.
- (B) Testing the software without any knowledge of its internal code or structure.
- (C) A security test to see if the software can be "hacked."
- (D) A performance test to see if the software can handle a heavy load.
- (E) Testing the software after all the code has been "locked" (boxed).
- *Correct Answer: (B)*

4. What is "White-Box Testing"?

- (A) Testing the software without any knowledge of its internal code.
- (B) Testing the software's functionality from a user's perspective.
- (C) Testing the internal code structure, logic, and paths of the software.
- (D) A test to ensure the user interface is visually "white" or clean.
- (E) Testing the software in a secure, isolated "white box."
- *Correct Answer: (C)*

5. What is a "test case"?

- (A) A document that describes the overall strategy for testing an application.
- (B) A set of inputs, execution steps, and expected results to verify a specific feature.
- (C) A "case" or "container" for storing all test data.
- (D) A single line of code in an automation script.
- (E) A report that describes a bug found in the software.
- *Correct Answer: (B)*

6. What is "Manual Testing"?

- (A) The process of using a software script to automatically test an application.
 - (B) The process of a human tester interacting with an application to find bugs.
 - (C) The process of writing a "user manual" for the software.
 - (D) The process of manually building and compiling the code.
 - (E) The process of manually configuring the test server.
 - *Correct Answer: (B)*
-

7. What is "Automation Testing"?

- (A) The process of using a human tester to find bugs.
 - (B) The process of writing a "user manual" for the software.
 - (C) The process of using software (scripts) to run tests and verify results.
 - (D) The process of manually writing a test plan.
 - (E) The process of designing the user interface.
 - *Correct Answer: (C)*
-

8. What is a "bug report"?

- (A) A document that lists all the new features to be built.
 - (B) A formal document that describes a defect, including steps to reproduce it.
 - (C) A test case that failed during execution.
 - (D) A list of all the automated tests.
 - (E) A log file generated by the server.
 - *Correct Answer: (B)*
-

9. What is "Functional Testing"?

- (A) Testing the software's performance, like its speed.
 - (B) Testing the software's security against attacks.
 - (C) Testing the software to ensure it "functions" and meets the specified requirements.
 - (D) Testing the internal code logic.
 - (E) Testing the usability and "feel" of the application.
 - *Correct Answer: (C)*
-

10. What is "Non-Functional Testing"?

- (A) Testing the software's core features, like a login button.
 - (B) Testing aspects *other* than features, such as performance, security, or usability.
 - (C) Testing a feature that is "non-functional" or broken.
 - (D) A type of test that does not require a human.
 - (E) A test that has not been written yet.
 - *Correct Answer: (B)*
-

11. What is "Regression Testing"?

- (A) Testing a new feature for the first time.
 - (B) Testing to ensure that new code changes have not broken existing features.
 - (C) A test to see if the software "regresses" or gets worse over time.
 - (D) A type of negative test case.
 - (E) Testing the software's performance.
 - *Correct Answer: (B)*
-

12. What does "SDLC" stand for?

- (A) Software Development Life Cycle
 - (B) Software Design and Logic Cycle
 - (C) Secure Development Logic Check
 - (D) System Deployment Life Cycle
 - (E) Software Defect Life Cycle
 - *Correct Answer: (A)*
-

13. What is a "Positive" test case?

- (A) A test that is expected to fail.
 - (B) A test that verifies the application works as expected with *valid* data.
 - (C) A test that checks the application's response to *invalid* data.
 - (D) A test that a tester is "positive" will pass.
 - (E) A test for a new feature.
 - *Correct Answer: (B)*
-

14. What is a "Negative" test case?

- (A) A test that is expected to pass.
 - (B) A test that verifies the application handles *invalid* data or errors correctly.
 - (C) A test that a tester is "negative" about.
 - (D) A test that has failed and produced a bug.
 - (E) A test of an old, deprecated feature.
 - *Correct Answer: (B)*
-

15. What is the "Severity" of a bug?

- (A) It defines how *soon* the bug needs to be fixed.
- (B) It defines the *impact* of the bug on the application's functionality.

- (C) It defines how *difficult* the bug will be to fix.
 - (D) It defines the *developer* who is assigned to fix the bug.
 - (E) It defines the *test case* that found the bug.
 - *Correct Answer: (B)*
-

16. What is the "Priority" of a bug?

- (A) It defines the *impact* of the bug on the application's functionality.
 - (B) It defines the *order* or "urgency" in which the bug should be fixed.
 - (C) It defines how *difficult* the bug is to fix.
 - (D) It defines the *component* where the bug is located.
 - (E) It defines the *version* of the software that has the bug.
 - *Correct Answer: (B)*
-

17. What is a "test environment"?

- (A) The specific hardware, software, and network setup used for testing.
 - (B) The "environment" or "mood" of the QA team.
 - (C) The document that outlines the test plan.
 - (D) The server that runs the *live* production application.
 - (E) The software used to write automation scripts.
 - *Correct Answer: (A)*
-

18. What is "Exploratory Testing"?

- (A) A type of testing where test cases are designed and executed simultaneously without a predefined script.
 - (B) A type of testing that "explores" the internal code.
 - (C) An automated test that "explores" all possible user paths.
 - (D) A test that is "exploring" a new, unreleased feature.
 - (E) A performance test that "explores" the limits of the system.
 - *Correct Answer: (A)*
-

19. What is "Acceptance Testing"?

- (A) Testing performed by the developer to "accept" the feature.
- (B) Testing performed by the end-user or client to "accept" the product.
- (C) Testing to see if the application "accepts" user input.

- (D) The very first test that is run in a project.
 - (E) A test of the "Terms and Conditions" acceptance box.
 - *Correct Answer: (B)*
-

20. What does "STLC" stand for?

- (A) Software Test Life Cycle
 - (B) System Test Logic Cycle
 - (C) Secure Test Life Cycle
 - (D) Software Technology Life Cycle
 - (E) System Technology Logic Cycle
 - *Correct Answer: (A)*
-

Level 2: Intermediate Application

1. A bug is "High Priority" and "Low Severity". What is the *most likely* scenario?

- (A) The app crashes when a user clicks a button, but only 1% of users see this button.
 - (B) The company logo on the homepage is the wrong color.
 - (C) The app crashes when the user clicks the "Buy Now" button, blocking all sales.
 - (D) A typo exists on the "Terms and Conditions" page.
 - (E) The login button does not work for any user.
 - *Correct Answer: (B)*
-

2. A bug is "Low Priority" and "High Severity". What is the *most likely* scenario?

- (A) The app crashes when a user performs a rare, complex sequence of 15 actions.
 - (B) The company logo on the homepage is the wrong color.
 - (C) The app crashes when the user clicks the "Buy Now" button.
 - (D) A typo exists on the login page.
 - (E) The login button does not work for any user.
 - *Correct Answer: (A)*
-

3. What is "Integration Testing"?

- (A) Testing individual functions or modules in isolation.
- (B) Testing how two or more different modules interact with each other.

- (C) Testing the entire application as a whole.
 - (D) Testing the application's performance.
 - (E) Testing the application from a user's perspective.
 - *Correct Answer: (B)*
-

4. What is "Unit Testing"?

- (A) Testing a small, isolated piece of code (a "unit," like a function) in isolation.
 - (B) Testing how two or more modules interact.
 - (C) Testing the entire application as a whole.
 - (D) Testing performed by the QA team.
 - (E) Testing the "unity" of the user interface.
 - *Correct Answer: (A)*
-

5. What is "System Testing"?

- (A) Testing individual functions in isolation.
 - (B) Testing how two or more modules interact.
 - (C) Testing the complete, integrated application to verify it meets all requirements.
 - (D) Testing the "system" (e.g., the operating system).
 - (E) Testing only the backend "system."
 - *Correct Answer: (C)*
-

6. What is a "Smoke Test"?

- (A) A deep, comprehensive test of every single feature.
 - (B) A quick, high-level test to ensure the *most critical* features are working after a new build.
 - (C) A test to see if the server can handle so much "smoke" (traffic) that it breaks.
 - (D) A test that is "all smoke and mirrors" and has no real value.
 - (E) A security test for firewalls.
 - *Correct Answer: (B)*
-

7. What is a "Sanity Test"?

- (A) A test of the *entire* application's functionality.
- (B) A test to see if the tester is "sane".

- (C) A test to ensure that a *specific bug fix* or *small change* has worked and not broken anything *related*.
 - (D) A test to ensure the "most critical" features are working (this is a smoke test).
 - (E) A performance test.
 - *Correct Answer: (C)*
-

8. What are the *minimum* components of a good bug report?

- (A) A title, a description, steps to reproduce, expected results, and actual results.
 - (B) A title, the name of the tester, and the date.
 - (C) A screenshot, the priority, and the developer's name.
 - (D) The expected result, the actual result, and a log file.
 - (E) A video of the bug, the severity, and the test case ID.
 - *Correct Answer: (A)*
-

9. What is "Use Case"?

- (A) A document that describes a user's interaction with the system to achieve a specific goal.
 - (B) A "case" or "container" for storing test data.
 - (C) A specific test case for a user.
 - (D) A bug report from a user.
 - (E) A document describing the internal logic of the code.
 - *Correct Answer: (A)*
-

10. What is "Acceptance Criteria"?

- (A) The conditions that a software product must meet to be *accepted* by a user, customer, or stakeholder.
 - (B) The "criteria" for a user to "accept" the terms of service.
 - (C) A list of all the bugs that are "acceptable" to have in a release.
 - (D) The "criteria" for passing a "unit test."
 - (E) The "criteria" for hiring a QA tester.
 - *Correct Answer: (A)*
-

11. What is the *main* purpose of test automation?

- (A) To find *all* bugs in the software.

- (B) To *replace* all manual testers.
 - (C) To run *repetitive* tests (like regression) quickly and reliably.
 - (D) To test the "usability" and "feel" of the application.
 - (E) To write the test plan.
 - *Correct Answer: (C)*
-

12. What is a "locator" in test automation?

- (A) A "strategy" (e.g., ID, XPath) used to "locate" and "find" a "UI element" on a "page".
 - (B) A "GPS" "locator" for "testing" "map" "applications".
 - (C) A "tool" for "locating" "bugs" "in" "code".
 - (D) A "file" "that" "stores" "the" "location" "of" "the" "test" "environment".
 - (E) A "variable" "that" "stores" "a" "memory" "location".
 - *Correct Answer: (A)*
-

13. What is an "ID" locator?

- (A) A "locator" "that" "finds" "an" "element" "by" "its" "text".
 - (B) A "locator" "that" "finds" "an" "element" "by" "its" "unique" **id** "attribute".
 - (C) A "locator" "that" "finds" "an" "element" "by" "its" "HTML" "tag" (e.g., **button**).
 - (D) A "locator" "that" "finds" "an" "element" "by" "its" "position" "on" "the" "page".
 - (E) A "locator" "that" "finds" "an" "element" "by" "its" "CSS" "class".
 - *Correct Answer: (B)*
-

14. What is an "XPath" locator?

- (A) A "locator" "that" "uses" a "path-like" "syntax" "to" "navigate" the "HTML" "structure".
 - (B) A "locator" "that" "finds" "an" "element" "by" "its" "unique" **id** "attribute".
 - (C) A "locator" "that" "is" "always" "the" "fastest" "and" "most" "reliable".
 - (D) A "locator" "that" "only" "works" "on" "XML" "files".
 - (E) A "locator" "that" "finds" "an" "element" "by" "its" "CSS" "class".
 - *Correct Answer: (A)*
-

15. What is the "Test Life Cycle" (STLC)?

- (A) The "process" of a "bug" "from" "being" "found" "to" "being" "fixed".
- (B) The "process" of "developing" "software" "from" "start" "to" "finish" (SDLC).

- (C) A "sequence" of "activities" (e.g., "planning", "design", "execution") "conducted" "during" "testing".
 - (D) The "life" "of" "a" "test" "server".
 - (E) The "amount" of "time" "a" "test" "is" "allowed" "to" "run".
 - *Correct Answer: (C)*
-

16. What is a "test suite"?

- (A) A "collection" of "test cases" "grouped" "together" (e.g., "for" "regression").
 - (B) A "single" "test case".
 - (C) A "document" "describing" the "test" "strategy".
 - (D) A "software" "tool" "used" "for" "testing".
 - (E) A "report" "of" "all" "failed" "tests".
 - *Correct Answer: (A)*
-

17. What is **git clone**?

- (A) A "command" to "create" a "new" "branch".
 - (B) A "command" to "create" a "local" "copy" of a "remote" "repository".
 - (C) A "command" to "upload" "your" "changes" "to" "the" "remote".
 - (D) A "command" to "merge" "two" "branches".
 - (E) A "command" to "delete" a "repository".
 - *Correct Answer: (B)*
-

18. What is **git pull**?

- (A) A "command" to "upload" "your" "local" "commits" "to" "the" "remote".
 - (B) A "command" to "fetch" "and" "merge" "changes" "from" "the" "remote" "repository".
 - (C) A "command" to "save" "your" "changes" "locally".
 - (D) A "command" to "switch" "branches".
 - (E) A "command" to "undo" "your" "last" "commit".
 - *Correct Answer: (B)*
-

19. What is **git push**?

- (A) A "command" to "fetch" "and" "merge" "changes" "from" "the" "remote".
- (B) A "command" to "upload" "your" "local" "commits" "to" "the" "remote" "repository".
- (C) A "command" to "save" "your" "changes" "locally".

- (D) A "command" to "switch" "branches".
 - (E) A "command" to "delete" "a" "remote" "branch".
 - *Correct Answer: (B)*
-

20. What is "Usability Testing"?

- (A) A "test" to "check" "if" "the" "code" "is" "reusable".
 - (B) A "test" "to" "evaluate" "how" "easy" "and" "intuitive" "a" "product" "is" "to" "use".
 - (C) A "test" "to" "check" "the" "database" "connections".
 - (D) A "test" "to" "check" "the" "server's" "performance".
 - (E) A "test" "to" "check" "the" "security" of "the" "login".
 - *Correct Answer: (B)*
-

Level 3: Advanced Scenarios & Tools

1. What is "Selenium"?

- (A) A "tool" "for" "testing" "backend" "APIs".
 - (B) A "browser" "automation" "library" "used" "for" "testing" "web" "applications".
 - (C) A "project" "management" "tool" "for" "QA" "teams".
 - (D) A "database" "for" "storing" "test" "results".
 - (E) A "programming" "language" "for" "writing" "tests".
 - *Correct Answer: (B)*
-

2. Which locator strategy is the *most* stable and recommended?

- (A) XPath (Absolute)
 - (B) CSS Selector
 - (C) ID
 - (D) Link Text
 - (E) Tag Name
 - *Correct Answer: (C)*
-

3. Which locator strategy is the *least* stable and most likely to break?

- (A) ID

- (B) Name
 - (C) CSS Selector
 - (D) Relative XPath
 - (E) Absolute XPath (e.g., `/html/body/div[1]/...`)
 - *Correct Answer: (E)*
-

4. What is "Equivalence Partitioning"?

- (A) A "test" "technique" "that" "divides" "input" "data" "into" "equivalent" "groups" (e.g., "valid" "and" "invalid").
 - (B) A "test" "technique" "that" "focuses" "only" "on" "the" "boundaries" "of" "input" "data".
 - (C) A "database" "technique" "for" "partitioning" "tables".
 - (D) A "performance" "test" "technique".
 - (E) A "way" "to" "make" "all" "tests" "have" "equal" "priority".
 - *Correct Answer: (A)*
-

5. What is "Boundary Value Analysis"?

- (A) A "test" "technique" "that" "focuses" "on" "the" "boundaries" (e.g., "min", "max") "of" "an" "input" "range".
 - (B) A "test" "technique" "that" "divides" "data" "into" "equivalent" "groups".
 - (C) A "security" "test" "for" "the" "network" "boundary".
 - (D) A "UI" "test" "for" "the" "border" "of" "an" "element".
 - (E) A "test" "to" "see" "if" "a" "value" "is" "within" "the" "database" "limits".
 - *Correct Answer: (A)*
-

6. You are testing a field that accepts ages 18-65. Using "Boundary Value Analysis", which values are *most* important to test?

- (A) 18, 65, 30
 - (B) 17, 18, 19, 64, 65, 66
 - (C) 0, 100, 200
 - (D) "eighteen", "sixty-five"
 - (E) 1, 2, 3, 4, 5
 - *Correct Answer: (B)*
-

7. What is an "API"?

- (A) "Application Performance Interface"
 - (B) "Advanced Programming Interface"
 - (C) "Application Programming Interface"
 - (D) "Automated Performance Integration"
 - (E) "Application Process Integration"
 - *Correct Answer: (C)*
-

8. Why is "API testing" often preferred over "UI testing"?

- (A) It is "slower" and "more" "thorough".
 - (B) It is "faster", "more" "stable", and "can" "be" "run" "earlier" "in" "the" "cycle".
 - (C) It "is" "the" "only" "way" "to" "test" "the" "database".
 - (D) It "is" "easier" "for" "manual" "testers" "to" "perform".
 - (E) It "replaces" "the" "need" "for" "unit" "testing".
 - *Correct Answer: (B)*
-

9. What is "Postman"?

- (A) A "tool" "for" "automating" "web" "browser" "tests".
 - (B) A "CI/CD" "server" "for" "running" "pipelines".
 - (C) A "popular" "tool" "for" "designing", "building", and "testing" "APIs".
 - (D) A "project" "management" "tool".
 - (E) A "performance" "testing" "tool".
 - *Correct Answer: (C)*
-

10. What does an HTTP "200 OK" status code mean?

- (A) The "request" "was" "successful".
 - (B) The "resource" "was" "not" "found".
 - (C) The "server" "had" "an" "error".
 - (D) The "request" "was" "invalid".
 - (E) The "user" "is" "not" "authorized".
 - *Correct Answer: (A)*
-

11. What does an HTTP "404 Not Found" status code mean?

- (A) The "request" "was" "successful".
- (B) The "server" "could" "not" "find" "the" "requested" "resource".

- (C) The "server" "had" "an" "internal" "error".
 - (D) The "user" "is" "not" "authorized" "to" "see" "the" "resource".
 - (E) The "request" "was" "successful", "and" "a" "new" "resource" "was" "created".
 - *Correct Answer: (B)*
-

12. What does an HTTP "500 Internal Server Error" status code mean?

- (A) The "request" "was" "successful".
 - (B) The "resource" "was" "not" "found".
 - (C) The "server" "encountered" "an" "unexpected" "error" "while" "processing" "the" "request".
 - (D) The "user's" "request" "was" "invalid".
 - (E) The "user" "is" "not" "authorized".
 - *Correct Answer: (C)*
-

13. What is "JSON"?

- (A) A "programming" "language" "for" "automation".
 - (B) A "database" "for" "storing" "test" "data".
 - (C) A "lightweight" "data-interchange" "format", "common" "in" "APIs".
 - (D) A "testing" "framework" "for" "Java".
 - (E) A "security" "protocol".
 - *Correct Answer: (C)*
-

14. What is "Continuous Integration" (CI)?

- (A) The "practice" of "automating" the "deployment" "of" "code" "to" "production".
 - (B) The "practice" of "merging" "code" "changes" "from" "all" "developers" "into" "a" "central" "repository" "and" "running" "automated" "builds" "and" "tests".
 - (C) The "practice" of "manually" "testing" "all" "new" "features".
 - (D) The "practice" of "integrating" "the" "QA" "team" "with" "the" "dev" "team".
 - (E) The "practice" of "continuously" "designing" "new" "features".
 - *Correct Answer: (B)*
-

15. What is "Continuous Delivery" (CD)?

- (A) The "practice" of "automating" the "build" "and" "test" "process" (this "is" "CI").

- (B) The "practice" of "automating" the "release" "of" "a" "build" "to" "a" "production-like" "environment".
 - (C) The "practice" of "manually" "deploying" "code" "every" "day".
 - (D) The "practice" of "delivering" "new" "feature" "ideas" "to" "developers".
 - (E) The "practice" of "automating" "the" "entire" "process" "up" "to" "production" (this "is" "Continuous" "Deployment").
 - *Correct Answer: (B)*
-

16. What is "Jira"?

- (A) A "popular" "tool" "for" "bug" "tracking", "issue" "tracking", and "project" "management".
 - (B) A "popular" "tool" "for" "automating" "web" "browser" "tests".
 - (C) A "popular" "tool" "for" "testing" "APIs".
 - (D) A "popular" "CI/CD" "server".
 - (E) A "popular" "code" "editor".
 - *Correct Answer: (A)*
-

17. What is "Cypress"?

- (A) An "all-in-one" "automation" "framework" "for" "web" "testing" "that" "runs" "directly" "in" "the" "browser".
 - (B) A "library" "for" "testing" "backend" "APIs".
 - (C) A "performance" "testing" "tool".
 - (D) A "Java-based" "automation" "library" "that" "uses" "WebDriver".
 - (E) A "project" "management" "tool".
 - *Correct Answer: (A)*
-

18. What is "Agile" methodology?

- (A) A "project" "management" "approach" "with" "long", "sequential" "phases" (this "is" "Waterfall").
 - (B) A "project" "management" "approach" "that" "is" "iterative", "flexible", and "values" "collaboration".
 - (C) A "programming" "language" "known" "for" "its" "speed".
 - (D) A "testing" "framework" "for" "automation".
 - (E) A "CI/CD" "pipeline" "tool".
 - *Correct Answer: (B)*
-

19. What is a "Sprint" in Agile/Scrum?

- (A) A "short", "time-boxed" "period" (e.g., "2" "weeks") "during" "which" "a" "team" "works" "to" "complete" "a" "set" "amount" of "work".
 - (B) A "meeting" "to" "plan" "the" "entire" "project".
 - (C) The "final" "phase" "of" "testing" "before" "a" "release".
 - (D) A "fast" "automated" "test".
 - (E) A "report" "of" "all" "bugs" "found".
 - *Correct Answer: (A)*
-

20. What is "TDD" (Test-Driven Development)?

- (A) A "process" "where" "you" "write" "tests" *after* "the" "code" "is" "complete".
 - (B) A "process" "where" "you" "write" "a" "failing" "test" *before* "you" "write" "the" "production" "code".
 - (C) A "process" "where" "the" "QA" "team" "drives" "all" "development".
 - (D) A "process" "of" "testing" "only" "the" "database".
 - (E) A "process" "of" "testing" "only" "the" "UI".
 - *Correct Answer: (B)*
-

Level 4: Expert & Frameworks

1. What is the "Page Object Model" (POM)?

- (A) A "design" "pattern" "that" "creates" "an" "object" "for" "each" "API" "endpoint".
 - (B) A "design" "pattern" "in" "automation" "that" "creates" "an" "object" "class" "for" "each" "page" "of" "the" "application".
 - (C) A "way" "to" "model" "the" "project's" "management" "structure".
 - (D) A "built-in" "feature" "of" "Selenium".
 - (E) A "type" of "test" "report".
 - *Correct Answer: (B)*
-

2. What is the *primary benefit* of using the Page Object Model (POM)?

- (A) It "makes" "tests" "run" "faster".
- (B) It "replaces" "the" "need" "for" "a" "test" "runner" "like" "TestNG".
- (C) It "makes" "tests" "more" "maintainable" "by" "separating" "locators" "and" "logic" "from" "the" "tests".
- (D) It "finds" "bugs" "without" "having" "to" "run" "the" "tests".

- (E) It "automatically" "writes" "the" "test" "cases".
 - *Correct Answer: (C)*
-

3. Your test fails with a **StaleElementReferenceException**. What is the *most likely* cause?

- (A) The "element" "was" "found", "but" "the" "DOM" "changed" (e.g., "AJAX" "refresh") "before" "the" "test" "could" "use" "it".
 - (B) The "locator" (e.g., "XPath") "is" "incorrect" "and" "was" "never" "valid".
 - (C) The "test" "is" "running" "too" "fast" "and" "needs" "a" `time.sleep()`.
 - (D) The "browser" "driver" (e.g., "chromedriver") "is" "out" "of" "date".
 - (E) The "element" "is" "hidden" "behind" "another" "element".
 - *Correct Answer: (A)*
-

4. What is an "Explicit Wait" in Selenium?

- (A) A "wait" "that" "is" "set" "once" "and" "applies" "to" "all" `findElement` "commands" (this "is" "an" "Implicit" "Wait").
 - (B) A "wait" "that" "polls" "for" a "specific" "condition" (e.g., "element" "is" "clickable") "before" "proceeding".
 - (C) A "hard-coded" "pause", "like" `time.sleep()`.
 - (D) A "wait" "that" "is" "defined" "in" "the" "test" "plan" "document".
 - (E) A "wait" "that" "is" "part" of "the" "Cypress" "framework" "only".
 - *Correct Answer: (B)*
-

5. Why is an "Explicit Wait" *better* than a `time.sleep()`?

- (A) "It" "is" "not" "better"; `time.sleep()` "is" "more" "reliable".
 - (B) `time.sleep()` "waits" "for" "a" "fixed" "time", "making" "tests" "slow" "and" "flaky"; "an" "Explicit" "Wait" "proceeds" "as" "soon" "as" "the" "condition" "is" "met".
 - (C) "Explicit" "Waits" "are" "easier" "to" "write" "and" "shorter".
 - (D) `time.sleep()` "can" "crash" "the" "test" "server".
 - (E) "Explicit" "Waits" "can" "be" "used" "for" "API" "testing", "while" `time.sleep()` "cannot".
 - *Correct Answer: (B)*
-

6. What is the difference between an "Implicit Wait" and an "Explicit Wait"?

- (A) "Implicit" "waits" "for" "a" "condition"; "Explicit" "waits" "for" "an" "element" "to" "exist".
 - (B) "Implicit" "is" "a" "global" "setting"; "Explicit" "is" "a" "specific", "local" "setting" "for" "a" "condition".
 - (C) "Implicit" "is" "a" "hard-coded" "pause"; "Explicit" "is" "a" "dynamic" "pause".
 - (D) "Implicit" "is" "unstable"; "Explicit" "is" "stable".
 - (E) "They" "are" "two" "names" "for" "the" "same" "thing".
 - *Correct Answer: (B)*
-

7. How do you automate a test for a file *download*?

- (A) "It" "is" "not" "possible"; "Selenium" "cannot" "interact" "with" "the" "file" "system".
 - (B) "You" "must" "use" "a" "GUI" "automation" "tool" "like" "Autolt" "to" "click" "the" "browser's" "save" "dialog".
 - (C) "You" "configure" "the" "browser" "driver" "to" "auto-download" "to" "a" "known" "folder" "and" "then" "use" "code" "to" "verify" "the" "file" "exists".
 - (D) "You" "use" `driver.download()` "and" "pass" "in" "the" "file" "name".
 - (E) "You" "must" "manually" "download" "the" "file" "before" "running" "the" "test".
 - *Correct Answer: (C)*
-

8. What is "BDD" (Behavior-Driven Development)?

- (A) A "development" "process" "that" "focuses" "on" "the" "technical" "implementation" "first".
 - (B) A "development" "process" "that" "uses" "a" "common" "language" (Gherkin) "to" "describe" "an" "application's" "behavior".
 - (C) A "testing" "process" "that" "only" "focuses" "on" "bugs" ("Bug-Driven" "Development").
 - (D) A "design" "process" "for" "creating" "UI" "mockups".
 - (E) A "project" "management" "style".
 - *Correct Answer: (B)*
-

9. What is "Gherkin"?

- (A) A "programming" "language" "for" "automation".
- (B) A "structured" "language" (Given/When/Then) "used" "in" "BDD" "to" "write" "test" "scenarios".
- (C) A "tool" "for" "testing" "APIs".
- (D) A "type" of "database".
- (E) A "reporting" "tool".
- *Correct Answer: (B)*

10. What is "Cucumber"?

- (A) A "tool" "that" "reads" "Gherkin" "files" "and" "executes" "the" "automation" "code" "linked" "to" "them".
 - (B) A "tool" "for" "designing" "user" "interfaces".
 - (C) A "CI/CD" "server".
 - (D) A "performance" "testing" "tool".
 - (E) A "programming" "language".
 - *Correct Answer: (A)*
-

11. What is an "assertion" in a test script?

- (A) A "command" "that" "clicks" "a" "button".
 - (B) A "command" "that" "logs" "a" "message" "to" "the" "console".
 - (C) A "command" "that" "verifies" "a" "condition" "is" "true" (e.g., `assert.equals(actual, expected)`).
 - (D) A "comment" "in" "the" "code" "explaining" "what" "it" "does".
 - (E) A "setup" "step" "that" "runs" "before" "the" "test".
 - *Correct Answer: (C)*
-

12. What is "TestNG" or "JUnit"?

- (A) "Automation" "libraries" "for" "controlling" "a" "browser".
 - (B) "Test" "running" "frameworks" "that" "provide" "features" "like" "assertions", "setup/teardown", and "reporting".
 - (C) "Programming" "languages" "for" "testing".
 - (D) "BDD" "tools" "for" "writing" "Gherkin".
 - (E) "Project" "management" "tools".
 - *Correct Answer: (B)*
-

13. What is "data-driven testing"?

- (A) "The" "practice" "of" "running" "the" "same" "test" "multiple" "times" "with" "different" "sets" of "input" "data" (e.g., "from" "a" "spreadsheet").
- (B) "The" "practice" "of" "testing" "the" "database".
- (C) "The" "practice" "of" "driving" "a" "car" "using" "data".
- (D) "The" "practice" "of" "making" "decisions" "based" "on" "test" "data".
- (E) "The" "practice" of "testing" "only" "data-heavy" "applications".
- *Correct Answer: (A)*

14. How do you handle an **iFrame** in Selenium?

- (A) "You" "cannot" "test" "elements" "inside" "an" **iFrame**.
- (B) "You" "must" "use" "a" "special" **iFrame** "locator" "to" "find" "the" "element".
- (C) "You" "must" "use" **driver.switchTo().frame()** "before" "you" "can" "interact" "with" "elements" "inside" "it".
- (D) "You" "must" "use" "JavaScript" "to" "delete" "the" **iFrame** "first".
- (E) "You" "must" "use" "a" "different" "tool", "like" "Cypress".
- *Correct Answer: (C)*

15. How do you handle a browser "alert" in Selenium?

- (A) "You" "must" "use" **driver.switchTo().alert()** "to" "get" "control" "of" "the" "alert".
- (B) "You" "must" "find" "the" "alert's" "OK" "button" "using" "an" "XPath" "locator".
- (C) "You" "cannot" "handle" "alerts"; "they" "must" "be" "disabled".
- (D) "You" "must" "use" "a" "separate" "tool" "like" "Autolt".
- (E) "You" "must" "use" "a" **try/catch** "block" "to" "ignore" "it".
- *Correct Answer: (A)*

16. What is a key **difference** between Selenium and Cypress?

- (A) "Selenium" "is" "free"; "Cypress" "is" "expensive".
- (B) "Selenium" "runs" "outside" "the" "browser" (using "WebDriver"); "Cypress" "runs" *inside* "the" "browser" "event" "loop".
- (C) "Selenium" "is" "for" "Chrome"; "Cypress" "is" "for" "Firefox".
- (D) "Selenium" "is" "new"; "Cypress" "is" "old".
- (E) "Selenium" "is" "for" "API" "testing"; "Cypress" "is" "for" "UI" "testing".
- *Correct Answer: (B)*

17. What is "parameterization" in testing?

- (A) "The" "process" of "passing" "parameters" (data) "into" "a" "test" "function" "to" "run" "it" "with" "different" "values".
- (B) "The" "process" of "defining" "the" "test" "environment" "parameters".
- (C) "The" "process" of "measuring" "the" "performance" "parameters" "of" "an" "app".
- (D) "The" "process" of "using" "the" "Page" "Object" "Model".

- (E) "The" "process" of "storing" "test" "data" "in" "a" "JSON" "file".
 - *Correct Answer: (A)*
-

18. What is the "single-responsibility principle" as it applies to test automation?

- (A) "The" "principle" "that" "only" "one" "person" "is" "responsible" "for" "all" "automation".
 - (B) "The" "principle" "that" "a" "test" "case" "should" "only" "test" "one" "single" "thing".
 - (C) "The" "principle" "that" "a" "test" "framework" "should" "only" "have" "one" "class".
 - (D) "The" "principle" "that" "a" "test" "should" "only" "have" "one" "assertion".
 - (E) "The" "principle" "that" "a" "test" "should" "only" "be" "run" "by" "one" "user".
 - *Correct Answer: (B)*
-

19. What is "Selenium Grid"?

- (A) A "tool" "for" "running" "tests" "in" "parallel" "across" "multiple" "machines", "browsers", and "OSs".
 - (B) A "UI" "element" "in" "Selenium" "for" "testing" "tables" "and" "grids".
 - (C) A "reporting" "tool" "that" "shows" "a" "grid" "of" "results".
 - (D) A "version" of "Selenium" "that" "is" "run" "from" "the" "command" "line".
 - (E) A "security" "feature" "of" "Selenium".
 - *Correct Answer: (A)*
-

20. What is "Mobile Automation" (e.g., Appium)?

- (A) "The" "process" of "automating" "tests" "for" "native" "mobile" "applications".
 - (B) "The" "process" of "automating" "a" "website" "on" "a" "mobile" "browser".
 - (C) "The" "process" of "automating" "the" "phone" "call" "and" "SMS" "features".
 - (D) "A" "type" "of" "test" "that" "is" "run" "while" "moving".
 - (E) "A" "manual" "test" "performed" "on" "a" "mobile" "phone".
 - *Correct Answer: (A)*
-

Level 5: Strategic & Architectural

1. What is the "Testing Pyramid"?

- (A) An "architectural" "model" "that" "shows" "the" "hierarchy" "of" "tests" (many "Unit", "fewer" "Integration", "fewest" "UI").

- (B) A "physical" "pyramid-shaped" "server" "for" "testing".
 - (C) A "reporting" "structure" "for" "a" "QA" "team".
 - (D) A "design" "pattern" "for" "test" "automation".
 - (E) A "model" "that" "puts" "UI" "tests" "at" "the" "base" "as" "the" "most" "important".
 - *Correct Answer: (A)*
-

2. What is the "Ice Cream Cone" anti-pattern in testing?

- (A) A "model" "where" "you" "have" "many" "fast" "unit" "tests".
 - (B) A "model" "where" "you" "rely" "on" "many" "slow", "flaky" "UI" "tests" "and" "few" "unit" "tests".
 - (C) A "model" "for" "testing" "food-delivery" "applications".
 - (D) A "model" "where" "all" "tests" "are" "run" "in" "parallel".
 - (E) A "model" "where" "testing" "is" "only" "done" "at" "the" "end".
 - *Correct Answer: (B)*
-

3. What is "Shift-Left Testing"?

- (A) The "practice" of "testing" "only" "the" "left" "side" "of" "the" "application".
 - (B) The "practice" of "moving" "testing" "activities" *earlier* "in" "the" "development" "lifecycle".
 - (C) The "practice" of "shifting" "all" "testing" "to" "the" "developers".
 - (D) The "practice" of "postponing" "testing" "until" "after" "the" "release".
 - (E) A "keyboard" "shortcut" "for" "running" "tests".
 - *Correct Answer: (B)*
-

4. A UI test fails 1 out of 10 times. What is this "flaky" test a symptom of, and what is the *best* action?

- (A) A "flaky" "app"; "the" "test" "is" "fine", "and" "should" "be" "ignored".
 - (B) A "race" "condition" "or" "a" "missing" "wait"; "the" "test" "should" "be" "quarantined" "and" "fixed".
 - (C) A "flaky" "server"; "the" "test" "should" "be" "re-run" "until" "it" "passes".
 - (D) A "flaky" "developer"; "the" "bug" "should" "be" "assigned" "to" "them".
 - (E) A "flaky" "test" "runner"; "the" "tool" "should" "be" "replaced".
 - *Correct Answer: (B)*
-

5. What is the strategic role of a "QA Automation Engineer" in a CI/CD pipeline?

- (A) To "manually" "approve" "every" "deployment" "to" "production".
 - (B) To "write" "the" "YAML" "code" "for" "the" "pipeline" "from" "scratch".
 - (C) To "build" and "maintain" the "automated" "tests" "that" "act" "as" "a" "quality" "gate".
 - (D) To "only" "write" "unit" "tests" "for" "the" "application".
 - (E) "QA" "has" "no" "role" "in" "CI/CD"; "it" "is" "a" "DevOps" "process".
 - *Correct Answer: (C)*
-

6. What is "Performance Testing"?

- (A) "Testing" "to" "see" "how" "well" "a" "developer" "performs" "their" "tasks".
 - (B) "Testing" "to" "measure" "the" "speed", "responsiveness", and "stability" of "an" "application" "under" "load".
 - (C) "Testing" "to" "ensure" "the" "application" "functions" "correctly".
 - (D) "Testing" "to" "ensure" "the" "application" "is" "secure".
 - (E) "Testing" "to" "ensure" "the" "UI" "looks" "good".
 - *Correct Answer: (B)*
-

7. What is the difference between "Load Testing" and "Stress Testing"?

- (A) "Load" "is" "normal" "traffic"; "Stress" "is" "overwhelming" "traffic".
 - (B) "Load" "is" "overwhelming" "traffic"; "Stress" "is" "normal" "traffic".
 - (C) "Load" "is" "for" "APIs"; "Stress" "is" "for" "UI".
 - (D) "Load" "is" "automated"; "Stress" "is" "manual".
 - (E) "They" "are" "the" "same" "thing".
 - *Correct Answer: (A)*
-

8. What is "JMeter"?

- (A) A "popular" "open-source" "tool" "for" "performance" "and" "load" "testing".
 - (B) A "popular" "tool" "for" "UI" "automation".
 - (C) A "popular" "tool" "for" "unit" "testing".
 - (D) A "popular" "tool" "for" "security" "testing".
 - (E) A "popular" "tool" "for" "project" "management".
 - *Correct Answer: (A)*
-

9. What is "containerization" (e.g., Docker) and why is it important for QA?

- (A) "It" "is" "not" "important"; "it" "only" "benefits" "developers".

- (B) "It" "creates" "consistent", "isolated" "test" "environments", "eliminating" "the" "works" "on" "my" "machine" "problem".
 - (C) "It" "is" "a" "tool" "for" "testing" "physical" "shipping" "containers".
 - (D) "It" "is" "a" "security" "tool" "that" "containers" "viruses".
 - (E) "It" "is" "a" "storage" "tool" "for" "containing" "test" "data".
 - *Correct Answer: (B)*
-

10. What is a "quality gate" in a CI/CD pipeline?

- (A) A "manual" "approval" "step" "by" "a" "manager".
 - (B) An "automated" "checkpoint" (e.g., "all" "tests" "pass", "code" "coverage" ">80%") "that" "must" "be" "met" "to" "proceed".
 - (C) A "physical" "gate" "at" "the" "data" "center".
 - (D) A "firewall" "that" "blocks" "unapproved" "code".
 - (E) A "meeting" "to" "decide" "if" "the" "quality" "is" "good" "enough".
 - *Correct Answer: (B)*
-

11. How do you measure the "Return on Investment" (ROI) of test automation?

- (A) "By" "the" "number" "of" "tests" "you" "have" "automated".
 - (B) "By" "comparing" "the" "cost" "of" "automation" "to" "the" "cost" "of" "manual" "testing" "it" "replaced" "and" "bugs" "it" "found".
 - (C) "By" "how" "fast" "the" "automation" "suite" "runs".
 - (D) "You" "cannot" "measure" "the" "ROI" "of" "automation".
 - (E) "By" "the" "number" "of" "automation" "engineers" "on" "the" "team".
 - *Correct Answer: (B)*
-

12. What is "Visual Regression Testing"?

- (A) "Testing" "to" "ensure" "the" "application" "has" "not" "regressed" "in" "functionality".
 - (B) "Testing" "that" "takes" "screenshots" "of" "the" "UI" "and" "compares" "them" "to" "a" "baseline" "to" "find" "visual" "bugs".
 - (C) "Testing" "that" "is" "done" "by" "visually" "inspecting" "the" "app".
 - (D) "Testing" "for" "people" "with" "visual" "impairments".
 - (E) "Testing" "that" "the" "app" "looks" "bad" ("regression").
 - *Correct Answer: (B)*
-

13. What is "Contract Testing"?

- (A) "Testing" "the" "legal" "contract" "of" "a" "software" "vendor".
 - (B) "A" "test" "that" "verifies" "the" "interaction" ("contract") "between" "two" "services" (e.g., "an" "API" "and" "its" "client").
 - (C) "A" "test" "that" "is" "outsourced" "to" "a" "contractor".
 - (D) "A" "test" "that" "is" "run" "by" "the" "development" "team".
 - (E) "A" "test" "for" "a" "smartphone" "contract".
 - *Correct Answer: (B)*
-

14. What is the role of an "SET" (Software Engineer in Test) vs. a "QA Analyst"?

- (A) "An" "SET" "only" "runs" "manual" "tests"; "a" "QA" "Analyst" "only" "writes" "automation".
 - (B) "They" "are" "the" "same" "job" "with" "a" "different" "title".
 - (C) "An" "SET" "is" "a" "developer" "who" "focuses" "on" "building" "test" "frameworks" "and" "tools"; "a" "QA" "Analyst" "focuses" "on" "test" "design" "and" "execution".
 - (D) "An" "SET" "works" "on" "the" "dev" "team"; "a" "QA" "Analyst" "works" "on" "the" "operations" "team".
 - (E) "An" "SET" "tests" "code"; "a" "QA" "Analyst" "tests" "the" "UI".
 - *Correct Answer: (C)*
-

15. What is "test parallelization"?

- (A) "The" "practice" "of" "running" "multiple" "tests" "at" "the" "same" "time" (in "parallel") "to" "reduce" "execution" "time".
 - (B) "The" "practice" "of" "writing" "tests" "that" "are" "parallel" "to" "the" "development" "code".
 - (C) "A" "design" "pattern" "for" "automation" "frameworks".
 - (D) "A" "method" "for" "testing" "parallel" "lines" "in" "a" "UI".
 - (E) "The" "practice" "of" "running" "tests" "one" "after" "another" (this "is" "serial").
 - *Correct Answer: (A)*
-

16. How would you design a test automation framework from scratch? What are the key components?

- (A) "A" "single" "large" "script" "file" "that" "contains" "all" "tests" "and" "locators".
- (B) "A" "spreadsheet" "of" "manual" "test" "cases" "and" "a" "folder" "of" "screenshots".
- (C) "Layers" "for" "Pages" (POM), "Test" "Scripts", "Utilities" (e.g., "Waits"), "a" "Test" "Runner", and "Reporting".
- (D) "A" "copy" "of" "the" "Selenium" "library" "and" "a" "text" "file" "with" "ideas".
- (E) "A" "CI/CD" "pipeline", "a" "server", "and" "a" "database".

- *Correct Answer: (C)*
-

17. What is "OWASP Top 10"?

- (A) "A" "list" of "the" "top" "10" "automation" "tools".
 - (B) "A" "list" of "the" "top" "10" "programming" "languages" "for" "QA".
 - (C) "A" "standard" "awareness" "document" "for" "the" "10" "most" "critical" "web" "application" "security" "risks".
 - (D) "A" "list" of "the" "top" "10" "QA" "companies".
 - (E) "A" "list" of "the" "top" "10" "bugs" "found" "last" "year".
 - *Correct Answer: (C)*
-

18. What is the goal of "Accessibility Testing"?

- (A) "To" "test" "if" "the" "application" "is" "accessible" "from" "the" "internet".
 - (B) "To" "test" "if" "the" "database" "is" "accessible".
 - (C) "To" "test" "if" "the" "application" "is" "usable" "by" "people" "with" "disabilities" (e.g., "screen" "readers").
 - (D) "To" "test" "if" "the" "QA" "team" "can" "access" "the" "code".
 - (E) "To" "test" "if" "the" "login" "page" "is" "accessible".
 - *Correct Answer: (C)*
-

19. A developer says, "It works on my machine." What is the *most productive* response from a QA engineer?

- (A) "No, "it" "doesn't"; "you" "are" "wrong".
 - (B) "Let's" "compare" "your" "environment" (e.g., "code" "branch", "DB", "config") "with" "the" "test" "environment" "to" "find" "the" "difference".
 - (C) "OK", "I" "will" "close" "the" "bug" "ticket".
 - (D) "You" "must" "have" "broken" "the" "test" "environment".
 - (E) "It" "doesn't" "matter"; "it" "fails" "in" "QA", "so" "you" "have" "to" "fix" "it".
 - *Correct Answer: (B)*
-

20. What is "code coverage"?

- (A) "A" "metric" "that" "measures" "how" "much" "of" "the" "test" "code" "is" "covered" "by" "tests".

- (B) "A" "metric" "that" "measures" "how" "much" of "the" "production" "code" "is" "executed" "by" "automated" "tests".
- (C) "A" "metric" "that" "measures" "how" "many" "comments" ("coverage") "are" "in" "the" "code".
- (D) "A" "metric" "that" "measures" "how" "many" "bugs" "are" "covered" (found).
- (E) "A" "metric" "that" "measures" "how" "many" "features" "are" "covered" "by" "the" "test" "plan".
- *Correct Answer: (B)*

BUSINESS SECTOR